

Classification of finite-dimensional Nichols algebras of rank two in twisted Yetter–Drinfeld categories

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Abstract

Let G be a finite non-abelian group, let $\Phi \in Z^3(G, \mathbb{C}^\times)$ be normalized, and let V, W be finite-dimensional simple objects of ${}^G_G\mathcal{YD}^\Phi$. We classify, up to interchange, the braided-indecomposable pairs (V, W) whose supports generate G and for which $\mathcal{B}(V \oplus W)$ is finite-dimensional. The classification consists of eight cases with five possible support quandles. In every case the Cartan graph is standard of type A_2 , B_2 , or G_2 , and the dimension is determined explicitly. A new phenomenon occurs for Γ_2 : the twisted setting admits a family of type G_2 absent from the ordinary Γ_2 classification and we construct an explicit example over a non-abelian group of order 16.

1 Introduction

Nichols algebras first appeared in the study of bialgebras of type one [37]. If V is an object in a braided tensor category, its Nichols algebra $\mathcal{B}(V)$ is the connected graded braided Hopf algebra generated by V , with V as its space of primitive elements. Nichols algebras play a central role in the lifting method for pointed Hopf algebras and are closely related to quantum groups and root systems, see [5, 6, 24]. For braided vector spaces of diagonal type, Weyl groupoids and arithmetic root systems led to the classification of finite-dimensional Nichols algebras and to explicit presentations by generators and relations [18, 19, 7, 8, 1].

For ordinary Yetter–Drinfeld modules over a non-abelian group, a finite-dimensional simple object is determined by a finite conjugacy class and an irreducible representation of the centralizer of a chosen element. The conjugation action on the class is part of the braiding and places strong restrictions on finite-dimensional Nichols algebras, see [2, 3, 17, 16]. Reflections and root systems for semisimple Yetter–Drinfeld modules were developed in [4, 21, 27, 13]. Finite-dimensional Nichols algebras $\mathcal{B}(V \oplus W)$, where V and W are simple Yetter–Drinfeld modules, were studied in [20, 25, 22] and completely classified in [26]. The corresponding higher-rank cases with finite root systems were classified in [23].

Quasi-Hopf algebras were introduced by Drinfeld [15], and twisted quantum doubles of finite groups were constructed in [14]. In this paper, G is finite and non-abelian, and we work in the twisted Yetter–Drinfeld category ${}^G_G\mathcal{YD}^\Phi$. A simple object is determined by a conjugacy class and an irreducible projective representation of the corresponding centralizer. The multiplier of this representation is determined by Φ . Yetter–Drinfeld modules and bosonization in quasi-Hopf and coquasi-Hopf settings were studied in [10, 9, 12]. Related classifications of finite quasi-quantum groups over abelian groups appear in [31, 32, 28, 29, 30].

The associator changes the projective representations and the braided adjoint maps. Thus the ordinary classification cannot be applied directly. Reflection theory for Nichols algebras over coquasi-Hopf algebras was established in [35, 36]. The Cartan graphs, tensor decomposition, and finite-dimensionality criterion used below are proved in [34].

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For a tuple $M = (V, W)$, we say that M is braided-indecomposable if $c_{W,V}c_{V,W} \neq \text{id}_{V \otimes W}$. The main classification is the following. Theorem 8.1 gives an equivalent formulation in terms of reflections.

Theorem A. *Let G be a finite non-abelian group, let $\Phi \in Z^3(G, \mathbb{C}^\times)$ be normalized, and let $M = (V, W)$ be a tuple of finite-dimensional simple objects in ${}^G_G\mathcal{YD}^\Phi$. Assume that $G = \langle \text{supp } V \cup \text{supp } W \rangle$ and that M is braided-indecomposable. Then $\mathcal{B}(V \oplus W)$ is finite-dimensional if and only if, after possibly interchanging V and W , the tuple satisfies one of the alternatives in statement (3) of Theorem 8.1, equivalently one of the eight cases in Theorems 4.1, 5.1, 6.1, and 7.1. In these cases, the Cartan graph is standard of type A_2 , B_2 , or G_2 , and the dimension is given in Table 1.*

The proof proceeds in two stages. Suppose first that $\mathcal{B}(V \oplus W)$ is finite-dimensional. The finite-dimensionality criterion gives all reflections and a finite Cartan graph. After finitely many reflections and, if necessary, interchanging the entries, one reaches a pair whose Cartan matrix is of type A_2 , B_2 , or G_2 . The corresponding adjoint vanishings force its support quandle to be one of

$$Z_T^{4,1}, \quad Z_2^{2,2}, \quad Z_3^{3,1}, \quad Z_3^{3,2}, \quad Z_4^{4,2}.$$

Their enveloping groups show that G is an epimorphic image of $T, \Gamma_2, \Gamma_3, \Gamma_3$, or Γ_4 , respectively. Thus necessity reduces to the four case classifications in Sections 4–7. These calculations yield exactly eight alternatives, and reflection transfers the appropriate alternative back to the original pair. Conversely, each alternative is covered by the corresponding case theorem, which proves finite-dimensionality and identifies the Cartan graph. The tensor decomposition gives the dimension formulas in Table 1.

Table 1 lists the eight cases. The notation in the dimension formulas is that of the corresponding case sections. The symbols $\mathcal{P}_i^\Phi$ denote the four Γ_3 cases defined in (7.5)–(7.8). The integers $N_\bullet$ in the Γ_2 - A_2 and Γ_4 - B_2 rows are defined in the corresponding case sections.

Case	Quandle	Cartan type	Dimension
Γ_2 - A_2	$Z_2^{2,2}$	A_2	$64N_V N_{X_1} N_W$ (Corollary 4.33)
Γ_2 - G_2	$Z_2^{2,2}$	G_2	2^{21}
Γ_3 - $\mathcal{P}_1^\Phi$	$Z_3^{3,2}$	B_2	10368
Γ_3 - $\mathcal{P}_2^\Phi$	$Z_3^{3,2}$	B_2	2304
Γ_3 - $\mathcal{P}_3^\Phi$	$Z_3^{3,1}$	B_2	2304
Γ_3 - $\mathcal{P}_4^\Phi$	$Z_3^{3,1}$	B_2	10368
Γ_4 - B_2	$Z_4^{4,2}$	B_2	$2^{16} N_V N_{Y_2} \prod_{\gamma \in \{0,01,1\}} N_{W,\gamma} \prod_{\gamma \in \{0,01,1\}} N_{Y_1,\gamma}$ (Corollary 5.32)
T - G_2	$Z_T^{4,1}$	G_2	$6^3 72^3 = 80\,621\,568$

Table 1: The eight finite-dimensional cases in Theorem A.

The main new result is the Γ_2 - G_2 case. When $\Phi = 1$, finite-dimensional Nichols algebras in the corresponding Γ_2 case have type A_2 [20, Theorem 4.6]. Thus the Γ_2 - G_2 case does not occur in the corresponding Hopf algebra setting. However, in the twisted Yetter–Drinfeld module category, the following case occurs.

Theorem B. *Let G be a finite non-abelian quotient of Γ_2 , let $\Phi \in Z^3(G, \mathbb{C}^\times)$ be normalized, and let*

$$V = M(g, \rho), \quad W = M(h, \sigma).$$

The inducing projective representations ρ and σ are automatically one-dimensional by Lemma 4.5. Assume that the tuple (V, W) is braided-indecomposable. Then $\mathcal{B}(V \oplus W)$ is finite-dimensional and the

tuple has a standard Cartan graph of type G_2 if and only if, after possibly interchanging (g, V, ρ) and (h, W, σ) , the parameters computed for the resulting order satisfy

$$\Delta = 1, \quad \lambda' = -1, \quad \lambda^2 = \Theta_\Phi(g, h) = -1.$$

The parameters are defined in (4.1), (4.2), and (4.15). In this case

$$\dim \mathcal{B}(V \oplus W) = 2^{21}.$$

This case is realized by the explicit example over a non-abelian group of order 16 in Section 4.11.

The Γ_2 - A_2 case is obtained from the same braided adjoint calculations. In the Γ_3 cases, the obstruction to removing the associator is forced to vanish. After inflation and twist, the ordinary classification applies. The Γ_4 and T cases require separate calculations of the projective representations and braided adjoint objects. In every case, reflections determine the Cartan graph. We then identify the simple object corresponding to each positive root and prove that its Nichols algebra is finite-dimensional. The tensor decomposition of $\mathcal{B}(V \oplus W)$ proves finite-dimensionality and gives its dimension.

The paper is organized as follows. Section 2 reviews simple objects, Nichols algebras, braided adjoint actions, reflections, and Cartan graphs. Section 3 describes the five quandles and their enveloping groups. Sections 4–7 treat the Γ_2 , Γ_4 , T , and Γ_3 cases, respectively. Section 8 proves the complete classification theorem. The appendices contain the cocycle identities used in the proofs, together with detailed coefficient calculations and computations in the bar complex.

2 Preliminaries

Throughout the paper, the base field is $\mathbb{k} = \mathbb{C}$.

2.1 Twisted Yetter–Drinfeld modules and simple objects

This subsection fixes the conventions for twisted Yetter–Drinfeld modules used throughout the paper. We recall the action, tensor product, associativity constraint, and braiding, and then describe simple objects in terms of projective representations of centralizers.

Let G be a finite group and let Φ be a normalized 3-cocycle on G . We first recall Yetter–Drinfeld modules over the coquasi-Hopf algebra $(\mathbb{k}G, \Phi)$. Let V be a left $\mathbb{k}G$ -comodule. Denote its structure map by $\delta_L: V \rightarrow \mathbb{k}G \otimes V$, and set

$$V_g := {}^g V := \{v \in V \mid \delta_L(v) = g \otimes v\}, \quad \text{supp } V := \{g \in G \mid V_g \neq 0\}.$$

Thus $V = \bigoplus_{g \in G} {}^g V$. An element of ${}^g V$ is homogeneous of degree g . Thus $\deg v = g$ for $v \in {}^g V$. The left $\mathbb{k}G$ -comodule (V, δ_L) is a left-left Yetter–Drinfeld module over $(\mathbb{k}G, \Phi)$ if there is a linear map $\triangleright: \mathbb{k}G \otimes V \rightarrow V$ such that, for all $e, f \in G$ and $v \in V_g$,

$$\begin{aligned} e \triangleright (f \triangleright v) &= \frac{\Phi(e, f, g)\Phi(efgf^{-1}e^{-1}, e, f)}{\Phi(e, fgf^{-1}, f)}(ef) \triangleright v, \\ 1_G \triangleright v &= v, \\ e \triangleright v &\in V_{ege^{-1}}. \end{aligned} \tag{2.1}$$

The category of left-left Yetter–Drinfeld modules over $(\mathbb{k}G, \Phi)$ is denoted by ${}^{\mathbb{C}}\mathcal{YD}^\Phi$. It is braided monoidal. For $M, N \in {}^{\mathbb{C}}\mathcal{YD}^\Phi$, the coaction and action on $M \otimes N$ are given by

$$\begin{aligned} \delta_L(m_g \otimes n_h) &:= gh \otimes m_g \otimes n_h, \\ x \triangleright (m_g \otimes n_h) &:= \frac{\Phi(x, g, h)\Phi(xgx^{-1}, xhx^{-1}, x)}{\Phi(xgx^{-1}, x, h)}x \triangleright m_g \otimes x \triangleright n_h, \end{aligned} \tag{2.2}$$

for all $x, g, h \in G$ and $m_g \in {}^g M$, $n_h \in {}^h N$. The associativity constraint a and the braiding c of ${}_G \mathcal{YD}^\Phi$ are given respectively by

$$\begin{aligned} a((u_e \otimes v_f) \otimes w_g) &= \Phi(e, f, g)^{-1} u_e \otimes (v_f \otimes w_g), \\ c(u_e \otimes v_f) &:= e \triangleright v_f \otimes u_e, \end{aligned}$$

for all $e, f, g \in G$, $u_e \in {}^e U$, $v_f \in {}^f V$, $w_g \in {}^g W$ and $U, V, W \in {}_G \mathcal{YD}^\Phi$. These conventions are the specialization of [35, Definition 2.5 and equations (2.9), (2.12)–(2.13)] to $(\mathbb{k}G, \Phi)$. Set

$$\begin{aligned} \Phi_x(g, h) &:= \frac{\Phi(g, h, x) \Phi(ghxh^{-1}g^{-1}, g, h)}{\Phi(g, h x h^{-1}, h)} \\ \Phi^x(g, h) &:= \frac{\Phi(x, g, h) \Phi(xgx^{-1}, xhx^{-1}, x)}{\Phi(xgx^{-1}, x, h)}. \end{aligned}$$

These functions satisfy the following relation.

Lemma 2.1. *For all $x, y, z, w \in G$, one has*

$$\Phi_x(z, w) \Phi_x(y, zw) = \Phi_{wxw^{-1}}(y, z) \Phi_x(yz, w). \quad (2.3)$$

Proof. Write ${}^a b = aba^{-1}$, and let $v = {}^w x$, $t = {}^z v = {}^{zw} x$, and $u = {}^y t = {}^{yzw} x$. We use the normalized cocycle identity in the form

$$\Phi(b, c, d) \Phi(a, bc, d) \Phi(a, b, c) = \Phi(ab, c, d) \Phi(a, b, cd). \quad (2.4)$$

After substituting the definition of the local cocycles, the quotient of the left-hand side by the right-hand side in (2.3) is AB , where

$$\begin{aligned} A &= \frac{\Phi(z, w, x) \Phi(y, zw, x) \Phi(yz, v, w)}{\Phi(yz, w, x) \Phi(z, v, w) \Phi(y, z, v)}, \\ B &= \frac{\Phi(t, z, w) \Phi(u, y, zw) \Phi(y, t, z)}{\Phi(y, t, zw) \Phi(u, y, z) \Phi(u, yz, w)}. \end{aligned}$$

Applying (2.4) to (y, z, w, x) and (y, z, v, w) , and using $vw = wx$, gives

$$A = \frac{\Phi(y, zv, w)}{\Phi(y, z, w)}.$$

Applying the same identity to (y, t, z, w) and (u, y, z, w) , and using $uy = yt$ and $tz = zv$, gives

$$B = \frac{\Phi(y, z, w)}{\Phi(y, tz, w)} = \frac{\Phi(y, z, w)}{\Phi(y, zv, w)}.$$

Thus $AB = 1$, which proves (2.3). □

Corollary 2.2. *If $a, b, c \in C_G(x)$, then*

$$\Phi_x(b, c) \Phi_x(a, bc) = \Phi_x(a, b) \Phi_x(ab, c).$$

Thus the restriction of Φ_x to $C_G(x)$ is a normalized 2-cocycle.

Proof. If $c \in C_G(x)$, then $cxc^{-1} = x$, and the assertion follows from Lemma 2.1. Normalization follows directly from the definition of Φ_x and the normalization of Φ . □

We shall use the following identity for duals.

Lemma 2.3. For every $a \in G$, one has

$$\Phi(a, a^{-1}, a)\Phi(a^{-1}, a, a^{-1}) = 1. \quad (2.5)$$

Proof. Apply the normalized 3-cocycle identity to (a, a^{-1}, a, a^{-1}) and remove the factors containing the identity element. $\square$

Recall that a Φ_x -projective representation of $C_G(x)$ on a vector space U is a map $\rho : C_G(x) \rightarrow \text{GL}(U)$ satisfying $\rho(a)\rho(b) = \Phi_x(a, b)\rho(ab)$ for $a, b \in C_G(x)$. The simple objects in ${}^G_G\mathcal{YD}^\Phi$ are described as follows.

Proposition 2.4. Let V be a simple object of ${}^G_G\mathcal{YD}^\Phi$, and let $x \in \text{supp } V$. Then $\text{supp } V = x^G$, where $x^G = \{gxg^{-1} \mid g \in G\}$, and V_x is an irreducible Φ_x -projective representation of $C_G(x)$.

Conversely, for every $x \in G$, each irreducible Φ_x -projective representation ρ of $C_G(x)$ determines a simple object $M(x, \rho)$ with support x^G . The Φ_x -projective representation of $C_G(x)$ on $M(x, \rho)_x$ is isomorphic to ρ . After fixing one representative of each conjugacy class, the pairs (x, ρ) , with ρ taken up to isomorphism, parametrize the isomorphism classes of simple objects in ${}^G_G\mathcal{YD}^\Phi$.

Proof. Let V be a simple object of ${}^G_G\mathcal{YD}^\Phi$. Since the sum of the homogeneous components belonging to a fixed conjugacy class is a subobject, the support of V is a single conjugacy class $\mathcal{O}$. Fix $x \in \mathcal{O}$. The component V_x is stable under $C_G(x)$, and (2.1) gives $a \triangleright (b \triangleright v) = \Phi_x(a, b)(ab) \triangleright v$ for $a, b \in C_G(x)$ and $v \in V_x$. Thus V_x is a Φ_x -projective representation of $C_G(x)$.

This representation is irreducible. Indeed, suppose that $0 \neq U \subsetneq V_x$ is $C_G(x)$ -stable and set $N = \sum_{a \in G} a \triangleright U$. The relation (2.1) implies that N is G -stable. It is also graded and hence is a subobject of V . Moreover, only elements of $C_G(x)$ contribute to the component of degree x . Therefore $N_x = U$. Thus N is a nonzero proper subobject of V , contrary to the simplicity of V .

Conversely, let ρ be an irreducible Φ_x -projective representation of $C_G(x)$, with underlying space W . Choose elements $g_y \in G$, $y \in \mathcal{O} = x^G$, such that $g_y x g_y^{-1} = y$ and $g_x = 1$, and set

$$M(x, \rho) = \bigoplus_{y \in \mathcal{O}} M_y, \quad M_y = g_y \otimes W.$$

For $w \in W$, the comodule structure on M_y is $\delta_L(g_y \otimes w) = y \otimes (g_y \otimes w)$. The left G -action on M is defined by $g \triangleright (g_y \otimes w) = \gamma(g, g_y)(g_{y g_y^{-1}} \otimes \rho(c)w)$, where $g g_y$ decomposes uniquely as $g_{y g_y^{-1}} c$ with $c \in C_G(x)$, and

$$\gamma(g, g_y) = \frac{\Phi_x(g, g_y)}{\Phi_x(g_{y g_y^{-1}}, c)} = \frac{\Phi(g, g_y, x)\Phi(g_y g_y^{-1}, g, g_y)}{\Phi(g, y, g_y)\Phi(g_{y g_y^{-1}}, c, x)} \frac{\Phi(g_{y g_y^{-1}}, x, c)}{\Phi(g_y g_y^{-1}, g_{y g_y^{-1}}, c)}. \quad (2.6)$$

These formulas define an object of ${}^G_G\mathcal{YD}^\Phi$. For $a \in G$ and $y \in \mathcal{O}$, let $c_{a, y} \in C_G(x)$ be determined by $a g_y = g_{a y a^{-1}} c_{a, y}$. Thus the action defined above takes M_y into $M_{a y a^{-1}}$. Moreover, $c_{1, y} = 1$, and the normalization of Φ_x gives $\gamma(1, g_y) = 1$. Hence $1 \triangleright m = m$ for all $m \in M$.

For the projective associativity relation, observe that $c_{h g, y} = c_{h, g y g^{-1}} c_{g, y}$. Since ρ is a Φ_x -projective representation of $C_G(x)$, we have

$$\rho(c_{h, g y g^{-1}})\rho(c_{g, y}) = \Phi_x(c_{h, g y g^{-1}}, c_{g, y})\rho(c_{h g, y}).$$

Let $z = g y g^{-1}$, $u = h z h^{-1} = h g y g^{-1} h^{-1}$. Thus

$$g g_y = g_z c_{g, y}, \quad h g_z = g_u c_{h, z}, \quad c_{h g, y} = c_{h, z} c_{g, y}.$$

Applying (2.3), respectively, to (h, g, g_y) , $(h, g_z, c_{g, y})$, $(g_u, c_{h, z}, c_{g, y})$, gives

$$\Phi_x(g, g_y)\Phi_x(h, g g_y) = \Phi_y(h, g)\Phi_x(h g, g_y), \quad (2.7)$$

$$\Phi_x(g_z, c_{g, y})\Phi_x(h, g g_y) = \Phi_x(h, g_z)\Phi_x(h g_z, c_{g, y}), \quad (2.8)$$

$$\Phi_x(c_{h, z}, c_{g, y})\Phi_x(g_u, c_{h, z} c_{g, y}) = \Phi_x(g_u, c_{h, z})\Phi_x(h g_z, c_{g, y}). \quad (2.9)$$

Here $c_{g,y}, c_{h,z} \in C_G(x)$, while $g_z c_{g,y} = g g_y$ and $g_u c_{h,z} = h g_z$. Solving (2.8) and (2.9) for $\Phi_x(h, g g_y)$, and then substituting the result into (2.7), yields

$$\frac{\Phi_x(g, g_y)}{\Phi_x(g_z, c_{g,y})} \frac{\Phi_x(h, g_z)}{\Phi_x(g_u, c_{h,z})} \Phi_x(c_{h,z}, c_{g,y}) = \Phi_y(h, g) \frac{\Phi_x(hg, g_y)}{\Phi_x(g_u, c_{h,z} c_{g,y})}.$$

By the definition of γ and the identity $c_{hg,y} = c_{h,z} c_{g,y}$, this is precisely

$$\gamma(g, g_y) \gamma(h, g_z) \Phi_x(c_{h,z}, c_{g,y}) = \Phi_y(h, g) \gamma(hg, g_y).$$

Consequently, for $w \in W$,

$$\begin{aligned} h \triangleright (g \triangleright (g_y \otimes w)) &= \gamma(g, g_y) \gamma(h, g_{gy} g^{-1}) g_{hgy} g^{-1} h^{-1} \otimes \rho(c_{hgy} g^{-1}) \rho(c_{g,y}) w \\ &= \Phi_y(h, g) \gamma(hg, g_y) g_{hgy} g^{-1} h^{-1} \otimes \rho(c_{hg,y}) w \\ &= \Phi_y(h, g) (hg) \triangleright (g_y \otimes w). \end{aligned}$$

Thus the action satisfies all the defining conditions of a left-left Yetter–Drinfeld module over $(\mathbb{k}G, \Phi)$. By construction, $\text{supp } M(x, \rho) = x^G$. If $c \in C_G(x)$, then $g_x = 1$, and normalization gives $\gamma(c, 1) = 1$. Hence the action of $C_G(x)$ on $M(x, \rho)_x$ is ρ .

To prove that $M(x, \rho)$ is simple, let $0 \neq N \subseteq M(x, \rho)$ be a subobject and choose $y \in \mathcal{O}$ such that $N_y \neq 0$. Then the action of g_y^{-1} is invertible by (2.1). Hence $g_y^{-1} \triangleright N_y$ is a nonzero subspace of N_x . Since N_x is $C_G(x)$ -stable and ρ is irreducible, one has $N_x = W$. The transitivity of the G -action on $\mathcal{O}$ now gives $N = M(x, \rho)$. Finally, let ρ be the Φ_x -projective representation of $C_G(x)$ on V_x . For every $y \in \mathcal{O}$, the action of g_y is an isomorphism from V_x to V_y . Therefore the componentwise map

$$\theta : M(x, \rho) \longrightarrow V, \quad \theta(g_y \otimes w) = g_y \triangleright w,$$

is an isomorphism of graded vector spaces. If $ag_y = g_{aya^{-1}} c_{a,y}$, then

$$\begin{aligned} \theta(a \triangleright (g_y \otimes w)) &= \gamma(a, g_y) g_{aya^{-1}} \triangleright (c_{a,y} \triangleright w) \\ &= \Phi_x(a, g_y) (ag_y) \triangleright w = a \triangleright \theta(g_y \otimes w). \end{aligned}$$

Hence θ is an isomorphism in ${}^G_G \mathcal{YD}^\Phi$.

If x, x' are fixed representatives and $M(x, \rho) \simeq M(x', \rho')$, equality of their supports gives $x = x'$. Restricting the isomorphism to degree x then gives $\rho \simeq \rho'$. This proves the parametrization. □

2.2 Dual objects

We record the duality conventions needed later. In particular, we identify the projective representation defining the dual of a simple object when the original projective representation is one-dimensional.

Let U be a finite-dimensional object of ${}^G_G \mathcal{YD}^\Phi$. For a homogeneous basis $\{u_j\}$, with $u_j \in U_{x_j}$, let $\{u_j^\vee\}$ be the dual basis, so that $u_j^\vee \in (U^*)_{x_j^{-1}}$. We use the evaluation and coevaluation maps

$$\text{ev}_U : U^* \otimes U \longrightarrow \mathbb{C}, \quad \text{coev}_U : \mathbb{C} \longrightarrow U \otimes U^*, \quad \text{coev}_U(1) = \sum_j \Phi(x_j, x_j^{-1}, x_j) u_j \otimes u_j^\vee.$$

The two triangle identities follow from (2.5).

Lemma 2.5. Let $U = M(x, \chi)$, where χ is one-dimensional, and choose $0 \neq u \in U_x$. Let $u^\vee \in (U^*)_{x^{-1}}$ satisfy $u^\vee(u) = 1$. For $g \in C_G(x)$, write $g \triangleright u = \chi(g)u$. Then $U^* \simeq M(x^{-1}, \chi^*)$, where

$$\chi^*(g) = \frac{1}{\Phi^g(x^{-1}, x)\chi(g)}. \quad (2.10)$$

In particular, $x^{-1} \triangleright u^\vee = \chi(x)u^\vee$.

Proof. Since the evaluation is a morphism in ${}^G\mathcal{YD}^\Phi$, its equivariance gives

$$\begin{aligned} 1 &= \text{ev}_U(u^\vee \otimes u) = \text{ev}_U(g \triangleright (u^\vee \otimes u)) \\ &= \Phi^g(x^{-1}, x) \text{ev}_U(g \triangleright u^\vee \otimes g \triangleright u). \end{aligned}$$

The component $(U^*)_{x^{-1}}$ is one-dimensional. Thus $g \triangleright u^\vee = \chi^*(g)u^\vee$, and the preceding equality gives (2.10). Duality preserves simplicity and reverses homogeneous degrees. Hence U^* is simple with support $(x^{-1})^G$, and Proposition 2.4 identifies it with $M(x^{-1}, \chi^*)$.

At x^{-1} , the relation (2.1) on U_x gives $\chi(x^{-1})\chi(x) = \Phi_x(x^{-1}, x)$. Hence

$$\chi^*(x^{-1}) = \frac{\chi(x)}{\Phi^{x^{-1}}(x^{-1}, x)\Phi_x(x^{-1}, x)}.$$

By definition, the product in the denominator is $\Phi(x^{-1}, x, x^{-1})\Phi(x, x^{-1}, x) = 1$ by (2.5). Therefore $\chi^*(x^{-1}) = \chi(x)$. $\square$

2.3 Nichols algebras

We recall the categorical definition of a Nichols algebra and its behavior under braided strong monoidal equivalences. These facts allow Nichols algebras and their dimensions to be compared after changing the braided category.

Let $\mathcal{C}$ be a twisted Yetter–Drinfeld category over a group, and let $V \in \mathcal{C}$. The tensor algebra $T(V) = \bigoplus_{n \geq 0} V^{\otimes n}$ is a graded braided Hopf algebra. Its largest graded Hopf ideal contained in $\bigoplus_{n \geq 2} T^n(V)$ is denoted by $I(V)$, and

$$\mathcal{B}(V) = T(V)/I(V).$$

Equivalently, $\mathcal{B}(V)$ is generated by $\mathcal{B}(V)_1 = V$, has $\mathcal{B}(V)_0 = \mathbf{1}_{\mathcal{C}}$, and its primitive subobject is V , see [34, Subsection 2.5].

If $F : \mathcal{C} \rightarrow \mathcal{D}$ is a $\mathbb{k}$ -linear braided strong monoidal equivalence of braided monoidal abelian categories, then its tensor structure induces braided Hopf algebra isomorphisms

$$T(F(V)) \simeq F(T(V)), \quad \mathcal{B}(F(V)) \simeq F(\mathcal{B}(V)). \quad (2.11)$$

See [35, Lemma 2.16].

2.4 Braided decomposability

We characterize when the Nichols algebra of a direct sum is the braided tensor product of the Nichols algebras of its two summands. This criterion gives the notion of braided decomposability used throughout the classification.

Let $V, W \in {}^G\mathcal{YD}^\Phi$ and let $U = V \oplus W$. Denote by

$$j_V : \mathcal{B}(V) \rightarrow \mathcal{B}(U), \quad j_W : \mathcal{B}(W) \rightarrow \mathcal{B}(U)$$

the canonical injective braided Hopf algebra morphisms.

Lemma 2.6. *The following conditions are equivalent:*

- (1) *The double braiding satisfies $c_{W,V}c_{V,W} = \text{id}_{V \otimes W}$.*
- (2) *The canonical morphisms j_V and j_W induce an isomorphism of graded braided Hopf algebras*

$$m_{V,W} : \mathcal{B}(V) \underline{\otimes} \mathcal{B}(W) \xrightarrow{\simeq} \mathcal{B}(V \oplus W), \quad a \otimes b \mapsto j_V(a)j_W(b).$$

Here $\underline{\otimes}$ denotes the braided tensor product.

Proof. Suppose first that $c_{W,V}c_{V,W} = \text{id}_{V \otimes W}$. By the hexagon axioms, the corresponding mixed double braidings are trivial on all tensor powers of V and W , and hence descend to $\mathcal{B}(V)$ and $\mathcal{B}(W)$. Consequently, $A = \mathcal{B}(V) \underline{\otimes} \mathcal{B}(W)$ is a connected $\mathbb{N}_0^2$ -graded braided Hopf algebra generated by $A_{(1,0)} \oplus A_{(0,1)} = V \oplus W$.

We claim that the primitive subobject of A is $V \oplus W$. Since the coproduct preserves the $\mathbb{N}_0^2$ -grading, every bihomogeneous component of a primitive element is primitive. Let $0 \neq a \in A_{(p,q)}$. If $p, q > 0$, write $a = \sum_i u_i \otimes w_i$, where $u_i \in \mathcal{B}(V)_p$ and $w_i \in \mathcal{B}(W)_q$. The projection of the reduced coproduct of a onto $A_{(p,0)} \otimes A_{(0,q)}$ is

$$\sum_i (u_i \otimes 1) \otimes (1 \otimes w_i).$$

This is the image of a under the natural identification $A_{(p,q)} \simeq A_{(p,0)} \otimes A_{(0,q)}$, so it is nonzero. Thus a is not primitive.

If $q = 0$, then a lies in the braided Hopf subalgebra $\mathcal{B}(V) \otimes 1$, whose primitive subobject is V . Similarly, the primitive subobject of $1 \otimes \mathcal{B}(W)$ is W . Hence the primitive subobject of A is $V \oplus W$. Since A is connected, graded, and generated by $V \oplus W$, the characterization of Nichols algebras recalled above gives

$$A \simeq \mathcal{B}(V \oplus W).$$

This isomorphism restricts to the identity on $V \oplus W$. Therefore the isomorphism induced by j_V and j_W is $m_{V,W}$.

Conversely, suppose that $m_{V,W}$ is an isomorphism. For $v \in V$ and $w \in W$, multiplication in the braided tensor product gives $(1 \otimes w)(v \otimes 1) = c_{W,V}(w \otimes v)$. It follows that $E = w \otimes v - c_{W,V}(w \otimes v)$ belongs to the degree-two component of the Nichols ideal $I(U) \subseteq T(U)$. Since $I(U)_2 = \ker(\text{id}_{U \otimes U} + c_{U,U})$, we obtain

$$0 = (\text{id}_{U \otimes U} + c_{U,U})(E) = (\text{id}_{W \otimes V} - c_{V,W}c_{W,V})(w \otimes v).$$

Hence $c_{V,W}c_{W,V} = \text{id}_{W \otimes V}$, which is equivalent to $c_{W,V}c_{V,W} = \text{id}_{V \otimes W}$. $\square$

Definition 2.7. *We call (V, W) braided-decomposable if it satisfies the equivalent conditions of Lemma 2.6. Otherwise, it is braided-indecomposable.*

2.5 The braided adjoint action in Nichols algebras

We describe the iterated braided adjoint objects by recursive morphisms in the Nichols algebra. We then reduce their computation to the images of a small number of homogeneous tensors.

Let $\mathcal{C} = {}^G\mathcal{YD}^\Phi$, let $V, W \in \mathcal{C}$, and let $\mathcal{B} = \mathcal{B}(V \oplus W)$. Denote the canonical inclusions by $\iota_V : V \rightarrow \mathcal{B}$ and $\iota_W : W \rightarrow \mathcal{B}$, and denote by $\mu_{\mathcal{B}}$ the multiplication of $\mathcal{B}$. Since the elements of V are primitive, the restriction of the left braided adjoint action of $\mathcal{B}$ to $V \otimes \mathcal{B}$ is the morphism

$$\text{ad}_V = \mu_{\mathcal{B}} \circ (\iota_V \otimes \text{id}_{\mathcal{B}}) - \mu_{\mathcal{B}} \circ (\text{id}_{\mathcal{B}} \otimes \iota_V) \circ c_{V,\mathcal{B}}.$$

Set $D_0 = W$ and, recursively, $D_n = V \otimes D_{n-1}$ for $n \geq 1$. Define morphisms $\text{ad}^{(n)} : D_n \rightarrow \mathcal{B}$ by

$$\text{ad}^{(0)} = \iota_W, \quad \text{ad}^{(n)} = \text{ad}_V \circ (\text{id}_V \otimes \text{ad}^{(n-1)}), \quad n \geq 1.$$

All the maps involved are morphisms in ${}^G\mathcal{YD}^\Phi$, hence so is $\text{ad}^{(n)}$. The n -th adjoint object of W with respect to V is $(\text{ad } V)^n(W) := \text{Im}(\text{ad}^{(n)}) \subseteq \mathcal{B}(V \oplus W)$. Define morphisms $\varphi_n^\Phi \in \text{End}_{\mathcal{C}}(D_n)$ recursively. For $n = 1$, set $\varphi_1^\Phi = \text{id}_{V \otimes W} - c_{W,V}c_{V,W}$. For $n \geq 2$, we define the operator

$$c_{1,2}^\Phi = a_{V,V,D_{n-2}}(c_{V,V} \otimes \text{id}_{D_{n-2}})a_{V,V,D_{n-2}}^{-1},$$

and set

$$\varphi_n^\Phi = \text{id}_{V \otimes D_{n-1}} - c_{D_{n-1},V}c_{V,D_{n-1}} + (\text{id}_V \otimes \varphi_{n-1}^\Phi)c_{1,2}^\Phi.$$

Starting with $X_0^{V,W} = W$, we regard $V \otimes X_{n-1}^{V,W}$ as a subobject of D_n and define

$$X_n^{V,W} = \text{Im} \left(\varphi_n^\Phi \Big|_{V \otimes X_{n-1}^{V,W}} \right), \quad n \geq 1.$$

Lemma 2.8. *For every $n \in \mathbb{N}_0$, one has $X_n^{V,W} \cong (\text{ad } V)^n(W)$ in ${}^G\mathcal{YD}^\Phi$.*

Proof. By braided strictification, choose a strict braided monoidal category $\mathcal{C}^{\text{str}}$ and a braided monoidal equivalence

$$F : \mathcal{C} \longrightarrow \mathcal{C}^{\text{str}}.$$

See [33, Section 1]. The braided structure is transported along the equivalence as in [33, Example 2.4]. We transport the $\mathbb{k}$ -linear abelian structure and countable coproducts of $\mathcal{C}$ to $\mathcal{C}^{\text{str}}$. The tensor product in $\mathcal{C}^{\text{str}}$ then preserves these coproducts, and F is $\mathbb{k}$ -linear and exact. Let $\xi_{A,B} : F(A) \otimes F(B) \rightarrow F(A \otimes B)$ denote its tensor structure.

Set $E_0 = F(W)$ and $E_n = F(V) \otimes E_{n-1}$. Define the canonical isomorphisms $\Xi_n : E_n \rightarrow F(D_n)$ recursively by $\Xi_0 = \text{id}_{F(W)}$ and $\Xi_n = \xi_{V,D_{n-1}}(\text{id}_{F(V)} \otimes \Xi_{n-1})$. Let φ_n^{str} be the operator on E_n defined by the same recursion as φ_n^Φ in the strict category. These maps satisfy

$$F(\varphi_n^\Phi) \circ \Xi_n = \Xi_n \circ \varphi_n^{\text{str}} \quad (n \geq 1). \quad (2.12)$$

For $n = 1$, this follows from the compatibility of F with the braiding, applied twice. Suppose that (2.12) holds for $n - 1$, and write c^{str} for the braiding of $\mathcal{C}^{\text{str}}$. Naturality of ξ , braidedness of F , and strictness give

$$\begin{aligned} \Xi_n^{-1} F(c_{D_{n-1},V} \circ c_{V,D_{n-1}}) \Xi_n &= c_{E_{n-1},F(V)}^{\text{str}} \circ c_{F(V),E_{n-1}}^{\text{str}}, \\ \Xi_n^{-1} F(c_{1,2}^\Phi) \Xi_n &= c_{F(V),F(V)}^{\text{str}} \otimes \text{id}_{E_{n-2}}, \\ \Xi_n^{-1} F(\text{id}_V \otimes \varphi_{n-1}^\Phi) \Xi_n &= \text{id}_{F(V)} \otimes \varphi_{n-1}^{\text{str}}. \end{aligned}$$

Substituting the three identities into the recursive formula for φ_n^Φ proves (2.12).

Let $X_n^{\text{str}} \subseteq E_n$ be defined by the same recursion in $\mathcal{C}^{\text{str}}$. An induction shows that Ξ_n identifies X_n^{str} with $F(X_n^{V,W})$. The claim holds for $n = 0$ by the definitions. If it holds for $n - 1$, then the tensor structure identifies $F(V) \otimes X_{n-1}^{\text{str}}$ with $F(V \otimes X_{n-1}^{V,W})$, while (2.12) identifies the two restricted morphisms whose images define X_n^{str} and $F(X_n^{V,W})$. Exactness of F therefore identifies their images. Thus Ξ_n restricts to an isomorphism $X_n^{\text{str}} \xrightarrow{\sim} F(X_n^{V,W})$.

The proof of [24, Theorem 13.3.1] uses only the strict braided structure. Applied in $\mathcal{C}^{\text{str}}$, it gives $X_n^{\text{str}} \cong (\text{ad } F(V))^n(F(W))$. Moreover, the universal property of Nichols algebras yields a braided Hopf algebra isomorphism $F(\mathcal{B}_{\mathcal{C}}(V \oplus W)) \cong \mathcal{B}_{\mathcal{C}^{\text{str}}}(F(V) \oplus F(W))$. Since F is monoidal and braided, it carries the multiplication and braiding in each adjoint morphism to the corresponding maps in $\mathcal{C}^{\text{str}}$. Since an equivalence preserves images,

$$F(X_n^{V,W}) \cong X_n^{\text{str}} \cong (\text{ad } F(V))^n(F(W)) \cong F((\text{ad } V)^n(W)).$$

Since F is an equivalence, the assertion follows. $\square$

Since $X_1^{V,W} = \text{Im}(\text{id}_{V \otimes W} - c_{W,V}c_{V,W})$, Lemma 2.8 gives

$$(V, W) \text{ is braided-indecomposable} \iff (\text{ad } V)(W) \neq 0. \quad (2.13)$$

For a subgroup $H \leq G$ and a homogeneous vector u , write

$$\mathbb{k}H \triangleright u := \text{span}_{\mathbb{k}}\{a \triangleright u \mid a \in H\}.$$

The relation (2.1) implies that $\mathbb{k}G \triangleright u$ is G -stable. The following results reduce the computations in Sections 4–7 to one or a few homogeneous vectors.

Lemma 2.9. *Let $V, X \in {}^G_G\mathcal{YD}^\Phi$, and let $0 \neq x \in X_s$ be homogeneous with $X = \mathbb{k}G \triangleright x$. Suppose that $H \leq G$ and that $0 \neq v_j \in V_{r_j}$, $j \in J$, are homogeneous vectors satisfying $H \triangleright x \subseteq \mathbb{k}x$ and $V = \sum_{j \in J} \mathbb{k}H \triangleright v_j$. Then*

$$V \otimes X = \sum_{j \in J} \mathbb{k}G \triangleright (v_j \otimes x).$$

In particular, if $V = \mathbb{k}H \triangleright v$, then $V \otimes X = \mathbb{k}G \triangleright (v \otimes x)$.

Proof. Set $L = \sum_{j \in J} \mathbb{k}G \triangleright (v_j \otimes x)$. It follows from (2.1) that L is G -stable. For $f \in H$, write $f \triangleright x = \chi_f x$, where $\chi_f \in \mathbb{k}^\times$. The relation (2.2) gives $f \triangleright (v_j \otimes x) = \Phi^f(r_j, s)\chi_f(f \triangleright v_j) \otimes x$. Thus $(f \triangleright v_j) \otimes x \in L$, and consequently $V \otimes x \subseteq L$.

Now let $u \in V_t$ be homogeneous and $g \in G$. Since $(g^{-1} \triangleright u) \otimes x \in L$ and L is G -stable,

$$g \triangleright ((g^{-1} \triangleright u) \otimes x) = \Phi^g(g^{-1}t, s)\Phi_t(g, g^{-1})u \otimes (g \triangleright x) \in L.$$

The coefficient is nonzero, so $u \otimes (g \triangleright x) \in L$. Since homogeneous vectors span V and $X = \mathbb{k}G \triangleright x$, this proves $V \otimes X \subseteq L$. Each summand defining L lies in $V \otimes X$, which gives the reverse inclusion. $\square$

Write $X_n = X_n^{V,W}$. Regard $V \otimes X_{n-1}$ as a subobject of D_n and denote the restriction of φ_n^Φ to this subobject by ψ_n . Thus $X_n = \text{Im } \psi_n$.

Proposition 2.10. *Let $n \geq 1$, and choose homogeneous vectors $0 \neq v \in V_r$ and $0 \neq x_{n-1} \in (X_{n-1})_s$. Suppose that, for some subgroup $H \leq G$, $V = \mathbb{k}H \triangleright v$, $X_{n-1} = \mathbb{k}G \triangleright x_{n-1}$, and $H \triangleright x_{n-1} \subseteq \mathbb{k}x_{n-1}$. Set $x_n := \psi_n(v \otimes x_{n-1})$. Then $X_n = \mathbb{k}G \triangleright x_n$. Consequently, $X_n = 0$ if and only if $x_n = 0$. Moreover, if $x_n \neq 0$, then $x_n \in (X_n)_{rs}$.*

Proof. Lemma 2.9 gives $V \otimes X_{n-1} = \mathbb{k}G \triangleright (v \otimes x_{n-1})$. Since ψ_n is a morphism in ${}^G_G\mathcal{YD}^\Phi$, it follows that

$$X_n = \psi_n(V \otimes X_{n-1}) = \mathbb{k}G \triangleright \psi_n(v \otimes x_{n-1}) = \mathbb{k}G \triangleright x_n.$$

Hence, $X_n = 0$ if and only if $x_n = 0$. Finally, $v \otimes x_{n-1}$ has degree rs , and ψ_n , being a comodule morphism, preserves the G -grading. $\square$

2.6 Racks and enveloping groups

We recall the relation between conjugation quandles and groups. In particular, the support of a nonzero twisted Yetter–Drinfeld module is a quandle under conjugation.

Definition 2.11. *A rack is a nonempty set X with a binary operation $\blacktriangleright$ such that each map $f_a(x) = a \blacktriangleright x$ is bijective and*

$$a \blacktriangleright (b \blacktriangleright c) = (a \blacktriangleright b) \blacktriangleright (a \blacktriangleright c)$$

for all $a, b, c \in X$. A quandle is a rack satisfying $a \blacktriangleright a = a$ for every $a \in X$.

For racks X and Y , a map $f : X \rightarrow Y$ is a rack morphism if $f(a \blacktriangleright b) = f(a) \blacktriangleright f(b)$ for all $a, b \in X$. For a finite quandle $X = \{1, \dots, n\}$, the notation $X : f_1 \cdots f_n$ records its left translations. The conjugacy class of g in a group G , with its conjugation quandle structure, is denoted by g^G . Its inner automorphism group is $\text{Inn}(X) = \langle f_a \mid a \in X \rangle$. The quandle is indecomposable if this group acts transitively on X .

Definition 2.12. *The enveloping group of a quandle X is*

$$G_X = \langle x \in X \mid xyx^{-1} = x \blacktriangleright y \text{ for all } x, y \in X \rangle.$$

Any group G defines the conjugation quandle $\text{Conj}(G)$, with $a \blacktriangleright b = aba^{-1}$. This gives the conjugation functor from groups to quandles, and the enveloping-group construction is its left adjoint.

Lemma 2.13. *There is a natural bijection*

$$\text{Hom}_{\mathbf{Qnd}}(X, \text{Conj}(G)) \cong \text{Hom}_{\mathbf{Grp}}(G_X, G).$$

Proof. This is the universal property of the displayed presentation of G_X . $\square$

Lemma 2.14. *Let $0 \neq V \in {}^G\mathcal{YD}^\Phi$. Then $\text{supp } V$ is a quandle under $g \blacktriangleright h = ghg^{-1}$.*

Proof. The Yetter–Drinfeld compatibility makes $\text{supp } V$ a union of conjugacy classes. The quandle axioms are the corresponding identities for conjugation in G . $\square$

2.7 Cartan graphs of Nichols algebras

We recall the rank-two reflection theory used throughout the classification. The main results collected here are the tensor-decomposition criterion, the positive-root factorization, and the invariance of finite-dimensionality and dimension under reflections. We use the notation of [34, Sections 2.1 and 2.6]. All references to that paper are to arXiv version 2. All statements in this subsection remain valid after replacing (G, Φ) by any finite group and any normalized 3-cocycle.

Let $\mathbb{I} = \{1, 2\}$, and let $P = (P_1, P_2)$ be a tuple of finite-dimensional simple objects in ${}^G\mathcal{YD}^\Phi$. Write $\mathcal{B}(P) = \mathcal{B}(P_1 \oplus P_2)$, with its standard $\mathbb{N}_0^2$ -grading $\deg P_i = \alpha_i$, where α_1, α_2 are the standard basis vectors of $\mathbb{Z}^2$. For tuples $Q = (Q_1, Q_2)$ and $Q' = (Q'_1, Q'_2)$, write $Q \simeq Q'$ if $Q_i \simeq Q'_i$ for every $i \in \mathbb{I}$, and denote the componentwise isomorphism class of Q by $[Q]$.

Definition 2.15. *For $i \neq j$, the tuple P admits the i -th reflection if there exists an integer $m_{ij}^P \geq 0$ such that $(\text{ad } P_i)^{m_{ij}^P}(P_j)$ is nonzero and $(\text{ad } P_i)^{m_{ij}^P+1}(P_j) = 0$. In this case, define*

$$R_i(P)_j = \begin{cases} P_i^*, & \text{if } j = i, \\ (\text{ad } P_i)^{m_{ij}^P}(P_j), & \text{if } j \neq i. \end{cases}$$

If P admits both reflections, define $a_{ii}^{[P]} = 2$, $a_{ij}^{[P]} = -m_{ij}^P$.

Lemma 2.16. *Suppose that P admits the i -th reflection. Then $R_i(P)$ is a tuple of finite-dimensional simple objects, and its isomorphism class depends only on $[P]$. Moreover, $R_i(P)$ admits the i -th reflection,*

$$R_i^2(P) \simeq P, \quad m_{ij}^{R_i(P)} = m_{ij}^P \quad (j \neq i).$$

Consequently, whenever both Cartan matrices are defined, $A^{[P]}$ and $A^{[R_i(P)]}$ have the same i -th row.

Proof. The terminal adjoint object is simple by [36, Lemma 5.8]. Functoriality of the adjoint construction gives the assertion about isomorphism classes. The remaining assertions are [36, Lemma 5.9(2) and Corollary 5.15(1)]. $\square$

Moreover, [34, Remark 2.26] shows that $A^{[P]} = (a_{ij}^{[P]})_{i,j \in \mathbb{I}}$ is a generalized Cartan matrix whenever P admits both reflections.

Definition 2.17. The tuple P admits the empty sequence. For $l \geq 1$, it admits $(i_1, \dots, i_l)$ if it admits the i_1 -th reflection and $R_{i_1}(P)$ admits $(i_2, \dots, i_l)$. The tuple P admits all reflections if it admits every finite sequence in $\mathbb{I}$. For such a tuple, let

$$\mathcal{F}_2(P) = \{R_{i_l}(\cdots R_{i_1}(P) \cdots) \mid l \in \mathbb{N}_0, i_1, \dots, i_l \in \mathbb{I}\}.$$

Define

$$\mathcal{X}(P) = \{[Q] \mid Q \in \mathcal{F}_2(P)\}, \quad r_i([Q]) = [R_i(Q)],$$

and set

$$\mathcal{G}(P) = (\mathbb{I}, \mathcal{X}(P), (r_i)_{i \in \mathbb{I}}, (A^{[Q]})_{[Q] \in \mathcal{X}(P)}).$$

Proposition 2.18. If P admits all reflections, then $\mathcal{G}(P)$ is a connected Cartan graph.

Proof. This follows from [34, Remark 2.26 and Theorem 4.6]. $\square$

Definition 2.19. Assume that P admits all reflections. For $[Q] \in \mathcal{X}(P)$, define $s_i^{[Q]}(\alpha_j) = \alpha_j - a_{ij}^{[Q]}\alpha_i$. The Weyl groupoid $\mathcal{W}(\mathcal{G}(P))$ is generated by the morphisms $s_i^{[Q]} : [Q] \rightarrow r_i([Q])$. The real roots at $[Q]$ and the positive real roots are

$$\Delta^{[Q],\text{re}} = \{w(\alpha_i) \mid i \in \mathbb{I}, w : [Q'] \rightarrow [Q] \text{ for some } [Q'] \in \mathcal{X}(P)\}, \quad \Delta_+^{[Q],\text{re}} = \Delta^{[Q],\text{re}} \cap \mathbb{N}_0^2.$$

The Cartan graph is finite if $\Delta^{[Q],\text{re}}$ is finite for every $[Q] \in \mathcal{X}(P)$. It is standard if its Cartan matrix is the same at every object.

Definition 2.20. An $\mathbb{N}_0^2$ -graded object is tensor decomposable if, for some $l \geq 0$, it is isomorphic as a graded object to

$$\begin{cases} \mathcal{B}(Q_l) \otimes (\cdots (\mathcal{B}(Q_2) \otimes \mathcal{B}(Q_1))), & l \geq 1, \\ \mathbf{1}, & l = 0, \end{cases}$$

where the Q_k are finite-dimensional simple objects, each concentrated in a single degree in $\mathbb{N}_0^2 \setminus \{0\}$, and these degrees are pairwise distinct.

Theorem 2.21. Let $P = (P_1, P_2)$ be a tuple of finite-dimensional simple objects in ${}^{\mathbb{C}}\mathcal{YD}^{\Phi}$. Then $\mathcal{B}(P)$ is tensor decomposable if and only if P admits all reflections and $\mathcal{G}(P)$ is finite.

Proof. This is [34, Theorem 5.8]. $\square$

Proposition 2.22. Assume that P admits all reflections and that $\mathcal{G}(P)$ is finite. Fix $Q \in \mathcal{F}_2(P)$. Let w_0 be a longest morphism with target $[Q]$, choose a reduced expression $w_0 = \text{id}_{[Q]} s_{i_1} \cdots s_{i_l}$, and let

$$\beta_k = \text{id}_{[Q]} s_{i_1} \cdots s_{i_{k-1}}(\alpha_{i_k}), \quad 1 \leq k \leq l.$$

Then $\{\beta_1, \dots, \beta_l\} = \Delta_+^{[Q],\text{re}}$, and there are finite-dimensional simple objects $Q_{\beta_1}, \dots, Q_{\beta_l}$ such that

$$\mathcal{B}(Q_{\beta_l}) \otimes (\cdots (\mathcal{B}(Q_{\beta_2}) \otimes \mathcal{B}(Q_{\beta_1}))) \xrightarrow{\sim} \mathcal{B}(Q)$$

as $\mathbb{N}_0^2$ -graded objects.

Proof. This is [34, Corollary 5.9]. $\square$

Lemma 2.23. Under the hypotheses and notation of Proposition 2.22, let $S \in \mathcal{F}_2(P)$, $i \in \mathbb{I}$, and let $w : [S] \rightarrow [Q]$ be a morphism in $\mathcal{W}(\mathcal{G}(P))$. If $\beta_k = w(\alpha_i)$, then $Q_{\beta_k} \simeq S_i$. In particular, every Q_{β_k} is isomorphic to a component of a tuple in $\mathcal{F}_2(P)$, and every such component is isomorphic to one of the Q_{β_k} or to a dual of one of them.

Proof. The first assertion is [34, Lemma 5.10]. Every β_k is real, so it also proves the first consequence. For the converse, choose a morphism $w : [S] \rightarrow [Q]$, which exists because $\mathcal{G}(P)$ is connected. Since $\mathcal{G}(P)$ is a Cartan graph, either $w(\alpha_i)$ or $-w(\alpha_i)$ belongs to $\Delta_+^{[Q], \text{re}}$. In the first case, $w(\alpha_i) = \beta_k$ for some k , and the first assertion applies. In the second case, let $S' = R_i(S)$. Then $s_i^{[S']} : [S'] \rightarrow [S]$, $ws_i^{[S']}(\alpha_i) = -w(\alpha_i) \in \Delta_+^{[Q], \text{re}}$, and $S'_i = S_i^*$. The first case identifies S_i^* with a factor, so S_i is a dual of that factor. $\square$

Lemma 2.24. *Under the hypotheses and notation of Proposition 2.22, let $i \neq j$ and $0 \leq t \leq -a_{ij}^{[Q]}$. Then there exists $k \in \{1, \dots, l\}$ such that*

$$\beta_k = \alpha_j + t\alpha_i, \quad (\text{ad } Q_i)^t(Q_j) \simeq Q_{\beta_k}.$$

In particular, $(\text{ad } Q_i)^t(Q_j)$ is simple.

Proof. This is [34, Proposition 5.11]. $\square$

Theorem 2.25. *Let $P = (P_1, P_2)$ be a tuple of finite-dimensional simple objects in ${}^{\mathbb{G}}\mathcal{YD}^{\Phi}$. Then $\mathcal{B}(P)$ is finite-dimensional if and only if the following conditions hold:*

- (1) *P admits all reflections,*
- (2) *$\mathcal{G}(P)$ is finite and $\mathcal{B}(Q_i)$ is finite-dimensional for every $Q \in \mathcal{F}_2(P)$ and $i \in \mathbb{I}$.*

Proof. This is [34, Theorem 5.12]. $\square$

Lemma 2.26. *Let $P = (P_1, P_2)$ be a tuple of finite-dimensional simple objects in ${}^{\mathbb{G}}\mathcal{YD}^{\Phi}$. If P admits the i -th reflection, then*

$$\dim \mathcal{B}(R_i(P)) < \infty \iff \dim \mathcal{B}(P) < \infty, \quad \dim \mathcal{B}(R_i(P)) = \dim \mathcal{B}(P) \text{ in this case.}$$

Proof. By [35, Lemma 6.2], if $\mathcal{B}(P)$ is finite-dimensional, then so is $\mathcal{B}(R_i(P))$, and their dimensions are equal. Moreover, Lemma 2.16 shows that $R_i(P)$ admits the i -th reflection and that $R_i^2(P) \simeq P$. Applying the same implication to $R_i(P)$ gives the converse. $\square$

2.8 Finite Cartan graphs of rank two

We record two consequences used in the case-by-case classification. If a pair is braided-indecomposable and its Nichols algebra is finite-dimensional, it can be reflected to one whose Cartan matrix is of type A_2 , B_2 , or G_2 , up to transpose. Reflections preserve the subgroup generated by the two supports.

Let G be a finite non-abelian group and let Φ be a normalized 3-cocycle on G . Let V and W be finite-dimensional simple Yetter–Drinfeld modules in ${}^{\mathbb{G}}\mathcal{YD}^{\Phi}$, and let $M = (V, W)$. If $\mathcal{B}(V \oplus W)$ is finite-dimensional, then Theorem 2.25 shows that M admits all reflections and $\mathcal{G}(M)$ is a finite Cartan graph. It is connected by Proposition 2.18.

A connected Cartan graph is called indecomposable if one of its Cartan matrices is indecomposable. In rank two, this is equivalent to every Cartan matrix being indecomposable. Indeed, the i -th row is unchanged along an i -edge, and $a_{ij} = 0$ if and only if $a_{ji} = 0$. If, in addition, M is braided-indecomposable, then by (2.13), $m_{12}^M > 0$. Since $A^{[M]}$ is a generalized Cartan matrix, also $m_{21}^M > 0$. Thus $A^{[M]}$, and hence $\mathcal{G}(M)$, is indecomposable.

Lemma 2.27. *Let $\mathcal{G}$ be a connected indecomposable finite Cartan graph. Then there exists $X \in \mathcal{X}$ such that A^X is of finite type.*

Proof. This is [23, Theorem 4.2]. $\square$

Proposition 2.28. *Let $M = (V, W)$ be a tuple of finite-dimensional simple objects in ${}^{\mathbb{G}}\mathcal{YD}^{\Phi}$. Assume that M is braided-indecomposable, admits all reflections, and has a finite Cartan graph $\mathcal{G}(M)$. Then there exists $N = (N_1, N_2) \in \mathcal{F}_2(M)$ such that*

$$1 \leq a_{12}^{[N]} a_{21}^{[N]} \leq 3.$$

Proof. Since M is braided-indecomposable, $\mathcal{G}(M)$ is indecomposable. Lemma 2.27 therefore gives an object $X \in \mathcal{X}(M)$ for which A^X is of finite type. By the definition of $\mathcal{X}(M)$, write $X = [N]$ for some $N \in \mathcal{F}_2(M)$. Connectedness and indecomposability imply that $A^{[N]}$ is indecomposable. The indecomposable rank-two generalized Cartan matrices of finite type are those of types A_2 , B_2 , and G_2 , together with their transposes. Hence

$$a_{12}^{[N]} a_{21}^{[N]} \in \{1, 2, 3\},$$

as claimed. $\square$

For a tuple $P = (P_1, P_2)$ in ${}^{\mathbb{G}}\mathcal{YD}^{\Phi}$, set $G_P = \langle \text{supp } P_1 \cup \text{supp } P_2 \rangle$.

Lemma 2.29. *Let $P = (P_1, P_2)$ be a tuple of finite-dimensional simple objects in ${}^{\mathbb{G}}\mathcal{YD}^{\Phi}$. If P admits the i -th reflection, then*

$$G_{R_i(P)} = G_P.$$

If P admits all reflections, then consequently $G_Q = G_P$ for every $Q \in \mathcal{F}_2(P)$.

Proof. Write $R_i(P) = (P'_1, P'_2)$. The reflected i -th component is $P'_i = P_i^*$, and therefore $\text{supp } P'_i = (\text{supp } P_i)^{-1} \subseteq G_P$. Let $j \neq i$ and let $m = m_{ij}^P$. Then $P'_j = (\text{ad } P_i)^m(P_j)$ is the image of a grading-preserving morphism from $P_i^{\otimes m} \otimes P_j$. Hence

$$\text{supp } P'_j \subseteq (\text{supp } P_i)^m \text{supp } P_j \subseteq G_P.$$

It follows that $G_{R_i(P)} \subseteq G_P$.

The tuple $R_i(P)$ again admits the i -th reflection, and $R_i^2(P) \simeq P$ by Lemma 2.16. Applying the preceding inclusion to $R_i(P)$ gives $G_P = G_{R_i^2(P)} \subseteq G_{R_i(P)}$. Thus $G_{R_i(P)} = G_P$. The equality $G_Q = G_P$ for every $Q \in \mathcal{F}_2(P)$ follows by induction on the length of a reflection sequence. $\square$

3 Support quandles and their enveloping groups

The five support quandles occurring in the classification are

$$\begin{aligned} Z_T^{4,1} &: (243) (134) (142) (123) \text{ id}, \\ Z_2^{2,2} &: (24) (13) (24) (13), \\ Z_3^{3,1} &: (23) (13) (12) \text{ id}, \\ Z_3^{3,2} &: (23)(45) (13)(45) (12)(45) (123) (132), \\ Z_4^{4,2} &: (24)(56) (13)(56) (24)(56) (13)(56) (1234) (1432). \end{aligned}$$

3.1 The quandle $Z_T^{4,1}$ and its enveloping group

Lemma 3.1. *The enveloping group of $Z_T^{4,1}$ has the presentation*

$$T = \langle x_1, x_2, z \mid zx_1 = x_1z, \quad zx_2 = x_2z, \quad x_1x_2x_1 = x_2x_1x_2, \quad x_1^3 = x_2^3 \rangle.$$

Set

$$x_3 = x_2x_1x_2^{-1}, \quad x_4 = x_1x_2x_1^{-1}.$$

Then z is central, $x_1^T = \{x_1, x_2, x_3, x_4\}$, and

$$z^T \cup x_1^T \cong Z_T^{4,1}$$

as quandles.

Proof. This is the presentation in [22, Section 2.1]. □

3.2 The groups Γ_n and their support quandles

Let $n \in \mathbb{N}_{\geq 2}$. Recall from [20, Section 3] that

$$\Gamma_n = \langle g, h, \varepsilon \mid hg = \varepsilon gh, \quad g\varepsilon = \varepsilon^{-1}g, \quad h\varepsilon = \varepsilon h, \quad \varepsilon^n = 1 \rangle.$$

Every element of Γ_n has a unique expression $\varepsilon^i h^j g^k$, where $0 \leq i < n$ and $j, k \in \mathbb{Z}$. For $z \in Z(\Gamma_n) = \langle \varepsilon^{-1}h^2, h^n, g^2 \rangle$ and $1 \leq j \leq \lfloor n/2 \rfloor$, the conjugacy classes are

$$\begin{aligned} z^{\Gamma_n} &= \{z\}, & (gz)^{\Gamma_n} &= \{\varepsilon^m gz \mid 0 \leq m < n\}, \\ (h^j z)^{\Gamma_n} &= \{h^j z, \varepsilon^{-j} h^j z\}, & (hgz)^{\Gamma_n} &= \{\varepsilon^m hgz \mid 0 \leq m < n\}. \end{aligned}$$

The centralizers of the noncentral representatives are

$$\begin{aligned} C_{\Gamma_n}(gz) &= \langle \varepsilon^{-1}h^2, g, h^n \rangle, & C_{\Gamma_n}(hgz) &= \langle \varepsilon^{-1}h^2, hg, h^n \rangle, \\ C_{\Gamma_n}(h^j z) &= \langle \varepsilon, h, g^2 \rangle. \end{aligned}$$

Proposition 3.2. *Let $Z_2^{2,2} = h^{\Gamma_2} \cup g^{\Gamma_2}$. Then $Z_2^{2,2} \cong D_4$, the dihedral quandle of four elements. Its enveloping group is*

$$\langle x_1, x_2, x_3, x_4 \mid x_i x_j = x_{2i-j \pmod{4}} x_i, \quad i, j \in \{1, 2, 3, 4\} \rangle \cong \Gamma_2.$$

The isomorphism sends x_1 to g and x_2 to h .

Let $Z_3^{3,1} = g^{\Gamma_3} \cup \{\varepsilon h\}$. The element εh is central, and the enveloping group of $Z_3^{3,1}$ is

$$\langle z \rangle \times \langle x_1, x_2, x_3 \mid x_i x_j = x_{2i-j \pmod{3}} x_i, \quad i, j \in \{1, 2, 3\} \rangle \cong \Gamma_3.$$

The latter isomorphism sends z to εh , x_1 to g , x_2 to εg , and x_3 to $\varepsilon^2 g$.

For $Z_3^{3,2} = g^{\Gamma_3} \cup h^{\Gamma_3}$, the enveloping group is Γ_3 .

For $Z_4^{4,2} = g^{\Gamma_4} \cup h^{\Gamma_4}$, the enveloping group is Γ_4 .

Proof. These are the enveloping-group identifications in [25, Examples 2.4–2.7]. □

3.3 Classification of the possible support quandles

Throughout this subsection, let G be a non-abelian group, let Φ be a normalized 3-cocycle on G , and let $V, W \in {}^G\mathcal{YD}^\Phi$. Set $D_0 = W$ and $D_m = V \otimes D_{m-1}$ for $m \geq 1$.

For $s \in \text{supp } W$, let $Q_0(s) = W_s$. If $m \geq 1$ and $r_1, \dots, r_m \in \text{supp } V$, define recursively

$$Q_m(r_1, \dots, r_m; s) := \varphi_m^\Phi(V_{r_1} \otimes Q_{m-1}(r_2, \dots, r_m; s)) \subseteq D_m.$$

Thus $X_m^{V,W}$ is the sum of the spaces $Q_m(r_1, \dots, r_m; s)$.

For a subspace $Q \subseteq D_m$, let $\text{supp } Q$ denote the union of the multidegrees occurring in the elements of Q . Since $X_m^{V,W} \cong (\text{ad } V)^m(W)$, we have

$$(\text{ad } V)^m(W) = 0 \iff Q_m(r_1, \dots, r_m; s) = 0 \tag{3.1}$$

for all $r_1, \dots, r_m \in \text{supp } V$ and $s \in \text{supp } W$.

Proposition 3.3. Let $m \in \mathbb{N}_0$. Suppose that $p_1, \dots, p_m, r_1, \dots, r_m \in \text{supp } V$, $p_{m+1}, s \in \text{supp } W$, and

$$(p_1, \dots, p_m, p_{m+1}) \in \text{supp } Q_m(r_1, \dots, r_m; s).$$

Let $p \in \text{supp } V$ and $i \in \{1, \dots, m+1\}$. Assume that

$$p_i \triangleright p \neq p, \quad p_j \triangleright p = p \text{ for } i < j \leq m+1, \quad (3.2)$$

$$p \notin \{p_j : 1 \leq j \leq m\} \cup \{(p_{j+1} \cdots p_{m+1})^{-1} \triangleright p_j : 1 \leq j < i\}. \quad (3.3)$$

Then

$$(p \triangleright p_1, \dots, p \triangleright p_{i-1}, p, p_i, \dots, p_m, p_{m+1}) \in \text{supp } Q_{m+1}(p, r_1, \dots, r_m; s).$$

Proof. Choose $t \in Q_m(r_1, \dots, r_m; s)$ whose component of multidegree $(p_1, \dots, p_m, p_{m+1})$ is nonzero, and take $0 \neq v \in V_p$. By the recursive definition of Q_{m+1} , the element $\varphi_{m+1}^\Phi(v \otimes t)$ belongs to $Q_{m+1}(p, r_1, \dots, r_m; s)$.

Associators preserve multidegrees and multiply homogeneous tensors by nonzero scalars. The braiding is an isomorphism and changes degrees only by conjugation. Hence each individual summand arising from a nonzero homogeneous component is nonzero. For $1 \leq j \leq m+1$, its multidegree has one of the forms

$$\begin{aligned} &(p \triangleright p'_1, \dots, p \triangleright p'_{j-1}, p, p'_j, \dots, p'_m, p'_{m+1}), \\ &(p \triangleright p'_1, \dots, p \triangleright p'_{j-1}, (pp'_j \cdots p'_{m+1}) \triangleright p, p \triangleright p'_j, \dots, p \triangleright p'_m, p \triangleright p'_{m+1}), \end{aligned}$$

where $(p'_1, \dots, p'_m, p'_{m+1}) \in \text{supp } t$.

The required multidegree occurs in the first family with $j = i$ and $(p'_1, \dots, p'_{m+1}) = (p_1, \dots, p_{m+1})$. In the first family, equality with the required multidegree gives $p = p_j$ if $j < i$, $p'_\ell = p_\ell$ for all $1 \leq \ell \leq m+1$ if $j = i$, and $p = p_{j-1}$ if $j > i$. In the second family, it gives $p = p_{j-1}$ if $j > i$, $p_i \triangleright p = p$ if $j = i$, and $p = (p_{j+1} \cdots p_{m+1})^{-1} \triangleright p_j$ if $j < i$. Conditions (3.2) and (3.3) exclude every case except the first family with $j = i$. This is the comparison used in [25, Proposition 5.5]. Thus the required component is the image of the chosen nonzero component under a single isomorphism and is therefore nonzero. $\square$

The preceding adjoint and multidegree results give the following support classification.

Theorem 3.4. Let G be a non-abelian group and let $V, W \in {}_G^G \mathcal{YD}^\Phi$ be simple objects. Assume that $G = \langle \text{supp } V \cup \text{supp } W \rangle$, that (V, W) is braided-indecomposable, and that

$$(\text{ad } V)^2(W) = 0, \quad (\text{ad } W)^4(V) = 0.$$

Then $\text{supp}(V \oplus W)$, with its conjugation quandle structure, is isomorphic to one of

$$Z_T^{4,1}, \quad Z_2^{2,2}, \quad Z_3^{3,1}, \quad Z_3^{3,2}, \quad Z_4^{4,2}.$$

Accordingly, G is an epimorphic image of $T, \Gamma_2, \Gamma_3, \Gamma_3$, or Γ_4 , respectively.

Proof. By Proposition 2.4, $\text{supp } V$ and $\text{supp } W$ are conjugacy classes. Moreover, (2.13) gives $(\text{ad } V)(W) \neq 0$. Thus the group-theoretic hypotheses in the proof of [25, Theorem 4.4] are satisfied.

The group-theoretic Lemma 5.1 of [25] applies unchanged. Lemma 2.8 and (3.1) replace [25, Lemmas 5.2–5.3 and Remark 5.4], while Proposition 3.3 replaces [25, Proposition 5.5]. Its case $m = 0$ also gives [25, Remark 5.6]. Associators do not change multidegrees, and Proposition 3.3 shows that the distinguished component cannot cancel. The arguments in [25, Corollaries 5.7–5.11, Lemmas 5.12–5.13 and 5.15–5.24] therefore remain valid, since they use only these facts and the conjugation relations in $\text{supp } V \cup \text{supp } W$.

If $\text{supp } V$ and $\text{supp } W$ commute elementwise, [25, Proposition 5.14] gives $\text{supp}(V \oplus W) \cong Z_3^{3,1}$ or $Z_T^{4,1}$. Otherwise, [25, Proposition 5.25] gives $\text{supp}(V \oplus W) \cong Z_2^{2,2}$, $Z_3^{3,2}$, or $Z_4^{4,2}$.

Let $X = \text{supp}(V \oplus W)$. The inclusion $X \hookrightarrow \text{Conj}(G)$ induces a homomorphism $G_X \rightarrow G$ from the enveloping group of X . It is surjective because $G = \langle X \rangle$. Lemma 3.1 and Proposition 3.2 identify G_X in the five cases and give the asserted epimorphic images. $\square$

4 The Γ_2 case

Let

$$\Gamma_2 = \langle g, h, \varepsilon \mid hg = \varepsilon gh, \quad \varepsilon^2 = 1, \quad \varepsilon g = g\varepsilon, \quad \varepsilon h = h\varepsilon \rangle,$$

and let G be a finite non-abelian quotient of Γ_2 . The images of g, h, ε in G are denoted by the same letters. Let Φ be a normalized 3-cocycle on G .

Throughout this section, let $V = M(g, \rho)$ be simple, and let $W = M(h, \sigma)$ be simple. The support of V is $\{g, \varepsilon g\}$, and the support of W is $\{h, \varepsilon h\}$. Assume that (V, W) is braided-indecomposable.

Lemma 4.5 below shows that ρ and σ are one-dimensional under the standing assumptions. The parameters used below are defined in (4.1), (4.2), and (4.15). Since their definitions depend on the order of V and W , they must be computed separately for the two orders. The classification is as follows.

Theorem 4.1. *Under the preceding assumptions, $\dim \mathcal{B}(V \oplus W) < \infty$ if and only if, for one of the two orders of V and W , one of the following two systems of equations holds:*

$$\Delta = 1, \quad \lambda = \lambda' = -1, \quad (A_2)$$

$$\Delta = 1, \quad \lambda' = -1, \quad \lambda^2 = \Theta_\Phi(g, h) = -1. \quad (\Gamma_2\text{-}G_2)$$

Outline of the proof.

- Subsections 4.1 and 4.2 compute the first adjoint objects and show that the relevant projective representations are one-dimensional.
- Subsection 4.3 computes the duals and the parameters after each reflection.
- Subsection 4.4 proves the A_2 case, and Subsection 4.5 shows that type B_2 cannot occur.
- Subsections 4.6 and 4.7 compute the remaining adjoint objects in the G_2 case and prove that its conditions are preserved by reflections.
- Subsection 4.8 shows that every finite-dimensional case has a standard Cartan graph of type A_2 or G_2 .
- Subsections 4.9 and 4.10 prove finite-dimensionality and compute the dimensions.
- Subsection 4.11 gives an explicit example of the G_2 case.

Parameters used in the theorem. The following table records where the parameters are defined and how they are used.

Parameter	Defined in	Use
λ	(4.1)	If $\Delta = 1$, then $X_2 = 0$ if and only if $\lambda = -1$.
λ'	(4.1)	If $\Delta' = 1$, then $Y_2 = 0$ if and only if $\lambda' = -1$.
Δ, Δ'	(4.2), (4.23)	The object X_1 is simple if and only if $\Delta = 1$, and Y_1 is simple if and only if $\Delta' = 1$. These two conditions are equivalent.
$\Theta_\Phi(g, h)$	(4.15)	If $\Delta = 1$ and $X_2 \neq 0$, then X_2 is simple if and only if $\lambda^2 = \Theta_\Phi(g, h)$. This condition occurs in the G_2 case.

4.1 Projective representations

The support representatives that arise below have the form $g^m h$ or $h^m g$. By Proposition 2.4, a simple object with support x^G is determined by an irreducible Φ_x -projective representation of $C_G(x)$. We therefore begin by computing the relevant centralizers. The corollary will allow us to use the same argument for the projective representations that occur in the later adjoint objects.

Lemma 4.2. *The center of G is $Z(G) = \langle \varepsilon, g^2, h^2 \rangle$. The centralizers needed below are*

$$\begin{aligned} C_G(g) &= \langle \varepsilon, g, h^2 \rangle, \\ C_G(h) &= \langle \varepsilon, h, g^2 \rangle, \\ C_G(gh) &= C_G(hg) = \langle \varepsilon, g^2, g^{-1}h \rangle = \langle \varepsilon, g^2, h^2, gh \rangle. \end{aligned}$$

More generally, for every $m \in \mathbb{Z}$,

$$C_G(g^m h) = \begin{cases} C_G(h), & m \text{ even}, \\ C_G(gh), & m \text{ odd}, \end{cases} \quad C_G(h^m g) = \begin{cases} C_G(g), & m \text{ even}, \\ C_G(gh), & m \text{ odd}. \end{cases}$$

Proof. Since G is non-abelian, $\varepsilon \neq 1$, and hence ε has order 2. Every element of G can be written, not necessarily uniquely, as $\varepsilon^e g^a h^b$, where $e \in \{0, 1\}$ and $a, b \in \mathbb{Z}$.

Let $u = \varepsilon^e g^a h^b$, $v = \varepsilon^f g^c h^d$. Since $h^b g^c = \varepsilon^{bc} g^c h^b$, one has $uv = \varepsilon^{e+f+bc} g^{a+c} h^{b+d}$, $vu = \varepsilon^{e+f+ad} g^{a+c} h^{b+d}$. Therefore $uv = \varepsilon^{bc-ad} vu$. Since $\varepsilon \neq 1$, the elements u and v commute if and only if $bc - ad$ is even. Taking $v = g$ shows that $u \in C_G(g)$ if and only if b is even. Taking $v = h$ shows that $u \in C_G(h)$ if and only if a is even. Thus

$$C_G(g) = \langle \varepsilon, g, h^2 \rangle, \quad C_G(h) = \langle \varepsilon, h, g^2 \rangle.$$

An element is central if and only if both a and b are even, which gives $Z(G) = \langle \varepsilon, g^2, h^2 \rangle$.

Taking $v = gh$ shows that $u \in C_G(gh)$ if and only if $a \equiv b \pmod{2}$. Hence

$$C_G(gh) = \langle \varepsilon, g^2, h^2, gh \rangle.$$

If $k = g^{-1}h$, then $k^2 = \varepsilon g^{-2}h^2$, $gh = g^2k$, and therefore

$$C_G(gh) = \langle \varepsilon, g^2, g^{-1}h \rangle.$$

The three centralizers are each generated by central elements and one additional element, so they are abelian. Finally, applying the same commutation criterion to $g^m h$ gives

$$C_G(g^m h) = \begin{cases} C_G(h), & m \text{ even}, \\ C_G(gh), & m \text{ odd}. \end{cases}$$

Applying it to $h^m g$ gives the second formula. □

Corollary 4.3. *For every $m \in \mathbb{Z}$, each of the pairs $(g^m h, g)$, $(h^m g, h)$ generates G . If (x, y) denotes either pair, then*

$$yx = \varepsilon xy, \quad C_G(x) = \langle \varepsilon, x, y^2 \rangle.$$

Thus x, y, ε satisfy the defining relations of Γ_2 .

Proof. First let $x = g^m h$ and $y = g$. Then $yx = \varepsilon xy$ and $h = y^{-m}x$. Next let $x = h^m g$ and $y = h$. Again $yx = \varepsilon xy$, and now $g = y^{-m}x$. In either case,

$$\varepsilon = yxy^{-1}x^{-1},$$

so x and y generate G . The formula for $C_G(x)$ follows from Lemma 4.2. □

The dimensions of the projective representations ρ and σ are controlled by the following alternator of Φ :

$$f_\Phi(a, b, c) := \frac{\Phi(a, b, c)\Phi(b, c, a)\Phi(c, a, b)}{\Phi(a, c, b)\Phi(c, b, a)\Phi(b, a, c)}.$$

Lemma 4.4. *Let $A \leq G$ be an abelian subgroup. Then $f_\Phi|_{A \times A \times A}$ is an alternating tricharacter. In particular,*

$$f_\Phi(ab, c, d) = f_\Phi(a, c, d)f_\Phi(b, c, d),$$

and similarly in the second and third variables, while $f_\Phi(a, a, b) = 1$, $f_\Phi(a, b, c) = f_\Phi(b, a, c)^{-1}$. Moreover, if $a, b \in Z(G)$, then the map

$$\chi : G \longrightarrow \mathbb{C}^\times, \quad x \longmapsto f_\Phi(x, a, b),$$

is a group character.

Proof. Suppose that $a, b \in G$ commute and that $x, y \in C_G(a) \cap C_G(b)$. Applying (2.4) to the six quadruples

$$(x, a, y, b), \quad (a, x, b, y), \quad (x, a, b, y), \quad (x, y, a, b), \quad (a, x, y, b), \quad (a, b, x, y),$$

and substituting the definition of the local cocycles gives

$$\Phi_{xy}(a, b) = \Phi_x(a, b)\Phi_y(a, b) \frac{\Phi_a(x, y)\Phi_b(x, y)}{\Phi_{ab}(x, y)}.$$

Interchanging a and b leaves the last factor unchanged. Taking the quotient of the two identities therefore gives

$$f_\Phi(xy, a, b) = f_\Phi(x, a, b)f_\Phi(y, a, b).$$

Taking $x, y, a, b \in A$ proves that $f_\Phi|_{A^3}$ is multiplicative in its first variable. Its defining formula gives

$$f_\Phi(a, b, c) = f_\Phi(b, c, a) = f_\Phi(b, a, c)^{-1}.$$

Hence it is multiplicative in all three variables. The same formula gives $f_\Phi(a, a, c) = 1$, so $f_\Phi|_{A^3}$ is an alternating tricharacter.

Finally, if $a, b \in Z(G)$, then $C_G(a) \cap C_G(b) = G$. The multiplicativity proved above, together with $f_\Phi(1, a, b) = 1$, shows that $x \mapsto f_\Phi(x, a, b)$ is a group character. $\square$

Lemma 4.5. *Every irreducible Φ_g -projective representation of $C_G(g)$ is one-dimensional. More generally, for every $m \in \mathbb{Z}$ and $x \in \{g^m h, h^m g\}$, every irreducible Φ_x -projective representation of $C_G(x)$ is one-dimensional.*

Proof. Put $\vartheta_g = f_\Phi(g, \varepsilon, h^2)$ and $c_g(a, b) = \Phi_g(a, b)/\Phi_g(b, a)$ for $a, b \in C_G(g)$. By Lemma 4.2, this centralizer is abelian. Writing $a = \varepsilon^{x_1} g^{y_1} (h^2)^{z_1}$ and $b = \varepsilon^{x_2} g^{y_2} (h^2)^{z_2}$, Lemma 4.4 gives $c_g(a, b) = f_\Phi(g, a, b) = \vartheta_g^{x_1 z_2 - x_2 z_1}$. Since $\varepsilon^2 = 1$, one has $\vartheta_g^2 = f_\Phi(g, \varepsilon^2, h^2) = 1$.

Assume, to the contrary, that $\vartheta_g = -1$. For $a \in C_G(g)$, one has $c_g(g, a) = f_\Phi(g, g, a) = 1$. The element corresponding to g is therefore central in the twisted group algebra $\mathbb{C}^{\Phi_g} C_G(g)$. Schur's lemma gives $g \triangleright v = \lambda v$ for every $v \in V_g$, where $\lambda \in \mathbb{C}^\times$.

Since

$$\varepsilon \triangleright (\varepsilon \triangleright v) = \Phi_g(\varepsilon, \varepsilon)(\varepsilon^2 \triangleright v) = \Phi_g(\varepsilon, \varepsilon)v,$$

there is a nonzero eigenvector, denoted by v , and a scalar $\eta \neq 0$ such that $\varepsilon \triangleright v = \eta v$. Let $v_- := h^2 \triangleright v$. Then

$$\varepsilon \triangleright v_- = \varepsilon \triangleright (h^2 \triangleright v) = \frac{\Phi_g(\varepsilon, h^2)}{\Phi_g(h^2, \varepsilon)} h^2 \triangleright (\varepsilon \triangleright v) = \vartheta_g \eta v_- = -\eta v_-.$$

Thus v and v_- are linearly independent. Set $\kappa := \lambda\eta/\Phi_g(\varepsilon, g)$. The projective action gives $(\varepsilon g) \triangleright v = \kappa v$ and $(\varepsilon g) \triangleright v_- = -\kappa v_-$. Since $gh = h\varepsilon g$, the action of g and the double braiding $c_{V,V}^2 := c_{V,V} \circ c_{V,V}$ satisfy

$$\begin{aligned} g \triangleright (hv) &= \frac{\Phi_g(g, h)}{\Phi_g(h, \varepsilon g)} \kappa hv, \\ c_{V,V}^2(v \otimes hv) &= \frac{\Phi_g(g, h)}{\Phi_g(h, \varepsilon g)} \kappa^2 v \otimes hv, \\ c_{V,V}^2(hv \otimes v_-) &= -\frac{\Phi_g(g, h)}{\Phi_g(h, \varepsilon g)} \kappa^2 hv \otimes v_-. \end{aligned}$$

The relation (2.2) gives

$$h \triangleright (v \otimes hv) = \Phi^h(g, \varepsilon g)(hv) \otimes (h \triangleright (hv)) = \Phi^h(g, \varepsilon g) \Phi_g(h, h)(hv) \otimes v_-.$$

Since the double braiding is a morphism in ${}^G\mathcal{YD}^\Phi$, it commutes with the G -action. Therefore,

$$c_{V,V}^2(h \triangleright (v \otimes hv)) = h \triangleright c_{V,V}^2(v \otimes hv) = \frac{\Phi_g(g, h)}{\Phi_g(h, \varepsilon g)} \kappa^2 h \triangleright (v \otimes hv).$$

The formula for $c_{V,V}^2(hv \otimes v_-)$ above gives instead

$$c_{V,V}^2(h \triangleright (v \otimes hv)) = -\frac{\Phi_g(g, h)}{\Phi_g(h, \varepsilon g)} \kappa^2 h \triangleright (v \otimes hv).$$

This is impossible because all displayed scalar factors and the vector $h \triangleright (v \otimes hv)$ are nonzero. Thus $\vartheta_g \neq -1$, and hence $\vartheta_g = 1$.

It follows that $c_g(a, b) = 1$ for all $a, b \in C_G(g)$. Thus $\mathbb{C}^{\Phi_g} C_G(g)$ is commutative, and every irreducible Φ_g -projective representation of $C_G(g)$ is one-dimensional.

Let $x \in \{g^m h, h^m g\}$, let y be g or h , respectively, and let τ be an irreducible Φ_x -projective representation of $C_G(x)$. Corollary 4.3 shows that (x, y, ε) satisfies the defining Γ_2 relations and that $C_G(x) = \langle \varepsilon, x, y^2 \rangle$. Replacing (g, h, ρ) by (x, y, τ) in the preceding argument shows that $\mathbb{C}^{\Phi_x} C_G(x)$ is commutative. Hence $\dim \tau = 1$. $\square$

Interchanging g and h gives the same conclusion for σ . Thus both ρ and σ are one-dimensional throughout the remainder of this section.

4.2 Adjoint objects

By the preceding subsection, ρ and σ are one-dimensional. Choose nonzero homogeneous vectors $v \in V_g$ and $w \in W_h$. Since $hgh^{-1} = \varepsilon g$ and $ghg^{-1} = \varepsilon h$, one has $hv \in V_{\varepsilon g}$ and $gw \in W_{\varepsilon h}$. Hence $V = \mathbb{k}v \oplus \mathbb{k}hv$, $W = \mathbb{k}w \oplus \mathbb{k}gw$ as G -graded vector spaces.

Define nonzero scalars

$$\begin{aligned} \varepsilon \triangleright v &= \zeta v, & g \triangleright v &= \lambda v, & h^2 \triangleright v &= \mu v, \\ \varepsilon \triangleright w &= \zeta' w, & h \triangleright w &= \lambda' w, & g^2 \triangleright w &= \mu' w. \end{aligned} \tag{4.1}$$

Thus ζ, λ, μ describe the one-dimensional Φ_g -projective action of $C_G(g)$ on V_g , while ζ', λ', μ' describe the one-dimensional Φ_h -projective action of $C_G(h)$ on W_h . Set

$$X_n = X_n^{V,W} \cong (\text{ad } V)^n(W), \quad Y_n = X_n^{W,V} \cong (\text{ad } W)^n(V).$$

Here X_n and Y_n are the restricted images defined using the operators φ_n^Φ .

For the higher adjoint objects, the following observation will be used repeatedly. Let $n \geq 2$, assume that X_{n-1} is nonzero and simple, and choose $0 \neq x_{n-1} \in (X_{n-1})_{g^{n-1}h}$. Lemma 4.5 shows that this homogeneous component is $\mathbb{C}x_{n-1}$. The subgroup $\langle g^{n-1}h \rangle$ preserves this line, and $(g^{n-1}h)g(g^{n-1}h)^{-1} = \varepsilon g$, so $V = \mathbb{C}\langle g^{n-1}h \rangle \triangleright v$. Since simplicity also gives $X_{n-1} = \mathbb{C}G \triangleright x_{n-1}$, Proposition 2.10 yields $X_n = \mathbb{C}G \triangleright x_n$, where $x_n = \varphi_n^\Phi(v \otimes x_{n-1})$. Thus $X_n = 0$ if and only if $x_n = 0$.

4.2.1 The first adjoint object

Lemma 4.6. *Let $x_1 := \varphi_1^\Phi(v \otimes w)$. Then*

$$x_1 = v \otimes w - \frac{\zeta}{\Phi_g(h, \varepsilon)} hv \otimes gw.$$

Moreover, $x_1 \neq 0$, $X_1 = \mathbb{C}G \triangleright x_1$.

Proof. Since $W = \mathbb{C}G \triangleright w$, $\langle h \rangle \triangleright w \subseteq \mathbb{C}w$, and $V = \mathbb{C}\langle h \rangle \triangleright v$, Proposition 2.10, applied with $n = 1$, $H = \langle h \rangle$, and $x_0 = w$, gives $X_1 = \mathbb{C}G \triangleright x_1$.

The calculation of x_1 starts from $\varphi_1^\Phi = \text{id}_{V \otimes W} - c_{W, V} c_{V, W}$. Applying the double braiding to $v \otimes w$ gives $c_{W, V} c_{V, W}(v \otimes w) = (\varepsilon h) \triangleright v \otimes gw$. To calculate $(\varepsilon h) \triangleright v$, we have $h \triangleright (\varepsilon \triangleright v) = \Phi_g(h, \varepsilon)(\varepsilon h) \triangleright v$, because $h\varepsilon = \varepsilon h$. Consequently, $(\varepsilon h) \triangleright v = \frac{\zeta}{\Phi_g(h, \varepsilon)} hv$. Substitution gives the asserted formula for x_1 .

Finally, $v \otimes w \in V_g \otimes W_h$, $hv \otimes gw \in V_{\varepsilon g} \otimes W_{\varepsilon h}$. These are distinct bi-homogeneous components. Hence the two summands cannot cancel, and therefore $x_1 \neq 0$. $\square$

For the simplicity of X_1 , set

$$\Delta := \frac{\zeta \zeta' \mu \mu' \Phi_h(\varepsilon g, g) \Phi_g(h, h)}{\Phi_g(h, \varepsilon) \Phi_h(\varepsilon, g^2)}. \quad (4.2)$$

Lemma 4.7. *The object X_1 is simple if and only if $\Delta = 1$. In this case, $X_1 \simeq M(gh, \sigma_1)$ for a one-dimensional projective representation σ_1 of $C_G(gh)$.*

Proof. Let $a := \frac{\zeta}{\Phi_g(h, \varepsilon)}$ and introduce the vectors

$$e_0 := v \otimes w, \quad e_1 := hv \otimes gw, \quad f_0 := v \otimes gw, \quad f_1 := hv \otimes w.$$

Thus $x_1 = e_0 - ae_1$.

The actions of g and h on x_1 are as follows. Relation (2.1) gives $(\varepsilon g) \triangleright v = \frac{\lambda \zeta}{\Phi_g(\varepsilon, g)} v$. Since $gh = h(\varepsilon g)$, we have $(gh) \triangleright v = \frac{1}{\Phi_g(h, \varepsilon g)} h \triangleright ((\varepsilon g) \triangleright v)$. It follows that

$$g \triangleright (hv) = \frac{\Phi_g(g, h) \lambda \zeta}{\Phi_g(\varepsilon, g) \Phi_g(h, \varepsilon g)} hv. \quad (4.3)$$

Similarly,

$$h \triangleright (gw) = \frac{\Phi_h(h, g) \lambda' \zeta'}{\Phi_h(\varepsilon, h) \Phi_h(g, \varepsilon h)} gw. \quad (4.4)$$

Moreover, $h \triangleright e_0 = \lambda' \Phi^h(g, h) f_1$. By (4.4) and $h \triangleright (hv) = \Phi_g(h, h) \mu v$, we also have $h \triangleright e_1 = A_1 f_0$, where

$$A_1 := \Phi^h(\varepsilon g, \varepsilon h) \Phi_g(h, h) \mu \frac{\Phi_h(h, g) \lambda' \zeta'}{\Phi_h(\varepsilon, h) \Phi_h(g, \varepsilon h)}. \quad (4.5)$$

Therefore

$$h \triangleright x_1 = \lambda' \Phi^h(g, h) f_1 - a A_1 f_0. \quad (4.6)$$

On the other hand, $g \triangleright e_0 = \lambda \Phi^g(g, h) f_0$. Using (4.3) and $g \triangleright (gw) = \Phi_h(g, g) \mu' w$, we get $g \triangleright e_1 = \lambda B_1 f_1$, where

$$B_1 := \Phi^g(\varepsilon g, \varepsilon h) \frac{\Phi_g(g, h) \zeta}{\Phi_g(\varepsilon, g) \Phi_g(h, \varepsilon g)} \Phi_h(g, g) \mu'. \quad (4.7)$$

Consequently,

$$g \triangleright x_1 = \lambda \Phi^g(g, h) f_0 - a \lambda B_1 f_1. \quad (4.8)$$

Both $h \triangleright x_1$ and $g \triangleright x_1$ are homogeneous of degree εgh . Comparing the f_0 -coefficients in (4.6) and (4.8), we see that $h \triangleright x_1 \in \mathbb{C}(g \triangleright x_1)$ if and only if

$$\frac{a^2 A_1 B_1}{\lambda' \Phi^h(g, h) \Phi^g(g, h)} = 1. \quad (4.9)$$

By Lemma B.1 in Appendix B, the left-hand side of (4.9) is precisely Δ . Hence $h \triangleright x_1 \in \mathbb{C}(g \triangleright x_1)$ if and only if $\Delta = 1$.

Suppose first that $\Delta = 1$. By the preceding coefficient comparison, there exists $r_1 \in \mathbb{C}^\times$ such that $h \triangleright x_1 = r_1 (g \triangleright x_1)$. Since both g and h conjugate gh to εgh , the element $g^{-1}h$ belongs to $C_G(gh)$. Applying g^{-1} to the preceding equality and using the relation (2.1), we obtain

$$(g^{-1}h) \triangleright x_1 = r_1 \frac{\Phi_{gh}(g^{-1}, g)}{\Phi_{gh}(g^{-1}, h)} x_1,$$

and hence the line $\mathbb{C}x_1$ is stable under $g^{-1}h$.

The line $\mathbb{C}x_1$ is also stable under ε and g^2 . By Lemma 4.2, $C_G(gh) = \langle \varepsilon, g^2, g^{-1}h \rangle$. It follows that $\mathbb{C}x_1$ is a one-dimensional Φ_{gh} -projective representation of $C_G(gh)$ and denote it by σ_1 . The assignment $1 \otimes x_1 \mapsto x_1$ induces a surjective morphism $M(gh, \sigma_1) \rightarrow X_1$. The induced object $M(gh, \sigma_1)$ is simple because σ_1 is one-dimensional. The above morphism is nonzero, and therefore it is an isomorphism. Hence $X_1 \simeq M(gh, \sigma_1)$ is simple.

Conversely, suppose that X_1 is simple. Then $X_1 \simeq M(gh, \tau)$ for an irreducible Φ_{gh} -projective representation τ of $C_G(gh)$. By Lemma 4.5, $\dim \tau = 1$. Therefore every homogeneous component of X_1 is one-dimensional. In particular, $\dim(X_1)_{\varepsilon gh} = 1$. Both $g \triangleright x_1$ and $h \triangleright x_1$ are nonzero vectors in this homogeneous component. They must therefore be proportional, that is, $h \triangleright x_1 \in \mathbb{C}(g \triangleright x_1)$. By the coefficient computation above, this proportionality is equivalent to $\Delta = 1$. $\square$

4.2.2 The second adjoint object

Assume that $\Delta = 1$. Therefore X_1 is simple.

Lemma 4.8. *Assume that $\Delta = 1$, and let $x_2 := \varphi_2^\Phi(v \otimes x_1)$. Then*

$$x_2 = (1 + \lambda) \left[v \otimes x_1 - \frac{1}{\Phi_g(h, g)} hv \otimes (g \triangleright x_1) \right].$$

Moreover, $X_2 = \mathbb{C}G \triangleright x_2$.

Proof. The orbit-generation argument at the beginning of this subsection gives $X_2 = \mathbb{C}G \triangleright x_2$.

For the explicit calculation, let $T_0 := v \otimes x_1$ and $T_1 := hv \otimes (g \triangleright x_1)$. Recall that $x_1 = v \otimes w - a hv \otimes gw$. By the definition of φ_2^Φ ,

$$\varphi_2^\Phi = \text{id}_{V \otimes X_1} - c_{X_1, V} c_{V, X_1} + (\text{id}_V \otimes \varphi_1^\Phi) c_{1, 2}^\Phi. \quad (4.10)$$

First, $c_{V, X_1}(v \otimes x_1) = (g \triangleright x_1) \otimes v$, and then $c_{X_1, V}((g \triangleright x_1) \otimes v) = (hg) \triangleright v \otimes (g \triangleright x_1)$. Thus $(hg) \triangleright v = \lambda hv / \Phi_g(h, g)$. Consequently,

$$c_{X_1, V} c_{V, X_1}(T_0) = \frac{\lambda}{\Phi_g(h, g)} T_1. \quad (4.11)$$

Consider $(\text{id}_V \otimes \varphi_1^\Phi) c_{1,2}^\Phi(T_0)$. For the first summand of T_0 , the definitions of $c_{1,2}^\Phi$ and φ_1^Φ give $(\text{id}_V \otimes \varphi_1^\Phi) c_{1,2}^\Phi(v \otimes (v \otimes w)) = \lambda T_0$. For the second summand, using (4.3) gives

$$c_{1,2}^\Phi(v \otimes (hv \otimes gw)) = \frac{\Phi(g, \varepsilon g, \varepsilon h) \Phi_g(g, h) \lambda \zeta}{\Phi(\varepsilon g, g, \varepsilon h) \Phi_g(\varepsilon, g) \Phi_g(h, \varepsilon g)} hv \otimes (v \otimes gw). \quad (4.12)$$

Since φ_1^Φ is a morphism in ${}^G_C \mathcal{YD}^\Phi$, $g \triangleright x_1 = \varphi_1^\Phi(g \triangleright (v \otimes w))$. Moreover, the relation (2.2) gives $g \triangleright (v \otimes w) = \lambda \Phi^g(g, h) v \otimes gw$. Thus

$$\varphi_1^\Phi(v \otimes gw) = \frac{1}{\lambda \Phi^g(g, h)} g \triangleright x_1. \quad (4.13)$$

Combining (4.12) and (4.13), and recalling that $T_0 = v \otimes (v \otimes w) - a v \otimes (hv \otimes gw)$, we obtain

$$\begin{aligned} (\text{id}_V \otimes \varphi_1^\Phi) c_{1,2}^\Phi(T_0) &= \lambda T_0 - \Xi_V T_1, \\ \Xi_V &:= \frac{\zeta^2 \Phi(g, \varepsilon g, \varepsilon h) \Phi_g(g, h)}{\Phi_g(h, \varepsilon) \Phi(\varepsilon g, g, \varepsilon h) \Phi_g(\varepsilon, g) \Phi_g(h, \varepsilon g) \Phi^g(g, h)}. \end{aligned} \quad (4.14)$$

Lemma B.2 gives $\Xi_V = \Phi_g(h, g)^{-1}$. Hence

$$(\text{id}_V \otimes \varphi_1^\Phi) c_{1,2}^\Phi(T_0) = \lambda T_0 - \frac{1}{\Phi_g(h, g)} T_1.$$

Substituting this identity and (4.11) into (4.10) gives $x_2 = (1 + \lambda)(T_0 - \Phi_g(h, g)^{-1} T_1)$, as required. $\square$

Lemma 4.9. Assume that $\Delta = 1$. Then $X_2 = 0$ if and only if $\lambda = -1$.

Proof. By Lemma 4.8, $X_2 = \mathbb{C} G \triangleright x_2$. Hence $X_2 = 0$ is equivalent to $x_2 = 0$. Now $v \otimes x_1 \in V_g \otimes (X_1)_{gh}$, whereas $hv \otimes (g \triangleright x_1) \in V_{\varepsilon g} \otimes (X_1)_{\varepsilon gh}$. Thus the two tensors inside the brackets in Lemma 4.8 belong to distinct bi-homogeneous components. In particular,

$$v \otimes x_1 - \frac{1}{\Phi_g(h, g)} hv \otimes (g \triangleright x_1) \neq 0.$$

It follows that $x_2 = 0$ if and only if $1 + \lambda = 0$, which proves the assertion. $\square$

For the simplicity criterion of a nonzero X_2 , define

$$\begin{aligned} \Theta_\Phi &:= \frac{\Phi_h(\varepsilon, \varepsilon) \Phi_h(\varepsilon g, g) \Phi^h(g, gh) \Phi^h(\varepsilon g, \varepsilon h) \Phi_g(h, g)^2}{\Phi_g(h, \varepsilon) \Phi_h(\varepsilon, g^2) \Phi_h(\varepsilon, h) \Phi_h(g, \varepsilon h) \Phi^g(g, h)} \\ &\quad \times \frac{\Phi_h(h, g) \Phi_g(\varepsilon, g) \Phi_g(h, \varepsilon g)}{\Phi^h(\varepsilon g, \varepsilon gh) \Phi^g(\varepsilon g, \varepsilon h) \Phi^h(\varepsilon g, h) \Phi_g(g, h) \Phi_h(g, g)}. \end{aligned} \quad (4.15)$$

Whenever $x, y \in G$ satisfy $yx = \varepsilon xy$, we write $\Theta_\Phi(x, y)$ for the expression in (4.15) after the simultaneous substitution $g \mapsto x, h \mapsto y$. Thus $\Theta_\Phi = \Theta_\Phi(g, h)$.

Lemma 4.10. Assume that $\lambda \neq -1$ and X_1 is simple. Then X_2 is simple if and only if $\lambda^2 = \Theta_\Phi$.

Proof. Let $p := \frac{1}{\Phi_g(h,g)}$ and $b_2 := T_0 - pT_1$. Since $\lambda \neq -1$, we have $x_2 = (1 + \lambda)b_2$, so that $\mathbb{C}x_2 = \mathbb{C}b_2$. We retain the notation a, A_1, B_1 from (4.5) and (4.7). Since $\Delta = 1$, equations (4.6), (4.8), and (4.9) give

$$h \triangleright x_1 = -\frac{aA_1}{\lambda\Phi^g(g,h)}(g \triangleright x_1). \quad (4.16)$$

The action $h \triangleright (g \triangleright x_1)$ is determined from

$$g \triangleright x_1 = \lambda\Phi^g(g,h)f_0 - a\lambda B_1f_1.$$

Using (4.4), we get $h \triangleright f_0 = D_0e_1$, where $D_0 := \Phi^h(g, \varepsilon h) \frac{\Phi_h(h,g)\lambda'\zeta'}{\Phi_h(\varepsilon,h)\Phi_h(g,\varepsilon h)}$. Similarly, $h \triangleright f_1 = D_1e_0$, where $D_1 := \Phi^h(\varepsilon g, h)\Phi_g(h,h)\mu\lambda'$. It follows that

$$h \triangleright (g \triangleright x_1) = \lambda\Phi^g(g,h)D_0e_1 - a\lambda B_1D_1e_0.$$

The equality $\Delta = 1$ is also equivalent to $\Phi^g(g,h)D_0 = a^2B_1D_1$. Therefore

$$h \triangleright (g \triangleright x_1) = -a\lambda B_1D_1x_1. \quad (4.17)$$

The action of h on T_0 and T_1 is determined next. By (4.16),

$$h \triangleright T_0 = \Phi^h(g, gh)(hv) \otimes (h \triangleright x_1) = -a \frac{\Phi^h(g, gh)A_1}{\lambda\Phi^g(g,h)}T_1.$$

On the other hand, $h \triangleright (hv) = \Phi_g(h,h)\mu v$. Together with (4.17), this gives

$$\begin{aligned} h \triangleright T_1 &= \Phi^h(\varepsilon g, \varepsilon gh)(h \triangleright (hv)) \otimes (h \triangleright (g \triangleright x_1)) \\ &= -a\lambda\Phi^h(\varepsilon g, \varepsilon gh)\Phi_g(h,h)\mu B_1D_1T_0. \end{aligned}$$

For brevity, let $r_0 := -a \frac{\Phi^h(g, gh)A_1}{\lambda\Phi^g(g,h)}$ and $r_1 := -a\lambda\Phi^h(\varepsilon g, \varepsilon gh)\Phi_g(h,h)\mu B_1D_1$. Then $h \triangleright T_0 = r_0T_1$, $h \triangleright T_1 = r_1T_0$, and hence $h \triangleright b_2 = r_0T_1 - pr_1T_0$.

Since T_0 and T_1 belong to distinct outer bi-homogeneous components, the condition $h \triangleright b_2 \in \mathbb{C}b_2$ is equivalent to the equality of the two coefficient ratios. Explicitly, $r_0 = p^2r_1$. After cancelling the common nonzero factor $-a$, this becomes

$$\lambda^2 = \frac{\Phi^h(g, gh)A_1}{p^2\Phi^g(g,h)\Phi^h(\varepsilon g, \varepsilon gh)\Phi_g(h,h)\mu B_1D_1}.$$

By Lemma B.3, the right-hand side is exactly Θ_Φ . We have therefore proved that $h \triangleright x_2 \in \mathbb{C}x_2$ if and only if $\lambda^2 = \Theta_\Phi$.

Finally, $\deg x_2 = g^2h$, and $C_G(g^2h) = C_G(h) = \langle \varepsilon, h, g^2 \rangle$ by Lemma 4.2. The line $\mathbb{C}x_2$ is automatically stable under ε and g^2 . The only remaining centralizer condition is stability under h . If $\lambda^2 = \Theta_\Phi$, the preceding calculation shows that $\mathbb{C}x_2$ is stable under $C_G(g^2h)$. It therefore defines a one-dimensional Φ_{g^2h} -projective representation σ_2 of $C_G(g^2h)$. Since $X_2 = \mathbb{C}G \triangleright x_2$, the assignment $1 \otimes x_2 \mapsto x_2$ induces a surjective morphism

$$M(g^2h, \sigma_2) \longrightarrow X_2.$$

The source is simple, and the morphism is nonzero. Hence it is an isomorphism. Thus X_2 is simple.

Conversely, if X_2 is simple, its g^2h -homogeneous component is the inducing representation τ in an isomorphism $X_2 \simeq M(g^2h, \tau)$. Lemma 4.5 gives $\dim \tau = 1$, and this component must be stable under h . The preceding calculation then forces $\lambda^2 = \Theta_\Phi$. $\square$

4.2.3 The third adjoint object

For the calculation of the third adjoint object, define

$$\kappa_3 := \frac{\zeta \Phi(g, \varepsilon g, \varepsilon gh) \Phi_g(g, h)}{\Phi_g(h, g) \Phi(\varepsilon g, g, \varepsilon gh) \Phi_g(\varepsilon, g) \Phi_g(h, \varepsilon g) \Phi^g(g, gh)}. \quad (4.18)$$

Lemma 4.11. Assume that X_1 and X_2 are simple and that $X_2 \neq 0$. Let $x_2 = \varphi_2^\Phi(v \otimes x_1)$ be as in Lemma 4.8, and set $x_3 := \varphi_3^\Phi(v \otimes x_2)$. Then

$$x_3 = (1 + \lambda + \lambda^2) (v \otimes x_2 - \kappa_3 hv \otimes (g \triangleright x_2)). \quad (4.19)$$

Moreover, $X_3 = \mathbb{C}G \triangleright x_3$. Consequently,

$$X_3 = 0 \iff 1 + \lambda + \lambda^2 = 0. \quad (4.20)$$

Proof. The same orbit-generation argument gives $X_3 = \mathbb{C}G \triangleright x_3$.

Set $S_0 := v \otimes x_2, S_1 := hv \otimes (g \triangleright x_2)$. On S_0 , the three terms in the recursive definition of φ_3^Φ are as follows. First,

$$c_{X_2, V} c_{V, X_2}(S_0) = d_3 S_1, \quad d_3 = \frac{\zeta \lambda^2}{\Phi_g(\varepsilon, g) \Phi_g(\varepsilon g, g) \Phi_g(h, \varepsilon g^2)}. \quad (4.21)$$

Write $p = \Phi_g(h, g)^{-1}$ and use the formula for x_2 in Lemma 4.8. The first summand gives $(\text{id}_V \otimes \varphi_2^\Phi) c_{1,2}^\Phi(v \otimes (v \otimes x_1)) = \lambda v \otimes x_2$. For the second summand, the associator corrections in $c_{1,2}^\Phi$ give

$$c_{1,2}^\Phi(v \otimes (hv \otimes (g \triangleright x_1))) = \frac{\Phi(g, \varepsilon g, \varepsilon gh)}{\Phi(\varepsilon g, g, \varepsilon gh)} (g \triangleright hv) \otimes (v \otimes (g \triangleright x_1)).$$

Since φ_2^Φ is a morphism and $g \triangleright (v \otimes x_1) = \lambda \Phi^g(g, gh) v \otimes (g \triangleright x_1)$, we have

$$\varphi_2^\Phi(v \otimes (g \triangleright x_1)) = \frac{1}{\lambda \Phi^g(g, gh)} g \triangleright x_2.$$

Using also $g \triangleright hv = \frac{\Phi_g(g, h) \lambda \zeta}{\Phi_g(\varepsilon, g) \Phi_g(h, \varepsilon g)} hv$, we obtain

$$(\text{id}_V \otimes \varphi_2^\Phi) c_{1,2}^\Phi(S_0) = \lambda(1 + \lambda) S_0 - (1 + \lambda) \kappa_3 S_1. \quad (4.22)$$

The cocycle identity in Lemma B.4 gives $d_3 = \lambda^2 \kappa_3$. Substituting (4.21), (4.22), and this equality into the recursive formula for φ_3^Φ proves (4.19).

Finally, S_0 and S_1 belong to the distinct outer bi-homogeneous components $V_g \otimes (X_2)_{g^2h}$ and $V_{\varepsilon g} \otimes (X_2)_{\varepsilon g^2h}$, respectively. Hence the vector in square brackets in (4.19) is nonzero. Equations $X_3 = \mathbb{C}G \triangleright \varphi_3^\Phi(v \otimes x_2)$ and (4.19) now give (4.20). $\square$

4.2.4 Adjoint objects in the opposite order

Since $hg = \varepsilon gh$ and $\varepsilon^2 = 1$, one also has $gh = \varepsilon hg$. The preceding formulas therefore apply to (W, V) , with support representatives (h, g) . Set

$$\Delta' := \frac{\zeta \zeta' \mu \mu' \Phi_g(\varepsilon h, h) \Phi_h(g, g)}{\Phi_h(g, \varepsilon) \Phi_g(\varepsilon, h^2)}. \quad (4.23)$$

Lemma 4.12. The object Y_1 is simple if and only if $\Delta' = 1$. In this case, $Y_1 \simeq M(hg, \tau_1)$ for a one-dimensional projective representation τ_1 of $C_G(hg)$, and $Y_2 = 0$ if and only if $\lambda' = -1$.

Proof. Apply Lemmas 4.7 and 4.9 to the pair (W, V) . Under the simultaneous substitutions $g \leftrightarrow h, \zeta \leftrightarrow \zeta', \lambda \leftrightarrow \lambda',$ and $\mu \leftrightarrow \mu'$, the parameter Δ becomes Δ' . $\square$

4.3 Duals and reflections

By Lemma 2.5, the self-braiding scalars of V^* and W^* on their homogeneous lines of degrees g^{-1} and h^{-1} are λ and λ' , respectively.

For the two reflections, let

$$m_{12} := \max\{m \geq 0 \mid X_m \neq 0\}, \quad m_{21} := \max\{n \geq 0 \mid Y_n \neq 0\}.$$

Suppose that m_{12} and m_{21} are finite. By Lemma 2.16, the last nonzero objects $X_{m_{12}}$ and $Y_{m_{21}}$ are simple, and the corresponding reflections are

$$R_1(V, W) = (V^*, X_{m_{12}}), \quad R_2(V, W) = (Y_{m_{21}}, W^*).$$

For brevity, write $m = m_{12}$ and $n = m_{21}$. Since $X_m \subseteq V^{\otimes m} \otimes W$, its support is contained in $\{g^m h, \varepsilon g^m h\}$. These two elements are conjugate by g . Since $X_m \neq 0$ and its support is stable under conjugation, its support equals this set. Similarly, $\text{supp } Y_n = \{h^n g, \varepsilon h^n g\}$.

Lemma 4.13. *Write $m = m_{12}$ and $n = m_{21}$. For the two reflected tuples, choose support representatives as follows:*

tuple	x	y
$R_1(V, W) = (V^*, X_m)$	g^{-1}	$g^m h$
$R_2(V, W) = (Y_n, W^*)$	$h^n g$	h^{-1}

For either row, the representatives x and y satisfy

$$yx = \varepsilon xy, \quad G = \langle x, y \rangle.$$

Moreover, the supports of the first and second entries are $\{x, \varepsilon x\}$, $\{y, \varepsilon y\}$, respectively. Thus both reflected tuples satisfy the same assumptions on their supports as (V, W) .

Proof. For $R_1(V, W)$, the relation $hg = \varepsilon gh$ gives $(g^m h)g^{-1} = \varepsilon g^{-1}(g^m h)$. Since $g = (g^{-1})^{-1}$, $h = g^{-m}(g^m h)$, the elements g^{-1} and $g^m h$ generate G .

For $R_2(V, W)$, one similarly obtains $h^{-1}(h^n g) = \varepsilon(h^n g)h^{-1}$. Since $h = (h^{-1})^{-1}$, $g = h^{-n}(h^n g)$, the elements $h^n g$ and h^{-1} also generate G .

The relation $yx = \varepsilon xy$, together with $\varepsilon^2 = 1$, gives

$$x^G = \{x, \varepsilon x\}, \quad y^G = \{y, \varepsilon y\}.$$

Since V^* , X_m , Y_n , and W^* are simple, their supports are the corresponding conjugacy classes. $\square$

Lemma 4.14. *Whenever the one-dimensional projective representations σ_1 and τ_1 are defined, one has*

$$\sigma_1(gh) = -\lambda\lambda', \quad \tau_1(hg) = -\lambda\lambda'. \quad (4.24)$$

Proof. Let $a = \zeta/\Phi_g(h, \varepsilon)$, so that $x_1 = v \otimes w - a hv \otimes gw$. Acting by gh on the first summand and comparing the coefficient of $hv \otimes gw$ gives $\sigma_1(gh) = -\lambda\lambda'K_1$, where

$$K_1 := \frac{\Phi^{gh}(g, h)\Phi_g(h, \varepsilon)}{\Phi_h(g, h)\Phi_g(h, \varepsilon g)\Phi_g(\varepsilon, g)}. \quad (4.25)$$

The opposite calculation gives $\tau_1(hg) = -\lambda\lambda'K_1^{\text{op}}$, with

$$K_1^{\text{op}} := \frac{\Phi^{hg}(h, g)\Phi_h(g, \varepsilon)}{\Phi_h(g, \varepsilon h)\Phi_h(\varepsilon, h)\Phi_g(h, g)}. \quad (4.26)$$

By Lemma B.5, $K_1 = K_1^{\text{op}} = 1$. This proves (4.24). $\square$

4.4 The standard case of type A_2

If $P = (P_1, P_2)$ is obtained from (V, W) by a sequence of reflections, we write

$$m_{ij}^P = \max\{r \geq 0 \mid (\text{ad } P_i)^r(P_j) \neq 0\}, \quad i \neq j.$$

Thus the generalized Cartan matrix at $[P]$ is

$$A^{[P]} = \begin{pmatrix} 2 & -m_{12}^P \\ -m_{21}^P & 2 \end{pmatrix}. \quad (4.27)$$

Rigid dual objects are denoted by P_i^* .

For any two objects U, T , the relation

$$c_{U,T}(\text{id}_{U \otimes T} - c_{T,U}c_{U,T}) = (\text{id}_{T \otimes U} - c_{U,T}c_{T,U})c_{U,T}$$

identifies the images of the two first adjoint maps and gives

$$(\text{ad } U)(T) \simeq (\text{ad } T)(U). \quad (4.28)$$

Proposition 4.15. *Assume that X_1 and Y_1 are simple and that $X_2 = Y_2 = 0$. Then (V, W) admits all reflections and $\mathcal{G}(V, W)$ is a standard Cartan graph of type A_2 . In particular, at every object $[P]$ one has*

$$A^{[P]} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \quad (4.29)$$

Proof. The vanishing criteria give $\lambda = \lambda' = -1$. The two reflections are $R_1(V, W) = (V^*, X_1)$ and $R_2(V, W) = (Y_1, W^*)$, and their supports satisfy the same assumptions by Lemma 4.13. For $R_1(V, W)$, [34, Lemma 3.8] gives $(\text{ad } V^*)(X_1) \simeq W$ and $(\text{ad } V^*)^2(X_1) = 0$. Equation (4.28) then gives $(\text{ad } X_1)(V^*) \simeq W$, so this object is simple. By Lemma 4.14, $\sigma_1(gh) = -\lambda\lambda' = -1$. Lemma 4.9, applied to (X_1, V^*) , therefore gives $(\text{ad } X_1)^2(V^*) = 0$.

The same argument applied to $R_2(V, W)$, using $\tau_1(hg) = -\lambda\lambda' = -1$, shows that its first adjoint objects are isomorphic to V and its second adjoint objects vanish. Thus both reflected tuples satisfy the hypotheses of the proposition. Induction on the length of a reflection sequence proves that all reflections are defined and that $m_{12}^P = m_{21}^P = 1$ at every object $[P]$. Equation (4.27) gives (4.29). $\square$

By Lemmas 4.7, 4.9, and 4.12, the hypotheses of Proposition 4.15 may equivalently be written as

$$\Delta = 1, \quad \Delta' = 1, \quad \lambda = \lambda' = -1. \quad (4.30)$$

Thus the conditions in (4.30) give a standard Cartan graph of type A_2 .

4.5 Exclusion of type B_2

The Sylow decomposition of G restricts the order of Θ_Φ and excludes type B_2 . For a positive integer N , let $\mu_N = \{z \in \mathbb{C}^\times \mid z^N = 1\}$.

Lemma 4.16. *Let P be the Sylow 2-subgroup of G , and let A be the product of the Sylow subgroups of odd order. Then $G = P \rtimes A$, $\varepsilon \in P$, and A is abelian. Every normalized 3-cocycle on G is cohomologous to one of the form*

$$\tilde{\Phi} = p_P^*(\Phi_P)p_A^*(\Phi_A), \quad (4.31)$$

where Φ_P may be chosen to take values in $\mu_{|P|}$. For this representative, Φ_A is a normalized 3-cocycle on the abelian group A .

Proof. The defining relations give $[G, G] = \langle \varepsilon \rangle \subseteq Z(G)$. Thus G is nilpotent. Since G is finite, it is the direct product of its Sylow subgroups, and hence $G = P \times A$. The group A has odd order, whereas $[G, G]$ has order two. Therefore $[A, A] \subseteq [G, G] \cap A = 1$, so A is abelian.

Restriction to P and A , together with inflation along the two projections, gives

$$H^3(G, \mathbb{C}^\times) \simeq H^3(P, \mathbb{C}^\times) \oplus H^3(A, \mathbb{C}^\times).$$

To prove this, let $\alpha \in H^3(G, \mathbb{C}^\times)$, and subtract from α the inflations of its restrictions to P and A . Call the resulting class β . Its restrictions to both P and A are zero. The restriction-corestriction formula then gives $|A|\beta = 0$ and $|P|\beta = 0$. Since $|A|$ and $|P|$ are coprime, $\beta = 0$. Thus every class is the sum of classes inflated from P and A . This decomposition is unique because restriction recovers the two classes. It gives the cocycle $\tilde{\Phi}$ in (4.31).

Finally, $H^3(P, \mathbb{C}^\times)$ is annihilated by $|P|$. The long exact sequence associated with

$$1 \longrightarrow \mu_{|P|} \longrightarrow \mathbb{C}^\times \xrightarrow{z \mapsto z^{|P|}} \mathbb{C}^\times \longrightarrow 1$$

therefore shows that every class in $H^3(P, \mathbb{C}^\times)$ has a representative with values in $\mu_{|P|}$. Hence Φ_P may be chosen to be $\mu_{|P|}$ -valued. $\square$

Lemma 4.17. *For the split representative in (4.31), one has*

$$\Theta_{\tilde{\Phi}} \in \mu_{|P|}. \quad (4.32)$$

In particular, the order of $\Theta_{\tilde{\Phi}}$ is a power of two.

Proof. The definitions of Φ_x , Φ^x , and Θ_Φ are multiplicative in the 3-cocycle. Hence $\Theta_{\tilde{\Phi}} = \Theta_{\Phi_P} \Theta_{\Phi_A}$. The first factor belongs to $\mu_{|P|}$. For the contribution of A , write g_A, h_A for the projections of g, h to A . Since A is abelian and the projection of ε to A is trivial, direct substitution in the definition of Θ_Φ gives

$$\Theta_{\Phi_A} = \left(\frac{(\Phi_A)_{g_A}(h_A, g_A)}{(\Phi_A)_{g_A}(g_A, h_A)} \right)^3 \frac{(\Phi_A)_{h_A}(h_A, g_A)}{(\Phi_A)_{h_A}(g_A, h_A)}.$$

By Lemma 4.4, the right-hand side is $f_{\Phi_A}(g_A, h_A, g_A)^3 f_{\Phi_A}(h_A, h_A, g_A) = 1$. Thus $\Theta_{\tilde{\Phi}} = \Theta_{\Phi_P}$, which proves (4.32). $\square$

Lemma 4.18. *Assume that X_1 and X_2 are simple and that $X_2 \neq 0$. Then $X_3 \neq 0$.*

Proof. Suppose that $X_3 = 0$. Replace Φ by the representative $\tilde{\Phi}$ from Lemma 4.16, and transport (V, W) through the induced braided monoidal equivalence. Write $\tilde{V}, \tilde{W}$ for the transported objects, $\tilde{X}_n = (\text{ad } \tilde{V})^n(\tilde{W})$, and $\tilde{\lambda}$ for the self-braiding scalar of $\tilde{V}$ on its chosen homogeneous line. The equivalence preserves adjoint objects, simplicity and vanishing by [34, Proposition 6.3], so

$$\tilde{X}_1, \tilde{X}_2 \text{ are simple,} \quad \tilde{X}_2 \neq 0, \quad \tilde{X}_3 = 0.$$

The second-adjoint criterion and the third-adjoint formula give

$$\tilde{\lambda}^2 = \Theta_{\tilde{\Phi}}, \quad 1 + \tilde{\lambda} + \tilde{\lambda}^2 = 0.$$

Consequently, $\tilde{\lambda}^2$ has order three. This contradicts Lemma 4.17. Hence $X_3 \neq 0$. $\square$

Proposition 4.19. *If $\mathcal{B}(V \oplus W)$ is finite-dimensional, then no object of $\mathcal{G}(V, W)$ has Cartan matrix of type B_2 , up to transposition.*

Proof. Suppose that $P = (P_1, P_2)$ is such an object. After interchanging the two components if necessary, its Cartan matrix is $\begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}$. By Lemma 4.13, the supports of P again have the form required in the Γ_2 case. Theorem 2.25, applied to (V, W) , shows that its Cartan graph is finite. Since P represents an object of this graph, Lemma 2.24 shows that $(\text{ad } P_1)(P_2)$ and $(\text{ad } P_1)^2(P_2)$ are simple and nonzero, whereas $(\text{ad } P_1)^3(P_2) = 0$. This contradicts Lemma 4.18, applied to P . $\square$

4.6 The third and fourth adjoint objects in type G_2

Define

$$\kappa_4 := \kappa_3 \frac{\Phi(g, \varepsilon g, \varepsilon g^2 h) \Phi_g(g, h) \zeta}{\Phi(\varepsilon g, g, \varepsilon g^2 h) \Phi_g(\varepsilon, g) \Phi_g(h, \varepsilon g) \Phi^g(g, g^2 h)}. \quad (4.33)$$

Lemma 4.20. Assume that X_1, X_2 , and X_3 are simple and nonzero. Let $x_3 = \varphi_3^\Phi(v \otimes x_2)$ be as in Lemma 4.11, and set $x_4 := \varphi_4^\Phi(v \otimes x_3)$. Then

$$x_4 = (1 + \lambda + \lambda^2 + \lambda^3) (v \otimes x_3 - \kappa_4 h v \otimes (g \triangleright x_3)). \quad (4.34)$$

Moreover, $X_4 = \mathbb{C}G \triangleright x_4$. Therefore $X_4 = 0$ if and only if $1 + \lambda + \lambda^2 + \lambda^3 = 0$. Since $X_2 \neq 0$ implies $\lambda \neq -1$, this is equivalent to $\lambda^2 = -1$.

Proof. By the orbit-generation argument in Subsection 4.2,

$$X_4 = \mathbb{C}G \triangleright \varphi_4^\Phi(v \otimes x_3).$$

Set $U_0 = v \otimes x_3$ and $U_1 = h v \otimes (g \triangleright x_3)$. Applying the formula φ_4^Φ gives

$$\begin{aligned} c_{X_3, V} c_{V, X_3}(U_0) &= d_4 U_1, \quad d_4 = \frac{\lambda^3}{\Phi_g(h, g^3) \Phi_g(g, g) \Phi_g(g^2, g)}, \\ (\text{id}_V \otimes \varphi_3^\Phi) c_{1,2}^\Phi(U_0) &= \lambda(1 + \lambda + \lambda^2) U_0 - (1 + \lambda + \lambda^2) \kappa_4 U_1. \end{aligned} \quad (4.35)$$

The cocycle reduction in Lemma B.6 gives $d_4 = \lambda^3 \kappa_4$. Substitution into the recursive formula yields (4.34), because $1 + \lambda(1 + \lambda + \lambda^2) = \lambda^3 + (1 + \lambda + \lambda^2) = 1 + \lambda + \lambda^2 + \lambda^3$. The first tensor factors have distinct degrees g and εg , so U_0 and U_1 are linearly independent. This proves the vanishing criterion. $\square$

Lemma 4.21. Assume that X_1, X_2 and X_3 are simple and nonzero, and write $X_3 \simeq M(g^3 h, \sigma_3)$. Then

$$\sigma_3(g^3 h) = -\lambda^6 \lambda'. \quad (4.36)$$

Proof. Let $q = g^2 h$. Acting by $g^3 h$ on the two components in (4.19) and using relation (2.1) gives $(g^3 h) \triangleright x_3 = -\lambda^6 \lambda' \mathcal{K}_3 x_3$, where

$$\begin{aligned} \mathcal{K}_3 &= \frac{\Phi_g(h, g)^2 \Phi_g(h, \varepsilon) \Phi(\varepsilon g, g, \varepsilon g h) \Phi^g(g, g h)}{\Phi_h(g, h) \Phi_g(h, g^2) \Phi_g(h, \varepsilon g^3) \Phi_g(g, h) \Phi_g(g, g) \Phi_g(\varepsilon, g)} \\ &\quad \times \frac{\Phi^{gh}(g, h) \Phi^q(g, g h) \Phi^{g^3 h}(g, q)}{\Phi_g(\varepsilon g, g) \Phi_g(\varepsilon g^2, g) \Phi_{gh}(g, g h) \Phi_q(g, q) \Phi(g, \varepsilon g, \varepsilon g h)}. \end{aligned} \quad (4.37)$$

Lemma B.6 gives $\mathcal{K}_3 = 1$, proving (4.36). $\square$

4.7 The standard case of type G_2

For $\mathfrak{s}_\Phi(x, y)$ defined in (B.3), Lemma B.7 gives $\mathfrak{s}_\Phi(x, y) = \Theta_\Phi(x, y)^2$.

Lemma 4.22. Assume $\Delta = 1$ and $\lambda^2 = \Theta_\Phi = -1$. Then X_1, X_2, X_3 are simple and nonzero.

Proof. Lemma 4.7 and 4.10 give the simplicity of X_1 and X_2 . Since $\lambda^2 = -1$, one has $1 + \lambda + \lambda^2 = \lambda \neq 0$. Hence the explicit third-adjoint formula gives $x_3 = (1 + \lambda + \lambda^2)(S_0 - \kappa_3 S_1)$, where $S_0 = v \otimes x_2$ and $S_1 = h v \otimes (g \triangleright x_2)$. Moreover, $X_3 = \mathbb{C}G \triangleright x_3$.

Write $q = g^2 h$. A direct application of the projective tensor action gives

$$g \triangleright x_3 = (1 + \lambda + \lambda^2) (A_g v \otimes (g \triangleright x_2) - \kappa_3 B_g h v \otimes x_2), \quad (4.38)$$

$$h \triangleright x_3 = (1 + \lambda + \lambda^2) (A_h h v \otimes x_2 - \kappa_3 B_h v \otimes (g \triangleright x_2)), \quad (4.39)$$

where

$$\begin{aligned} A_g &= \Phi^g(g, q)\lambda, & A_h &= \Phi^h(g, q)\sigma_2(h), \\ B_g &= \Phi^g(\varepsilon g, \varepsilon q) \frac{\Phi_g(g, h)\lambda\zeta}{\Phi_g(\varepsilon, g)\Phi_g(h, \varepsilon g)} \Phi_q(g, g)\sigma_2(g^2), \\ B_h &= \Phi^h(\varepsilon g, \varepsilon q) \Phi_g(h, h)\mu \frac{\Phi_q(h, g)\sigma_2(h)\sigma_2(\varepsilon)}{\Phi_q(\varepsilon g, h)\Phi_q(g, \varepsilon)}. \end{aligned}$$

Here $X_2 \simeq M(q, \sigma_2)$ and

$$\sigma_2(\varepsilon) = \Phi^\varepsilon(g, gh)\Phi^\varepsilon(g, h)\zeta^2\zeta', \quad (4.40)$$

$$\sigma_2(g^2) = \Phi^{g^2}(g, gh)\Phi^{g^2}(g, h) \frac{\lambda^4\mu'}{\Phi_g(g, g)^2}. \quad (4.41)$$

For the degree g^3h , one has $C_G(g^3h) = \langle \varepsilon, g^2, h^2, g^{-1}h \rangle$ by Lemma 4.2. The elements ε, g^2, h^2 belong to $C_G(g) \cap C_G(q)$. The line $\mathbb{C}(v \otimes x_2)$ is therefore stable under these three elements. The map φ_3^Φ commutes with the G -action. Hence, for $a \in \{\varepsilon, g^2, h^2\}$,

$$a \triangleright x_3 = \varphi_3^\Phi(a \triangleright (v \otimes x_2)) \in \mathbb{C}x_3.$$

Thus $\mathbb{C}x_3$ is stable under these three elements. It remains to check stability under $g^{-1}h$. By the invertibility of the projective action, this is equivalent to $\mathbb{C}(g \triangleright x_3) = \mathbb{C}(h \triangleright x_3)$. Equations (4.38) and (4.39) show that this holds if and only if

$$\kappa_3^2 B_g B_h = A_g A_h. \quad (4.42)$$

Lemma 4.5 now shows that the invariance of $\mathbb{C}x_3$ under $C_G(g^3h)$ is equivalent to the simplicity of the object $X_3 = \mathbb{C}G \triangleright x_3$.

Substituting (4.40) and (4.41) into $\kappa_3^2 B_g B_h / (A_g A_h)$ reduces its representation factors to $\zeta^5 \zeta' \mu \mu' \lambda^4$. Using $\lambda^4 = 1$, $\zeta^2 = \Phi_g(\varepsilon, \varepsilon)$, $\Delta = 1$, and the definition of Δ , the quotient becomes $\mathfrak{s}_\Phi(g, h)$. Thus (4.42) is equivalent to $\mathfrak{s}_\Phi(g, h) = 1$.

Lemma B.7 gives $\mathfrak{s}_\Phi(g, h) = \Theta_\Phi(g, h)^2 = 1$. Hence the determinant condition holds and X_3 is simple. $\square$

Lemma 4.23. *The parameters defined above satisfy*

$$\Delta = 1 \iff \Delta' = 1, \quad (4.43)$$

$$\Theta_\Phi(hg, h^{-1}) = \Theta_\Phi(g, h). \quad (4.44)$$

Here hg and h^{-1} are the support representatives chosen for the second reflection.

Proof. Equation (4.28) gives $X_1 \simeq Y_1$. The two established first-adjoint simplicity criteria now give (4.43). Equation (4.44) is the cocycle identity proved in Lemma B.8. $\square$

Proposition 4.24. *Assume*

$$\Delta = 1, \quad \lambda' = -1, \quad \lambda^2 = \Theta_\Phi(g, h) = -1. \quad (4.45)$$

Then (V, W) admits all reflections and its Cartan graph is standard of type G_2 .

Proof. At the initial object, the first two conditions and (4.43) make X_1 and Y_1 simple and give $Y_2 = 0$. Lemma 4.10 and Lemma 4.22 make X_2 and X_3 simple and nonzero. Since $1 + \lambda + \lambda^2 + \lambda^3 = 0$, (4.34) gives $X_4 = 0$. Thus both initial reflections are defined and the initial Cartan matrix is of type G_2 .

Let $Q_1 = R_1(V, W)$, so that $Q_1 = (V^*, X_3)$. Use the support representatives (g^{-1}, g^3h) from Lemma 4.13. By [34, Lemma 3.8],

$$(\operatorname{ad} V^*)(X_3) \simeq X_2, \quad (\operatorname{ad} V^*)^2(X_3) \simeq X_1.$$

Since X_2 is simple, Lemma 4.7, applied to Q_1 , gives $\Delta_{Q_1} = 1$. Lemma 2.5 gives $\lambda_{Q_1} = \lambda$. In particular, $\lambda_{Q_1} \neq -1$, because $\lambda^2 = -1$. The second displayed isomorphism and Lemma 4.10 therefore give $\Theta_{Q_1} = \lambda_{Q_1}^2 = \lambda^2 = -1$. On the other hand, Lemma 4.21 gives $\lambda'_{Q_1} = \sigma_3(g^3h) = -\lambda^6\lambda' = -1$. Thus Q_1 satisfies all three equations in (4.45).

Let $Q_2 = R_2(V, W) = (Y_1, W^*)$. By [34, Lemma 3.8] and (4.28), $(\operatorname{ad} Y_1)(W^*) \simeq V$, so $\Delta_{Q_2} = 1$. Lemma 4.14 and the cocycle identity in Lemma 4.23 give $\lambda_{Q_2} = \tau_1(hg) = -\lambda\lambda' = \lambda$, $\lambda'_{Q_2} = \lambda' = -1$, and $\Theta_{Q_2} = \Theta_\Phi(g, h) = -1$. Thus Q_2 also satisfies (4.45).

After either reflection, the support assumptions and all three parameter equations are preserved. The first paragraph of the proof therefore applies to each reflected tuple. Induction proves that all reflections are defined and that the Cartan graph is standard of type G_2 . $\square$

Corollary 4.25. *The following conditions are equivalent:*

- (1) *The objects X_1, X_2, X_3, Y_1 are simple and nonzero, while $X_4 = Y_2 = 0$.*
- (2) *The equations in (4.45) hold.*

Whenever these conditions hold, the tuple admits all reflections and its Cartan graph is standard of type G_2 .

Proof. Assume (1). Lemmas 4.7 and 4.12 give $\Delta = \Delta' = 1$. Since $Y_2 = 0$, Lemma 4.12 gives $\lambda' = -1$. Since $X_2 \neq 0$, Lemma 4.9 gives $\lambda \neq -1$. The simplicity of X_2 and Lemma 4.10 then give $\lambda^2 = \Theta_\Phi(g, h)$. Finally, Lemma 4.20 and $X_4 = 0$ give $(1 + \lambda)(1 + \lambda^2) = 0$. Since $\lambda \neq -1$, it follows that $\lambda^2 = -1$. Thus $\lambda^2 = \Theta_\Phi(g, h) = -1$, and (2) holds.

The implication (2) $\Rightarrow$ (1) was proved in the first paragraph of the proof of Proposition 4.24. The same proposition gives the assertion about reflections and the Cartan graph. $\square$

4.8 Finite Cartan graphs

Proposition 4.26. *Assume that (V, W) is braided-indecomposable and that $\dim \mathcal{B}(V \oplus W) < \infty$. Then its Cartan graph is standard of type A_2 or standard of type G_2 , up to interchanging the two indices.*

Proof. By Theorem 2.25, finite-dimensionality gives all reflections and a finite Cartan graph. It is connected by Proposition 2.18. By Lemma 2.27, there exists an object $P = (P_1, P_2)$ whose Cartan matrix is of finite type. The tuple P again satisfies the standing Γ_2 support assumptions, and all nonzero objects obtained by the two iterated braided adjoint actions are simple by Lemma 2.24. The possible indecomposable rank-two Cartan matrices of finite type are A_2, B_2 , and G_2 , up to transposition.

If the matrix at P is of type A_2 , Proposition 4.15, applied to P , shows that the entire connected graph is standard of type A_2 . Type B_2 is impossible by Proposition 4.19. Finally, if the matrix at P is of type G_2 , interchange the entries of P , if necessary, so that its matrix is $\begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$. Corollary 4.25, applied to P , shows that the entire graph is standard of type G_2 . $\square$

4.9 Nichols algebras of the positive-root factors of type G_2

Let $R = M(x, \tau)$ be a simple object considered below, with $\operatorname{supp} R = \{x, \varepsilon x\}$ and one-dimensional homogeneous components. Choose nonzero vectors $u_0 \in R_x$ and $u_1 \in R_{\varepsilon x}$, and write

$$c_{R,R}(u_i \otimes u_j) = q_{ij}^R u_j \otimes u_i.$$

Thus $q_{01}^R q_{10}^R$ is the edge label of the generalized Dynkin diagram associated with R .

We use the following special case of the abelian reduction in [28], which applies to any object with two commuting support elements and one-dimensional homogeneous components. Restrict the action to the subgroup generated by the support. Give the two homogeneous lines the degrees of the standard generators of $\mathbb{Z}^2$, and let $\mathbb{Z}^2$ act through the homomorphism sending these generators to the two original degrees. Pulling back Φ along this homomorphism gives a Yetter–Drinfeld object on the same vector space, since its defining identities follow from those of the original object. On every tensor power, the associators and braidings are unchanged as linear maps.

The classifying space of $\mathbb{Z}^2$ is a two-dimensional torus, so $H^3(\mathbb{Z}^2, \mathbb{C}^\times) = 0$. A normalized two-cochain therefore trivializes the pulled-back cocycle and gives a braided monoidal equivalence to the ordinary Yetter–Drinfeld category. Its tensor constraints conjugate the braid operators, and hence the quantum symmetrizers, in every degree. Thus the resulting ordinary diagonal Nichols algebra has the same Hilbert series. Since these constraints are scalar on each ordered tensor line, q_{ii}^R and $q_{01}^R q_{10}^R$ are unchanged. We may therefore use the classification of ordinary Nichols algebras of diagonal type.

Lemma 4.27. *Let $P = (R, S)$ be an ordered Γ_2 -tuple with chosen support representatives (x, y) , and write $R = M(x, \tau)$. If Θ_P denotes the scalar obtained from (4.15) using the support representatives (x, y) in this order, then*

$$q_{00}^R = q_{11}^R = \tau(x), \quad q_{01}^R q_{10}^R = \frac{\tau(x)^2}{\Theta_P}. \quad (4.46)$$

Proof. Choose $0 \neq u_0 \in R_x$ and let $u_1 = y \triangleright u_0 \in R_{\varepsilon x}$. Then u_0, u_1 is a homogeneous basis of R . Since $x \triangleright u_0 = \tau(x)u_0$, the definition of the braiding gives $q_{00}^R = \tau(x)$. The braiding commutes with the G -action. Apply y to $c_{R,R}(u_0 \otimes u_0) = \tau(x)u_0 \otimes u_0$. Since $y \triangleright (u_0 \otimes u_0)$ is a nonzero scalar multiple of $u_1 \otimes u_1$, it follows that $c_{R,R}(u_1 \otimes u_1) = \tau(x)u_1 \otimes u_1$. Thus $q_{11}^R = q_{00}^R = \tau(x)$. The one-dimensional relation (2.1) gives

$$q_{10}^R = \frac{\tau(x)\tau(\varepsilon)}{\Phi_x(\varepsilon, x)}, \quad q_{01}^R = \frac{\tau(x)\tau(\varepsilon)\Phi_x(x, y)}{\Phi_x(\varepsilon, x)\Phi_x(y, \varepsilon x)}.$$

Since $\tau(\varepsilon)^2 = \Phi_x(\varepsilon, \varepsilon)$, their product is

$$q_{01}^R q_{10}^R = \tau(x)^2 \frac{\Phi_x(\varepsilon, \varepsilon)\Phi_x(x, y)}{\Phi_x(\varepsilon, x)^2 \Phi_x(y, \varepsilon x)}.$$

Lemma B.10 identifies the last fraction with Θ_P^{-1} , proving (4.46). $\square$

Lemma 4.28. *Assume that X_1 and X_2 are simple and nonzero, and write $X_2 \simeq M(g^2 h, \sigma_2)$. Then*

$$\sigma_2(g^2 h) = \lambda^3 \lambda'. \quad (4.47)$$

Proof. Acting by $g^2 h$ on the two outer bi-homogeneous components in the formula for x_2 gives

$$\sigma_2(g^2 h) = \lambda^3 \lambda' K_2,$$

where

$$K_2 := \frac{\Phi_g(h, g)\Phi_g(h, \varepsilon)\Phi^{g^h}(g, h)\Phi^{g^2 h}(g, gh)}{\Phi_h(g, h)\Phi_g(h, g^2)\Phi_g(h, \varepsilon g)\Phi_g(g, g)\Phi_g(\varepsilon, g)\Phi_{gh}(g, gh)}. \quad (4.48)$$

Lemma B.9 gives $K_2 = 1$, proving (4.47). $\square$

The simple factors in the tensor decomposition are given explicitly below.

Proposition 4.29. Assume that (V, W) admits all reflections and that its Cartan graph is standard with Cartan matrix

$$\begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}.$$

Set $H = (\text{ad } Y_1)^3(W^*)$. The simple objects corresponding to the six positive roots of G_2 have the following support representatives and braiding coefficients:

root	object	support representative	$q_{00} = q_{11}$	$q_{01}q_{10}$
α_1	V	g	λ	1
α_2	W	h	-1	-1
$\alpha_1 + \alpha_2$	X_1	gh	λ	1
$2\alpha_1 + \alpha_2$	X_2	g^2h	λ	1
$3\alpha_1 + \alpha_2$	X_3	g^3h	-1	-1
$3\alpha_1 + 2\alpha_2$	H	g^3h^2	-1	-1

(4.49)

Consequently,

$$\begin{aligned} \dim \mathcal{B}(R) &= 16 \quad \text{for } R \in \{V, X_1, X_2\}, \\ \dim \mathcal{B}(R) &= 8 \quad \text{for } R \in \{W, X_3, H\}. \end{aligned} \tag{4.50}$$

Proof. Proposition 2.22 gives a tensor decomposition of $\mathcal{B}(V \oplus W)$ indexed by the six positive roots of G_2 . The corresponding simple objects are identified by Lemmas 2.23 and 2.24.

For the chosen Cartan matrix, X_1, X_2, X_3 , and Y_1 are simple and nonzero, while $X_4 = Y_2 = 0$. Corollary 4.25 therefore gives $\lambda^2 = \Theta_\Phi(g, h) = -1$ and $\lambda' = -1$. In particular, λ is a primitive fourth root of unity.

We use gh , g^2h , and g^3h as support representatives of X_1, X_2 , and X_3 . Equations (4.24), (4.47), and (4.36) give, respectively, $\sigma_1(gh) = -\lambda\lambda' = \lambda$, $\sigma_2(g^2h) = \lambda^3\lambda' = \lambda$, and $\sigma_3(g^3h) = -\lambda^6\lambda' = -1$. Lemma 4.27 shows that each of these values is equal to both diagonal coefficients q_{00} and q_{11} . Together with the values for V and W , this proves the fourth column of (4.49), except for the highest root.

Let $Q = R_2(V, W) = (Y_1, W^*)$. By Lemma 4.13, we may use (hg, h^{-1}) as the support representatives of Q . Let λ_Q and λ'_Q denote the corresponding values of the two projective representations. Equation (4.24) and Lemma 2.5 give $\lambda_Q = \tau_1(hg) = \lambda$ and $\lambda'_Q = -1$.

At Q , the third adjoint object in the first direction is $H = (\text{ad } Y_1)^3(W^*)$, and $(hg)^3h^{-1} = g^3h^2$ is a support representative of H . Applying (4.36) to Q gives $\sigma_3^Q(g^3h^2) = -\lambda_Q^6\lambda'_Q = -1$. Moreover, $s_2(3\alpha_1 + \alpha_2) = 3\alpha_1 + 2\alpha_2$. Thus H corresponds to the highest root. Lemma 4.27 gives $q_{00} = q_{11} = -1$ for H .

It remains to compute the last column of the table. Let $P = (P_s, P_\ell)$ be any tuple obtained from (V, W) by reflections, where P_s and P_ℓ correspond to a short and a long simple root, respectively. By Lemma 4.13, the results proved above for the Γ_2 case also apply to P . Write λ_P, λ'_P , and Θ_P for its corresponding parameters. Corollary 4.25 gives $\lambda_P^2 = \Theta_P = -1$ and $\lambda'_P = -1$. Lemma B.11 shows that Θ_P is still -1 when the two entries are interchanged. Formula (4.46), applied in the two orders, now gives $q_{01}^{P_s}q_{10}^{P_s} = 1$ and $q_{01}^{P_\ell}q_{10}^{P_\ell} = -1$. Every object in the table occurs in such a reflected tuple by Lemmas 2.23 and 2.24. This proves the last column.

Finally, for V, X_1 , and X_2 , the diagonal braiding has two disconnected vertices labelled by λ . Since λ has order four, the corresponding Nichols algebra has dimension $4^2 = 16$. For W, X_3 , and H , both vertex labels and the product of the off-diagonal coefficients are -1 . This is the diagonal case of type A_2 . Its three root vectors square to zero, so the dimension is $2^3 = 8$. This proves (4.50). $\square$

Corollary 4.30. Under the hypotheses of Proposition 4.29, one has

$$\dim \mathcal{B}(V \oplus W) = 16^3 8^3 = 2^{21}.$$

Proof. A standard Cartan graph of type G_2 is finite. Proposition 2.22 therefore gives a tensor decomposition with the six factors in (4.49). Their Nichols algebra dimensions are given by (4.50). Multiplying these dimensions gives $\dim \mathcal{B}(V \oplus W) = 16^3 8^3 = 2^{21}$. $\square$

Remark 4.31. The Γ_2 - G_2 case obtained above does not occur when $\Phi = 1$. Indeed, in this case all the cocycle terms in (4.15) are equal to 1, and hence $\Theta_1(g, h) = 1$. On the other hand, the G_2 conditions require $\Theta_\Phi(g, h) = -1$. Thus this case has no counterpart in the ordinary Yetter–Drinfeld category ${}^G_G\mathcal{YD}$ over the Hopf algebra $\mathbb{C}G$. In fact, in the ordinary Γ_2 case, a finite-dimensional Nichols algebra under the standing assumptions has Cartan type A_2 , see [20, Theorem 4.6].

4.10 The A_2 factors and completion of the classification

For type A_2 , finite-dimensionality reduces to the following three simple factors.

Lemma 4.32. *If $\Delta = \Delta' = 1$ and $\lambda = \lambda' = -1$, then $\mathcal{B}(V)$, $\mathcal{B}(W)$ and $\mathcal{B}(X_1)$ are finite-dimensional.*

Proof. Let $R \in \{V, W, X_1\}$, and write $\text{supp } R = \{x, \varepsilon x\}$. Both homogeneous components of R are one-dimensional. Since $\langle \varepsilon, x \rangle$ is abelian, the reduction in Subsection 4.9 gives an ordinary diagonal Nichols algebra with the same diagonal coefficients, edge label, and Hilbert series.

The two diagonal coefficients are both -1 . For V and W , this follows from $\lambda = \lambda' = -1$. For X_1 , it follows from $\sigma_1(gh) = -\lambda\lambda' = -1$. Let $Q_R = q_{01}^R q_{10}^R$.

We first show that Q_R is a root of unity. Let $n = |G|$ and $\mu_n = \{z \in \mathbb{C}^\times \mid z^n = 1\}$. Since G is finite, $H^3(G, \mathbb{C}^\times)$ is annihilated by n . The exact sequence

$$1 \longrightarrow \mu_n \longrightarrow \mathbb{C}^\times \xrightarrow{z \mapsto z^n} \mathbb{C}^\times \longrightarrow 1$$

therefore shows that the cohomology class of Φ has a normalized representative $\tilde{\Phi}$ with values in μ_n . The braided monoidal equivalence associated with the change from Φ to $\tilde{\Phi}$ preserves the eigenvalues of the double braiding and the dimension of the Nichols algebra. It is therefore enough to work with $\tilde{\Phi}$.

Let $\tilde{\chi}$ be the one-dimensional $(\tilde{\Phi})_x$ -projective representation on the homogeneous component of degree x . If $c \in C_G(x)$ has order m , repeated use of the projective multiplication rule gives

$$\tilde{\chi}(c)^m = \prod_{j=1}^{m-1} (\tilde{\Phi})_x(c, c^j).$$

Every factor on the right belongs to μ_n . Hence $\tilde{\chi}(c)^{mm} = 1$, so $\tilde{\chi}(c)$ is a root of unity. The coefficients of the diagonal braiding are products of values of $\tilde{\Phi}$ and values of these projective representations. Thus Q_R is a root of unity.

If $Q_R = 1$, the diagram is of type $A_1 \times A_1$. Both generators square to zero, and hence $\dim \mathcal{B}(R) = 2^2 = 4$. Suppose that $Q_R \neq 1$, and let $N_R = \text{ord}(Q_R)$. Since both diagonal coefficients are -1 , the generalized Cartan matrix is of type A_2 . The braiding coefficients corresponding to its three positive roots are -1 , -1 , and Q_R . Their orders are therefore 2, 2, and N_R , respectively. Hence $\dim \mathcal{B}(R) = 2 \cdot 2 \cdot N_R = 4N_R$.

Thus $\mathcal{B}(R)$ is finite-dimensional in both cases. $\square$

Corollary 4.33. *Assume (A₂), and let $X_1 = (\text{ad } V)(W)$. For $R \in \{V, X_1, W\}$, let Q_R be the edge label of the two-dimensional diagonal braiding associated with R , and set $N_R = \text{ord}(Q_R)$. Then*

$$\dim \mathcal{B}(V \oplus W) = 64N_V N_{X_1} N_W.$$

Proof. The proof of Lemma 4.32 gives $\dim \mathcal{B}(R) = 4N_R$ for $R \in \{V, X_1, W\}$. Equation (4.43) gives $\Delta' = 1$, so (4.30) holds. Proposition 4.15 gives the standard A_2 Cartan graph, while Proposition 2.22 and Lemma 2.24 give the corresponding tensor decomposition with factors V, X_1, W . Multiplying their dimensions proves the formula. $\square$

Proof of Theorem 4.1. Assume that $\mathcal{B}(V \oplus W)$ is finite-dimensional. By Proposition 4.26, after interchanging the two entries if necessary, its Cartan graph is standard of type A_2 or G_2 .

Suppose first that it is of type A_2 . Then X_1 and Y_1 are simple, while $X_2 = Y_2 = 0$. Lemmas 4.7 and 4.12 give $\Delta = \Delta' = 1$. Lemmas 4.9 and 4.12 give $\lambda = \lambda' = -1$. Hence (A₂) holds.

Suppose next that the Cartan graph is of type G_2 . Interchange (g, V, ρ) and (h, W, σ) , if necessary, so that the Cartan matrix is $\begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$. Then X_1, X_2, X_3 , and Y_1 are simple and nonzero, while $X_4 = Y_2 = 0$. Corollary 4.25 now gives (Γ_2 - G_2).

Conversely, assume first that (A₂) holds. Corollary 4.33 shows that $\mathcal{B}(V \oplus W)$ is finite-dimensional.

Finally, assume that (Γ_2 - G_2) holds. Proposition 4.24 shows that the tuple admits all reflections and that its Cartan graph is standard of type G_2 . Corollary 4.30 then gives $\dim \mathcal{B}(V \oplus W) = 2^{21}$. In particular, $\mathcal{B}(V \oplus W)$ is finite-dimensional. $\square$

4.11 An explicit G_2 example over a group of order 16

The following data give an explicit finite quotient of Γ_2 , a normalized 3-cocycle, and two simple objects satisfying (Γ_2 - G_2). The resulting Cartan graph is standard of type G_2 , and the Nichols algebra has dimension 2^{21} .

For $b \in \mathbb{Z}/4\mathbb{Z}$, let $\bar{b} \in \mathbb{Z}/2\mathbb{Z}$ denote its reduction modulo two. For later use, also let $\hat{b} = (b - \bar{b})/2 \in \mathbb{Z}/2\mathbb{Z}$, using the representatives $b \in \{0, 1, 2, 3\}$ and $\bar{b} \in \{0, 1\}$.

Let

$$G = \{[e, a, b] \mid e, a \in \mathbb{Z}/2\mathbb{Z}, b \in \mathbb{Z}/4\mathbb{Z}\}.$$

Define multiplication by

$$[e, a, b][f, c, d] = [e + f + \bar{b}c, a + c, b + d], \quad (4.51)$$

where the first two coordinates are computed modulo two and the third modulo four.

This multiplication is associative. Indeed, only the first coordinate requires verification. For a third element $[r, u, v]$, associativity follows from $\bar{b}c + \bar{b} + \bar{d}u = \bar{d}u + \bar{b}(c + u)$ in $\mathbb{F}_2$, since $\bar{b} + \bar{d} = \bar{b} + \bar{d}$. The identity element is $[0, 0, 0]$, and $[e, a, b]^{-1} = [e + \bar{b}a, a, -b]$. Thus G is a group of order $2 \cdot 2 \cdot 4 = 16$.

Set $\varepsilon = [1, 0, 0]$, $g = [0, 1, 0]$, and $h = [0, 0, 1]$. A direct calculation using (4.51) gives $hg = \varepsilon gh$, $\varepsilon^2 = g^2 = h^4 = 1$, and $\varepsilon \in Z(G)$. Moreover, $\varepsilon = hgh^{-1}g^{-1}$, and every element of G can be written as $[e, a, b] = \varepsilon^e g^a h^b$. Hence g and h generate G . Since $\varepsilon \neq 1$, the relation $hg = \varepsilon gh$ also shows that G is non-abelian. Therefore the assignment of the generators of Γ_2 to ε, g, h defines a surjective homomorphism $\Gamma_2 \twoheadrightarrow G$.

For $x = [e, a, b]$, $y = [f, c, d]$ and $z = [r, u, t]$, define

$$\Phi(x, y, z) = (-1)^{\omega(x, y, z)}, \quad \omega(x, y, z) = c(au + \bar{t}(e + a\bar{b} + a\bar{d}) + a\hat{t}). \quad (4.52)$$

All exponents in the definition of ω are evaluated in $\mathbb{F}_2$.

Lemma 4.34. *The map Φ in (4.52) is a normalized 3-cocycle on G .*

Proof. All exponents in this proof are computed in $\mathbb{F}_2$. If one of the three arguments is the identity element, every term in the formula for ω is zero. Hence $\Phi = (-1)^\omega$ is normalized.

For $x = [e_x, a_x, b_x]$ and $y = [e_y, a_y, b_y]$, write $xy = [e_{xy}, a_{xy}, b_{xy}]$. The multiplication in (4.51) gives

$$\begin{aligned} e_{xy} &= e_x + e_y + \bar{b}_x a_y, & a_{xy} &= a_x + a_y, \\ \bar{b}_{xy} &= \bar{b}_x + \bar{b}_y, & \hat{b}_{xy} &= \hat{b}_x + \hat{b}_y + \bar{b}_x \bar{b}_y. \end{aligned} \quad (4.53)$$

The last equality records the carry that occurs when b_x and b_y are added modulo four.

Write $x_j = [e_j, a_j, b_j]$ for $0 \leq j \leq 3$. By (4.52), one has $\omega = \omega_1 + \cdots + \omega_5$, where

$$\begin{aligned} \omega_1(x_0, x_1, x_2) &= a_0 a_1 a_2, & \omega_2(x_0, x_1, x_2) &= e_0 a_1 \bar{b}_2, \\ \omega_3(x_0, x_1, x_2) &= a_0 \bar{b}_0 a_1 \bar{b}_2, & \omega_4(x_0, x_1, x_2) &= a_0 a_1 \bar{b}_1 \bar{b}_2, \\ \omega_5(x_0, x_1, x_2) &= a_0 a_1 \hat{b}_2. \end{aligned}$$

Substitution of (4.53) into the inhomogeneous coboundary formula gives

$$\begin{aligned} \delta\omega_1 &= 0, \\ \delta\omega_2 &= a_1 a_2 \bar{b}_0 \bar{b}_3, \\ \delta\omega_3 &= a_0 a_2 \bar{b}_1 \bar{b}_3 + a_1 a_2 \bar{b}_0 \bar{b}_3, \\ \delta\omega_4 &= a_0 a_1 \bar{b}_2 \bar{b}_3 + a_0 a_2 \bar{b}_1 \bar{b}_3, \\ \delta\omega_5 &= a_0 a_1 \bar{b}_2 \bar{b}_3, \end{aligned}$$

where each coboundary is evaluated at (x_0, x_1, x_2, x_3) . Adding these identities gives $\delta\omega = 0$. Thus $\Phi = (-1)^\omega$ is a normalized 3-cocycle. $\square$

The relevant centralizers are

$$C_G(g) = \{[e, a, 2j]\}, \quad C_G(h) = \{[e, 0, b]\}.$$

Here $e, a, j \in \mathbb{Z}/2\mathbb{Z}$ and $b \in \mathbb{Z}/4\mathbb{Z}$. Define

$$\rho([e, a, 2j]) = i^a, \quad \sigma([e, 0, b]) = (-1)^{e+\bar{b}}. \quad (4.54)$$

Direct substitution into the fixed definition of Φ_x gives

$$\begin{aligned} \Phi_g([e, a, 2j], [f, c, 2k]) &= (-1)^{ac}, \\ \Phi_h([e, 0, b], [f, 0, d]) &= 1. \end{aligned}$$

It follows at once that ρ and σ are respectively one-dimensional Φ_g - and Φ_h -projective representations. Set $V = M(g, \rho)$ and $W = M(h, \sigma)$.

The parameters of the initial pair and the resulting scalar values are

$$(\zeta, \lambda, \mu) = (1, i, 1), \quad (\zeta', \lambda', \mu') = (-1, -1, 1), \quad (4.55)$$

$$\Delta = \Delta' = 1, \quad \Theta_\Phi = -1 = \lambda^2. \quad (4.56)$$

Thus the three equations in $(\Gamma_2\text{-}G_2)$ hold, and the following proposition applies.

Proposition 4.35. *The pair (V, W) defined by (4.52) and (4.54) admits all reflections and generates a standard Cartan graph of type G_2 .*

Proof. Equations (4.55) and (4.56) give precisely the three conditions in (4.45). The assertion follows from Proposition 4.24. $\square$

Corollary 4.30 now gives

$$\dim \mathcal{B}(V \oplus W) = 16^3 8^3 = 2^{21}.$$

Hence this example realizes the G_2 alternative of Theorem 4.1.

5 The Γ_4 case

Let

$$\Gamma_4 = \langle g, h, \varepsilon \mid hg = \varepsilon gh, \quad g\varepsilon = \varepsilon^{-1}g, \quad h\varepsilon = \varepsilon h, \quad \varepsilon^4 = 1 \rangle,$$

and let G be a finite non-abelian quotient of Γ_4 in which the image of ε has order four. The images of the generators in G are denoted by the same letters. Let Φ be a normalized 3-cocycle on G .

Throughout this section, let $V = M(h, \rho)$ be simple, and let $W = M(g, \sigma)$ be simple. The support of V is $\{h, \varepsilon^{-1}h\}$, and the support of W is $\{g, \varepsilon g, \varepsilon^2 g, \varepsilon^3 g\}$. Assume that (V, W) is braided-indecomposable. Lemma 5.3 below shows that ρ is one-dimensional under these standing assumptions. When σ is one-dimensional, we use the scalar Δ_4 and the seven scalars $\Theta_{2,j}$, defined in (5.3) and (5.9), respectively. They are determined by (Φ, ρ, σ) . The classification for quotients of Γ_4 is as follows.

Theorem 5.1. *Under the preceding assumptions, $\dim \mathcal{B}(V \oplus W) < \infty$ if and only if*

$$\begin{aligned} \dim \sigma &= 1, & \rho(h) &= \sigma(g) = -1, & \Delta_4 &= 1, \\ \Theta_{2,j} &= 1 & (1 \leq j \leq 7). \end{aligned} \tag{B_2}$$

In this case, the Cartan graph is standard of type B_2 , with matrix $\begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}$ at every object.

Outline of the proof.

- Subsection 5.1 proves that ρ is one-dimensional under the standing assumptions and that σ is one-dimensional in every finite-dimensional case.
- Subsection 5.2 computes X_1, X_2, Y_1, Y_2 , and Y_3 , and obtains the equations in (B₂).
- Subsection 5.3 proves that these equations are preserved by both reflections.
- Subsection 5.4 shows that every finite-dimensional case has a standard Cartan graph of type B_2 and proves the necessity of the equations in (B₂).
- Subsection 5.5 proves that the Nichols algebras of V, Y_1, Y_2 , and W are finite-dimensional and computes their dimensions. The final subsection proves the classification theorem.

Notation used in this section.

Notation	Defined in	Used to prove
$\rho(h), \sigma(g)$	Values of the inducing projective representations at the chosen support representatives	The equalities $\rho(h) = -1$ and $\sigma(g) = -1$ give $X_2 = 0$ and $Y_3 = 0$, respectively, once the preceding adjoint objects are simple.
Δ_4	(5.3)	The equality $\Delta_4 = 1$ is equivalent to the simplicity of X_1 .
$\Theta_{2,j}, 1 \leq j \leq 7$	(5.9)	Once $\Delta_4 = 1$ and $\sigma(g) = -1$, the seven equalities $\Theta_{2,j} = 1$ are exactly the remaining conditions for the simplicity of Y_2 .

5.1 Projective representations

Since ρ and σ are projective representations of $C_G(h)$ and $C_G(g)$, respectively, we first determine these two centralizers.

Lemma 5.2. *The centralizers $C_G(h)$ and $C_G(g)$ are abelian. More precisely, $C_G(h) = \langle \varepsilon, h, g^2 \rangle$ and $C_G(g) = \langle \varepsilon^2, \varepsilon^{-1}h^2, g \rangle$.*

Proof. The defining relations show that the two subgroups in the statement are abelian and are contained in $C_G(h)$ and $C_G(g)$, respectively. Every element of G has a representative of the form $\varepsilon^i h^j g^r$. Hence the first subgroup has index at most two. The second has index at most four, since G is the union of its cosets represented by $1, \varepsilon, h, \varepsilon h$. Since $|h^G| = 2$ and $|g^G| = 4$, the centralizer index formula gives the asserted equalities. $\square$

For commuting $x, a, b \in G$, let $c_x(a, b) := \Phi_x(a, b) / \Phi_x(b, a)$. By the definition of Φ_x and Lemma 4.4, $c_x(a, b) = f_\Phi(x, a, b)$.

Put $\eta_h = f_\Phi(h, \varepsilon, g^2)$ and $\eta_g = f_\Phi(g, \varepsilon^2, \varepsilon^{-1}h^2)$. For $a = \varepsilon^i h^j g^{2r}$ and $b = \varepsilon^{i'} h^{j'} g^{2r'}$ in $C_G(h)$, and for $a' = (\varepsilon^2)^i (\varepsilon^{-1}h^2)^j g^r$ and $b' = (\varepsilon^2)^{i'} (\varepsilon^{-1}h^2)^{j'} g^{r'}$ in $C_G(g)$, Lemma 4.4 gives

$$c_h(a, b) = \eta_h^{ir' - i'r}, \quad c_g(a', b') = \eta_g^{ij' - i'j}. \quad (5.1)$$

Since $\varepsilon^4 = 1$, the same lemma gives $\eta_g^2 = f_\Phi(g, \varepsilon^4, \varepsilon^{-1}h^2) = 1$. Thus $\eta_g \in \{1, -1\}$.

We first show that ρ is one-dimensional.

Lemma 5.3. *One has $\eta_h = 1$, and hence $\dim \rho = 1$.*

Proof. The defining relations give $ghg^{-1} = \varepsilon^{-1}h$, $g(\varepsilon^{-1}h)g^{-1} = h$, and $hg = g(\varepsilon^{-1}h)$, while $h, \varepsilon^{-1}h, g^2 \in C_G(h)$. Since $\varepsilon^{-1}h \in C_G(h)$, its action preserves V_h . If $m = |\varepsilon^{-1}h|$, repeated use of (2.1) shows that the m -th power of this action is a nonzero scalar multiple of the identity. Hence it is diagonalizable. Choose $0 \neq v \in V_h$ and $\alpha \in \mathbb{C}^\times$ such that $(\varepsilon^{-1}h) \triangleright v = \alpha v$, and let $v_- = g^2 \triangleright v$ and $v_1 = g \triangleright v$.

By (5.1), $c_h(\varepsilon^{-1}h, g^2) = \eta_h^{-1}$. Since $\varepsilon^{-1}h$ and g^2 commute, (2.1) gives

$$(\varepsilon^{-1}h) \triangleright v_- = c_h(\varepsilon^{-1}h, g^2) \alpha v_- = \eta_h^{-1} \alpha v_-.$$

To determine the action on v_1 , compare

$$\begin{aligned} h \triangleright (g \triangleright v) &= \Phi_h(h, g)(hg) \triangleright v, \\ g \triangleright ((\varepsilon^{-1}h) \triangleright v) &= \Phi_h(g, \varepsilon^{-1}h)(g\varepsilon^{-1}h) \triangleright v. \end{aligned}$$

Since $hg = g(\varepsilon^{-1}h)$, this gives $h \triangleright v_1 = \alpha' v_1$, where $\alpha' = \alpha \Phi_h(h, g) / \Phi_h(g, \varepsilon^{-1}h) \in \mathbb{C}^\times$.

Let $c_{V,V}^2 = c_{V,V} \circ c_{V,V}$. The vectors v and v_- have degree h , whereas v_1 has degree $\varepsilon^{-1}h$. The braiding formula therefore gives

$$\begin{aligned} c_{V,V}^2(v \otimes v_1) &= \alpha \alpha' v \otimes v_1, \\ c_{V,V}^2(v_1 \otimes v_-) &= \eta_h^{-1} \alpha \alpha' v_1 \otimes v_-. \end{aligned}$$

Moreover, (2.1) gives $g \triangleright v_1 = \Phi_h(g, g)v_-$. Hence (2.2) gives $g \triangleright (v \otimes v_1) = \kappa v_1 \otimes v_-$, where $\kappa = \Phi^g(h, \varepsilon^{-1}h) \Phi_h(g, g) \in \mathbb{C}^\times$.

Since $c_{V,V}^2$ is a morphism in ${}^G\mathcal{YD}^\Phi$, it commutes with the action of g . The preceding formulas give

$$\begin{aligned} c_{V,V}^2(g \triangleright (v \otimes v_1)) &= \eta_h^{-1} \kappa \alpha \alpha' v_1 \otimes v_-, \\ g \triangleright c_{V,V}^2(v \otimes v_1) &= \kappa \alpha \alpha' v_1 \otimes v_-. \end{aligned}$$

Since $v_1, v_- \neq 0$ and $\kappa \alpha \alpha' \neq 0$, equality of the two expressions gives $\eta_h = 1$. Equation (5.1) then shows that $\mathbb{C}^{\Phi_h} C_G(h)$ is commutative. Its irreducible representation ρ is therefore one-dimensional. $\square$

For σ , we use the finite-dimensionality of $\mathcal{B}(W)$.

Lemma 5.4. *If $\mathcal{B}(W)$ is finite-dimensional, then $\eta_g = 1$, and hence $\dim \sigma = 1$.*

Proof. Assume, to the contrary, that $\eta_g = -1$. Let $K = \langle \varepsilon, g \rangle$. Then $\varepsilon^4 = 1$, $\varepsilon^2 \in Z(K)$, $(\varepsilon g)g = \varepsilon^2 g(\varepsilon g)$, $g^2 = (\varepsilon g)^2$, and $K = \langle g, \varepsilon g \rangle$. Since $\varepsilon^2 \neq 1$, the group K is a finite non-abelian quotient of Γ_2 . Moreover,

$$\begin{aligned} g^K &= \{g, \varepsilon^2 g\}, & C_K(g) &= \langle \varepsilon^2, g \rangle, \\ (\varepsilon g)^K &= \{\varepsilon g, \varepsilon^{-1} g\}, & C_K(\varepsilon g) &= \langle \varepsilon^2, \varepsilon g \rangle. \end{aligned}$$

Since $c_g(\varepsilon^2, \varepsilon^{-1} h^2) = -1$, the projective actions of ε^2 and $\varepsilon^{-1} h^2$ anticommute. The square of the action of ε^2 is a nonzero scalar multiple of the identity, so this action is diagonalizable. Choose $0 \neq v_+ \in W_g$ such that $\varepsilon^2 \triangleright v_+ = \zeta_+ v_+$, and let $v_- = \varepsilon^{-1} h^2 \triangleright v_+$. Direct computation gives $\varepsilon^2 \triangleright v_- = -\zeta_+ v_-$. Thus v_+ and v_- are linearly independent. Set $\zeta_- = -\zeta_+$.

For every $q \in C_G(g)$, one has $c_g(g, q) = f_\Phi(g, g, q) = 1$. Hence g is central in $\mathbb{C}^{\Phi_s} C_G(g)$, and Schur's lemma gives $g \triangleright v_\pm = \lambda v_\pm$ for some $\lambda \in \mathbb{C}^\times$.

The lines $\mathbb{C}v_+$ and $\mathbb{C}v_-$ generate simple subobjects A_+ and A_- of $W|_K$, respectively, supported on g^K . Since $C_K(\varepsilon g) = \langle \varepsilon^2, \varepsilon g \rangle$ and $c_{\varepsilon g}(\varepsilon^2, \varepsilon g) = f_\Phi(\varepsilon g, \varepsilon^2, \varepsilon g) = 1$, the alternating and multiplicative properties of f_Φ show that the commutator bicharacter of $\Phi_{\varepsilon g}$ on $C_K(\varepsilon g)$ is trivial. Therefore $\mathbb{C}^{\Phi_{\varepsilon g}} C_K(\varepsilon g)$ is commutative and semisimple. Choose a one-dimensional projective subrepresentation $\mathbb{C}w \subseteq W_{\varepsilon g}$. It generates a simple subobject $B = \mathbb{C}K \triangleright w$ of $W|_K$, supported on $(\varepsilon g)^K$.

Write

$$\varepsilon^2 \triangleright w = \zeta' w, \quad g^2 \triangleright w = \mu' w.$$

Since $g^2 = (\varepsilon g)^2$, (2.1) gives $(\varepsilon g)^2 \triangleright v_\pm = \lambda^2 v_\pm / \Phi_g(g, g)$. By (4.2), the scalar associated with the tuple $(A_\pm, B)$ is

$$\Delta_\pm = \frac{\zeta_\pm \zeta' \lambda^2 \mu' \Phi_{\varepsilon g}(\varepsilon^2 g, g) \Phi_g(\varepsilon g, \varepsilon g)}{\Phi_g(g, g) \Phi_g(\varepsilon g, \varepsilon^2) \Phi_{\varepsilon g}(\varepsilon^2, g^2)}.$$

All factors except $\zeta_\pm$ are independent of the sign. Since $\zeta_- = -\zeta_+$, one has

$$\Delta_- = -\Delta_+. \quad (5.2)$$

Since K is finite, its twisted Yetter–Drinfeld category is semisimple, so each $A_\pm \oplus B$ is a braided direct summand of $W|_K$. Restriction to K leaves the braidings and associators on tensor powers of W unchanged. Hence $\mathcal{B}(A_\pm \oplus B)$ embeds into $\mathcal{B}(W)$, so both Nichols algebras are finite-dimensional.

On $A_{\pm, g} \otimes B_{\varepsilon g}$, the double braiding has image in $A_{\pm, \varepsilon^2 g} \otimes B_{\varepsilon^{-1} g}$. Since $\varepsilon^2 \neq 1$, these are distinct bi-homogeneous components. Thus both tuples $(A_\pm, B)$ are braided-indecomposable. Theorem 4.1, applied to each tuple, gives either $\Delta_\pm = 1$ or $\Delta'_\pm = 1$. In the second case, (4.43) also gives $\Delta_\pm = 1$. Consequently, $\Delta_+ = \Delta_- = 1$, contrary to (5.2). Hence $\eta_g \neq -1$. Since $\eta_g \in \{1, -1\}$, it follows that $\eta_g = 1$. Equation (5.1) then shows that $\mathbb{C}^{\Phi_s} C_G(g)$ is commutative, and hence $\dim \sigma = 1$. $\square$

Corollary 5.5. *If $\mathcal{B}(V \oplus W)$ is finite-dimensional, then $\dim \rho = \dim \sigma = 1$. Consequently, $\dim V = 2$ and $\dim W = 4$.*

Proof. The Nichols algebra $\mathcal{B}(W)$ embeds into $\mathcal{B}(V \oplus W)$, and hence it is finite-dimensional. Lemmas 5.3 and 5.4 give $\dim \rho = \dim \sigma = 1$. Finally, $|h^G| = 2$ and $|g^G| = 4$, so $\dim V = 2$ and $\dim W = 4$. $\square$

5.2 Adjoint objects

Throughout this subsection, assume that ρ and σ are one-dimensional. The elements ε^2 , g^2 , and $\varepsilon^{-1}h^2$ are central in G . Choose $0 \neq v \in V_h$ and $0 \neq w \in W_g$, and set $v_1 = g \triangleright v$, $w_1 = h \triangleright w$, and $w_2 = \varepsilon \triangleright w$.

5.2.1 The objects X_1 and X_2

We first record the centralizer fact needed to decide when X_1 is simple.

Lemma 5.6. *One has $C_G(hg) = \langle \varepsilon^2, \varepsilon^{-1}h^2, hg \rangle$, and this group is abelian. Moreover, every irreducible Φ_{hg} -projective representation of $C_G(hg)$ is one-dimensional.*

Proof. The defining relations give $(hg)^G = \{\varepsilon^i hg \mid 0 \leq i < 4\}$ and show that $\langle \varepsilon^2, \varepsilon^{-1}h^2, hg \rangle$ is an abelian subgroup of $C_G(hg)$. The identities $g^2 = \varepsilon h^{-2}(hg)^2$, $g = h^{-1}(hg)$, and $h^2 = \varepsilon(\varepsilon^{-1}h^2)$, together with $\varepsilon^2 \in \langle \varepsilon^2, \varepsilon^{-1}h^2, hg \rangle$, show that every normal-form representative lies in one of its four cosets represented by $1, \varepsilon, h, \varepsilon h$. Hence $[G : \langle \varepsilon^2, \varepsilon^{-1}h^2, hg \rangle] \leq 4$, whereas $[G : C_G(hg)] = |(hg)^G| = 4$. Since $\langle \varepsilon^2, \varepsilon^{-1}h^2, hg \rangle \subseteq C_G(hg)$, equality follows, and therefore $C_G(hg) = \langle \varepsilon^2, \varepsilon^{-1}h^2, hg \rangle$.

Since ε^2 and $\varepsilon^{-1}h^2$ are central, Lemma 4.4 and (5.1) give

$$\begin{aligned} c_{hg}(\varepsilon^2, \varepsilon^{-1}h^2) &= f_\Phi(hg, \varepsilon^2, \varepsilon^{-1}h^2) = f_\Phi(h, \varepsilon^2, \varepsilon^{-1}h^2) f_\Phi(g, \varepsilon^2, \varepsilon^{-1}h^2) \\ &= c_h(\varepsilon^2, \varepsilon^{-1}h^2) c_g(\varepsilon^2, \varepsilon^{-1}h^2) = 1. \end{aligned}$$

Alternation also gives $c_{hg}(hg, \varepsilon^2) = c_{hg}(hg, \varepsilon^{-1}h^2) = 1$. Since c_{hg} is multiplicative in both variables and $\varepsilon^2, \varepsilon^{-1}h^2, hg$ generate $C_G(hg)$, it is trivial on $C_G(hg)$. Thus $\mathbb{C}^{\Phi_{hg}} C_G(hg)$ is commutative, and every irreducible Φ_{hg} -projective representation of $C_G(hg)$ is one-dimensional. $\square$

Define

$$\begin{aligned} \alpha &= \frac{\rho(\varepsilon^{-1})}{\Phi_h(g, \varepsilon^{-1})}, \quad \beta = \alpha \Phi_g(\varepsilon^{-1}h, h) \sigma(\varepsilon^{-1}h^2), \\ \Delta_4 &= \beta \Phi_h(g, g) \rho(g^2) = \Phi_h(g, \varepsilon g) \Phi_g(\varepsilon^{-1}h, h) \rho(\varepsilon^{-1}g^2) \sigma(\varepsilon^{-1}h^2). \end{aligned} \tag{5.3}$$

Lemma 5.7. *The vector*

$$x_1 := \varphi_1^\Phi(v \otimes w) = v \otimes w - \alpha v_1 \otimes w_1 \tag{5.4}$$

is nonzero, and $X_1 = (\text{ad } V)(W) = \mathbb{C}G \triangleright x_1$, with $\text{supp } X_1 = (hg)^G$. Moreover,

$$X_1 \text{ is simple} \iff \Delta_4 = 1.$$

When these conditions hold, $X_1 \simeq M(hg, \tau_1)$ for a one-dimensional Φ_{hg} -projective representation τ_1 , and $\dim X_1 = 4$.

Proof. Let $T = c_{W,V} c_{V,W}$. Since $hgh^{-1} = \varepsilon g$, the braiding formula gives $T(v \otimes w) = ((\varepsilon g) \triangleright v) \otimes (h \triangleright w)$. The relation $g\varepsilon^{-1} = \varepsilon g$, together with (2.1), gives

$$(\varepsilon g) \triangleright v = \frac{\rho(\varepsilon^{-1})}{\Phi_h(g, \varepsilon^{-1})} v_1.$$

Thus $x_1 = (\text{id} - T)(v \otimes w)$ is given by (5.4). Its two summands have distinct bidegrees (h, g) and $(\varepsilon^{-1}h, \varepsilon g)$, and hence $x_1 \neq 0$.

The group $C_G(g)$ stabilizes $\mathbb{C}w$, while g sends the degree h to $\varepsilon^{-1}h$. Since both homogeneous components of V are one-dimensional, it follows that $V = \mathbb{C}C_G(g) \triangleright v$. Proposition 2.10, applied with $X_0 = W$, now gives

$$X_1 = \mathbb{C}G \triangleright x_1.$$

Both summands of x_1 have total degree hg . Therefore $\text{supp } X_1 = (hg)^G$.

It remains to determine when X_1 is simple. The component $(V \otimes W)_{hg}$ has basis $v \otimes w, v_1 \otimes w_1$, and T interchanges the two corresponding basis lines. The projective-action relation and Lemma 2.1 give

$$\begin{aligned}\rho(\varepsilon^{-1})\rho(g^2) &= \Phi_h(\varepsilon^{-1}, g^2)\rho(\varepsilon^{-1}g^2), \\ \Phi_h(g, \varepsilon^{-1})\Phi_h(g, \varepsilon g) &= \Phi_h(g, g)\Phi_h(g^2, \varepsilon^{-1}).\end{aligned}$$

Using also $c_h(g^2, \varepsilon^{-1}) = 1$, a second application of T gives

$$T^2(v \otimes w) = \Phi_h(g, \varepsilon g)\Phi_g(\varepsilon^{-1}h, h)\rho(\varepsilon^{-1}g^2)\sigma(\varepsilon^{-1}h^2)v \otimes w = \Delta_4 v \otimes w.$$

Since T interchanges the two basis lines, the same scalar occurs on $v_1 \otimes w_1$. Hence $T^2 = \Delta_4 \text{id}$ on $(V \otimes W)_{hg}$. Since $\varphi_1^\Phi = \text{id} - T$, one has $(X_1)_{hg} = \text{Im}((\text{id} - T)|_{(V \otimes W)_{hg}})$. If $\Delta_4 \neq 1$, then $(\text{id} - T)(\text{id} + T) = (1 - \Delta_4) \text{id}$, so $\dim(X_1)_{hg} = 2$. This is incompatible with simplicity, because Lemma 5.6 shows that every irreducible Φ_{hg} -projective representation of $C_G(hg)$ is one-dimensional.

Suppose now that $\Delta_4 = 1$. Then $T^2 = \text{id}$, and, since T interchanges the two basis lines, $\text{id} - T$ has rank one. Thus $(X_1)_{hg} = \mathbb{C}x_1$. This line is stable under $C_G(hg)$ and affords a one-dimensional projective representation τ_1 . Since $X_1 = \mathbb{C}G \triangleright x_1$, there is a nonzero surjection

$$M(hg, \tau_1) \longrightarrow X_1.$$

The source is simple, so this map is an isomorphism. Therefore X_1 is simple and $\dim X_1 = |(hg)^G| = 4$. $\square$

For the second adjoint object, let

$$\kappa_X = \frac{\rho(\varepsilon^2)}{\Phi_h(\varepsilon^2, h)\Phi_h(g, \varepsilon^2 h)}.$$

Lemma 5.8. *Assume that $\Delta_4 = 1$. Then*

$$x_2 := \varphi_2^\Phi(v \otimes x_1) = (1 + \rho(h))(v \otimes x_1 - \kappa_X v_1 \otimes (h \triangleright x_1)). \quad (5.5)$$

Moreover, $X_2 = (\text{ad } V)^2(W) = \mathbb{C}G \triangleright x_2$, and hence $X_2 = 0$ if and only if $\rho(h) = -1$.

Proof. Let $T_0 = v \otimes x_1$ and $T_1 = v_1 \otimes (h \triangleright x_1)$. The two nonidentity terms in the recursive formula (4.10) are

$$\begin{aligned}c_{X_1, V}c_{V, X_1}(T_0) &= \rho(h)\kappa_X T_1, \\ (\text{id}_V \otimes \varphi_1^\Phi)c_{1, 2}^\Phi(T_0) &= \rho(h)T_0 - \Xi_X T_1,\end{aligned}$$

where Ξ_X is the scalar defined in Lemma C.1. That lemma proves $\Xi_X = \kappa_X$. Therefore

$$x_2 = T_0 - \rho(h)\kappa_X T_1 + \rho(h)T_0 - \Xi_X T_1 = (1 + \rho(h))(T_0 - \kappa_X T_1),$$

which is (5.5).

Since $\Delta_4 = 1$, the line $\mathbb{C}x_1$ is stable under $C_G(hg)$. The element $hg \in C_G(hg)$ conjugates h to $\varepsilon^{-1}h$, and hence $V = \mathbb{C}C_G(hg) \triangleright v$. Proposition 2.10 now gives $X_2 = \mathbb{C}G \triangleright x_2$. The degrees of the first V -factors in T_0 and T_1 are h and $\varepsilon^{-1}h$, respectively, so $T_0 - \kappa_X T_1 \neq 0$. The last assertion follows from (5.5). $\square$

5.2.2 The objects Y_1 and Y_2

Lemma 5.9. *The vector*

$$y_1 := c_{V,W}(x_1) = w_1 \otimes v - \beta w \otimes v_1 = \varphi_1^\Phi(w_1 \otimes v) \quad (5.6)$$

generates $Y_1 = (\text{ad } W)(V)$. Moreover, $Y_1 \simeq X_1$ and $\text{supp } Y_1 = (hg)^G$. In particular,

$$X_1 \text{ is simple} \iff Y_1 \text{ is simple} \iff \Delta_4 = 1.$$

If these equivalent conditions hold, then $X_1 \simeq Y_1 \simeq M(hg, \tau_1)$, for the same one-dimensional Φ_{hg} -projective representation τ_1 , and $\dim X_1 = \dim Y_1 = 4$.

Proof. By (5.4) and the braiding formula, $c_{V,W}(v \otimes w) = w_1 \otimes v$, while

$$c_{V,W}(v_1 \otimes w_1) = ((\varepsilon^{-1}h) \triangleright w_1) \otimes v_1 = \Phi_g(\varepsilon^{-1}h, h)\sigma(\varepsilon^{-1}h^2)w \otimes v_1.$$

Since $\beta = \alpha\Phi_g(\varepsilon^{-1}h, h)\sigma(\varepsilon^{-1}h^2)$, it follows that $c_{V,W}(x_1) = w_1 \otimes v - \beta w \otimes v_1$. The identity preceding (4.28) gives $c_{V,W}(x_1) = \varphi_1^\Phi(w_1 \otimes v)$ and identifies X_1 with Y_1 . Thus y_1 generates Y_1 . Since this identification is graded and G -linear, it preserves the support and, when $\Delta_4 = 1$, the inducing representation τ_1 . The remaining assertions follow from Lemma 5.7. $\square$

Lemma 5.10. *For every homogeneous vector $a \in W_x$, one has*

$$\begin{aligned} \varphi_2^\Phi(a \otimes y_1) &= a \otimes y_1 - (x(hg)x^{-1} \triangleright a) \otimes (x \triangleright y_1) \\ &\quad + \frac{\Phi(x, \varepsilon g, h)}{\Phi(x(\varepsilon g)x^{-1}, x, h)}(x \triangleright w_1) \otimes \varphi_1^\Phi(a \otimes v) \\ &\quad - \beta \frac{\Phi(x, g, \varepsilon^{-1}h)}{\Phi(xgx^{-1}, x, \varepsilon^{-1}h)}(x \triangleright w) \otimes \varphi_1^\Phi(a \otimes v_1). \end{aligned} \quad (5.7)$$

Proof. Here the maps φ_n^Φ are those associated with the ordered pair (W, V) . Since

$$w_1 \otimes v \in (W \otimes V)_{(\varepsilon g)h}, \quad w \otimes v_1 \in (W \otimes V)_{g(\varepsilon^{-1}h)},$$

and $(\varepsilon g)h = g(\varepsilon^{-1}h) = hg$, the vector y_1 is homogeneous of degree hg . Hence, for $a \in W_x$, the braiding formula gives

$$c_{Y_1, W}c_{W, Y_1}(a \otimes y_1) = (x(hg)x^{-1} \triangleright a) \otimes (x \triangleright y_1).$$

Here $x \triangleright y_1$ denotes the tensor-product action on $W \otimes V$.

Let $b \in W_y$ and $r \in V_\ell$ be homogeneous. By the definition of $c_{1,2}^\Phi$ and the associator convention,

$$c_{1,2}^\Phi(a \otimes (b \otimes r)) = \frac{\Phi(x, y, \ell)}{\Phi(xy x^{-1}, x, \ell)}(x \triangleright b) \otimes (a \otimes r).$$

Now substitute $y_1 = w_1 \otimes v - \beta w \otimes v_1$ into the recursive formula $\varphi_2^\Phi = \text{id}_{W \otimes Y_1} - c_{Y_1, W}c_{W, Y_1} + (\text{id}_W \otimes \varphi_1^\Phi)c_{1,2}^\Phi$. Applying the preceding formula for $c_{1,2}^\Phi$ to the two summands of y_1 gives (5.7). $\square$

Let $z_0 = \varphi_2^\Phi(w \otimes y_1)$, and $z_1 = \varphi_2^\Phi(w_2 \otimes y_1)$.

Lemma 5.11. *Assume that $\Delta_4 = 1$. Then*

$$\begin{aligned} Y_2 &= \mathbb{C}G \triangleright z_0 \oplus \mathbb{C}G \triangleright z_1, \\ z_1 &\neq 0. \end{aligned} \quad (5.8)$$

Moreover, $z_0 \in (Y_2)_{\varepsilon^{-1}hg^2}$ and $0 \neq z_1 \in (Y_2)_{\varepsilon hg^2}$. The conjugacy classes $(\varepsilon^{-1}hg^2)^G = \{\varepsilon^{-1}hg^2, hg^2\}$ and $(\varepsilon hg^2)^G = \{\varepsilon hg^2, \varepsilon^2 hg^2\}$ are disjoint, so the two summands have disjoint supports.

Proof. Lemma 5.9 shows that Y_1 is simple and that $C_G(hg)$ stabilizes $\mathbb{C}y_1$. This centralizer has two orbits on $\text{supp } W$, represented by g and ε^2g . The conjugation by hg interchanges g with εg , and ε^2g with $\varepsilon^{-1}g$. Lemma 2.9, followed by the G -linearity of φ_2^Φ , therefore gives $Y_2 = \mathbb{C}G \triangleright z_0 + \mathbb{C}G \triangleright z_1$. The map φ_2^Φ preserves the grading, so $z_0 \in (Y_2)_{\varepsilon^{-1}hg^2}$ and $z_1 \in (Y_2)_{\varepsilon hg^2}$. The two displayed conjugacy classes are disjoint, and hence the sum is direct.

In Lemma 5.10 with $a = w_2$, the W -degrees of the four terms are $\varepsilon^2g, \varepsilon g, \varepsilon^{-1}g, g$, respectively. Thus the component of multidegree $(\varepsilon^2g, \varepsilon g, h)$ occurs only in $w_2 \otimes y_1$, where it is $w_2 \otimes (w_1 \otimes v) \neq 0$. Hence $z_1 \neq 0$. $\square$

5.2.3 Simplicity of Y_2 and vanishing of Y_3

Lemma 5.12. *Assume that $\Delta_4 = 1$. Then*

$$z_0 = 0 \iff \sigma(g) = -1.$$

Proof. Lemma C.2 shows that the coefficient of $w \otimes (w \otimes v_1)$ in z_0 is $-\beta(1 + \sigma(g))$. Since $\beta \neq 0$, the equality $z_0 = 0$ implies $\sigma(g) = -1$. Conversely, if $\sigma(g) = -1$, the same lemma gives $z_0 = 0$, since $\Delta_4 = 1$. $\square$

Lemma 5.13. *One has*

$$C_G(\varepsilon hg^2) = \langle \varepsilon, h, g^2 \rangle = \langle \varepsilon^2, \varepsilon^{-1}h^2, g^2, h \rangle.$$

Moreover, every irreducible $\Phi_{\varepsilon hg^2}$ -projective representation of this centralizer is one-dimensional.

Proof. Let $H = \langle \varepsilon, h, g^2 \rangle$. The defining relations show that g^2 commutes with both ε and h . Thus H is abelian and $H \subseteq C_G(\varepsilon hg^2)$. Every element of G has a representative $\varepsilon^i h^j g^r$. According as r is even or odd, this representative belongs to H or Hg . Hence $[G : H] \leq 2$. Moreover, conjugation by g interchanges εhg^2 and $\varepsilon^2 hg^2$. Since ε has order four, these elements are distinct. Hence $[G : C_G(\varepsilon hg^2)] = 2$, and consequently $C_G(\varepsilon hg^2) = H$.

Clearly, $\langle \varepsilon^2, \varepsilon^{-1}h^2, g^2, h \rangle \subseteq H$. The reverse inclusion follows from $\varepsilon = h^2(\varepsilon^{-1}h^2)^{-1}$. Hence $H = \langle \varepsilon^2, \varepsilon^{-1}h^2, g^2, h \rangle$. Lemma 5.3 gives $\eta_h = 1$. By Lemma 4.4,

$$c_{\varepsilon hg^2}(\varepsilon, h) = \eta_h^{-1}, \quad c_{\varepsilon hg^2}(\varepsilon, g^2) = \eta_h, \quad c_{\varepsilon hg^2}(h, g^2) = \eta_h^{-1}.$$

Since $c_{\varepsilon hg^2}$ is a bicharacter and ε, h, g^2 generate H , it is trivial on H . Therefore $\mathbb{C}^{\Phi_{\varepsilon hg^2}} H$ is commutative, and every irreducible $\Phi_{\varepsilon hg^2}$ -projective representation of H is one-dimensional. $\square$

Lemma 5.14. *Write $z_1 = \sum_{j=0}^7 z_{1,j}$, where $z_{1,j}$ is the projection of z_1 onto the following multidegree:*

j	multidegree of $z_{1,j}$	j	multidegree of $z_{1,j}$
0	$(g, \varepsilon^2g, \varepsilon^{-1}h)$	4	$(g, \varepsilon^{-1}g, h)$
1	$(\varepsilon g, \varepsilon^{-1}g, \varepsilon^{-1}h)$	5	$(\varepsilon g, g, h)$
2	$(\varepsilon^2g, g, \varepsilon^{-1}h)$	6	$(\varepsilon^2g, \varepsilon g, h)$
3	$(\varepsilon^{-1}g, \varepsilon g, \varepsilon^{-1}h)$	7	$(\varepsilon^{-1}g, \varepsilon^2g, h)$

All eight components are nonzero. If $\pi = (0 \ 1 \ 2 \ 3)(4 \ 5 \ 6 \ 7)$, then there are unique scalars $\lambda_j \in \mathbb{C}^\times$ such that

$$h \triangleright z_{1,j} = \lambda_j z_{1,\pi(j)} \quad (0 \leq j \leq 7).$$

For $1 \leq j \leq 7$, define

$$\Theta_{2,j} = \frac{\lambda_j}{\lambda_0}. \tag{5.9}$$

If $\Delta_4 = 1$, then

$$\mathbb{C}z_1 \text{ is } C_G(\varepsilon hg^2)\text{-stable} \iff \Theta_{2,j} = 1 \quad (1 \leq j \leq 7).$$

Proof. Lemma C.3 gives the eight nonzero projections displayed above and shows that h permutes their lines according to π . This proves the existence and uniqueness of the λ_j . Replacing v and w by av and bw , respectively, multiplies every $z_{1,j}$ by ab^2 . Hence the scalars λ_j and the ratios $\Theta_{2,j}$ are unchanged.

Assume now that $\Delta_4 = 1$. Then $\mathbb{C}y_1$ is stable under $C_G(hg)$. The elements $\varepsilon^2, \varepsilon^{-1}h^2, g^2$ belong to this centralizer and are central in G . Hence each of them acts by a scalar on both $W_{\varepsilon^2g} = \mathbb{C}w_2$ and $(Y_1)_{hg} = \mathbb{C}y_1$. Since φ_2^Φ is a morphism, $\varepsilon^2, \varepsilon^{-1}h^2, g^2$ stabilize $\mathbb{C}z_1$.

By Lemma 5.13, $C_G(\varepsilon hg^2) = \langle \varepsilon^2, \varepsilon^{-1}h^2, g^2, h \rangle$. It remains only to decide whether $h \triangleright z_1$ belongs to $\mathbb{C}z_1$. The eight multidegrees are distinct, and therefore

$$h \triangleright z_1 \in \mathbb{C}z_1 \iff \lambda_0 = \lambda_1 = \dots = \lambda_7.$$

By (5.9), this is equivalent to $\Theta_{2,j} = 1$ for $1 \leq j \leq 7$. $\square$

Lemma 5.15. *Assume that ρ and σ are one-dimensional and that $\Delta_4 = 1$. Then*

$$Y_2 \text{ is simple} \iff \sigma(g) = -1 \text{ and } \Theta_{2,j} = 1 \quad (1 \leq j \leq 7).$$

If these conditions hold, then $Y_2 \simeq M(\varepsilon hg^2, \tau_2)$, where τ_2 is one-dimensional, and $\dim Y_2 = 2$.

Proof. By Lemma 5.11, Y_2 is the direct sum of the subobjects generated by z_0 and z_1 , whose supports are disjoint, and $z_1 \neq 0$. Hence, if Y_2 is simple, then $z_0 = 0$. By Lemma 5.12, this is equivalent to $\sigma(g) = -1$.

Assume that $\sigma(g) = -1$. Then $z_0 = 0$, so $Y_2 = \mathbb{C}G \triangleright z_1$, with $0 \neq z_1 \in (Y_2)_{\varepsilon hg^2}$. If Y_2 is simple, Proposition 2.4 shows that $(Y_2)_{\varepsilon hg^2}$ is an irreducible $\Phi_{\varepsilon hg^2}$ -projective representation of $C_G(\varepsilon hg^2)$. By Lemma 5.13, it is one-dimensional. Hence $(Y_2)_{\varepsilon hg^2} = \mathbb{C}z_1$, and this line is $C_G(\varepsilon hg^2)$ -stable.

Conversely, suppose that $\mathbb{C}z_1$ is $C_G(\varepsilon hg^2)$ -stable. It then affords a one-dimensional $\Phi_{\varepsilon hg^2}$ -projective representation τ_2 . The standard induced construction gives a nonzero surjective morphism $M(\varepsilon hg^2, \tau_2) \rightarrow Y_2$, whose image is $\mathbb{C}G \triangleright z_1 = Y_2$. Its source is simple by Proposition 2.4. Hence its kernel is zero, and the morphism is an isomorphism.

Lemma 5.14 now gives the stated conditions on the $\Theta_{2,j}$. Finally, the conjugacy class $(\varepsilon hg^2)^G = \{\varepsilon hg^2, \varepsilon^2 hg^2\}$ has two elements. Since τ_2 is one-dimensional, $\dim Y_2 = 2$. $\square$

Lemma 5.16. *Assume that $\Delta_4 = 1$ and that Y_2 is simple, and let $y_3 = \varphi_3^\Phi(w \otimes z_1)$. Then*

$$Y_3 = (\text{ad } W)^3(V) = \mathbb{C}G \triangleright y_3 = 0.$$

Proof. Lemmas 5.11 and 5.15 give $Y_2 = \mathbb{C}G \triangleright z_1$ and $(Y_2)_{\varepsilon hg^2} = \mathbb{C}z_1$. This line is stable under $C_G(\varepsilon hg^2) = \langle \varepsilon, h, g^2 \rangle$. Conjugation by h cyclically permutes the four homogeneous components of W , so $W = \mathbb{C}C_G(\varepsilon hg^2) \triangleright w$. Proposition 2.10 therefore gives $Y_3 = \mathbb{C}G \triangleright y_3$. Lemma C.5 gives $y_3 = 0$, and hence $Y_3 = 0$. $\square$

We can now determine the initial Cartan matrix. Whenever the relevant reflections are defined, write $m_{ij} = m_{ij}^{(V,W)}$ for the initial tuple.

Corollary 5.17. *Assume that ρ and σ are one-dimensional and that*

$$\rho(h) = \sigma(g) = -1, \quad \Delta_4 = 1, \quad \Theta_{2,j} = 1 \quad (1 \leq j \leq 7).$$

Then X_1, Y_1, Y_2 are nonzero simple objects, $X_2 = Y_3 = 0$, $\dim X_1 = \dim Y_1 = 4$, and $\dim Y_2 = 2$. In particular, $m_{12} = 1$ and $m_{21} = 2$ for (V, W) .

Proof. Lemmas 5.7 and 5.9 give the assertions about X_1 and Y_1 . Lemma 5.8 gives $X_2 = 0$, while Lemma 5.15 gives the assertion about Y_2 . Finally, Lemma 5.16 gives $Y_3 = 0$. $\square$

Proposition 5.18. *Assume that $\mathcal{B}(V \oplus W)$ is finite-dimensional and that the Cartan matrix of (V, W) is indecomposable and of finite type. Then*

$$\rho(h) = \sigma(g) = -1, \quad \Delta_4 = 1, \quad \Theta_{2,j} = 1 \quad (1 \leq j \leq 7).$$

The Cartan matrix of (V, W) is

$$\begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}.$$

Proof. Corollary 5.5 shows that ρ and σ are one-dimensional. By Theorem 2.25, the pair (V, W) admits all reflections and has a finite Cartan graph.

Lemma 5.7 gives $x_1 \neq 0$, and hence $X_1 \neq 0$ and $m_{12} \geq 1$. Therefore Lemma 2.24 shows that X_1 is simple. Lemma 5.7 then gives $\Delta_4 = 1$. Under this condition, (5.8) gives $0 \neq z_1 \in Y_2$. Consequently, $Y_2 \neq 0$ and $m_{21} \geq 2$. Another application of Lemma 2.24 shows that Y_2 is simple.

Since the Cartan matrix of (V, W) is an indecomposable rank-two generalized Cartan matrix of finite type, one has $m_{12}m_{21} \in \{1, 2, 3\}$. Together with $m_{12} \geq 1$ and $m_{21} \geq 2$, this forces $m_{12} = 1$. By the definition of m_{12} , it follows that $X_2 = 0$. Lemma 5.8, which applies because $\Delta_4 = 1$, now gives $\rho(h) = -1$.

Since Y_2 is simple, Lemma 5.15 gives $\sigma(g) = -1$ and $\Theta_{2,j} = 1$ for $1 \leq j \leq 7$. Corollary 5.17 now gives $(m_{12}, m_{21}) = (1, 2)$, as asserted. $\square$

5.3 Reflections

The following lemma gives support representatives for the two reflected tuples.

Lemma 5.19. *Assume that ρ and σ are one-dimensional, that $m_{12} = 1$, $m_{21} = 2$, and that X_1 and Y_2 are simple, so that both reflections are defined. Let $(\varepsilon_1, h_1, g_1) = (\varepsilon^{-1}, h^{-1}, hg)$, $(\varepsilon_2, h_2, g_2) = (\varepsilon^{-1}, \varepsilon hg^2, g^{-1})$. Then*

$$R_1(V, W) = (V^*, X_1), \quad R_2(V, W) = (Y_2, W^*)$$

have supports of the same form as (V, W) , with support representatives h_1, g_1 and h_2, g_2 , respectively. In both cases the chosen representatives satisfy the defining relations of Γ_4 and generate the same group G .

Proof. Since $m_{12} = 1$, $m_{21} = 2$, and X_1 and Y_2 are simple, the definition of the reflections gives $R_1(V, W) = (V^*, X_1)$ and $R_2(V, W) = (Y_2, W^*)$. Since X_1 is simple, Lemma 5.7 gives $\Delta_4 = 1$ and $\text{supp } X_1 = \{\varepsilon^i hg \mid 0 \leq i < 4\}$. The simplicity of Y_2 , together with Lemma 5.15, then gives $\text{supp } Y_2 = \{\varepsilon hg^2, \varepsilon^2 hg^2\}$. Recall that g^2 is central. The defining relations of Γ_4 give

$$\begin{aligned} h_1 g_1 &= g = \varepsilon^{-1}(g_1 h_1), & g_1 \varepsilon^{-1} &= \varepsilon g_1, & h_1 \varepsilon^{-1} &= \varepsilon^{-1} h_1, \\ h_2 g_2 &= \varepsilon hg = \varepsilon^{-1}(g_2 h_2), & g_2 \varepsilon^{-1} &= \varepsilon g_2, & h_2 \varepsilon^{-1} &= \varepsilon^{-1} h_2. \end{aligned}$$

For the second row, one uses $g_2 h_2 = g^{-1} \varepsilon hg^2 = \varepsilon^2 hg$. Thus both triples satisfy $h_i g_i = \varepsilon_i g_i h_i$, $g_i \varepsilon_i = \varepsilon_i^{-1} g_i$, and $h_i \varepsilon_i = \varepsilon_i h_i$. Since ε has order four, $\varepsilon_1 = \varepsilon_2 = \varepsilon^{-1}$ also has order four.

For both triples, the first relation gives $\varepsilon^{-1} = h_i g_i h_i^{-1} g_i^{-1}$, so $\varepsilon^{-1} \in \langle h_i, g_i \rangle$. For the first triple, $h = h_1^{-1}$ and $g = h_1 g_1$. For the second triple, $g = g_2^{-1}$ and $h = \varepsilon^{-1} h_2 g_2^2$. Consequently, both pairs h_i, g_i generate G . The grading on a dual object is obtained by taking inverses of the original degrees. In particular, $(\varepsilon^i g)^{-1} = g^{-1} \varepsilon^{-i} = \varepsilon^i g^{-1}$. Therefore

$$\begin{aligned} \text{supp } V^* &= \{h^{-1}, \varepsilon h^{-1}\} = \{h_1, \varepsilon_1^{-1} h_1\}, \\ \text{supp } X_1 &= \{\varepsilon^i hg \mid 0 \leq i < 4\} = \{\varepsilon_1^i g_1 \mid 0 \leq i < 4\}, \\ \text{supp } Y_2 &= \{\varepsilon hg^2, \varepsilon^2 hg^2\} = \{h_2, \varepsilon_2^{-1} h_2\}, \\ \text{supp } W^* &= \{\varepsilon^i g^{-1} \mid 0 \leq i < 4\} = \{\varepsilon_2^i g_2 \mid 0 \leq i < 4\}. \end{aligned}$$

Thus both reflected tuples have supports of the required form. $\square$

Lemma 5.20. *Under the assumptions of Lemma 5.19, one has*

$$\begin{aligned} (\mathrm{ad} V^*)(X_1) &\simeq W, & (\mathrm{ad} V^*)^2(X_1) &= 0, \\ (\mathrm{ad} W^*)(Y_2) &\simeq Y_1, & (\mathrm{ad} W^*)^2(Y_2) &\simeq V, & (\mathrm{ad} W^*)^3(Y_2) &= 0. \end{aligned}$$

Consequently, $m_{12}^{R_1(V,W)} = 1$ and $m_{21}^{R_2(V,W)} = 2$.

Proof. Apply [34, Lemma 3.8] with $m_{12} = 1$ for the first reflection and with $m_{21} = 2$ for the second. Taking the corresponding $\mathbb{Z}$ -graded components gives the asserted isomorphisms and vanishing. $\square$

The next two lemmas determine the values of the inducing projective representations needed for the two reflected tuples.

Lemma 5.21. *Assume that $\Delta_4 = 1$. Then*

$$\tau_1(hg) = -\rho(h)\sigma(g).$$

Proof. The computation in the proof of Lemma 5.7 gives $(c_{W,V}c_{V,W})^2 = \Delta_4 \mathrm{id}$ on $(V \otimes W)_{hg}$. Since $x_1 = (\mathrm{id} - c_{W,V}c_{V,W})(v \otimes w)$ and $v \otimes w \in (V \otimes W)_{hg}$, the assumption $\Delta_4 = 1$ implies $c_{W,V}c_{V,W}(x_1) = -x_1$.

For a homogeneous vector, the canonical twist of ${}^{\mathbb{C}}\mathcal{YD}^{\Phi} \simeq \mathcal{Z}(\mathrm{Vec}_G^{\Phi})$ is given by the action of its degree. Since V and W are simple, the naturality of the twist and Schur's lemma show that their twists are scalar. Evaluating them on $v \in V_h$ and $w \in W_g$ gives $\theta_V = \rho(h) \mathrm{id}_V$ and $\theta_W = \sigma(g) \mathrm{id}_W$. The balancing identity $\theta_{V \otimes W} = c_{W,V}c_{V,W}(\theta_V \otimes \theta_W)$ therefore gives

$$(hg) \triangleright x_1 = \theta_{V \otimes W}(x_1) = -\rho(h)\sigma(g)x_1.$$

By Lemma 5.7, $(X_1)_{hg} = \mathbb{C}x_1$, and this line affords τ_1 . Since $x_1 \neq 0$, it follows that $\tau_1(hg) = -\rho(h)\sigma(g)$. $\square$

Lemma 5.22. *Assume that ρ and σ are one-dimensional, $\Delta_4 = 1$, $\rho(h) = \sigma(g) = -1$, and that Y_2 is simple. Write $Y_2 \simeq M(\epsilon hg^2, \tau_2)$. Then $\tau_2(\epsilon hg^2) = -1$.*

Proof. The line $\mathbb{C}z_1 = (Y_2)_{\epsilon hg^2}$ is stable under $C_G(\epsilon hg^2)$. Lemma C.6 gives

$$(\epsilon hg^2) \triangleright (z_{1,3} + z_{1,7}) = -(z_{1,2} + z_{1,6}).$$

The first W -factors of $z_{1,i}$ and $z_{1,i+4}$ have degree $\epsilon^i g$, and $(\epsilon hg^2)(\epsilon^i g)(\epsilon hg^2)^{-1} = \epsilon^{i-1} g$. Thus the left-hand side is the entire part of $(\epsilon hg^2) \triangleright z_1$ whose first W -factor has degree $\epsilon^2 g$. The corresponding part of $\tau_2(\epsilon hg^2)z_1$ is $\tau_2(\epsilon hg^2)(z_{1,2} + z_{1,6})$. By Lemma 5.14, the two summands are nonzero and have distinct multidegrees, so their sum is nonzero. Comparing with $(\epsilon hg^2) \triangleright z_1 = \tau_2(\epsilon hg^2)z_1$ gives $\tau_2(\epsilon hg^2) = -1$. $\square$

Proposition 5.23. *Assume (B₂). Then (V, W) admits all reflections. Furthermore, with support representatives chosen successively as in Lemma 5.19, every reflected tuple satisfies the parameter conditions. The Cartan graph of (V, W) is standard of type B_2 , with*

$$A^{[P]} = \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}$$

at every object $[P]$.

Proof. Corollary 5.17 gives $m_{12} = 1$, $m_{21} = 2$, and the simplicity of X_1 and Y_2 , so both initial reflections are defined. Lemma 5.19 provides the support representatives $(\varepsilon_i, h_i, g_i)$ for $P^{(i)} = R_i(V, W)$, $i = 1, 2$.

For $P^{(1)} = (V^*, X_1)$, write $V^* \simeq M(h_1, \rho_1)$ and $X_1 \simeq M(g_1, \sigma_1)$. The inducing representations are one-dimensional by Lemmas 2.5 and 5.7. Lemma 5.20 gives $(\text{ad } V^*)(X_1) \simeq W$. Since W is simple, Lemma 5.7, applied to $P^{(1)}$, gives $\Delta_4^{(1)} = 1$. Lemma 2.5 gives $\rho_1(h_1) = \rho(h) = -1$, while Lemma 5.21 gives $\sigma_1(g_1) = \tau_1(hg) = -\rho(h)\sigma(g) = -1$. The hypotheses of Lemma C.7 are therefore satisfied, and it gives $\Theta_{2,j}^{(1)} = 1$ for $1 \leq j \leq 7$. Thus $P^{(1)}$ satisfies (B₂).

For $P^{(2)} = (Y_2, W^*)$, both inducing representations are one-dimensional by Lemmas 5.15 and 2.5. Lemma 5.20 gives $(\text{ad } W^*)(Y_2) \simeq Y_1$ and $(\text{ad } W^*)^2(Y_2) \simeq V$. By (4.28), the first of these isomorphisms also gives $X_1^{P^{(2)}} \simeq Y_1$. Hence Lemma 5.7 gives $\Delta_4^{(2)} = 1$. Lemmas 5.22 and 2.5 show that these representations take the value -1 at h_2 and g_2 , respectively. Since $Y_2^{P^{(2)}} \simeq V$ is simple, Lemma 5.15 gives $\Theta_{2,j}^{(2)} = 1$ for $1 \leq j \leq 7$. Thus $P^{(2)}$ also satisfies (B₂).

The reflected support representatives satisfy the defining relations of Γ_4 and generate G by Lemma 5.19. The preceding argument therefore applies to every tuple satisfying (B₂). Corollary 5.17 gives both reflections and the same Cartan matrix for each such tuple. Induction on the length of a reflection sequence proves that all reflections are defined, that the parameter conditions are preserved, and that the Cartan graph is standard of type B_2 . $\square$

5.4 Finite Cartan graphs

Proposition 5.24. *Assume that (V, W) admits all reflections and that its Cartan graph is finite. Let $P = (P_1, P_2) \in \mathcal{F}_2(V, W)$ have an indecomposable Cartan matrix of finite type. After interchanging P_1 and P_2 if necessary, there exist $\varepsilon_P, h_P, g_P \in G$ such that*

$$h_P g_P = \varepsilon_P g_P h_P, \quad g_P \varepsilon_P = \varepsilon_P^{-1} g_P, \quad h_P \varepsilon_P = \varepsilon_P h_P, \quad \text{ord}(\varepsilon_P) = 4,$$

and

$$\text{supp } P_1 = \{h_P, \varepsilon_P^{-1} h_P\}, \quad \text{supp } P_2 = \{\varepsilon_P^i g_P \mid 0 \leq i < 4\}.$$

Moreover, $G = \langle \varepsilon_P, h_P, g_P \rangle$. Thus, after this interchange, P has the supports and generators specified at the beginning of this section.

Proof. The relation $hg = \varepsilon gh$ gives $\varepsilon = hgh^{-1}g^{-1}$, so $G = \langle h, g \rangle$. Since h and g belong to the initial supports, Lemma 2.29 gives $G = \langle \text{supp } P_1 \cup \text{supp } P_2 \rangle$.

An indecomposable Cartan matrix of rank two and finite type is of type A_2 , B_2 , or G_2 , possibly with the two indices interchanged. Thus, in one of the two orderings, $-a_{12}^{[P]} = 1$ and $1 \leq -a_{21}^{[P]} \leq 3$. In this ordering,

$$(\text{ad } P_1)^2(P_2) = 0, \quad (\text{ad } P_2)^4(P_1) = 0.$$

Moreover, $(\text{ad } P_1)(P_2) \neq 0$, so P is braided-indecomposable. Therefore Theorem 3.4 applies.

If a union of conjugacy classes generates G , then its inner automorphism group is isomorphic to $G/Z(G)$. The initial support is $Z_4^{4,2}$ and generates G . Hence $G/Z(G)$ is isomorphic to $\text{Inn}(Z_4^{4,2})$, which has order eight. The support of $P_1 \oplus P_2$ also generates G , so its inner automorphism group has order eight as well.

The permutation descriptions in Section 3 show that the inner automorphism groups of $Z_T^{4,1}$, $Z_2^{2,2}$, $Z_3^{3,1}$, $Z_3^{3,2}$, and $Z_4^{4,2}$ have orders 12, 4, 6, 6, 8, respectively. Hence Theorem 3.4 implies that $\text{supp}(P_1 \oplus P_2)$ is isomorphic to $Z_4^{4,2}$.

The two supports are the two orbits of $Z_4^{4,2}$. After interchanging P_1 and P_2 if necessary, assume that $\text{supp } P_1$ is the two-element orbit and $\text{supp } P_2$ is the four-element orbit. The corresponding quandle isomorphism induces an epimorphism $\Gamma_4 \twoheadrightarrow G$. Let ε_P, h_P, g_P be the images

of the standard generators. The defining relations and conjugacy classes of Γ_4 give the relations and supports in the statement.

The four elements of $\text{supp } P_2$ are distinct, while $\varepsilon_P^4 = 1$. Therefore $\text{ord}(\varepsilon_P) = 4$. Finally, $\varepsilon_P = h_P g_P h_P^{-1} g_P^{-1}$, and the epimorphism above shows that $G = \langle h_P, g_P \rangle$. $\square$

Proposition 5.25. *Assume that $\mathcal{B}(V \oplus W)$ is finite-dimensional. Then the Cartan graph of (V, W) is standard of type B_2 . Moreover,*

$$\dim \rho = \dim \sigma = 1, \quad \rho(h) = \sigma(g) = -1, \quad \Delta_4 = 1, \quad \Theta_{2,j} = 1 \quad (1 \leq j \leq 7).$$

With the original ordering, the Cartan matrix at every object $[Q]$ is

$$A^{[Q]} = \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}.$$

Proof. Theorem 2.25 shows that (V, W) admits all reflections and has a finite Cartan graph. It is connected by Proposition 2.18. By Proposition 2.28, choose $P \in \mathcal{F}_2(V, W)$ whose Cartan matrix is indecomposable and of finite type. Proposition 5.24 shows that, after interchanging its two entries if necessary, the resulting tuple $\tilde{P}$ has the supports and generators fixed at the beginning of this section, with a distinguished element of order four.

Repeated use of Lemma 2.26 shows that $\mathcal{B}(P_1 \oplus P_2)$ is finite-dimensional. Interchanging the two entries only reorders the direct summands, so $\mathcal{B}(\tilde{P}_1 \oplus \tilde{P}_2)$ is also finite-dimensional. The Cartan matrix of $\tilde{P}$ remains indecomposable and of finite type. Hence Corollary 5.5 and Proposition 5.18, applied to $\tilde{P}$, give all the conditions in (B_2) . Proposition 5.23 then shows that the Cartan graph of $\tilde{P}$ is standard of type B_2 .

Since $P \in \mathcal{F}_2(V, W)$, it lies in the Cartan graph of (V, W) . Interchanging the two entries only interchanges the labels 1 and 2 throughout the graph. Thus the Cartan graph of (V, W) is standard of type B_2 , possibly with the two indices interchanged.

For the original tuple, Corollary 5.5 gives $\dim \rho = \dim \sigma = 1$. The Cartan matrix of the original tuple is now indecomposable and of finite type. Proposition 5.18, applied to (V, W) , gives $\rho(h) = \sigma(g) = -1$, $\Delta_4 = 1$, and $\Theta_{2,j} = 1$ for $1 \leq j \leq 7$. It also identifies the initial Cartan matrix with the matrix displayed in the statement. Since the Cartan graph is standard, the same matrix occurs at every object. $\square$

5.5 Nichols algebras of the positive-root factors of type B_2

The positive real roots correspond to V, Y_1, Y_2 , and W . We shall prove that the Nichols algebra of each of these simple objects is finite-dimensional.

Lemma 5.26. *Let $H \leq G$, and let $R \in {}^G_G \mathcal{YD}^\Phi$ satisfy $\text{supp } R \subseteq H$. Restricting the action to H and viewing the grading as an H -grading gives an object $R|_H \in {}^H_H \mathcal{YD}^{\Phi|_H}$. After restriction to H , the identity maps on the tensor powers identify $\mathcal{B}(R)$ with $\mathcal{B}(R|_H)$. In particular, the two Nichols algebras have the same dimension.*

Proof. The defining identities for $R|_H$ are obtained by restricting those for R , so $R|_H$ is well defined. On every tensor power of R , the grading and the H -action agree in the two categories. Since all homogeneous degrees belong to H , the associativity constraints and braidings also agree. Hence the quantum symmetrizers, and therefore the Nichols ideals, coincide in every degree. The identity maps on the tensor powers induce the asserted isomorphism. $\square$

The supports of V and Y_2 are contained in abelian subgroups of G .

Lemma 5.27. Assume that ρ and σ are one-dimensional and that

$$\rho(h) = \sigma(g) = -1, \quad \Delta_4 = 1, \quad \Theta_{2,j} = 1 \quad (1 \leq j \leq 7).$$

Write $Y_2 \simeq M(\varepsilon h g^2, \tau_2)$. Then $\mathcal{B}(V)$ and $\mathcal{B}(Y_2)$ are finite-dimensional. More precisely, set

$$\kappa_V = \frac{\Phi_h(h, g)}{\Phi_h(g, \varepsilon^{-1}h)} \rho(\varepsilon^{-1}h)^2, \quad \kappa_{Y_2} = \frac{\Phi_{\varepsilon h g^2}(\varepsilon h g^2, g^{-1})}{\Phi_{\varepsilon h g^2}(g^{-1}, \varepsilon^2 h g^2)} \tau_2(\varepsilon^2 h g^2)^2.$$

For $R \in \{V, Y_2\}$, let $N_R = \text{ord}(\kappa_R)$. One has

$$\dim \mathcal{B}(R) = 4N_R. \quad (5.10)$$

Proof. Lemma 5.15 shows that Y_2 is simple, and Lemma 5.22 gives $\tau_2(\varepsilon h g^2) = -1$.

The supports of V and Y_2 are contained in the abelian subgroups $\langle \varepsilon, h \rangle$ and $\langle \varepsilon, \varepsilon h g^2 \rangle$, respectively. The bases (v, v_1) and $(z_1, g^{-1} \triangleright z_1)$ have degrees $h, \varepsilon^{-1}h$ and $\varepsilon h g^2, \varepsilon^2 h g^2$, respectively. Using $hg = g(\varepsilon^{-1}h)$, $(\varepsilon h g^2)g^{-1} = g^{-1}(\varepsilon^2 h g^2)$, and the projective action relation, their braiding matrices are

$$\begin{pmatrix} -1 & \frac{\Phi_h(h, g)}{\Phi_h(g, \varepsilon^{-1}h)} \rho(\varepsilon^{-1}h) \\ \rho(\varepsilon^{-1}h) & -1 \end{pmatrix}, \quad \begin{pmatrix} -1 & \frac{\Phi_{\varepsilon h g^2}(\varepsilon h g^2, g^{-1})}{\Phi_{\varepsilon h g^2}(g^{-1}, \varepsilon^2 h g^2)} \tau_2(\varepsilon^2 h g^2) \\ \tau_2(\varepsilon^2 h g^2) & -1 \end{pmatrix}.$$

The first diagonal entries are $\rho(h) = \tau_2(\varepsilon h g^2) = -1$. Since the canonical twist is scalar on each simple object, the second diagonal entries have the same value. The two edge labels are therefore κ_V and κ_{Y_2} .

By Lemma 5.26, these Nichols algebras may be computed after restriction to the two abelian subgroups above. The argument in the proof of Lemma 4.32 applies to both restricted objects. Namely, Φ may be replaced by a cohomologous root-of-unity-valued cocycle. The corresponding one-dimensional projective action values are then roots of unity. The associated braided monoidal equivalence preserves Nichols algebras and the eigenvalues of the double braiding. Hence κ_V and κ_{Y_2} have finite order.

The reduction in Subsection 4.9 preserves these diagonal coefficients and edge labels, as well as the Hilbert series. We may therefore use the ordinary diagonal classification. If $\kappa_R = 1$, the diagonal braiding is of type $A_1 \times A_1$, and its two PBW generators have height two. If $\kappa_R \neq 1$, its generalized Cartan matrix is of type A_2 . The three positive roots have diagonal coefficients $-1, \kappa_R, -1$, and hence PBW heights $2, N_R, 2$. Therefore, in both cases, $\dim \mathcal{B}(R) = 4N_R$. $\square$

It remains to consider W and Y_1 , whose supports have four elements. For $R \in \{W, Y_1\}$, let

$$(x_R, \tau_R) = \begin{cases} (g, \sigma), & R = W, \\ (hg, \tau_1), & R = Y_1, \end{cases} \quad K_R = \langle \varepsilon, x_R \rangle, \quad b_R = \varepsilon x_R.$$

Lemma 5.28. Assume that ρ and σ are one-dimensional, that $\Delta_4 = 1$, and that $\rho(h) = \sigma(g) = -1$. For $R \in \{W, Y_1\}$, restriction to K_R gives $R|_{K_R} = R^{(0)} \oplus R^{(1)}$, where $R^{(0)} = R_{x_R} \oplus R_{\varepsilon^2 x_R}$ and $R^{(1)} = R_{b_R} \oplus R_{\varepsilon^2 b_R}$. The two summands are simple, and the tuple $(R^{(0)}, R^{(1)})$ is braided-indecomposable. If χ_R and ψ_R denote the one-dimensional projective representations inducing $R^{(0)}$ and $R^{(1)}$, respectively, then

$$\chi_R(x_R) = \psi_R(b_R) = -1.$$

Proof. For both choices of R , one has $x_R \varepsilon = \varepsilon^{-1} x_R$. Since ε^2 is a nontrivial central involution and $b_R = \varepsilon x_R$, it follows that

$$K_R = \langle x_R, b_R \rangle, \quad b_R x_R = \varepsilon^2 x_R b_R, \quad b_R^2 = x_R^2.$$

Thus K_R is a quotient of Γ_2 . Since $K_R \leq G$, it is finite. Since $\varepsilon^2 \neq 1$, it is non-abelian.

The same relations give

$$b_R x_R b_R^{-1} = \varepsilon^2 x_R, \quad x_R b_R x_R^{-1} = \varepsilon^2 b_R.$$

Hence the two K_R -conjugacy classes in $\text{supp } R$ are $\{x_R, \varepsilon^2 x_R\}$ and $\{b_R, \varepsilon^2 b_R\}$.

For $R = W$, these two classes partition $\text{supp } W$. For $R = Y_1$, the same assertion follows from Lemma 5.9. The corresponding homogeneous components are one-dimensional. Therefore the two orbit sums are K_R -subobjects and give the stated decomposition. Their representative components afford one-dimensional, hence irreducible, projective representations of the corresponding centralizers. Proposition 2.4 then shows that both summands are simple.

The double braiding maps $R_{x_R} \otimes R_{b_R}$ onto $R_{\varepsilon^2 x_R} \otimes R_{\varepsilon^2 b_R}$. These are distinct nonzero bihomogeneous components because $\varepsilon^2 \neq 1$. Hence the double braiding is not the identity, and the tuple is braided-indecomposable.

Finally, $\chi_R(x_R) = \tau_R(x_R)$. The canonical twist is scalar on the simple object R , so its value on R_{b_R} is the same. Thus $\psi_R(b_R) = \tau_R(x_R)$. For $R = W$, this common value is $\sigma(g) = -1$. For $R = Y_1$, Lemma 5.21 gives $\tau_1(hg) = -\rho(h)\sigma(g) = -1$. $\square$

For the ordered tuple $(R^{(0)}, R^{(1)})$, let Δ_{01}^R be the scalar from (4.2). Its explicit expression is recorded in (C.8).

Lemma 5.29. *Assume the conditions in (B₂). Then*

$$\Delta_{01}^W = \Delta_{01}^{Y_1} = 1.$$

Proof. Corollary 5.17 shows that Y_2 is nonzero and simple and that $Y_3 = 0$. Together with $\Delta_4 = 1$, Lemma C.8 gives $\Delta_{01}^W = 1$.

For $R_1(V, W) = (V^*, X_1)$, Proposition 5.23 gives the same parameter conditions with support representatives $(\varepsilon^{-1}, h^{-1}, hg)$. Applying the same corollary and Lemma C.8 to this tuple, followed by Lemma 4.7, shows that the first adjoint of the two local summands of X_1 is simple. The local subgroup is $\langle \varepsilon^{-1}, hg \rangle = K_{Y_1}$, and the two supports are $\{hg, \varepsilon^2 hg\}$ and $\{\varepsilon^{-1} hg, \varepsilon hg\}$. The graded G -isomorphism $X_1 \simeq Y_1$ from Lemma 5.9 identifies these summands with $Y_1^{(0)}$ and $Y_1^{(1)}$, respectively, and hence identifies their first adjoints. Applying Lemma 4.7 with the representatives $hg, \varepsilon hg$ now gives $\Delta_{01}^{Y_1} = 1$. $\square$

Lemma 5.30. *Under the conditions in (B₂), the Nichols algebras of V , $Y_1 \simeq X_1$, Y_2 , and W are finite-dimensional.*

Proof. Lemma 5.27 proves the assertion for V and Y_2 . Let $R \in \{W, Y_1\}$. By Lemma 5.28, one has $R|_{K_R} = R^{(0)} \oplus R^{(1)}$, where the two summands are simple, their inducing projective representations are one-dimensional, and the ordered tuple $(R^{(0)}, R^{(1)})$ is braided-indecomposable. Moreover, K_R is a finite non-abelian quotient of Γ_2 . Since $b_R = \varepsilon x_R$, one has $K_R = \langle x_R, b_R \rangle$, so the two supports generate K_R .

Apply Theorem 4.1 to $(R^{(0)}, R^{(1)})$, with x_R, b_R, ε^2 in place of g, h, ε , respectively. Under this identification, the two self-action parameters are $\chi_R(x_R)$ and $\psi_R(b_R)$, respectively, and the parameter Δ is Δ_{01}^R . Lemmas 5.28 and 5.29 give $\chi_R(x_R) = \psi_R(b_R) = -1$ and $\Delta_{01}^R = 1$. Hence the ordered tuple satisfies the A_2 conditions in Theorem 4.1. Its Nichols algebra is therefore finite-dimensional.

Since $R|_{K_R} = R^{(0)} \oplus R^{(1)}$, Lemma 5.26 now gives $\dim \mathcal{B}(R) < \infty$. This proves the assertion for W and Y_1 . Finally, Lemma 5.9 gives $Y_1 \simeq X_1$, so the same conclusion holds for X_1 . $\square$

Proposition 5.31. Assume (B₂). Then $\dim \mathcal{B}(V \oplus W) < \infty$. Moreover, there is an isomorphism of $\mathbb{N}_0^2$ -graded objects in ${}^G\mathcal{YD}^\Phi$,

$$\mathcal{B}(V \oplus W) \simeq \mathcal{B}(V) \otimes (\mathcal{B}(Y_1) \otimes (\mathcal{B}(Y_2) \otimes \mathcal{B}(W))). \quad (5.11)$$

Proof. Proposition 5.23 shows that (V, W) admits all reflections and has a standard Cartan graph of type B_2 . The positive roots in the associated convex order are $\alpha_2, \alpha_1 + 2\alpha_2, \alpha_1 + \alpha_2, \alpha_1$, with corresponding objects W, Y_2, Y_1, V by Lemma 2.24. Proposition 2.22 therefore gives (5.11). Each factor is finite-dimensional by Lemma 5.30, so $\mathcal{B}(V \oplus W)$ is finite-dimensional. $\square$

For $R \in \{V, Y_2\}$, retain the notation N_R from Lemma 5.27. For $R \in \{W, Y_1\}$, the tuple $(R^{(0)}, R^{(1)})$ has Cartan type A_2 . The indices 0, 01, 1 correspond, respectively, to

$$R^{(0)}, \quad (\text{ad } R^{(0)})(R^{(1)}), \quad R^{(1)}.$$

For $\gamma \in \{0, 01, 1\}$, let $N_{R,\gamma}$ be the order of the edge label of the diagonal braiding of the corresponding simple object, with order one when that label is one.

Corollary 5.32. Under the hypotheses of Proposition 5.31,

$$\dim \mathcal{B}(V \oplus W) = 2^{16} N_V N_{Y_2} \prod_{\gamma \in \{0, 01, 1\}} N_{W,\gamma} \prod_{\gamma \in \{0, 01, 1\}} N_{Y_1,\gamma}.$$

Proof. Equation (5.10) gives $\dim \mathcal{B}(R) = 4N_R$ for $R \in \{V, Y_2\}$. Let $R \in \{W, Y_1\}$. Lemmas 5.28 and 5.29 show that $(R^{(0)}, R^{(1)})$ is in the A_2 case of Theorem 4.1. Applying Corollary 4.33 over K_R , and then Lemma 5.26, gives

$$\dim \mathcal{B}(R) = 64 N_{R,0} N_{R,01} N_{R,1}.$$

Taking dimensions in (5.11) proves the formula. $\square$

5.6 Proof of the classification theorem

Proof of Theorem 5.1. If $\mathcal{B}(V \oplus W)$ is finite-dimensional, then Proposition 5.25 gives (B₂) and the asserted standard Cartan graph of type B_2 .

Conversely, assume (B₂). Proposition 5.23 shows that the Cartan graph is standard of type B_2 , with the matrix in the statement at every object. Proposition 5.31 gives $\dim \mathcal{B}(V \oplus W) < \infty$. $\square$

Remark 5.33. When $\Phi = 1$, the category ${}^G\mathcal{YD}^\Phi$ is the ordinary Yetter–Drinfeld category over the Hopf algebra $\mathbb{C}G$. In this case, Theorem 5.1 specializes to the Γ_4 case of [22, Theorem 5.5]. In particular, the Cartan graph is standard of type B_2 , with Cartan matrix $\begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}$ at every object. Moreover, over $\mathbb{C}$, the corresponding Nichols algebra has dimension 2^{18} , in agreement with the ordinary Hopf algebra case.

6 The T case

Let G be a finite non-abelian quotient of

$$T = \langle z, x_1, x_2 \mid zx_1 = x_1z, \quad zx_2 = x_2z, \quad x_1x_2x_1 = x_2x_1x_2, \quad x_1^3 = x_2^3 \rangle.$$

Write $\pi : T \twoheadrightarrow G$ for the quotient map, and denote the images of the generators by the same letters. The defining relations imply that $z \in Z(G)$. Let $x_3 = x_2x_1x_2^{-1}$ and $x_4 = x_1x_2x_1^{-1}$. Throughout this section, the elements x_1, x_2, x_3, x_4 are distinct, $x_1^G = \{x_1, x_2, x_3, x_4\}$, and $G =$

$\langle z, x_1^G \rangle$. Let Φ be a normalized 3-cocycle on G . Let $V = M(z, \rho)$ and $W = M(x_1, \sigma)$ be simple, and assume that (V, W) is braided-indecomposable.

The rack table gives $2 \blacktriangleright 3 = 4$, $4 \blacktriangleright 2 = 3$, and $3 \blacktriangleright 4 = 2$. Hence $c_{W,W}^3$ preserves $W_{x_2} \otimes W_{x_3}$. When $\dim \sigma = 1$, this space is one-dimensional, and we define $\varepsilon_\Phi(W)$ by

$$c_{W,W}^3|_{W_{x_2} \otimes W_{x_3}} = \sigma(x_1)^{-1} \varepsilon_\Phi(W) \text{id}. \quad (6.1)$$

This normalization is chosen so that, when $\sigma(x_1) = -1$, the condition $\varepsilon_\Phi(W) = 1$ gives the cyclic product -1 of the constant tetrahedral braiding. This definition is independent of all basis choices. A normalized gauge transformation conjugates $c_{W,W}^3$ on $W_{x_2} \otimes W_{x_3}$ and $c_{W,W}$ on $W_{x_1} \otimes W_{x_1}$. Each source equals its target and is one-dimensional. Hence both defining scalars, and therefore $\varepsilon_\Phi(W)$, are preserved.

The classification for quotients of T is as follows.

Theorem 6.1. *Under the preceding assumptions, $\dim \mathcal{B}(V \oplus W) < \infty$ if and only if*

$$\begin{aligned} \dim \rho = \dim \sigma = 1, & \quad \sigma(x_1) = -1, \\ (\rho(x_1)\sigma(z))^2 - \rho(x_1)\sigma(z) + 1 = 0, & \\ \rho(x_1)\rho(z)\sigma(z) = 1, & \quad \varepsilon_\Phi(W) = 1, \\ \Phi(x_1x_4^{-1}, x_1x_3^{-1}, x_1x_2^{-1})\Phi(x_1x_3^{-1}, x_1x_2^{-1}, x_1x_4^{-1})\Phi(x_1x_2^{-1}, x_1x_4^{-1}, x_1x_3^{-1}) = 1. & \end{aligned} \quad (\text{T-G}_2)$$

Whenever these conditions hold, (V, W) admits all reflections and its Cartan graph is standard of type G_2 , with matrix $\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$ at every object. Moreover, $\dim \mathcal{B}(V \oplus W) = 6^3 72^3 = 80\,621\,568$.

Outline of the proof.

- Subsection 6.1 computes the third homology of T and reduces the pullback of Φ to eight classes. It also compares the adjoint maps after a gauge transformation.
- Subsection 6.2 describes the centralizers and the projective representations that occur.
- Subsection 6.3 proves $\dim \rho = 1$ at a finite Cartan matrix of type G_2 and computes the adjoint objects. It gives $\sigma(x_1) = -1$, the quadratic condition in (T-G₂), and $\varepsilon_\Phi(W) = 1$. Under the last equality in (T-G₂), it also gives $\dim Y_3 = 1$ and $Y_4 = 0$.
- Subsections 6.4 and 6.5 derive $\rho(x_1)\rho(z)\sigma(z) = 1$, exclude the remaining nontrivial cocycle class, and prove that all parameter conditions are preserved by both reflections.
- Subsection 6.6 proves necessity and shows that the Cartan graph is standard of type G_2 .
- Subsections 6.7 and 6.8 prove sufficiency. There are three simple factors of dimension 1 and three of dimension 4. Their Nichols algebras have dimensions 6 and 72, respectively.

6.1 The pullback of the cocycle to T

We first relate the last equality in (T-G₂) to the pullback of Φ .

Lemma 6.2. *One has $H_3(T, \mathbb{Z}) \simeq \mathbb{Z}/8$. The class of $\pi^*\Phi$ is trivial if and only if the last equality in (T-G₂) holds. The left-hand side of this equality is unchanged when Φ is replaced by a cohomologous normalized cocycle.*

Proof. In T , put

$$i = x_1x_4^{-1}, \quad j = x_1x_3^{-1}, \quad k = x_1x_2^{-1}.$$

The braid relation gives $j = x_2^{-1}x_1$. Since the common cube $x_1^3 = x_2^3$ is central,

$$x_1ix_1^{-1} = j, \quad x_1jx_1^{-1} = k, \quad x_1kx_1^{-1} = i.$$

Substituting $x_2 = k^{-1}x_1$ into the braid relation gives $i^{-1} = k^{-1}j^{-1}$. Its cyclic conjugates therefore give $ij = k$, $jk = i$, and $ki = j$. These relations present Q_8 : eliminating k gives $i^4 = 1$, $i^2 = j^2$, and $jij^{-1} = i^{-1}$. Conversely, in $Q_8 \rtimes \langle x_1 \rangle$, with the indicated cyclic action, one has $(k^{-1}x_1)^3 = k^{-1}i^{-1}j^{-1}x_1^3 = x_1^3$ and $x_1(k^{-1}x_1)x_1 = i^{-1}x_1^3 = (k^{-1}x_1)x_1(k^{-1}x_1)$. Thus the substitutions $k = x_1x_2^{-1}$ and $x_2 = k^{-1}x_1$ give mutually inverse homomorphisms. The central generator z gives the direct factor, so $T \simeq (Q_8 \rtimes \langle x_1 \rangle) \times \langle z \rangle$, where both displayed cyclic groups are infinite.

The periodic free resolution in [38, Proposition 3.2], after tensoring with the trivial module $\mathbb{Z}$, gives

$$H_1(Q_8, \mathbb{Z}) \simeq (\mathbb{Z}/2)^2, \quad H_2(Q_8, \mathbb{Z}) = 0, \quad H_3(Q_8, \mathbb{Z}) \simeq \mathbb{Z}/8.$$

Conjugation by x_1 cyclically permutes the three nonzero elements of $H_1(Q_8, \mathbb{Z})$, so its invariant subgroup is zero. It acts trivially on $H_3(Q_8, \mathbb{Z})$, since its order divides three and $\text{Aut}(\mathbb{Z}/8)$ has order four.

The Lyndon–Hochschild–Serre spectral sequence for $Q_8 \rtimes \langle x_1 \rangle$ has only columns zero and one, see [11, Chapter VII, Section 6]. In total degrees two and three, the preceding computations leave only $H_0(\langle x_1 \rangle, H_3(Q_8, \mathbb{Z})) \simeq \mathbb{Z}/8$, in bidegree $(0, 3)$. Consequently, $H_2(Q_8 \rtimes \langle x_1 \rangle, \mathbb{Z}) = 0$, and the inclusion of Q_8 induces an isomorphism in degree three. The Künneth formula for the direct factor $\langle z \rangle \simeq \mathbb{Z}$ then shows that $Q_8 \hookrightarrow T$ also induces an isomorphism $H_3(Q_8, \mathbb{Z}) \simeq H_3(T, \mathbb{Z})$. Since $\mathbb{C}^\times$ is divisible, the universal coefficient theorem identifies $H^3(T, \mathbb{C}^\times)$ with $\text{Hom}(H_3(T, \mathbb{Z}), \mathbb{C}^\times)$ by evaluation on cycles.

In the normalized bar complex with trivial integer coefficients, the chain $[i|j|k] + [j|k|i] + [k|i|j]$ is a cycle. Indeed, $\partial[i|j|k] = [j|k] - [k|k] + [i|i] - [i|j]$, and adding its two cyclic permutations gives zero. Exponentiating the additive cocycle in Lemma D.4 gives the value $e^{(3+3+3)\pi i/4} = e^{\pi i/4}$ on this cycle. Thus it represents a generator of $H_3(Q_8, \mathbb{Z})$, hence also of $H_3(T, \mathbb{Z})$. Under the preceding universal coefficient isomorphism, $[\pi^*\Phi]$ is therefore trivial if and only if its value on this cycle is one. This value is precisely the left-hand side of the last equality in (T-G₂). Finally, replacing Φ by a cohomologous cocycle multiplies its pullback by a coboundary, which evaluates to one on every cycle. This proves the last assertion. $\square$

For the rest of this subsection, retain i, j, k from the preceding proof, and let A be the automorphism of Q_8 cycling them. Let $F : Q_8^3 \rightarrow \mathbb{Z}/8$ be the normalized A -invariant additive cocycle of Lemma D.4. Define

$$\Omega(ux_1^m z^a, vx_1^n z^b, wx_1^r z^c) = \exp\left(\frac{\pi i}{4} F(u, A^m v, A^{m+n} w)\right). \quad (6.2)$$

The cocycle identity for F , applied to $u, A^m v, A^{m+n} w, A^{m+n+r} t$, and its A -invariance show that Ω is a normalized cocycle on T . Its value on the cycle in the preceding proof is $e^{\pi i/4}$. Thus there are a unique $h \in \{0, \dots, 7\}$ and a normalized two-cochain J on T such that

$$\pi^*\Phi = \Omega^h \delta J, \quad \delta J(a, b, c) = \frac{J(b, c)J(a, bc)}{J(ab, c)J(a, b)}. \quad (6.3)$$

In particular, the last equality in (T-G₂) is equivalent to $h = 0$.

Lemma 6.3. *The pair (V, W) can be replaced by a pair in ${}_T\mathcal{YD}^{\Omega^h}$, with h as in (6.3), preserving the dimensions of its Nichols algebra and all its iterated adjoint objects. The scalars $\rho(z)$ and $\sigma(x_1)$ are preserved. When $\dim \sigma = 1$, so is $\varepsilon_\Phi(W)$. When $\dim \rho = \dim \sigma = 1$, so is $\rho(x_1)\sigma(z)$.*

Proof. The map π is bijective from x_1^T onto x_1^G . Lifting the support degrees and letting T act through π therefore gives simple objects in ${}^T_T\mathcal{YD}^{\pi^*\Phi}$. On every tensor word, the associators and braidings agree with the original ones.

By (6.3), the two-cochain J gives a braided monoidal equivalence to ${}^T_T\mathcal{YD}^{\Omega^h}$, with tensor structure and action

$$u_a \otimes v_b \longmapsto J(a, b)^{-1} u_a \otimes v_b, \quad g \triangleright_J v_x = \frac{J(gxg^{-1}, g)}{J(g, x)} g \triangleright v_x. \quad (6.4)$$

The tensor structure intertwines quantum symmetrizers and adjoint maps, so the asserted dimensions are preserved. The scalar assertions follow from the self-braidings, the mixed double braiding, and (6.1). Both operations preserve the dimensions of the homogeneous spaces.

Retain ρ, σ for the inducing representations after this replacement. By (6.2), the multipliers at z and x_1 are trivial on T and $C_T(x_1) = \langle -1, x_1, z \rangle \simeq C_2 \times \mathbb{Z}^2$, respectively. Thus ρ and σ are ordinary irreducible representations. $\square$

Lemma 6.4. *Assume that $\dim \sigma = 1$, $\sigma(x_1) = -1$, and $\varepsilon_\Phi(W) = 1$. After the replacement in Lemma 6.3, denote the inducing character of $C_T(x_1)$ again by σ . Then*

$$(h, \sigma((x_1 x_2^{-1})^2)) = (0, 1) \quad \text{or} \quad (4, -1).$$

The first possibility holds exactly when the last equality in (T-G₂) holds. In the second possibility, the left-hand side of that equality equals -1 .

Proof. Work over T with cocycle Ω^h . Choose the induced basis w_1, w_2, w_3, w_4 corresponding to the representatives $1, i, k, j$ and the ordered support x_1, x_2, x_3, x_4 . Write $x_a \triangleright w_b = \beta_{ab} w_{a \blacktriangleright b}$, $q = \sigma(x_1)$, and $\eta = \sigma((x_1 x_2^{-1})^2)$. Since σ is an ordinary character after the replacement and $(x_1 x_2^{-1})^2$ has order two, one has $\eta^2 = 1$. Substituting the values of F from Lemma D.4 into (2.6) gives

$$\beta_{23} = \beta_{42} = \beta_{34} = q\eta \exp(7h\pi i/4).$$

For each of these three entries the remaining centralizer element is $-x_1$, and the quotient of the two multiplier factors in (2.6) is $\exp(7h\pi i/4)$. Equation (6.1) therefore gives

$$\varepsilon_{\Omega^h}(W) = q\beta_{23}\beta_{42}\beta_{34} = q^4\eta \exp(5h\pi i/4).$$

Since $q = -1$, its value is one exactly when $(h, \eta) = (0, 1)$ or $(4, -1)$. Evaluation of Ω^h on the generating cycle gives $\exp(h\pi i/4)$, proving the remaining assertions. $\square$

6.2 Projective representations

Lemma 6.5. *One has*

$$Z(G) = \langle z, x_1^3, x_1 x_2 x_3 \rangle, \quad G/Z(G) \simeq A_4, \quad C_G(x_1) = \langle x_1, x_2 x_3, z \rangle.$$

The group $C_G(x_1)$ is abelian.

Proof. The conjugation action of x_1, x_2, x_3, x_4 on their common conjugacy class is

$$(243), \quad (134), \quad (142), \quad (123).$$

These permutations lie in A_4 . The subgroup generated by the first two is transitive. Its order is divisible by both 4 and 3, so it is A_4 . An element lies in the kernel of this action exactly when it commutes with all four x_i . Since z is central and G is generated by z, x_1, x_2, x_3, x_4 , the kernel is $Z(G)$. Hence $G/Z(G) \simeq A_4$.

It is direct that the elements $z, x_1^3, x_1x_2x_3$ are central, and $G/\langle z, x_1^3, x_1x_2x_3 \rangle$ is a quotient of A_4 . This quotient maps onto $G/Z(G) \simeq A_4$. Its order is therefore at most 12 and at least 12. Thus $Z(G) = \langle z, x_1^3, x_1x_2x_3 \rangle$.

The centralizer of (243) in A_4 is $\langle (243) \rangle$. Its inverse image in G is $\langle x_1, Z(G) \rangle = \langle x_1, x_2x_3, z \rangle$. Every element of $C_G(x_1)$ lies in this inverse image, while its generators commute with x_1 . Hence it is $C_G(x_1)$. The group is abelian: z and $x_1x_2x_3$ are central, and $x_2x_3 = x_1^{-1}(x_1x_2x_3)$. $\square$

Proposition 6.6. *One has $\dim \sigma = 1$.*

Proof. The defining relations of T give $x_1x_2x_3 = x_2x_3x_1$, and z is central. Thus x_1, x_2x_3, z commute in T . As before, the same symbols denote their images in G .

For any commuting triple a, b, c , substituting the formula for δJ in (6.3) into the definition of the alternator gives $f_{\delta J}(a, b, c) = 1$. Indeed, all two-cochain factors cancel because $ab = ba$, $bc = cb$, and $ca = ac$. Moreover, (6.2) and the normalization of F give $\Omega(a, b, z) = \Omega(a, z, b) = \Omega(z, a, b) = 1$ for all $a, b \in T$. Hence all six cocycle values occurring in $f_{\Omega^h}(x_1, x_2x_3, z)$ are one. Using $\pi^*\Phi = \Omega^h\delta J$, we therefore obtain

$$\begin{aligned} f_\Phi(x_1, x_2x_3, z) &= f_{\pi^*\Phi}(x_1, x_2x_3, z) \\ &= f_{\Omega^h}(x_1, x_2x_3, z)f_{\delta J}(x_1, x_2x_3, z) = 1. \end{aligned}$$

By Lemma 6.5, $C_G(x_1) = \langle x_1, x_2x_3, z \rangle$ is abelian. Lemma 4.4 shows that the restriction of f_Φ to this group is an alternating tricharacter, determined by its value on (x_1, x_2x_3, z) . It is therefore trivial. Thus $\Phi_{x_1}(a, b)/\Phi_{x_1}(b, a) = f_\Phi(x_1, a, b) = 1$ for all $a, b \in C_G(x_1)$. The corresponding twisted group algebra is commutative, so its irreducible module W_{x_1} is one-dimensional. $\square$

Lemma 6.7. *The homomorphism $g \mapsto [\rho(g)] \in \text{PGL}(V_z)$ factors through $G/Z(G) \simeq A_4$, and the resulting projective representation is irreducible. In particular, $\dim \rho \in \{1, 2, 3\}$.*

Proof. The proof of Proposition 6.6 gives $f_\Phi(x_1, x_2x_3, z) = 1$.

For each $c \in \{x_1^3, x_1x_2x_3\}$, Lemma 4.4 shows that $g \mapsto f_\Phi(g, z, c)$ is a character of G . Its value at z is 1 by alternation. For $c = x_1^3$, its value at x_1 is $f_\Phi(x_1, z, x_1)^3 = 1$. For $c = x_1x_2x_3$, its value at x_1 is $f_\Phi(x_1, x_2x_3, z)^{-1} = 1$. A character has the same value on conjugate elements. Since $G = \langle z, x_1^G \rangle$, both characters are trivial. For $c = z$, alternation gives $f_\Phi(g, z, z) = 1$ directly.

Let $c \in \{z, x_1^3, x_1x_2x_3\}$ and $g \in G$. Since z and c are central, Lemma 4.4 gives $\frac{\Phi_z(c, g)}{\Phi_z(g, c)} = f_\Phi(z, c, g) = f_\Phi(g, z, c) = 1$. The projective multiplication rule therefore yields

$$\rho(c)\rho(g) = \Phi_z(c, g)\rho(cg) = \Phi_z(g, c)\rho(gc) = \rho(g)\rho(c).$$

By Schur's lemma, $\rho(c)$ is a scalar. These three elements generate $Z(G)$, so projective multiplication shows that every element of $Z(G)$ acts by a scalar. The projective multiplication rule also shows that $g \mapsto [\rho(g)]$ is a homomorphism from G to $\text{PGL}(V_z)$. It is trivial on $Z(G)$, so it factors through $G/Z(G) \simeq A_4$. The resulting projective representation is irreducible, since any invariant subspace would be invariant under every $\rho(g)$.

The corresponding twisted group algebra of A_4 is semisimple and has dimension 12. Hence the matrix block belonging to this irreducible representation has dimension $(\dim \rho)^2 \leq 12$. Therefore $\dim \rho \in \{1, 2, 3\}$. $\square$

6.3 Adjoint objects

Write $Y_m = (\text{ad } W)^m(V)$. Let $\varphi_m^{Y, \Phi}$ be the recursive adjoint map from Lemma 2.8, with V and W interchanged. Let $Q_m^Y(r_1, \dots, r_m; z)$ denote the spaces Q_m from Subsection 3.3, with V and W interchanged. We retain the notation $\text{supp } Q$ from that subsection for multidegree support. By Lemma 2.8, the sum of the spaces $Q_m^Y(r_1, \dots, r_m; z)$, over all $r_1, \dots, r_m \in \text{supp } W$, is isomorphic to Y_m .

6.3.1 Dimension constraints from the adjoint objects

Lemma 6.8. *Assume that $Y_3 = (\text{ad } W)^3(V)$ is nonzero and simple. Then*

$$x_1 \triangleright w = -w \quad (w \in W_{x_1}).$$

Proof. If $Y_1 = 0$, then the recursive definition gives $Y_3 = 0$. Hence $Y_1 \neq 0$. Lemma 2.8, the G -linearity of the first adjoint map, and the transitive G -action on $\{x_1, x_2, x_3, x_4\}$ give $Q_1^Y(x_4; z) \neq 0$, with multidegree (x_4, z) . Since

$$x_4 \blacktriangleright x_1 = x_2 \neq x_1, \quad z \blacktriangleright x_1 = x_1, \quad x_1 \neq x_4,$$

Proposition 3.3, applied by inserting x_1 in the first position, gives

$$(x_1, x_4, z) \in \text{supp } Q_2^Y(x_1, x_4; z).$$

Insert x_2 in the second position. Here $x_2 \blacktriangleright x_1 = x_3$, $x_4 \blacktriangleright x_2 = x_3 \neq x_2$, and $z \blacktriangleright x_2 = x_2$, while the exclusion set in (3.3) is $\{x_1, x_4, x_3\}$. Therefore

$$(x_3, x_2, x_4, z) \in \text{supp } Q_3^Y(x_2, x_1, x_4; z). \quad (6.5)$$

The relations $x_3x_2 = x_1x_3$ and $x_3x_4 = x_2x_3$ give $x_3x_2x_4 = x_1x_2x_3$. Hence the component in (6.5) has degree $x_1x_2x_3z$, which is central by Lemma 6.5. Since Y_3 is simple, its support is one conjugacy class, and therefore $\text{supp } Y_3 = \{x_1x_2x_3z\}$. On the other hand, every vector in $Q_3^Y(x_1, x_1, x_4; z)$ has degree $x_1^2x_4z$. This degree is not central: since $x_1^2x_4 = x_1^3x_2x_1^{-1}$ and x_1^3 is central,

$$(x_1^2x_4) \blacktriangleright x_1 = x_2x_1x_2^{-1} = x_3 \neq x_1.$$

Consequently,

$$Q_3^Y(x_1, x_1, x_4; z) = 0. \quad (6.6)$$

The degree formulas in the proof of Proposition 3.3 show that the possible multidegrees of $Q_2^Y(x_1, x_4; z)$ are (x_1, x_4, z) , (x_3, x_1, z) , (x_4, x_3, z) . Consider the (x_1, x_1, x_4, z) -component of $\phi_3^{Y, \Phi}$ on $W_{x_1} \otimes Q_2^Y(x_1, x_4; z)$. The identity term contributes only from (x_1, x_4, z) . The middle double-braiding term begins in degree $(x_1^2x_4z) \blacktriangleright x_1 = x_3$, and the other two multidegrees begin, after $c_{1,2}^\Phi$, in degrees x_2 and x_3 . Thus the required component has exactly two contributions, acting on the first two factors as $\text{id}_{W \otimes W}$ and $c_{W, W}$. The two associativity factors in $c_{1,2}^\Phi$ cancel.

Choose an element of $Q_2^Y(x_1, x_4; z)$ with a nonzero (x_1, x_4, z) -component. Applying a suitable linear functional on $W_{x_4} \otimes V_z$ gives $0 \neq w_0 \in W_{x_1}$. Equation (6.6) then yields

$$w \otimes w_0 + (x_1 \triangleright w_0) \otimes w = 0 \quad (w \in W_{x_1}).$$

Taking $w = w_0$ gives $x_1 \triangleright w_0 = -w_0$. By Proposition 6.6, W_{x_1} is one-dimensional, so x_1 acts by -1 on all of W_{x_1} . $\square$

Proposition 6.9. *Assume that $\mathcal{B}(V \oplus W)$ is finite-dimensional and that (V, W) has Cartan matrix*

$$\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}.$$

Then $\dim \rho = 1$.

Proof. Proposition 6.6 gives $\dim \sigma = 1$.

By Theorem 2.25, the tuple (V, W) admits all reflections and has a finite Cartan graph. The Cartan matrix gives $X_1 \neq 0$, $X_2 = 0$, and $Y_3 \neq 0$. Since $-a_{12} = 1$ and $-a_{21} = 3$, Lemma 2.24, applied with $t = 1$ and $t = 3$, shows that X_1 and Y_3 are simple. Lemma 6.8 therefore gives

$\sigma(x_1) = -1$. Lemma 6.7 gives $\dim \rho \in \{1, 2, 3\}$. Its proof also shows that every element of $Z(G)$ acts on V_z by a scalar and that the projective class of ρ gives an irreducible projective representation of $G/Z(G) \simeq A_4$. In particular, we regard $\rho(z)$ as a nonzero scalar.

Let $T_0 = \sigma(z)\rho(x_1) \in \text{End}(V_z)$. By the proof of Proposition 6.6, $f_\Phi(x_1, x_2x_3, z) = 1$. Now $C_G(zx_1) = C_G(x_1) = \langle x_1, x_2x_3, z \rangle$. By Lemma 4.4, the restriction of f_Φ to this abelian group is an alternating tricharacter. It is determined by its value on x_1, x_2x_3, z , and is therefore trivial. Thus, for $a, b \in C_G(x_1)$,

$$\frac{\Phi_{zx_1}(a, b)}{\Phi_{zx_1}(b, a)} = f_\Phi(zx_1, a, b) = 1.$$

The corresponding twisted group algebra of $C_G(x_1)$ is therefore commutative.

The support of the nonzero simple object X_1 is $\{zx_1, zx_2, zx_3, zx_4\}$. By Proposition 2.4, its component $(X_1)_{zx_1}$ is an irreducible Φ_{zx_1} -projective representation of $C_G(x_1)$. Consequently, $\dim(X_1)_{zx_1} = 1$. For $v \in V_z$ and $0 \neq w \in W_{x_1}$, the original-category double braiding gives $\varphi_1^\Phi(v \otimes w) = (\text{id}_{V_z} - T_0)v \otimes w$. Lemma 2.8 identifies its image with $(X_1)_{zx_1}$. Since W_{x_1} is one-dimensional, it follows that

$$\text{rank}(\text{id}_{V_z} - T_0) = 1. \quad (6.7)$$

Suppose first that $\dim \rho = 3$. The associated twisted group algebra of A_4 is semisimple of dimension 12, and the simple module V_z contributes a matrix block of dimension 9. The remaining blocks have total dimension 3. Since every matrix block has square dimension, these must be three one-dimensional blocks. In particular, a one-dimensional projective representation exists. Its multiplication rule expresses the defining 2-cocycle as a coboundary. Hence the quotient projective representation on V_z can be rescaled to an ordinary irreducible representation of A_4 .

The three-dimensional irreducible representation of A_4 is realized on the sum-zero subspace of its permutation representation on four letters. On this subspace, a three-cycle has eigenvalues $1, e^{2\pi i/3}, e^{4\pi i/3}$. Since $x_1Z(G)$ is a three-cycle and T_0 is a nonzero scalar multiple of its operator, every eigenspace of T_0 is one-dimensional. However, (6.7) gives a two-dimensional eigenspace for the eigenvalue 1, a contradiction.

Assume now that $\dim \rho = 2$. Since $x_1^3 \in Z(G)$, projective multiplication shows that $\rho(x_1)^3$, and hence T_0^3 , is a nonzero scalar. Thus T_0 is diagonalizable. Equation (6.7) gives a one-dimensional eigenspace for the eigenvalue 1. Therefore the scalar T_0^3 is the identity. We may choose a basis e_0, e_r of V_z such that

$$T_0 e_0 = e_0, \quad T_0 e_r = r e_r, \quad r^2 + r + 1 = 0.$$

Choose $0 \neq w \in W_{x_1}$ and let $\xi = (1 - r)e_r \otimes w$. Then $\xi = \varphi_1^\Phi(e_r \otimes w)$, so ξ spans $(X_1)_{zx_1}$. We compute φ_2^Φ directly in the original category. For $v \in V_z$, the tensor-product action gives $z \triangleright \xi = \Phi^z(z, x_1)\rho(z)\sigma(z)\xi$, while projective multiplication gives

$$(zx_1) \triangleright v = \Phi_z(z, x_1)^{-1}\rho(z)\rho(x_1)(v).$$

Since z is central, $\Phi^z(z, x_1) = \Phi_z(z, x_1) = \Phi(z, x_1, z)$. The two factors therefore cancel, and $c_{x_1, V}c_{V, x_1}(v \otimes \xi) = \rho(z)^2 T_0(v) \otimes \xi$.

Moreover, $c_{1,2}^\Phi = a_{V,V,W}(c_{V,V} \otimes \text{id}_W)a_{V,V,W}^{-1}$. The two associativity constraints contribute $\Phi(z, z, x_1)$ and its inverse, while $c_{V,V}$ contributes $\rho(z)$. Thus the last term in the recursive formula contributes $\rho(z)(\text{id}_{V_z} - T_0)e_r \otimes ((\text{id}_{V_z} - T_0)v \otimes w)$. Together with the identity and double-braiding terms, this gives

$$\begin{aligned} \varphi_2^\Phi(e_0 \otimes \xi) &= (1 - \rho(z)^2)e_0 \otimes \xi, \\ \varphi_2^\Phi(e_r \otimes \xi) &= (1 - \rho(z)^2 r + \rho(z)(1 - r))e_r \otimes \xi. \end{aligned}$$

By Lemma 2.8, $X_2 = 0$ forces both expressions to vanish. The first gives $\rho(z)^2 = 1$. If $\rho(z) = 1$, the coefficient in the second expression is $2(1 - r) \neq 0$. Hence $\rho(z) = -1$.

Consider the reflected tuple $R_1(V, W) = (V^*, X_1)$. Lemma 2.26 shows that its Nichols algebra is finite-dimensional. Since $m_{12} = 1$, [34, Lemma 3.8] gives $(\text{ad } V^*)(X_1) \simeq W$. Equation (4.28) then gives $(\text{ad } X_1)(V^*) \simeq W \neq 0$. In particular, the reflected tuple is braided-indecomposable. Its supports are $\{z^{-1}\}$ and $\{zx_1, zx_2, zx_3, zx_4\}$. The element z^{-1} is central. The elements z^{-1}, zx_1, zx_2 satisfy the defining relations of T : the two braid words and the two cubes acquire the same factor z^3 . They generate G , and the four elements zx_i are distinct and have the same conjugation relations as the x_i .

The degree- x_4 component of $(\text{ad } X_1)(V^*)$ is nonzero and comes from the ordered degree (zx_4, z^{-1}) . The two applications of Proposition 3.3 used in the proof of Lemma 6.8 give successively (zx_1, zx_4, z^{-1}) and $(zx_3, zx_2, zx_4, z^{-1})$. Hence the corresponding third multidegree space is nonzero, and Lemma 2.8 gives $(\text{ad } X_1)^3(V^*) \neq 0$. Thus the corresponding reflected Cartan entry satisfies $a_{21} \leq -3$. The reflected tuple lies in the same finite Cartan graph, so Lemma 2.24, applied with $t = 3$, shows that $(\text{ad } X_1)^3(V^*)$ is simple. Lemma 6.8, applied to the reflected pair, therefore gives $(zx_1) \triangleright \xi = -\xi$.

Finally, note that $\rho(z)\rho(x_1) = \Phi_z(z, x_1)\rho(zx_1)$ and $\sigma(z)\sigma(x_1) = \Phi_{x_1}(z, x_1)\sigma(zx_1)$. The 3-cocycle identity applied to (z, x_1, z, x_1) gives $\Phi^{zx_1}(z, x_1) = \Phi_z(z, x_1)\Phi_{x_1}(z, x_1)$. Using these identities, $T_0 e_r = r e_r$, and the tensor-product action, we obtain

$$(zx_1) \triangleright \xi = \frac{\Phi^{zx_1}(z, x_1)}{\Phi_z(z, x_1)\Phi_{x_1}(z, x_1)} \rho(z)\sigma(x_1)r\xi = r\xi,$$

because $\rho(z) = \sigma(x_1) = -1$. Comparison with $(zx_1) \triangleright \xi = -\xi$ gives $r = -1$, contrary to $r^2 + r + 1 = 0$.

Thus neither $\dim \rho = 2$ nor $\dim \rho = 3$ is possible. Therefore $\dim \rho = 1$. $\square$

6.3.2 The first and second adjoint objects

Lemma 6.10. *Assume $\dim \rho = \dim \sigma = 1$. Then $\rho(x_1)\sigma(z) \neq 1$, and $X_1 = (\text{ad } V)(W)$ is simple of dimension four. Moreover,*

$$X_2 = 0 \iff (1 + \rho(z))(1 - \rho(x_1)\rho(z)\sigma(z)) = 0. \quad (6.8)$$

Proof. Lemma 2.8 identifies X_1 and X_2 with the images of the corresponding recursive maps. Choose $0 \neq v \in V_z$ and $0 \neq w_1 \in W_{x_1}$. On $V_z \otimes W_{x_1}$, the double braiding sends $v \otimes w_1$ to $(x_1 \triangleright v) \otimes (z \triangleright w_1) = \rho(x_1)\sigma(z)v \otimes w_1$. The double braiding is G -linear, and the four homogeneous components of $V \otimes W$ form one G -orbit. Hence $c_{W,V}c_{V,W} = \rho(x_1)\sigma(z)\text{id}_{V \otimes W}$. Since (V, W) is braided-indecomposable, $\rho(x_1)\sigma(z) \neq 1$. Therefore $\varphi_1^\Phi = (1 - \rho(x_1)\sigma(z))\text{id}_{V \otimes W}$, and its image is $V \otimes W$. This image has four one-dimensional homogeneous components, of degrees zx_1, zx_2, zx_3, zx_4 . Since z is central, these degrees form one conjugacy class. Every nonzero subobject is graded and G -stable, so it contains all four components. Thus X_1 is simple and $\dim X_1 = 4$.

Since φ_1^Φ is a nonzero scalar multiple of the identity, $X_1 = V \otimes W$ as a subobject of D_1 . Since φ_2^Φ is G -linear, it is enough to evaluate it on $v \otimes (v \otimes w_1)$. In $c_{1,2}^\Phi = a_{V,V,W}(c_{V,V} \otimes \text{id}_W)a_{V,V,W}^{-1}$, the two associativity constraints contribute $\Phi(z, z, x_1)$ and its inverse, while $c_{V,V}$ contributes $\rho(z)$. Hence $c_{1,2}^\Phi$ acts on this vector by $\rho(z)$. For the middle double braiding, the tensor-product action contributes $\Phi^z(z, x_1)$, while projective multiplication on V_z contributes $\Phi_z(z, x_1)^{-1}$. Since z is central, $\Phi^z(z, x_1) = \Phi_z(z, x_1) = \Phi(z, x_1, z)$, so these factors cancel. The remaining actions contribute $\rho(z)^2\rho(x_1)\sigma(z)$. The last term in the recursive formula contributes $\rho(z)(1 - \rho(x_1)\sigma(z))$. Consequently,

$$\begin{aligned} \varphi_2^\Phi(v \otimes (v \otimes w_1)) &= (1 - \rho(x_1)\rho(z)^2\sigma(z) + \rho(z)(1 - \rho(x_1)\sigma(z)))v \otimes (v \otimes w_1) \\ &= (1 + \rho(z))(1 - \rho(x_1)\rho(z)\sigma(z))v \otimes (v \otimes w_1). \end{aligned}$$

The four one-dimensional homogeneous components of $V \otimes X_1$, of degrees $z^2 x_i$, form one G -orbit. Since φ_2^Φ is G -linear, the same scalar occurs on all four components. Lemma 2.8 therefore gives the equivalence in (6.8). $\square$

Lemma 6.11. *Assume $\dim \rho = \dim \sigma = 1$ and $\sigma(x_1) = -1$. Then the homogeneous component $(Y_2)_{x_2 x_3 z}$ is one-dimensional if and only if $(\rho(x_1)\sigma(z))^2 - \rho(x_1)\sigma(z) + 1 = 0$ and $\varepsilon_\Phi(W) = 1$.*

Proof. Lemma 2.8 identifies Y_1 and Y_2 with the images of the corresponding recursive maps. Choose $0 \neq w_i \in W_{x_i}$ and $0 \neq v \in V_z$. On $W_{x_1} \otimes V_z$, the double braiding sends $w_1 \otimes v$ to $(z \triangleright w_1) \otimes (x_1 \triangleright v) = \rho(x_1)\sigma(z)w_1 \otimes v$. It is G -linear, so it acts by $\rho(x_1)\sigma(z)$ on all four homogeneous components of $W \otimes V$. Braided indecomposability gives $\rho(x_1)\sigma(z) \neq 1$. Hence $\varphi_1^{Y,\Phi} = (1 - \rho(x_1)\sigma(z)) \text{id}_{W \otimes V}$, and we identify Y_1 with $W \otimes V$. Define $\beta_{ij} \in \mathbb{C}^\times$ by $c_{W,W}(w_i \otimes w_j) = \beta_{ij} w_{i \blacktriangleright j} \otimes w_i$, and let $E_{ij} = \Phi(x_i, x_j, z)^{-1} w_i \otimes (w_j \otimes v)$.

Using the definition of $c_{1,2}^\Phi$ and the associativity constraint, we obtain

$$\begin{aligned} c_{1,2}^\Phi(E_{ij}) &= \Phi(x_i, x_j, z)^{-1} a_{W,W,V}(c_{W,W} \otimes \text{id}_V) a_{W,W,V}^{-1}(w_i \otimes (w_j \otimes v)) \\ &= \Phi(x_i, x_j, z)^{-1} \Phi(x_i, x_j, z) \beta_{ij} \Phi(x_{i \blacktriangleright j}, x_i, z)^{-1} w_{i \blacktriangleright j} \otimes (w_i \otimes v) \\ &= \beta_{ij} E_{i \blacktriangleright j, i}. \end{aligned}$$

For the middle double braiding, the W - V double braiding contributes $\rho(x_1)\sigma(z)$, and the two W - W braidings contribute $\beta_{ij}\beta_{i \blacktriangleright j, i}$. Since z is central, the definitions of Φ^{x_i} and Φ_{x_i} give

$$\frac{\Phi^{x_i}(x_j, z)}{\Phi_{x_i}(x_{i \blacktriangleright j}, z)} \Phi(x_i, x_j, z)^{-1} = \Phi(x_{(i \blacktriangleright j) \blacktriangleright i}, x_{i \blacktriangleright j}, z)^{-1}.$$

Hence

$$c_{W \otimes V, W} c_{W, W \otimes V}(E_{ij}) = \rho(x_1)\sigma(z) \beta_{ij} \beta_{i \blacktriangleright j, i} E_{(i \blacktriangleright j) \blacktriangleright i, i \blacktriangleright j}.$$

The recursive formula now gives

$$\varphi_2^{Y,\Phi}(E_{ij}) = E_{ij} - \rho(x_1)\sigma(z) \beta_{ij} \beta_{i \blacktriangleright j, i} E_{(i \blacktriangleright j) \blacktriangleright i, i \blacktriangleright j} + (1 - \rho(x_1)\sigma(z)) \beta_{ij} E_{i \blacktriangleright j, i}. \quad (6.9)$$

Since $c_{W,W}$ is G -linear and the diagonal homogeneous lines are conjugate, $\beta_{ii} = \beta_{11} = \sigma(x_1) = -1$ for every i . Thus (6.9) gives $\varphi_2^{Y,\Phi}(E_{ii}) = 0$. The relations of T give $x_2 x_3 z = x_3 x_4 z = x_4 x_2 z$, and the non-diagonal input tensors of this degree are E_{23}, E_{34}, E_{42} . With rows and columns ordered by these tensors, their images are the columns of

$$M = \begin{pmatrix} 1 & (1 - \rho(x_1)\sigma(z))\beta_{34} & -\rho(x_1)\sigma(z)\beta_{42}\beta_{34} \\ -\rho(x_1)\sigma(z)\beta_{23}\beta_{42} & 1 & (1 - \rho(x_1)\sigma(z))\beta_{42} \\ (1 - \rho(x_1)\sigma(z))\beta_{23} & -\rho(x_1)\sigma(z)\beta_{34}\beta_{23} & 1 \end{pmatrix}.$$

The vectors E_{23}, E_{34}, E_{42} form a basis of the off-diagonal subspace of this degree, and (6.9) maps this subspace into itself. Any diagonal input of this degree maps to zero. Lemma 2.8 therefore gives $\dim(Y_2)_{x_2 x_3 z} = \text{rank } M$. Since the diagonal entries of M are one, this component is one-dimensional if and only if $\text{rank } M = 1$.

Suppose that $\text{rank } M = 1$. The following two minors must vanish:

$$\det M_{\{1,3\},\{1,2\}} = -\beta_{23}\beta_{34}(\rho(x_1)\sigma(z) + (1 - \rho(x_1)\sigma(z))^2), \quad (6.10)$$

$$\begin{aligned} \det M_{\{1,2\},\{1,2\}} &= 1 + \rho(x_1)\sigma(z)(1 - \rho(x_1)\sigma(z))\beta_{23}\beta_{34}\beta_{42} \\ &= 1 - \rho(x_1)\sigma(z)(1 - \rho(x_1)\sigma(z))\varepsilon_\Phi(W). \end{aligned} \quad (6.11)$$

Indeed, three successive braidings on $W_{x_2} \otimes W_{x_3}$ contribute $\beta_{23}\beta_{42}\beta_{34}$. Since $\sigma(x_1) = -1$, (6.1) gives $\beta_{23}\beta_{34}\beta_{42} = -\varepsilon_\Phi(W)$. As all β_{ij} are nonzero, (6.10) gives $(\rho(x_1)\sigma(z))^2 - \rho(x_1)\sigma(z) + 1 = 0$, and hence $\rho(x_1)\sigma(z)(1 - \rho(x_1)\sigma(z)) = 1$. Equation (6.11) then gives $\varepsilon_\Phi(W) = 1$.

Conversely, assume $(\rho(x_1)\sigma(z))^2 - \rho(x_1)\sigma(z) + 1 = 0$ and $\varepsilon_\Phi(W) = 1$. Then the quadratic equality and (6.1) give $\beta_{23}\beta_{34}\beta_{42} = -1$. Substitution in M shows that its second column is $(1 - \rho(x_1)\sigma(z))\beta_{34}$ times its first. Its third column is $-\rho(x_1)\sigma(z)\beta_{42}\beta_{34}$ times its first. Since the first column is nonzero, $\text{rank } M = 1$. Thus $(Y_2)_{x_2x_3z}$ is one-dimensional. $\square$

Lemma 6.12. *Assume $\dim \rho = \dim \sigma = 1$ and $\sigma(x_1) = -1$. Then*

$$\text{supp } Y_2 = \{(x_1x_2x_3z)x_i^{-1} \mid 1 \leq i \leq 4\}.$$

The four corresponding homogeneous components are one-dimensional if and only if

$$(\rho(x_1)\sigma(z))^2 - \rho(x_1)\sigma(z) + 1 = 0, \quad \varepsilon_\Phi(W) = 1. \quad (6.12)$$

In this case, $\dim Y_2 = 4$ and Y_2 is simple.

Proof. By Lemma 2.8, Y_2 is the image of the second recursive map computed in (6.9). This map vanishes on $W_{x_i} \otimes (W_{x_i} \otimes V_z)$. If $i \neq j$, its identity term lies in $W_{x_i} \otimes (W_{x_j} \otimes V_z)$, whereas the other two terms lie in two other summands of $\bigoplus_{a,b=1}^4 W_{x_a} \otimes (W_{x_b} \otimes V_z)$. Hence its restriction to every off-diagonal summand is nonzero.

The relations of T give

$$\begin{aligned} x_2x_3 &= x_3x_4 = x_4x_2 = (x_1x_2x_3)x_1^{-1}, & x_1x_4 &= x_4x_3 = x_3x_1 = (x_1x_2x_3)x_2^{-1}, \\ x_1x_2 &= x_2x_4 = x_4x_1 = (x_1x_2x_3)x_3^{-1}, & x_1x_3 &= x_3x_2 = x_2x_1 = (x_1x_2x_3)x_4^{-1}. \end{aligned}$$

Thus the twelve off-diagonal summands give precisely the four degrees in the displayed support. By Lemma 6.5, $x_1x_2x_3z$ is central. Since the x_i^{-1} form one conjugacy class, the G -action is transitive on these four degrees and identifies the corresponding homogeneous components. One of them has degree $x_2x_3z = (x_1x_2x_3z)x_1^{-1}$, so Lemma 6.11 gives the equivalence in (6.12). Under these conditions the four components are one-dimensional, and hence $\dim Y_2 = 4$. Moreover, every nonzero subobject contains one of these four lines and therefore, by transitivity, all four. Thus Y_2 is simple. $\square$

6.3.3 The tetrahedral braiding and higher adjoint objects

Lemma 6.13. *Assume that $\dim \sigma = 1$, $\sigma(x_1) = -1$, and $\varepsilon_\Phi(W) = 1$, and that the last equality in (T-G₂) holds. There are nonzero vectors $e_i^W \in W_{x_i}$ and, simultaneously for every $m \geq 2$ and $i_1, \dots, i_m \in \{1, 2, 3, 4\}$, nonzero vectors*

$$e_{i_1 \dots i_m}^W \in W_{x_{i_1}} \otimes (W_{x_{i_2}} \otimes \dots \otimes (W_{x_{i_{m-1}}} \otimes W_{x_{i_m}}))$$

with the following property. For $1 \leq r < m$, let $B_r^{(m), \Phi}$ denote the endomorphism of the fixed right-associated tensor power $W^{\otimes m}$ representing the r -th standard braid generator, under the action induced by $c_{W,W}$ and the associativity constraints determined by Φ . Then

$$B_r^{(m), \Phi}(e_{i_1 \dots i_m}^W) = -e_{i_1 \dots (i_r \blacktriangleright i_{r+1}) i_{r+2} \dots i_m}^W. \quad (6.13)$$

Consequently, the quantum symmetrizers on $W^{\otimes m}$ have the same matrices as those of the braided vector space with basis $u_1, \dots, u_4$ and braiding $c(u_i \otimes u_j) = -u_{i \blacktriangleright j} \otimes u_i$.

Proof. By Lemma 6.2, the last equality in (T-G₂) gives $\pi^* \Phi = \delta J$. Lift W to T and apply the cochain equivalence of Lemma 6.3, retaining σ for the resulting ordinary character of $C_T(x_1)$. Lemma 6.4 gives $\sigma((x_1x_2^{-1})^2) = 1$. Since $C_T(x_1) = \langle (x_1x_2^{-1})^2, x_1, z \rangle$, the character

$$\chi(ux_1^m z^n) = (-1)^m \sigma(z)^n \quad (u \in Q_8; m, n \in \mathbb{Z})$$

of T extends σ .

Choose representatives t_i with $t_i x_1 t_i^{-1} = x_i$, and put $\tilde{e}_i = \chi(t_i)^{-1}(t_i \otimes 1)$ in the induced module. If $gt_i = t_j c$, where $c \in C_T(x_1)$, then $g \triangleright \tilde{e}_i = \chi(g)\tilde{e}_j$. Since $\chi(x_i) = -1$, its braiding is $c(\tilde{e}_i \otimes \tilde{e}_j) = -\tilde{e}_{i \blacktriangleright j} \otimes \tilde{e}_i$.

Transport these homogeneous vectors and their tensor-product bases back by the iterated tensorator of the same equivalence. It preserves every ordered homogeneous line and intertwines all adjacent braidings. Thus the vectors $e_{i_1 \dots i_m}^W$ are defined simultaneously in all degrees and satisfy (6.13). The same tensor structure intertwines the tensor products occurring in the adjoint recursion and the restrictions to preceding images. Every braid generator, and therefore every quantum symmetrizer, has the asserted matrix. $\square$

Assume that $\dim \rho = \dim \sigma = 1$, $\sigma(x_1) = -1$, and $\varepsilon_\Phi(W) = 1$, together with the last equality in (T-G₂). Fix $0 \neq v \in V_z$ and put $w_i = e_i^W$. Apply the cochain equivalence of Lemma 6.3, retaining v, w_i for the corresponding homogeneous vectors in the ordinary category. With the basis chosen in the proof of Lemma 6.13, one has $c_{W,W}(w_i \otimes w_j) = -w_{i \blacktriangleright j} \otimes w_i$. Define the mixed braiding coefficients by

$$c_{W,V}(w_i \otimes v) = r_i v \otimes w_i, \quad c_{V,W}(v \otimes w_i) = \ell_i w_i \otimes v.$$

The braid relation on $W_{x_i} \otimes W_{x_j} \otimes V_z$ gives $-r_i r_{i \blacktriangleright j} = -r_i r_j$, and hence $r_{i \blacktriangleright j} = r_j$. Similarly, on $V_z \otimes W_{x_i} \otimes W_{x_j}$, it gives $-\ell_i \ell_j = -\ell_{i \blacktriangleright j} \ell_i$, so $\ell_{i \blacktriangleright j} = \ell_j$. Here all coefficients are nonzero because the braidings are invertible. Transitivity of the conjugation action on $\{x_1, x_2, x_3, x_4\}$ therefore shows that both families are constant.

Transport the tensor-product bases of all tensor words containing exactly one V -factor back to the original tensor spaces using the iterated tensorator of the same equivalence. In these bases, the braidings between adjacent factors and between tensor products of factors, as well as the restricted adjoint maps, have the same matrices as in the ordinary category. In particular, the mixed double braiding has coefficient $r_i \ell_i = r_1 \ell_1$. Its coefficient on the original space $W_{x_1} \otimes V_z$ is $\rho(x_1)\sigma(z)$, which is preserved by Lemma 6.3. Consequently,

$$r_i \ell_i = \rho(x_1)\sigma(z) \quad (1 \leq i \leq 4). \quad (6.14)$$

For $m \geq 1$, write $e_{i_1 \dots i_m}$ for the chosen basis vector with ordered degrees $(x_{i_1}, \dots, x_{i_m}, z)$. Its total degree is $x_{i_1} \cdots x_{i_m} z$. We compute the recursive adjoint maps in these bases.

The middle term of the defining recursion is the double braiding between the first W -factor and the tensor product of the remaining factors. The first braiding changes the ordered degrees to $(x_{i_1} \blacktriangleright x_{i_2}, \dots, x_{i_1} \blacktriangleright x_{i_m}, z, x_{i_1})$. After the second braiding, the first factor has degree $x_j = (x_{i_1} \cdots x_{i_m} z) \blacktriangleright x_{i_1}$, while the other degrees remain in the indicated order. Since $x_1^G = \{x_1, x_2, x_3, x_4\}$, the index j is uniquely determined. For $m \geq 2$, write $\mathfrak{h}_m(i_1, \dots, i_m) = (j, i_1 \blacktriangleright i_2, \dots, i_1 \blacktriangleright i_m)$ for the resulting W -indices.

Each braiding contains $m-1$ braidings between W -factors and one mixed braiding. The former have total coefficient $(-1)^{2(m-1)} = 1$, and the latter have product $\rho(x_1)\sigma(z)$ by (6.14). Thus the double-braiding term sends $e_{i_1 \dots i_m}$ to $\rho(x_1)\sigma(z)e_{\mathfrak{h}_m(i_1, \dots, i_m)}$. Finally, $c_{1,2}^\Phi$ sends this basis vector to $-e_{i_1 \blacktriangleright i_2, i_1, i_3, \dots, i_m}$. Applying $\text{id}_W \otimes \varphi_{m-1}^{Y, \Phi}$ gives the last term below. Hence the defining recursion gives

$$\varphi_m^{Y, \Phi}(e_{i_1 \dots i_m}) = e_{i_1 \dots i_m} - \rho(x_1)\sigma(z)e_{\mathfrak{h}_m(i_1, \dots, i_m)} - w_{i_1 \blacktriangleright i_2} \otimes \varphi_{m-1}^{Y, \Phi}(e_{i_1 i_3 \dots i_m}), \quad (6.15)$$

with initial map $\varphi_1^{Y, \Phi}(e_i) = (1 - \rho(x_1)\sigma(z))e_i$. In the last term, $w_j \otimes (\cdot)$ denotes the linear operation defined in these coordinates by $w_j \otimes e_{k_1 \dots k_{m-1}} = e_{jk_1 \dots k_{m-1}}$. By Lemma 2.8, the image of the restriction of $\varphi_m^{Y, \Phi}$ to $W \otimes Y_{m-1}$ realizes Y_m .

To describe the third adjoint object, define

$$\begin{aligned} y = & e_{123} + e_{432} + e_{214} + e_{341} \\ & + (\rho(x_1)\sigma(z) - 1)(e_{142} + e_{413} + e_{231} + e_{324}) \\ & - \rho(x_1)\sigma(z)(e_{421} + e_{134} + e_{243} + e_{312}). \end{aligned} \quad (6.16)$$

The next lemma shows that this vector spans Y_3 when the additional relation $(\rho(x_1)\sigma(z))^2 - \rho(x_1)\sigma(z) + 1 = 0$ holds.

Lemma 6.14. *Assume $\dim \rho = \dim \sigma = 1$, $\sigma(x_1) = -1$, $(\rho(x_1)\sigma(z))^2 - \rho(x_1)\sigma(z) + 1 = 0$, and $\varepsilon_\Phi(W) = 1$. Assume also the last equality in (T-G₂). Then*

$$Y_3 = \mathbb{k}y \neq 0, \quad \text{supp } Y_3 = \{x_1x_2x_3z\} \subseteq Z(G), \quad Y_4 = 0.$$

Proof. Lemma D.2 gives $Y_3 = \mathbb{k}y \neq 0$. Each index triple in (6.16) is obtained from $(1, 2, 3)$ by successive replacements $(i, j, k) \mapsto (i \blacktriangleright j, i, k)$ and $(i, j, k) \mapsto (i, j \blacktriangleright k, j)$. The relation $x_i x_j = x_{i \blacktriangleright j} x_i$ shows that both replacements preserve the product of the corresponding group elements. Hence every summand of y has degree $x_1 x_2 x_3 z$. This element is central by Lemma 6.5, and therefore $\text{supp } Y_3 = \{x_1 x_2 x_3 z\} \subseteq Z(G)$.

Lemma D.3 gives $\varphi_4^{Y, \Phi}(w_1 \otimes y) = 0$. Since $Y_3 = \mathbb{k}y$ is stable under the action of G , $G \triangleright y \subseteq \mathbb{k}y$. Since W is simple and $w_1 \neq 0$, one has $W = \mathbb{k}G \triangleright w_1$. Lemma 2.9, applied with $H = G$, gives $W \otimes Y_3 = \mathbb{k}G \triangleright (w_1 \otimes y)$. The restriction of $\varphi_4^{Y, \Phi}$ to $W \otimes Y_3$ is a morphism in ${}^G_C \mathcal{YD}^\Phi$, so it vanishes on all of $W \otimes Y_3$. By Lemma 2.8, its image realizes Y_4 . Hence $Y_4 = 0$. $\square$

6.4 Parameter conditions for type G_2

At a finite Cartan matrix of type G_2 , the adjoint calculations give the first five conditions in (T-G₂). We first exclude the nontrivial class left by Lemma 6.4.

Retain the quaternion elements i, j, k from Lemma 6.2, and let A cyclically permute them. Write $u = (-1)^s i^{a(u)} j^{b(u)}$, with $a(u), b(u), s \in \mathbb{F}_2$, and put

$$E(u, v, w) = \sum_{r=0}^2 a(A^r u) a(A^r v) b(A^r w) \in \mathbb{F}_2.$$

By Lemma D.4, $F \equiv E \pmod{2}$. Thus E is a normalized A -invariant additive three-cocycle. Its formula shows that multiplying any argument by the central involution of Q_8 leaves its value unchanged.

Let

$$H = Q_8 \rtimes \langle t \mid t^3 = 1 \rangle, \quad G_* = H \times \langle z \mid z^6 = 1 \rangle,$$

where t acts by A , so that $H \simeq \text{SL}_2(3)$. The normalized cocycle

$$\Phi_*(ut^r z^a, vt^s z^b, wt^l z^c) = (-1)^{E(u, A^r v, A^{r+s} w)} \quad (6.17)$$

is independent of the z -coordinate. The homomorphism $T \rightarrow G_*$ sending x_1 to $-t$ and z to z pulls Φ_* back to Ω^4 : the powers of x_1 introduce only central quaternion signs, which do not affect E . We also write x_i for the images of the support elements in G_* .

For each $p \in \mathbb{k}^\times$ satisfying $p^2 - p + 1 = 0$, define

$$\begin{aligned} V_0 &= M(z, \rho_0), & \rho_0|_H &= 1, & \rho_0(z) &= p^{-1}, \\ W_0 &= M(-t, \sigma_0), & \sigma_0(-t) &= -1, & \sigma_0(z) &= p. \end{aligned} \quad (6.18)$$

Here $C_{G_*}(-t) = \langle -t \rangle \times \langle z \rangle \simeq C_6 \times C_6$. The multipliers at z and $-t$ are trivial on G_* and $C_{G_*}(-t)$, respectively. Since $p^3 = -1$ and $p^6 = 1$, the displayed values define ordinary characters and hence simple objects V_0, W_0 .

Lemma 6.15. Assume that $\dim \rho = \dim \sigma = 1$, $\sigma(x_1) = -1$, $p^2 - p + 1 = 0$, $p\rho(z) = 1$, and $\varepsilon_\Phi(W) = 1$, where $p = \rho(x_1)\sigma(z)$. If the product in the last equality of (T-G₂) is -1 , then $\dim \mathcal{B}(V \oplus W) = \dim \mathcal{B}(V_0 \oplus W_0)$.

Proof. By Lemmas 6.4 and 6.3, the pair can be replaced by one in ${}^T_T\mathcal{YD}^{\Omega^4}$, preserving its Nichols dimension. Retain ρ, σ for the inducing characters after this replacement. The character ρ is trivial on Q_8 , since $T_{\text{ab}} \simeq \mathbb{Z}^2$, and $\sigma((x_1x_2^{-1})^2) = -1$.

The normalized two-cocycle

$$J(ux_1^m z^n, vx_1^r z^s) = \rho(x_1)^{ms} \quad (u, v \in Q_8; m, n, r, s \in \mathbb{Z})$$

is pulled back from T_{ab} . By (6.4), its associated equivalence replaces $\rho(x_1)$ by 1 and $\sigma(z)$ by p . It leaves Ω^4 , $\rho(z) = p^{-1}$, $\sigma(x_1) = -1$, and $\sigma((x_1x_2^{-1})^2) = -1$ unchanged.

These are precisely the inducing characters obtained by lifting (V_0, W_0) along $T \rightarrow G_*$. Indeed, $(-t)^3$ is the image of $(x_1x_2^{-1})^2$, and $\sigma_0((-t)^3) = -1$. The map is bijective on the four-element conjugacy class. Using the induction representatives $1, i, k, j$, the action coefficients therefore agree with those of the normalized pair over T . The tensor actions and associators also agree because the pulled-back cocycle is Ω^4 .

Consequently, the braid operators and quantum symmetrizers agree on the corresponding tensor spaces. The Nichols algebras have equal dimensions in every $\mathbb{N}_0^2$ -degree. Together with the two cochain equivalences, this proves the assertion. $\square$

Lemma 6.16. Assume that the pair (V_0, W_0) in (6.18) admits all reflections. Let (V_r, W_r) , $r \geq 0$, be obtained by successively applying $R_2, R_1, R_2, R_1, \dots$. Then $A^{[(V_{r+4}, W_{r+4})]} = A^{[(V_r, W_r)]}$ for every $r \geq 0$.

Proof. Lemma D.5 gives

$$(-a_{12}^{[(V_r, W_r)]}, -a_{21}^{[(V_r, W_r)]}) = \begin{cases} (1, 3), & r = 0, 3, \\ (2, 3), & r = 1, 2. \end{cases}$$

To compare all subsequent reflections, we identify the fourth pair with the initial pair after lifting to T and applying a cochain equivalence.

The same lemma gives $\text{supp } V_4 = \{z\}$ and $\text{supp } W_4 = \{z^3x_i\}$. The H -action on V_4 is trivial, and, after rescaling the homogeneous basis vectors, the H -action on W_4 agrees with that on W_0 . The actions of z on V_4, W_4 are $p^{-1}, -p$, respectively.

For the initial pair, use the support map $z \mapsto z$, $x_i \mapsto x_i$ from T to G_* . For the fourth pair, use $z \mapsto z$, $x_i \mapsto z^3x_i$. Multiplying all x_i by the same central element preserves the defining relations of T , so both maps are homomorphisms. They have the same restriction to Q_8 and are bijective on the respective four-element conjugacy classes. Since Φ_* is independent of the z -coordinate, both maps pull it back to Ω^4 . For each map, the lift comparison in Lemma 6.3 preserves the recursive adjoint maps and is compatible with duals.

For this comparison, write ρ, σ for the inducing characters of each lifted pair. For the fourth pair, the value of ρ at x_1 is $(p^{-1})^3 = -1$, while $\sigma(z) = -p$. The value of σ at x_1 is $(-p)^3(-1) = -1$, since the support map sends x_1 to z^3x_1 . In both lifted pairs, $\rho|_{Q_8} = 1$, $\rho(z) = p^{-1}$, $\sigma(x_1) = -1$, and $\sigma((x_1x_2^{-1})^2) = -1$. Thus the remaining inducing character values $(\rho(x_1), \sigma(z))$ are $(-1, -p)$ for the fourth pair and $(1, p)$ for the initial pair.

The normalized two-cocycle

$$J(ux_1^m z^n, vx_1^r z^s) = (-1)^{ms} \quad (u, v \in Q_8; m, n, r, s \in \mathbb{Z})$$

is pulled back from T_{ab} . By (6.4), it multiplies both $\rho(x_1)$ and $\sigma(z)$ by -1 , leaving the other displayed inducing character values unchanged. The transformed lift of the fourth pair is therefore isomorphic to the lift of the initial pair in ${}^T_T\mathcal{YD}^{\Omega^4}$.

This braided monoidal equivalence preserves adjoint objects and duals. Together with the two lift comparisons, it therefore identifies the adjoint dimensions obtained after any further finite sequence of reflections. The alternating reflection sequences starting at steps zero and four consequently have the same Cartan matrices at every step. This proves the assertion. $\square$

Lemma 6.17. *Assume that $\dim \rho = \dim \sigma = 1$, $\sigma(x_1) = -1$, $p^2 - p + 1 = 0$, $p\rho(z) = 1$, and $\varepsilon_\Phi(W) = 1$, where $p = \rho(x_1)\sigma(z)$. If the product in the last equality of (T-G₂) is -1 , then $\mathcal{B}(V \oplus W)$ is infinite-dimensional.*

Proof. Suppose, to the contrary, that $\mathcal{B}(V \oplus W)$ is finite-dimensional. By Lemma 6.15, $\mathcal{B}(V_0 \oplus W_0)$ is finite-dimensional. Theorem 2.25 therefore implies that (V_0, W_0) admits all reflections and has a finite Cartan graph.

Starting from (V_0, W_0) , denote the successive pairs under $R_2, R_1, R_2, R_1, \dots$ by (V_r, W_r) . By Lemma 6.16, their Cartan matrices repeat after every four reflections. Using the Cartan entries computed in Lemma D.5, the composite of the first four reflections transports the root lattice at step four to that at step zero by

$$P = \begin{pmatrix} 1 & 0 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} -1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -5 & 3 \\ -12 & 7 \end{pmatrix},$$

where coordinates are written as columns.

The four-step periodicity implies that P^n transports roots from step $4n$ to the initial object. Since $(P - I)^2 = 0$, one has $P^n = I + n(P - I)$, and hence

$$P^n \alpha_2 = 3n\alpha_1 + (6n + 1)\alpha_2 \quad (n \geq 0).$$

Each vector is the image of the simple root α_2 at step $4n$, so each is a real root at the initial object. These roots are pairwise distinct, contradicting the finiteness of its Cartan graph. $\square$

Proposition 6.18. *Assume (T-G₂). Then X_1 and Y_3 are nonzero and simple, $\dim X_1 = 4$, $\dim Y_3 = 1$, and $X_2 = Y_4 = 0$. Consequently, the Cartan matrix of (V, W) is $\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$.*

Proof. The quadratic condition in (T-G₂) gives $\rho(x_1)\sigma(z) \neq 1$, while the next condition gives $\rho(x_1)\rho(z)\sigma(z) = 1$. Lemma 6.10 therefore shows that X_1 is simple of dimension four and that $X_2 = 0$. Lemma 6.14 gives $\dim Y_3 = 1$ and $Y_4 = 0$. Thus Y_3 is nonzero and simple. Therefore $m_{12} = 1$ and $m_{21} = 3$, which gives the asserted Cartan matrix. $\square$

Proposition 6.19. *Assume that $\mathcal{B}(V \oplus W)$ is finite-dimensional and that (V, W) has Cartan matrix $\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$. Then all conditions in (T-G₂) hold.*

Proof. Propositions 6.6 and 6.9 give $\dim \sigma = \dim \rho = 1$. By Theorem 2.25, the pair admits all reflections and $\mathcal{G}(V, W)$ is finite. The Cartan matrix gives $X_1, Y_2, Y_3 \neq 0$, and Lemma 2.24, applied in the two directions with $t = 1$ and $t = 2, 3$, shows that these objects are simple. Therefore Lemma 6.8 gives $\sigma(x_1) = -1$, while Lemma 6.10 gives $\rho(x_1)\sigma(z) \neq 1$.

By Lemma 6.12,

$$\text{supp } Y_2 = \{(x_1 x_2 x_3 z) x_i^{-1} \mid 1 \leq i \leq 4\}.$$

Since $x_2 x_3 z = (x_1 x_2 x_3 z) x_1^{-1}$ and $x_1 x_2 x_3 z$ is central, one has $C_G(x_2 x_3 z) = C_G(x_1) = \langle x_1, x_2 x_3, z \rangle$. The proof of Proposition 6.6 shows that f_Φ is trivial on $C_G(x_1)^3$. Hence, for $a, b \in C_G(x_1)$,

$$\frac{\Phi_{x_2 x_3 z}(a, b)}{\Phi_{x_2 x_3 z}(b, a)} = f_\Phi(x_2 x_3 z, a, b) = 1.$$

The corresponding twisted group algebra of $C_G(x_1)$ is commutative. Since Y_2 is simple, Proposition 2.4 identifies its component $(Y_2)_{x_2 x_3 z}$ with an irreducible $\Phi_{x_2 x_3 z}$ -projective representation

of this centralizer. Thus $\dim(Y_2)_{x_2x_3z} = 1$. The G -action identifies the four homogeneous components, so Lemma 6.12 gives (6.12).

It remains to prove $\rho(x_1)\rho(z)\sigma(z) = 1$. By Lemma 2.26, $\mathcal{B}(V^* \oplus X_1)$ is finite-dimensional. Since $m_{12} = 1$, [34, Lemma 3.8] and (4.28) give

$$(\text{ad } V^*)(X_1) \simeq W \simeq (\text{ad } X_1)(V^*).$$

Thus $R_1(V, W)$ is braided-indecomposable, with supports $\{z^{-1}\}$ and $\{zx_i \mid 1 \leq i \leq 4\}$. The elements z^{-1}, zx_1, zx_2 satisfy the defining relations of T and generate G , while the zx_i have the same conjugation table as the x_i . Two applications of Proposition 3.3 give

$$(zx_4, z^{-1}) \mapsto (zx_1, zx_4, z^{-1}) \mapsto (zx_3, zx_2, zx_4, z^{-1}).$$

Hence $(\text{ad } X_1)^3(V^*) \neq 0$, so $-a_{21}^{[R_1(V, W)]} \geq 3$. Since the reflected pair lies in the same finite graph, Lemma 2.24 shows that this third adjoint object is simple. Lemma 6.8, applied to the reflected pair, therefore shows that zx_1 acts by -1 on $(X_1)_{zx_1}$.

Choose $0 \neq v \in V_z$, $0 \neq w \in W_{x_1}$, and let $\xi = (1 - \rho(x_1)\sigma(z))v \otimes w$. This vector spans $(X_1)_{zx_1}$, and

$$\begin{aligned} (zx_1) \triangleright \xi &= \frac{\Phi^{zx_1}(z, x_1)}{\Phi_z(z, x_1)\Phi_{x_1}(z, x_1)} \sigma(x_1)\rho(x_1)\rho(z)\sigma(z) \xi \\ &= -\rho(x_1)\rho(z)\sigma(z) \xi. \end{aligned}$$

The cocycle identity at (z, x_1, z, x_1) identifies $\Phi^{zx_1}(z, x_1)$ with $\Phi_z(z, x_1)\Phi_{x_1}(z, x_1)$. Together with $\sigma(x_1) = -1$, this gives the second equality. Comparison with $(zx_1) \triangleright \xi = -\xi$ yields $\rho(x_1)\rho(z)\sigma(z) = 1$. Together with (6.12), this proves the first five conditions in (T-G₂). If its last equality failed, Lemma 6.4 would make its left-hand side equal to -1 . Lemma 6.17 would then imply $\dim \mathcal{B}(V \oplus W) = \infty$, contrary to the hypothesis. Thus all conditions in (T-G₂) hold. $\square$

6.5 Reflections

Lemma 6.20. Assume (T-G₂). The supports of $R_1(V, W) = (V^*, X_1)$ are

$$\text{supp } V^* = \{z^{-1}\}, \quad \text{supp } X_1 = \{zx_1, zx_2, zx_3, zx_4\},$$

and the supports of $R_2(V, W) = (Y_3, W^*)$ are

$$\text{supp } Y_3 = \{x_1x_2x_3z\}, \quad \text{supp } W^* = \{x_1^{-1}, x_2^{-1}, x_4^{-1}, x_3^{-1}\}.$$

The assignments

$$\begin{aligned} (z, x_1, x_2) &\mapsto (z^{-1}, zx_1, zx_2) && \text{for } R_1(V, W), \\ (z, x_1, x_2) &\mapsto (x_1x_2x_3z, x_1^{-1}, x_2^{-1}) && \text{for } R_2(V, W) \end{aligned}$$

extend to epimorphisms $T \twoheadrightarrow G$.

Proof. Proposition 6.18 shows that both reflections exist. The proof of Lemma 6.10 identifies X_1 with $V \otimes W$, so $\text{supp } X_1 = \{zx_1, zx_2, zx_3, zx_4\}$. Lemma 6.14 gives $\text{supp } Y_3 = \{x_1x_2x_3z\}$, and duality replaces every degree by its inverse. This proves the displayed support formulas.

The elements z^{-1} and $x_1x_2x_3z$ are central by Lemma 6.5. For the first assignment, the two sides of the braid relation and the two cubes are obtained from the original ones by multiplication by z^3 . For the second assignment, the braid and cube relations follow by taking inverses. Thus both assignments define homomorphisms from T to G .

Under the first assignment, the elements x_3 and x_4 defined in T map to zx_3 and zx_4 . Under the second assignment, the braid relation gives

$$x_2^{-1}x_1^{-1}x_2 = x_4^{-1}, \quad x_1^{-1}x_2^{-1}x_1 = x_3^{-1},$$

so they map to the third and fourth displayed elements. Lemma 3.1 therefore gives the stated tetrahedral conjugation tables.

Finally, the first image contains $z = (z^{-1})^{-1}$ and $x_i = (zx_i)z^{-1}$ for every i . The second image contains all x_i and also

$$z = (x_1x_2x_3)^{-1}(x_1x_2x_3z).$$

Both homomorphisms are therefore surjective. $\square$

Lemma 6.21. *Assume (T-G₂), and use the ordered supports in Lemma 6.20. The pair $R_1(V, W) = (V^*, X_1)$ satisfies the first five conditions in (T-G₂). Moreover, z^{-1} acts by $\rho(z)$ on $(V^*)_{z^{-1}}$.*

Proof. By Proposition 6.18, X_1 is simple of dimension four. Lemma 6.20 gives its four-element support, so its homogeneous spaces are one-dimensional. Since V^* is also one-dimensional, both inducing projective representations have dimension one.

The quadratic condition in (T-G₂) gives $\rho(x_1)\sigma(z) \neq 1$. The first recursive map therefore identifies X_1 with $V \otimes W$. Use the cochain equivalence of Lemma 6.3, together with its tensorator and the induced identification of duals. Choose the bases of X_1 and its tensor products by transporting those induced by the vectors $v \otimes w_i$ in the ordinary category, with the bases fixed before (6.15). Write ξ_i for the resulting vector of degree zx_i , and ξ_{ij} for the resulting tensor basis vector with ordered degrees (zx_i, zx_j) .

The hexagon identities express the braiding between two copies of $V \otimes W$ as one V - V braiding, two mixed braidings, and one W - W braiding. Their coefficients are $\rho(z)$, $\rho(x_1)\sigma(z)$, and -1 , respectively. Hence

$$c_{X_1, X_1}(\xi_{ij}) = -\rho(x_1)\rho(z)\sigma(z)\xi_{i\blacktriangleright j, i} = -\xi_{i\blacktriangleright j, i}. \quad (6.19)$$

The second equality uses $\rho(x_1)\rho(z)\sigma(z) = 1$. Taking $i = j = 1$ shows that zx_1 acts by -1 on $(X_1)_{zx_1}$. Applying (6.19) three times to the component indexed by $(2, 3)$ gives the scalar -1 . Together with the self-braiding scalar -1 , this gives $\varepsilon_\Phi(X_1) = 1$ by (6.1).

By Lemma 2.5, z^{-1} acts by $\rho(z)$ on $(V^*)_{z^{-1}}$. Since V is one-dimensional, evaluation gives $V^* \otimes V \simeq \mathbb{k}$. The twists of V^* and V are both $\rho(z)$, so the balancing identity gives the double-braiding coefficient $\rho(z)^{-2}$ between V^* and V . The evaluation and coevaluation identities show that the double-braiding coefficient between V^* and W is the inverse of that between V and W . Using $X_1 \simeq V \otimes W$, the hexagon identities therefore give the mixed double-braiding coefficient

$$\rho(z)^{-2}(\rho(x_1)\sigma(z))^{-1} = \rho(x_1)\sigma(z). \quad (6.20)$$

Again, the equality follows from $\rho(x_1)\rho(z)\sigma(z) = 1$.

Thus the mixed double-braiding coefficient and the central self-braiding scalar are the original $\rho(x_1)\sigma(z)$ and $\rho(z)$. Their quadratic and product relations follow from (T-G₂). Together with the preceding dimension, self-braiding, and ε_Φ calculations, this proves the first five conditions. $\square$

Lemma 6.22. *Assume (T-G₂), and use the ordered supports in Lemma 6.20. The pair $R_2(V, W) = (Y_3, W^*)$ satisfies the first five conditions in (T-G₂). Moreover, $x_1x_2x_3z$ acts by $\rho(z)$ on $(Y_3)_{x_1x_2x_3z}$.*

Proof. By Lemma 6.14, $Y_3 = \mathbb{k}y \neq 0$ has degree $x_1x_2x_3z$. The homogeneous spaces of W^* are one-dimensional, so both inducing projective representations have dimension one. Throughout the calculation, use the cochain equivalence of Lemma 6.3, its tensorator, and the induced identification of duals. We use the tensor bases fixed before (6.15), in which the braiding coefficients are those of (6.13) and (6.14).

The canonical twists of W and V are $\sigma(x_1)\text{id}_W$ and $\rho(z)\text{id}_V$. Apply the iterated balancing identity to $Y_3 \subseteq W \otimes (W \otimes (W \otimes V))$. The twists of the individual factors contribute $\sigma(x_1)^3\rho(z)$. The six braidings between the W -factors conjugate their degrees by the product of

those degrees. Every ordered triple occurring in y has product $x_1x_2x_3 \in Z(G)$, so these braidings fix each ordered triple and have coefficient $(-1)^6 = 1$. The three mixed double braidings contribute $(\rho(x_1)\sigma(z))^3$. Consequently, $x_1x_2x_3z$ acts on Y_3 by

$$\sigma(x_1)^3(\rho(x_1)\sigma(z))^3\rho(z) = \rho(z).$$

Here we used $\sigma(x_1) = -1$ and $(\rho(x_1)\sigma(z))^3 = -1$.

We next compute the double braiding between W and Y_3 . Since Y_3 has central degree, this operator preserves the one-dimensional space $W_{x_1} \otimes Y_3$. It therefore suffices to compare the coefficient of $w_1 \otimes e_{123}$ in its value on $w_1 \otimes y$. The two braidings send $w_1 \otimes e_{134}$ to $\rho(x_1)\sigma(z)w_1 \otimes e_{123}$: conjugation by x_1 sends the indices 134 to 123, the six braidings between W -factors contribute 1, and the two mixed braidings contribute $\rho(x_1)\sigma(z)$. The operator permutes the ambient tensor basis up to nonzero scalars, so no other basis vector contributes to this coefficient.

By (6.16), the coefficients of e_{134} and e_{123} in y are $-\rho(x_1)\sigma(z)$ and 1, respectively. The required scalar is therefore $-(\rho(x_1)\sigma(z))^2 = 1 - \rho(x_1)\sigma(z)$, and hence

$$c_{Y_3, W} c_{W, Y_3}(w_1 \otimes y) = (1 - \rho(x_1)\sigma(z))w_1 \otimes y. \quad (6.21)$$

The GAP program [GAP source](#) also verifies this identity by computing the eight adjacent braidings on every term of $w_1 \otimes y$. Lemma 6.13 and (6.14) identify the coefficients used by the program with the categorical braidings.

Since Y_3 is one-dimensional and the G -action on the four homogeneous components of W is transitive, G -linearity extends (6.21) to all of $W \otimes Y_3$. The evaluation and coevaluation identities invert this coefficient when W is replaced by W^* . Since $\rho(x_1)\sigma(z)(1 - \rho(x_1)\sigma(z)) = 1$, the double braiding between Y_3 and W^* has coefficient

$$(1 - \rho(x_1)\sigma(z))^{-1} = \rho(x_1)\sigma(z). \quad (6.22)$$

By Lemma 2.5, x_1^{-1} acts by $\sigma(x_1) = -1$ on $(W^*)_{x_1^{-1}}$. In the ordinary category, the action of x_i^{-1} on the dual basis is the transpose of the action of x_i on W . Thus its coefficients are again -1 , with the inverse conjugation permutations. Using the ordering of $\text{supp } W^*$ in Lemma 6.20 and transporting the dual tensor bases back, the braiding coefficients of W^* are therefore all -1 . Its cube on the component indexed by $(2, 3)$ has coefficient -1 , and (6.1) gives $\varepsilon_\Phi(W^*) = 1$.

The mixed double-braiding coefficient is $\rho(x_1)\sigma(z)$, and the central self-braiding scalar is $\rho(z)$. Their quadratic and product relations are precisely those assumed in (T-G₂). This proves the remaining two conditions and the assertion. $\square$

Lemma 6.23. *Assume (T-G₂), and use the ordered supports in Lemma 6.20. Both reflected pairs $R_1(V, W) = (V^*, X_1)$ and $R_2(V, W) = (Y_3, W^*)$ satisfy (T-G₂). Moreover, z^{-1} and $x_1x_2x_3z$ act by $\rho(z)$ on $(V^*)_{z^{-1}}$ and $(Y_3)_{x_1x_2x_3z}$, respectively.*

Proof. Lemmas 6.21 and 6.22 give the first five conditions and the asserted central actions. It remains to verify the last equality in (T-G₂).

By Lemma 6.2, $[\pi^*\Phi] = 0$. The two support maps in Lemma 6.20 are obtained by composing π with the following endomorphisms of T :

$$\begin{aligned} z &\longmapsto z^{-1}, & x_i &\longmapsto zx_i, \\ z &\longmapsto x_1x_2x_3z, & (x_1, x_2, x_3, x_4) &\longmapsto (x_1^{-1}, x_2^{-1}, x_4^{-1}, x_3^{-1}). \end{aligned}$$

Indeed, $x_1x_2x_3$ is central already in T . For the first assignment, both sides of the braid relation and of the cube relation are multiplied by z^3 . For the second assignment, these relations follow

by taking inverses. The images of x_3, x_4 follow from the braid relation as in Lemma 6.20. Thus both assignments preserve the defining relations.

By functoriality of group cohomology, the class pulled back along either reflected support map is therefore zero. Applying Lemma 6.2 to each reflected pair gives the last equality in (T-G₂). $\square$

Proposition 6.24. *Assume (T-G₂). Then (V, W) admits all reflections. Choose the ordered support representatives successively as in Lemma 6.20. Every $Q \in \mathcal{F}_2(V, W)$ satisfies the assumptions at the beginning of this section and (T-G₂), with its one-element support in the first coordinate. The element of this support acts on the corresponding homogeneous line by the original scalar $\rho(z)$. Moreover, every such Q has Cartan matrix $\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$. Consequently, the Cartan graph is finite and standard of type G₂.*

Proof. We induct on the length of a reflection word that has already been defined. The empty word has the required properties. Suppose that such a word gives a pair Q , and use the support representatives and notation induced by Lemma 6.20. Proposition 6.18 gives $m_{12}^Q = 1$ and $m_{21}^Q = 3$. Hence both reflections of Q are defined, their components are simple by the same proposition and rigid duality, and the Cartan matrix at Q is $\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$. Lemma 6.20 gives the required support relations and generation of G for both reflected pairs. Lemma 6.23 shows that both pairs satisfy (T-G₂) and that the element of their one-element support acts by the same scalar as for Q , hence by $\rho(z)$.

By (6.20) and (6.22), the mixed double braidings of the two reflected pairs have coefficient $\rho(x_1)\sigma(z) \neq 1$, where the inequality follows from the quadratic relation in (T-G₂). Thus both pairs are braided-indecomposable and satisfy all the assumptions at the beginning of this section. The induction proves that every finite reflection word is defined and establishes the assertions for every $Q \in \mathcal{F}_2(V, W)$. Since the Cartan matrix is constant, every real root belongs to the finite root system of type G₂. Hence the Cartan graph is finite and standard. $\square$

6.6 Finite Cartan graphs

Throughout this subsection, we retain the group G , the cocycle Φ , and the pair (V, W) fixed at the beginning of this section. The support classification determines the finite Cartan graph of a finite-dimensional pair.

Proposition 6.25. *Assume that (V, W) admits all reflections, and let $P = (P_1, P_2) \in \mathcal{F}_2(V, W)$ have an indecomposable Cartan matrix of finite type. After interchanging the two entries if necessary, there is a quandle isomorphism*

$$Z_T^{4,1} \longrightarrow \text{supp}(P_1 \oplus P_2)$$

that sends the one-element orbit to $\text{supp } P_1$ and the four-element orbit to $\text{supp } P_2$, where $\text{supp } P_1$ consists of a single central element. The induced homomorphism of enveloping groups is an epimorphism $T \twoheadrightarrow G$.

Proof. Lemma 2.29 gives $G = \langle \text{supp } P_1 \cup \text{supp } P_2 \rangle$. For the moment, order the entries so that $-a_{12}^{[P]} = 1$ and $-a_{21}^{[P]} \in \{1, 2, 3\}$, as permitted by the classification of indecomposable rank-two Cartan matrices of finite type. By the definition of the Cartan entries, $(\text{ad } P_1)(P_2) \neq 0$, $(\text{ad } P_1)^2(P_2) = 0$, and $(\text{ad } P_2)^4(P_1) = 0$. Hence P is braided-indecomposable by (2.13). Its entries are simple, so Theorem 3.4 applies to $X = \text{supp}(P_1 \oplus P_2)$.

Because X generates G , the conjugation action of G on X has image $\text{Inn}(X)$ and kernel $C_G(X) = Z(G)$. Thus Lemma 6.5 gives $\text{Inn}(X) \simeq A_4$. For the five quandles in Theorem 3.4, the permutations displayed in Section 3 generate groups of orders 12, 4, 6, 6, 8, respectively. Only $Z_T^{4,1}$ has inner automorphism group of order 12, and therefore $X \simeq Z_T^{4,1}$. By Proposition 2.4, the supports of P_1 and P_2 are G -orbits, hence the two $\text{Inn}(X)$ -orbits of X . After interchanging the entries if necessary, these are the singleton and the four-element orbit, in that order. The

singleton commutes with every element of X , and is therefore central because X generates G . Finally, a quandle isomorphism $Z_T^{4,1} \rightarrow X$ induces a homomorphism from its enveloping group to G . Its image contains X , hence equals G . Lemma 3.1 identifies the source with T and gives the required support labeling. $\square$

Proposition 6.26. *Assume that $\mathcal{B}(V \oplus W)$ is finite-dimensional. Then (T-G₂) holds.*

Proof. By Theorem 2.25, the pair (V, W) admits all reflections and has a finite Cartan graph. Proposition 2.28 gives $P \in \mathcal{F}_2(V, W)$ such that $1 \leq a_{12}^{[P]} a_{21}^{[P]} \leq 3$. Thus the Cartan matrix of P is indecomposable and of finite type. Proposition 6.25 provides $\tau \in S_2$ and support representatives for $P^\tau = (P_{\tau(1)}, P_{\tau(2)})$, where the support of the first component consists of a single central element and that of the second has four elements. Repeated use of Lemma 2.26, together with invariance under interchanging the two entries, shows that $\mathcal{B}((P^\tau)_1 \oplus (P^\tau)_2)$ is finite-dimensional. Moreover, $a_{12}^{[P^\tau]} a_{21}^{[P^\tau]} = a_{12}^{[P]} a_{21}^{[P]}$.

The Cartan matrix of P^τ is indecomposable, so both first adjoint objects are nonzero. Lemma 2.8, G -linearity, and the transitivity of the tetrahedral support give a nonzero component with ordered degrees (x_4, z) . For the successive insertions of x_1 and x_2 , the relations $x_4 \blacktriangleright x_1 = x_2 \neq x_1$, $x_2 \blacktriangleright x_1 = x_3$, and $x_4 \blacktriangleright x_2 = x_3 \neq x_2$, together with the centrality of z , verify (3.2). The corresponding exclusion sets in (3.3) are $\{x_4\}$ and $\{x_1, x_4, x_3\}$, respectively. Proposition 3.3 therefore gives the successive nonzero multidegrees

$$(x_4, z) \longmapsto (x_1, x_4, z) \longmapsto (x_3, x_2, x_4, z).$$

Hence $(\text{ad}(P^\tau)_2)^3((P^\tau)_1) \neq 0$, so $-a_{21}^{[P^\tau]} \geq 3$. Since $-a_{12}^{[P^\tau]} \geq 1$ and $a_{12}^{[P^\tau]} a_{21}^{[P^\tau]} \leq 3$, the Cartan matrix of P^τ is $\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$. Proposition 6.19 now gives (T-G₂) for P^τ .

The identity $(R_i Q)^\tau \simeq R_{\tau(i)}(Q^\tau)$, together with the involutivity of reflections [35, Corollary 5.8], gives $(V, W)^\tau \in \mathcal{F}_2(P^\tau)$. Proposition 6.24, applied to P^τ , shows that $(V, W)^\tau$ satisfies (T-G₂) and that the support of its first component consists of a single central element. If τ were the transposition, this first support would be $\text{supp } W$, which has four elements. Thus $\tau = \text{id}$, and the parameter conditions hold for (V, W) with the support representatives obtained along this reflection word. Their ordering differs from the original tetrahedral ordering by a rack automorphism. It extends to an automorphism of the enveloping group T fixing z , so the pulled-back class is also zero for the original map π . Lemma 6.2 gives the last equality in (T-G₂) for the original representatives. The other parameters are preserved by this relabelling, and therefore (T-G₂) holds for (V, W) . $\square$

6.7 Nichols algebras of the positive-root factors of type G_2

Under (T-G₂), Proposition 6.24 gives a finite standard Cartan graph of type G_2 . Its six positive real roots index the six simple factors. We first determine the dimensions of their Nichols algebras and then describe their supports and tensor decomposition.

Lemma 6.27. *Assume (T-G₂). Every simple factor indexed by a positive real root of G_2 has a Nichols algebra of dimension 6 when its support consists of a single central element, and of dimension 72 when its support consists of four elements.*

Proof. Let R be such a factor. By Lemma 2.23, R is isomorphic to a component of some pair in $\mathcal{F}_2(V, W)$. Proposition 6.24 shows that this pair again satisfies (T-G₂). Its central component is one-dimensional, and its support representative acts by $\rho(z)$. If R is this component, its self-braiding is multiplication by $\rho(z)$. The quadratic relation in (T-G₂) gives $(\rho(x_1)\sigma(z))^3 = -1$ and $\rho(x_1)\sigma(z) \neq -1$, while the other scalar relation gives $\rho(z) = (\rho(x_1)\sigma(z))^{-1}$. Thus $\rho(z)$ is a primitive sixth root of unity. On a fixed parenthesization of $R^{\otimes n}$, every braid generator acts by a nonzero scalar. One of them acts by $\rho(z)$, and the braid relations force adjacent scalar

generators to agree. Hence the degree- n quantum symmetrizer acts by $\prod_{j=1}^n (1 + \rho(z) + \cdots + \rho(z)^{j-1})$. This scalar is nonzero for $n < 6$ and zero for $n \geq 6$. Consequently, $\mathcal{B}(R)$ is one-dimensional in degrees $0, \dots, 5$ and zero in higher degrees, so $\dim \mathcal{B}(R) = 6$.

If R has tetrahedral support, Lemma 6.13, applied to the corresponding reflected pair, identifies the matrices of its quantum symmetrizers in every degree with those of the tetrahedral rack with constant coefficient -1 . Their ranks therefore agree in every degree, so the two Nichols algebras have the same Hilbert series. By [3, Theorem 6.15], the latter has dimension 72. $\square$

Proposition 6.28. *Assume (T- G_2). For the reduced expression $(2, 1, 2, 1, 2, 1)$ of a longest morphism with target $[(V, W)]$, let $\beta_1, \dots, \beta_6$ and $M_{\beta_1}, \dots, M_{\beta_6}$ be the corresponding positive real roots and simple factors. Multiplication induces an isomorphism of $\mathbb{N}_0^2$ -graded objects in ${}^G\mathcal{YD}^\Phi$,*

$$\mathcal{B}(M_{\beta_6}) \otimes (\cdots (\mathcal{B}(M_{\beta_2}) \otimes \mathcal{B}(M_{\beta_1}))) \xrightarrow{\sim} \mathcal{B}(V \oplus W).$$

Proof. Write $(i_1, \dots, i_6) = (2, 1, 2, 1, 2, 1)$. Proposition 6.24 gives all reflections and the common Cartan matrix $\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$. Using $s_i(\alpha_j) = \alpha_j - a_{ij}\alpha_i$, one obtains

$$(\beta_1, \dots, \beta_6) = (\alpha_2, \alpha_1 + 3\alpha_2, \alpha_1 + 2\alpha_2, 2\alpha_1 + 3\alpha_2, \alpha_1 + \alpha_2, \alpha_1).$$

These are precisely the six positive roots of G_2 , so the chosen word is reduced and represents a longest morphism.

For each k , the prefix $\text{id}_{[(V, W)]} s_{i_1} \cdots s_{i_{k-1}}$ maps α_{i_k} to β_k . Hence Lemma 2.23 identifies M_{β_k} with the i_k -th component of the tuple obtained by the successive reflections $i_1, \dots, i_{k-1}$. Proposition 6.24 shows that the first component of every such tuple is supported on a single central element and the second on four elements. Since $i_k = 2$ for odd k and $i_k = 1$ for even k , there are three factors of each kind. Finally, since all reflections exist and the Cartan graph is finite, Proposition 2.22 gives the displayed multiplication isomorphism. $\square$

Corollary 6.29. *Under (T- G_2),*

$$\dim \mathcal{B}(V \oplus W) = 6^3 72^3 = 80\,621\,568.$$

Proof. Among the six simple objects M_{β_k} in Proposition 6.28, three are one-dimensional and three are four-dimensional. By Lemma 6.27, their Nichols algebras have dimensions 6 and 72, respectively. The tensor decomposition gives the asserted dimension. $\square$

6.8 Proof of the classification theorem

Proof of Theorem 6.1. If $\mathcal{B}(V \oplus W)$ is finite-dimensional, then Proposition 6.26 gives (T- G_2).

Conversely, assume (T- G_2). Proposition 6.24 shows that (V, W) admits all reflections and that its Cartan graph is standard of type G_2 , with the stated matrix at every object. Corollary 6.29 gives the stated dimension. In particular, $\mathcal{B}(V \oplus W)$ is finite-dimensional. $\square$

Remark 6.30. When $\Phi = 1$, the category ${}^G\mathcal{YD}^\Phi$ is the ordinary Yetter–Drinfeld category over the Hopf algebra $\mathbb{C}G$. Under the conditions of Theorem 6.1, one has $\varepsilon_1(W) = \sigma(x_2 x_3)$ and $\rho(x_1)\rho(z) = \rho(x_1 z)$. Hence Theorem 6.1 specializes to [22, Theorem 2.9]. In particular, the Cartan graph is standard of type G_2 , with Cartan matrix $\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$ at every object. Moreover, over $\mathbb{C}$, the corresponding Nichols algebra has dimension $6^3 72^3 = 80\,621\,568$, in agreement with the ordinary Hopf algebra case.

7 The Γ_3 case

In this section, Γ_3 is given by the presentation

$$\Gamma_3 = \langle \varepsilon, g, h \mid \varepsilon^3 = 1, \quad g\varepsilon g^{-1} = \varepsilon^{-1}, \quad [h, g] = [h, \varepsilon] = 1 \rangle. \quad (7.1)$$

This is the presentation from Subsection 3.2 after replacing its generator h by $\varepsilon^{-1}h$. Let $\pi : \Gamma_3 \rightarrow G$ be an epimorphism onto a finite non-abelian group. The images in G are denoted by the same letters ε, g, h . Let $\Phi \in Z^3(G, \mathbb{C}^\times)$ be normalized.

Throughout this section, $V, W \in {}^G_G\mathcal{YD}^\Phi$ are finite-dimensional simple objects satisfying

$$G = \langle \text{supp}(V \oplus W) \rangle, \quad (V, W) \text{ is braided-indecomposable}. \quad (7.2)$$

We use the following notation for the two ordered support pairs that occur in the classification:

$$(3, 2) : \quad V = M(g, \rho), \quad W = M(\varepsilon h, \sigma), \quad (7.3)$$

and

$$(3, 1) : \quad V = M(g, \rho), \quad W = M(h, \tau). \quad (7.4)$$

Their support quandles are $Z_3^{3,2}$ and $Z_3^{3,1}$, respectively.

Lemma 7.3 below shows that the inducing representations in (7.3) are one-dimensional and that $\dim \rho = 1$ in (7.4). These automatic dimensions are omitted from the conditions below. $\dim \tau$ is the only nonautomatic dimension condition and separates the two (3, 1) dimension branches.

7.1 The four parameter families

The parameters occurring in the four families below, namely Δ_1, s, Δ_3 , and $\kappa_\Phi(\varepsilon)$, are computed from the fixed cocycle Φ and the indicated inducing projective representations. They are defined in (7.19), (7.23), and (7.14), respectively. In the two-dimensional case, $I = \text{id}_{W_h}$. The four parameter families are as follows.

$$\mathcal{P}_1^\Phi : \quad \begin{aligned} & V = M(g, \rho), \quad W = M(\varepsilon h, \sigma), \\ & \rho(g) = \sigma(\varepsilon h) = -1, \quad \Delta_1 = 1, \quad 1 + s + s^2 = 0; \end{aligned} \quad (7.5)$$

$$\mathcal{P}_2^\Phi : \quad \begin{aligned} & V = M(g, \rho), \quad W = M(\varepsilon h, \sigma), \\ & \rho(g) = \sigma(\varepsilon h) = -1, \quad \Delta_1 = 1, \quad s = 1; \end{aligned} \quad (7.6)$$

$$\mathcal{P}_3^\Phi : \quad \begin{aligned} & V = M(g, \rho), \quad W = M(h, \tau), \quad \dim \tau = 2, \\ & \kappa_\Phi(\varepsilon) = 1, \quad \rho(g) = -1, \quad \tau(h) = -I, \quad \Delta_3 = I; \end{aligned} \quad (7.7)$$

$$\mathcal{P}_4^\Phi : \quad \begin{aligned} & V = M(g, \rho), \quad W = M(h, \tau), \quad \dim \tau = 1, \\ & \kappa_\Phi(\varepsilon) = 1, \quad \rho(g) = -1, \\ & 1 - \rho(h)\tau(g) + \rho(h)^2\tau(g)^2 = 0, \quad \rho(h)\tau(g)\tau(h) = 1. \end{aligned} \quad (7.8)$$

In $\mathcal{P}_1^\Phi$ and $\mathcal{P}_2^\Phi$, the displayed conditions and Lemma 7.8 give $\kappa_\Phi(\varepsilon) = s^3 = 1$. Thus every pair in the four families satisfies $\kappa_\Phi(\varepsilon) = 1$, although this condition is explicit only in $\mathcal{P}_3^\Phi$ and $\mathcal{P}_4^\Phi$. The classification for finite quotients of Γ_3 is as follows.

Theorem 7.1. *Under the standing assumptions (7.1)–(7.2), the Nichols algebra $\mathcal{B}(V \oplus W)$ is finite-dimensional if and only if, after possibly interchanging V and W and retaining the notation (V, W) for the resulting ordered pair, there is a choice of generators $\varepsilon, g, h \in G$ inducing the epimorphism from*

(7.1) and realizing one of the two ordered pairs in (7.3) or (7.4), for which the parameters computed with the fixed cocycle Φ place the tuple in one of

$$\mathcal{P}_1^\Phi, \quad \mathcal{P}_2^\Phi, \quad \mathcal{P}_3^\Phi, \quad \mathcal{P}_4^\Phi.$$

In every case the Cartan graph is standard of type B_2 , with matrix

$$\begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix} \quad (7.9)$$

at every object. Moreover,

$$\dim \mathcal{B}(V \oplus W) = \begin{cases} 10368, & (V, W) \in \mathcal{P}_1^\Phi \cup \mathcal{P}_4^\Phi, \\ 2304, & (V, W) \in \mathcal{P}_2^\Phi \cup \mathcal{P}_3^\Phi. \end{cases} \quad (7.10)$$

Outline of the proof.

- Subsection 7.2 determines the possible supports and projective dimensions. This reduces the proof to the $(3, 2)$ case and the two $(3, 1)$ cases with $\dim \tau = 2$ or 1 .
- Subsection 7.3 identifies $\kappa_\Phi(\varepsilon)$ as the obstruction to the triviality of the inflated cocycle. Subsections 7.4 and 7.5 compute the required adjoint objects, derive the parameter conditions, and show that this obstruction vanishes in every finite-dimensional case.
- Subsection 7.6 constructs a braided equivalence with the ordinary Yetter–Drinfeld setting and identifies the four families $\mathcal{P}_i^\Phi$ with their ordinary counterparts.
- Subsection 7.7 proves necessity and shows that the four families are closed under reflections. It also proves that their Cartan graphs are standard of type B_2 .
- Subsection 7.8 transfers finite-dimensionality and the dimension formulas from the ordinary classification.

Notation used in this section.

Notation	Defined in	Used to prove
Δ_1	(7.19)	In the $(3, 2)$ case, X_1 is simple if and only if $\Delta_1 = 1$.
s	(7.19)	The conditions $1 + s + s^2 = 0$ and $s = 1$ distinguish $\mathcal{P}_1^\Phi$ from $\mathcal{P}_2^\Phi$.
Δ_3	(7.23)	In the $(3, 1)$ case with $\dim \tau = 2$, X_1 is simple if and only if $\Delta_3 = I$.
$\kappa_\Phi(\varepsilon)$	(7.14)	The equality $\kappa_\Phi(\varepsilon) = 1$ is equivalent to the triviality of $[\pi^* \Phi]$. It follows from the other conditions in $\mathcal{P}_1^\Phi, \mathcal{P}_2^\Phi$ and is imposed in $\mathcal{P}_3^\Phi, \mathcal{P}_4^\Phi$.

7.2 Supports and projective dimensions

Lemma 7.2. *For every $x \in \Gamma_3$,*

$$\pi^{-1}(C_G(\pi(x))) := \{a \in \Gamma_3 : \pi(a)\pi(x) = \pi(x)\pi(a)\} = C_{\Gamma_3}(x). \quad (7.11)$$

Consequently, π induces a bijection $x^{\Gamma_3} \simeq \pi(x)^G$. Moreover, $C_G(g) = \langle g, h \rangle$, $C_G(\varepsilon h) = \langle \varepsilon, g^2, h \rangle$, and $C_G(h) = G$. Every conjugacy class has one of the following forms, for some $\delta \in Z(G)$:

$$\{\delta\} \quad \{\varepsilon\delta, \varepsilon^2\delta\} \quad \{g\delta, g\varepsilon\delta, g\varepsilon^2\delta\}. \quad (7.12)$$

Proof. The presentation is that of $(\langle \varepsilon \rangle \rtimes \langle g \rangle) \times \langle h \rangle$, where g acts on $\langle \varepsilon \rangle$ by inversion. Hence every element has a unique normal form $\varepsilon^i g^m h^n$, with $i \in \{0, 1, 2\}$, and $[\Gamma_3, \Gamma_3] = \langle \varepsilon \rangle$. Such an element commutes with ε exactly when m is even and with g exactly when $i = 0$. Hence $Z(\Gamma_3) = \langle g^2, h \rangle$. If $\pi(\varepsilon) = 1$, then G is generated by the commuting images of g and h , a contradiction. Since $\langle \varepsilon \rangle$ has order three, $\ker \pi \cap \langle \varepsilon \rangle = 1$. For $a \in \Gamma_3$ and $n \in \ker \pi$, the commutator $[a, n]$ belongs to $\ker \pi \cap [\Gamma_3, \Gamma_3]$, so it is trivial. If $\pi(a)$ centralizes $\pi(x)$, the same intersection contains $[a, x]$, and therefore $[a, x] = 1$. The reverse inclusion is immediate, proving (7.11).

Surjectivity of π makes the induced map on conjugacy classes surjective. If $\pi(axa^{-1}) = \pi(bxb^{-1})$, then $\pi(b^{-1}a)$ centralizes $\pi(x)$. By (7.11), $b^{-1}a$ centralizes x , so the two conjugates are equal. Since h is central, the element $\varepsilon^i g^m h^n$ commutes with g exactly when $i = 0$. Hence $C_{\Gamma_3}(g) = \langle g, h \rangle$. It commutes with εh exactly when m is even. Hence $C_{\Gamma_3}(\varepsilon h) = \langle \varepsilon, g^2, h \rangle$. Finally, h is central, so $C_{\Gamma_3}(h) = \Gamma_3$. Equation (7.11) now gives the stated centralizers in G . Finally, split the element $\varepsilon^i g^m h^n$ according to the parity of m . If m is even, $g^m h^n$ is central, and conjugation by g interchanges its two nontrivial ε -multiples. If m is odd, $g^{m-1} h^n$ is central, and conjugation by ε cycles the three possible ε -multiples of $g^m h^n$. The bijection between conjugacy classes now gives (7.12). $\square$

For commuting a, b, c , recall that $f_\Phi(a, b, c)$ is the quotient of $\Phi(a, b, c)\Phi(b, c, a)\Phi(c, a, b)$ by $\Phi(a, c, b)\Phi(c, b, a)\Phi(b, a, c)$. On an abelian subgroup this is an alternating tricharacter, and

$$\frac{\Phi_x(a, b)}{\Phi_x(b, a)} = f_\Phi(x, a, b) \quad (a, b \in C_G(x), ab = ba). \quad (7.13)$$

It is invariant under simultaneous conjugation. This follows from the standard prism homotopy between an inner automorphism of the bar resolution and the identity, together with the fact that the alternator of a coboundary is one on a commuting triple.

Lemma 7.3. *The restriction of f_Φ to each of $\langle g, h \rangle$ and $\langle \varepsilon, g^2, h \rangle$ is trivial. Consequently, in the case (7.3) one has $\dim \rho = \dim \sigma = 1$, whereas in the case (7.4) one has $\dim \rho = 1$ and $\dim \tau \leq 2$.*

Proof. The abelian group $\langle g, h \rangle$ is generated by two elements, so every alternating tricharacter on it is trivial. Conjugation by g sends ε to ε^2 and fixes g^2, h . Hence simultaneous-conjugation invariance and trilinearity give

$$f_\Phi(\varepsilon, g^2, h) = f_\Phi(\varepsilon^2, g^2, h) = f_\Phi(\varepsilon, g^2, h)^2.$$

Thus $f_\Phi(\varepsilon, g^2, h) = 1$. Since the abelian subgroup $\langle \varepsilon, g^2, h \rangle$ is generated by these three elements, alternation and trilinearity show that f_Φ is trivial on $\langle \varepsilon, g^2, h \rangle$. By Lemma 7.2 and (7.13), the twisted group algebras $\mathbb{C}^{\Phi_g} C_G(g)$ and $\mathbb{C}^{\Phi_{\varepsilon h}} C_G(\varepsilon h)$ are commutative. Since G is finite, their irreducible modules are one-dimensional, proving $\dim \rho = \dim \sigma = 1$.

Finally, consider (7.4), and let

$$A = \langle \varepsilon, g^2, h \rangle.$$

The subgroup A is abelian and normal in G . Moreover, $G = \langle A, g \rangle$, $g^2 \in A$, and $A \neq G$, since G is non-abelian. Hence $[G : A] = 2$. Since $h \in A$ and f_Φ is trivial on A^3 , (7.13) shows that $\mathbb{C}^{\Phi_h} A$ is commutative. Therefore the operators $\tau(a)$, $a \in A$, have a common eigenline $L \subseteq W_h$. For $a \in A$, normality gives $g^{-1}ag \in A$, so the projective multiplication law shows that a preserves $g \triangleright L$. Moreover,

$$g \triangleright (g \triangleright L) = g^2 \triangleright L \subseteq L$$

as subspaces. Thus $L + g \triangleright L$ is a nonzero G -submodule of W_h . Since W_h is irreducible, $W_h = L + g \triangleright L$, and therefore $\dim \tau \leq 2$. $\square$

Corollary 7.4. *Let $U, Z \in {}^G \mathcal{YD}^\Phi$ be simple, and suppose that $(\text{ad } U)(Z)$ is nonzero and simple. Then the supports of U and Z cannot both have three elements.*

Proof. Assume otherwise. By (7.12), choose $\delta_1, \delta_2 \in Z(G)$ such that $x = g\delta_1 \in \text{supp } U$ and $y = g\varepsilon\delta_2 \in \text{supp } Z$, and take nonzero $u \in U_x, w \in Z_y$. By Lemma 2.8, identify $X = (\text{ad } U)(Z)$ with the image of $\text{id} - c_{Z,U}c_{U,Z}$, and let

$$\xi = (\text{id} - c_{Z,U}c_{U,Z})(u \otimes w) \in X_{xy}.$$

Its two summands are nonzero and have ordered degrees $(g\delta_1, g\varepsilon\delta_2)$ and $(g\varepsilon\delta_1, g\varepsilon^2\delta_2)$, respectively. Since $xy = \varepsilon g^2\delta_1\delta_2$, conjugation by xy cycles these two pairs and $(g\varepsilon^2\delta_1, g\delta_2)$, in that order. Thus $(xy) \triangleright \xi$ has a nonzero component of the last ordered degree, whereas ξ does not, so $(xy) \triangleright \xi \notin \mathbb{C}\xi$.

On the other hand, X_{xy} is an irreducible Φ_{xy} -projective $C_G(xy)$ -module by Proposition 2.4. For every $a \in C_G(xy)$, (7.13) gives $\Phi_{xy}(xy, a)/\Phi_{xy}(a, xy) = f_\Phi(xy, xy, a) = 1$. Hence the action of xy commutes with that of $C_G(xy)$, and Schur's lemma shows that xy acts by a scalar on X_{xy} . This is a contradiction. $\square$

Proposition 7.5. *Assume that $\mathcal{B}(V \oplus W)$ is finite-dimensional. After possibly interchanging V and W , there is a choice of generators $\varepsilon, g, h \in G$ satisfying the relations in (7.1) and inducing an epimorphism $\Gamma_3 \rightarrow G$, for which the pair has one of the forms (7.3) or (7.4). In the case (3, 2), one has $\dim \rho = \dim \sigma = 1$. In the case (3, 1), one has $\dim \rho = 1$ and $\dim \tau \in \{1, 2\}$.*

Proof. Braided indecomposability and (2.13) give $X_1 \neq 0$. By Theorem 2.25, the pair admits all reflections and has a finite Cartan graph. Lemma 2.24 therefore shows that X_1 is simple. Corollary 7.4 excludes two three-element supports.

Equation (7.11) gives $Z(G) = \langle g^2, h \rangle$. Hence (7.12) shows that every conjugacy class with at most two elements is contained in the abelian subgroup $\langle \varepsilon, g^2, h \rangle$. Since both supports generate the non-abelian group G , at least one of them has three elements. Thus exactly one support has three elements.

After interchanging the two entries if necessary, (7.12) therefore gives, for some $\delta_1, \delta_2 \in Z(G)$,

$$\text{supp } V = \{g\delta_1, g\varepsilon\delta_1, g\varepsilon^2\delta_1\}, \quad \text{supp } W = \begin{cases} \{\varepsilon\delta_2, \varepsilon^2\delta_2\}, & (3, 2), \\ \{\delta_2\}, & (3, 1). \end{cases}$$

The three-point support generates $g\delta_1$ and ε , and the other support then gives δ_2 . Since δ_1, δ_2 are central, replacing g by $g\delta_1$ and h by δ_2 preserves the relations in (7.1). The resulting elements generate G , so they induce the required epimorphism and give (7.3) or (7.4). The dimension assertions follow from Lemma 7.3, with $\tau \neq 0$ giving $\dim \tau \in \{1, 2\}$. $\square$

7.3 The third homology of Γ_3

We use the normalized integral bar complex of Γ_3 . Its degree-three chains are integral sums of symbols $[a|b|c]$, with any symbol containing the identity understood to be zero. For a normalized $\mathbb{C}^\times$ -valued three-cochain Ψ on Γ_3 , our conventions are

$$\begin{aligned} \partial[a|b|c] &= [b|c] - [ab|c] + [a|bc] - [a|b], \\ \left\langle \Psi, \sum_j n_j [a_j|b_j|c_j] \right\rangle &= \prod_j \Psi(a_j, b_j, c_j)^{n_j}. \end{aligned}$$

Recall that $\pi : \Gamma_3 \rightarrow G$ is an epimorphism onto a finite non-abelian group. The pullback of Φ is $(\pi^*\Phi)(a, b, c) = \Phi(\pi(a), \pi(b), \pi(c))$. Define

$$\kappa_\Phi(\varepsilon) = \Phi(\varepsilon, \varepsilon, \varepsilon)\Phi(\varepsilon, \varepsilon^2, \varepsilon). \quad (7.14)$$

In the bar complex, let

$$K = [\varepsilon|\varepsilon|\varepsilon] + [\varepsilon|\varepsilon^2|\varepsilon]. \quad (7.15)$$

The boundaries of the two summands are $-\lceil \varepsilon^2 | \varepsilon \rceil + \lceil \varepsilon | \varepsilon^2 \rceil$ and its negative. Hence K is a cycle, and $\langle \pi^* \Phi, K \rangle = \kappa_\Phi(\varepsilon)$.

Lemma 7.6. *One has*

$$H_3(\Gamma_3, \mathbb{Z}) \simeq \mathbb{Z}/3,$$

and K represents a generator. Moreover, for

$$\omega : \Gamma_3 \longrightarrow S_3, \quad \omega(\varepsilon) = (123), \quad \omega(g) = (12), \quad \omega(h) = 1,$$

the map $\omega_ : H_3(\Gamma_3, \mathbb{Z}) \rightarrow H_3(S_3, \mathbb{Z})$ is injective.*

Proof. Let $D = \langle \varepsilon, g \rangle$. The presentation shows that $D \simeq C_3 \rtimes \mathbb{Z}$, where the generator of $\mathbb{Z}$ acts on C_3 by inversion, and that $\Gamma_3 = D \times \langle h \rangle$. For the extension $1 \rightarrow C_3 \rightarrow D \rightarrow \mathbb{Z} \rightarrow 1$, the Lyndon–Hochschild–Serre homology spectral sequence is

$$E_{p,q}^2 = H_p(\mathbb{Z}, H_q(C_3, \mathbb{Z})) \implies H_{p+q}(D, \mathbb{Z}),$$

see [11, Chapter VII, Section 6]. The positive-degree homology of C_3 , computed from its standard two-periodic resolution, is $\mathbb{Z}/3$ in positive odd degrees and zero in positive even degrees. Inversion acts as -1 on H_1 and as $+1$ on H_3 . Since $\mathbb{Z}$ has homological dimension one, only the columns $p = 0, 1$ occur. In total degrees two and three, the only possible nonzero terms are

$$E_{1,1}^2 = H_1(\mathbb{Z}, H_1(C_3, \mathbb{Z})) = 0, \quad E_{0,3}^2 = H_0(\mathbb{Z}, H_3(C_3, \mathbb{Z})) = \mathbb{Z}/3.$$

No differential enters or leaves these terms. Thus $H_2(D, \mathbb{Z}) = 0$ and $H_3(D, \mathbb{Z}) \simeq \mathbb{Z}/3$. Under the standard comparison map from the two-periodic resolution of $\mathbb{Z}$ over $\mathbb{Z}C_3$ to the normalized bar resolution, the degree-three generator is sent to

$$\lceil \varepsilon | \varepsilon | \varepsilon \rceil + \lceil \varepsilon | \varepsilon^2 | \varepsilon \rceil = K.$$

Thus $[K]$ generates $H_3(C_3, \mathbb{Z})$. The edge map in the above spectral sequence is induced by the inclusion $C_3 \hookrightarrow D$. Since $E_{0,3}^\infty \cong \mathbb{Z}/3$ is the only nonzero term of total degree 3, this edge map is an isomorphism. Hence $[K]$ also generates $H_3(D, \mathbb{Z})$.

The Künneth formula has no Tor terms because the homology of $\langle h \rangle \simeq \mathbb{Z}$ is free. It follows that $H_3(\Gamma_3, \mathbb{Z}) \simeq H_3(D, \mathbb{Z}) \oplus H_2(D, \mathbb{Z}) \simeq \mathbb{Z}/3$, and K represents a generator.

It remains to prove the assertion about ω_* . Let $i : C_3 \hookrightarrow S_3$ send ε to (123) , and let $\text{tr} : H_3(S_3, \mathbb{Z}) \rightarrow H_3(C_3, \mathbb{Z})$ be transfer. Since C_3 is normal of index two in S_3 , the transfer formula [11, Chapter III, pp. 80–83] says that $\text{tr} \circ i_*$ is the sum of the endomorphisms of $H_3(C_3, \mathbb{Z})$ induced by conjugation by the coset representatives 1 and (12) . The first is the identity, while the second is induced by $\varepsilon \mapsto \varepsilon^{-1}$. As computed above from the standard two-periodic resolution, inversion also induces the identity on $H_3(C_3, \mathbb{Z})$. Hence $\text{tr} \circ i_* = 2 \text{id}_{H_3(C_3, \mathbb{Z})}$. Since $H_3(C_3, \mathbb{Z}) = \mathbb{Z}/3$, multiplication by 2 is invertible, so i_* is injective. Finally, ω_* sends $[K]$ to $i_*[K]$. Since $[K]$ generates $H_3(\Gamma_3, \mathbb{Z})$, the map ω_* is injective. $\square$

With the above pairing, coboundaries evaluate to 1 on cycles and cocycles evaluate to 1 on boundaries. Since $[K]$ has order three, $\kappa_\Phi(\varepsilon)$ is unchanged when Φ is replaced by a cohomologous normalized cocycle, and $\kappa_\Phi(\varepsilon)^3 = 1$.

7.4 Adjoint calculations in the three cases

We now compute the parts of $X_1 = (\text{ad } V)(W)$ and $X_2 = (\text{ad } V)^2(W)$ that determine simplicity and support. Appendix E contains the longer coefficient and bar-complex calculations.

7.4.1 The (3,2) case

Let $V = M(g, \rho)$ and $W = M(\varepsilon h, \sigma)$, with $\dim \rho = \dim \sigma = 1$. Define

$$a = \frac{\Phi_g(\varepsilon^2 h, \varepsilon^2)}{\Phi_g(\varepsilon, h) \Phi_{\varepsilon h}(g, \varepsilon^2)} \rho(h) \sigma(\varepsilon^2), \quad (7.16)$$

$$b_0 = \Phi^g(g \varepsilon^2, \varepsilon h) \frac{\Phi_g(g, \varepsilon^2)}{\Phi_g(\varepsilon, g)} \rho(g), \quad (7.17)$$

$$b_1 = \Phi^g(g \varepsilon, \varepsilon^2 h) \frac{\Phi_g(g, \varepsilon)}{\Phi_g(\varepsilon^2, g)} \rho(g) \Phi_{\varepsilon h}(g, g) \sigma(g^2). \quad (7.18)$$

Set

$$\Delta_1 = \frac{a^2 b_1}{b_0}, \quad \chi = -\frac{b_0}{a}, \quad s = -\chi \rho(gh). \quad (7.19)$$

Choose nonzero vectors $v_0 \in V_g$ and $w_0 \in W_{\varepsilon h}$, and let

$$v_i = \varepsilon^i \triangleright v_0 \quad (i \in \mathbb{Z}/3\mathbb{Z}), \quad w_1 = g \triangleright w_0.$$

Lemma 7.7. *Let*

$$E_0 = v_2 \otimes w_0, \quad E_1 = v_1 \otimes w_1, \quad u_0 = \varphi_1^\Phi(E_0).$$

Then $u_0 = E_0 - aE_1 \neq 0$, and X_1 is simple if and only if $\Delta_1 = 1$. Under this condition, its support is $(gh)^G$, and g acts by χ on the line $\mathbb{C}u_0 = (X_1)_{gh}$.

Proof. Equation (2.1) gives

$$(g \varepsilon^2) \triangleright w_0 = \frac{\sigma(\varepsilon^2)}{\Phi_{\varepsilon h}(g, \varepsilon^2)} w_1, \quad (\varepsilon^2 h) \triangleright v_2 = \frac{\Phi_g(\varepsilon^2 h, \varepsilon^2)}{\Phi_g(\varepsilon, h)} \rho(h) v_1.$$

Using $\varphi_1^\Phi = \text{id} - c_{W,V} c_{V,W}$ therefore gives $u_0 = E_0 - aE_1$. The tensors E_0 and E_1 lie in the distinct direct summands $V_{g\varepsilon^2} \otimes W_{\varepsilon h}$ and $V_{g\varepsilon} \otimes W_{\varepsilon^2 h}$, respectively, so $u_0 \neq 0$. Equations (2.2) and (2.1) also give

$$g \triangleright E_0 = b_0 E_1, \quad g \triangleright E_1 = b_1 E_0.$$

Consequently, $g \triangleright u_0 = b_0 E_1 - ab_1 E_0$. This vector is proportional to u_0 if and only if $a^2 b_1 = b_0$, or equivalently $\Delta_1 = 1$. In that case

$$g \triangleright u_0 = -\frac{b_0}{a} u_0 = \chi u_0.$$

The degree- gh component of $V \otimes W$ is $\mathbb{C}E_0 \oplus \mathbb{C}E_1$. Its image is $\mathbb{C}u_0 + \mathbb{C}(g \triangleright u_0)$. This space has dimension one exactly when $\Delta_1 = 1$. Since $u_0 \neq 0$ and X_1 is G -stable, while the homogeneous degrees of $V \otimes W$ are $gh, g\varepsilon h, g\varepsilon^2 h$, its support is exactly $(gh)^G$. By Lemma 7.2, $C_G(gh) = \langle g, h \rangle$. This group is abelian and generated by two elements, so f_Φ is trivial on $C_G(gh)^3$. Equation (7.13) therefore shows that the Φ_{gh} -twisted group algebra of $C_G(gh)$ is commutative. In particular, all its irreducible modules are one-dimensional. The description of simple objects in Proposition 2.4 now shows that X_1 is simple exactly when $(X_1)_{gh}$ is one-dimensional. This proves the assertions. $\square$

We next record the part of the second-adjoint calculation used below. The full coefficient calculation is given in Appendix E.2.

Lemma 7.8. Assume $\Delta_1 = 1$, and let $u_i = \varepsilon^i \triangleright u_0$ and $P_i = v_i \otimes u_i$. Then the following statements hold.

(1) There are nonzero scalars c_1, c_2, t_0, t_1, t_2 such that

$$\varphi_2^\Phi(P_0) = (1+s)P_0 + c_1P_1 + c_2P_2, \quad \varepsilon \triangleright P_i = t_iP_{i+1}, \quad (7.20)$$

where the indices are taken modulo three.

(2) The component $(X_2)_{g^2h}$ is nonzero. If $(X_2)_{\varepsilon g^2h} = 0$, then $\rho(g) = -1$.

(3) For the scalars in (1), if $\rho(g) = -1$, let

$$\lambda = \frac{c_1t_1}{c_2}, \quad r = \frac{c_1^2t_1}{c_2t_0}. \quad (7.21)$$

Then

$$s^3 = \kappa_\Phi(\varepsilon), \quad \lambda^3 = t_0t_1t_2, \quad r^2 = s. \quad (7.22)$$

Proof. (1) Lemma E.1 gives the first formula in (7.20) and $c_1c_2 \neq 0$. Equations (E.4)–(E.6) give the second formula and show that t_0, t_1, t_2 are nonzero.

(2) The tensors P_0, P_1, P_2 lie in distinct direct summands. Hence the first formula in (7.20), together with $c_1c_2 \neq 0$, shows that $\varphi_2^\Phi(P_0) \neq 0$. Since this vector has degree g^2h , we obtain $(X_2)_{g^2h} \neq 0$. Moreover, $\varphi_2^\Phi(v_2 \otimes u_0)$ belongs to $(X_2)_{\varepsilon g^2h}$. If this component is zero, then $\varphi_2^\Phi(v_2 \otimes u_0) = 0$. By (E.3), its two terms lie in distinct direct summands, and the coefficient of $v_2 \otimes u_0$ is $1 + \rho(g)$. Thus $1 + \rho(g) = 0$, so $\rho(g) = -1$.

(3) If $\rho(g) = -1$, then Lemma E.2 applies and gives precisely the three identities in (7.22). $\square$

7.4.2 The $(3, 1)$ case with $\dim \tau = 2$

Let $V = M(g, \rho)$ and $W = M(h, \tau)$, with $\dim \rho = 1$ and $\dim \tau = 2$. Define

$$\Delta_3 = \rho(h)^2 \Phi_h(g, g) \tau(g^2) \in \text{End}(W_h). \quad (7.23)$$

Lemma 7.9. Define

$$a_\varepsilon = \frac{\Phi_h(\varepsilon, g)}{\Phi_h(g, \varepsilon^2) \Phi_h(\varepsilon, \varepsilon)}. \quad (7.24)$$

There exist $b \in \mathbb{C}^\times$, a primitive third root of unity ζ_3 , and a basis $w, \bar{w}$ of W_h , with $\Delta_3 = \rho(h)^2 bI$, such that

$$\begin{array}{c|cc} & w & \bar{w} \\ \hline \varepsilon & \zeta_3 a_\varepsilon^{-1} w & \zeta_3^2 a_\varepsilon^{-1} \bar{w} \\ g & \bar{w} & bw \end{array}. \quad (7.25)$$

Proof. Equation (7.13) and Lemma 7.3 give $\Phi_h(h, a) = \Phi_h(a, h)$ and $\Phi_h(g^2, a) = \Phi_h(a, g^2)$ for $a \in \{\varepsilon, g, h\}$. Hence $\tau(h)$ and $\tau(g^2)$ commute with $\tau(G)$, so both are scalar by Schur's lemma. In particular, Δ_3 is scalar.

Let $E_\varepsilon = a_\varepsilon \tau(\varepsilon)$. Equation (2.1), applied in the orders $\varepsilon, g, \varepsilon, \varepsilon$, and g, ε^2 , gives $E_\varepsilon \tau(g) = \tau(g) E_\varepsilon^2$. Since $\tau(g)^2 = \Phi_h(g, g) \tau(g^2)$ is scalar, conjugating twice by $\tau(g)$ gives $E_\varepsilon = E_\varepsilon^4$. As E_ε is invertible, it follows that

$$E_\varepsilon^3 = I. \quad (7.26)$$

Choose an eigenvector $0 \neq w \in W_h$ of E_ε , write $E_\varepsilon w = \mu w$, and let $\bar{w} = g \triangleright w$. If $\bar{w} \in \mathbb{C}w$, then $\mathbb{C}w$ is G -stable, contrary to the irreducibility of τ . Hence $w, \bar{w}$ is a basis. Equation (2.1) gives $g \triangleright \bar{w} = \Phi_h(g, g) \tau(g^2) w = bw$ for a nonzero scalar b . Hence $\Delta_3 = \rho(h)^2 bI$. The relation above also gives $E_\varepsilon \bar{w} = \mu^2 \bar{w}$, while (7.26) gives $\mu^3 = 1$. If $\mu = 1$, then $\langle \varepsilon, g^2, h \rangle$ acts by scalars. An eigenline of $\tau(g)$ would then be G -stable, contrary to irreducibility. Hence $\mu \neq 1$, so μ is a primitive third root of unity. Let $\zeta_3 = \mu$. Then the action is given by (7.25). $\square$

Lemma 7.10. *The object X_1 is simple if and only if $\Delta_3 = I$. Under this condition, for $0 \neq v \in V_g$, one has*

$$(X_1)_{gh} = \mathbb{C}x_1, \quad x_1 = v \otimes (w - \rho(h)\bar{w}).$$

One also has $X_2 \neq 0$. If, in addition, X_2 is simple, then $\rho(g) = -1$.

Proof. The braiding formula and (7.25) give

$$\begin{aligned} \varphi_1^\Phi(v \otimes w) &= v \otimes (w - \rho(h)\bar{w}), \\ \varphi_1^\Phi(v \otimes \bar{w}) &= v \otimes (\bar{w} - \rho(h)bw). \end{aligned}$$

These vectors span the same line if and only if $\rho(h)^2b = 1$, which is the equation $\Delta_3 = I$. Since the first vector is nonzero, X_1 is G -stable, and all homogeneous degrees of $V \otimes W$ belong to $(gh)^G$, the support of X_1 is $(gh)^G$. By Lemma 7.2, $C_G(gh) = \langle g, h \rangle$. Lemma 7.3 and (7.13) show that the Φ_{gh} -twisted group algebra of $C_G(gh)$ is commutative. Proposition 2.4 therefore shows that X_1 is simple exactly when its degree- gh component is one-dimensional. This proves the first assertion and the formula for x_1 .

Assume $\Delta_3 = I$. Equation (2.2) gives

$$g \triangleright x_1 = -\Phi^g(g, h)\rho(g)\rho(h)^{-1}x_1.$$

Because g and h commute, $\Phi^g(g, h) = \Phi_g(g, h)$, and $\rho(gh) = \rho(g)\rho(h)/\Phi_g(g, h)$. Hence the summand $-c_{X_1, V}c_{V, X_1}(P_0)$, where $P_0 = v \otimes x_1$, is $\rho(g)^2P_0$. Moreover, $c_{1,2}^\Phi$ contributes $\rho(g)$ when it interchanges the first two copies of v , and $\varphi_1^\Phi(v \otimes (w - \rho(h)\bar{w})) = 2x_1$. Thus the last recursive summand is $2\rho(g)P_0$. Together with the identity summand, this gives

$$\varphi_2^\Phi(P_0) = (1 + \rho(g))^2P_0. \quad (7.27)$$

For $(\varepsilon^2 \triangleright v) \otimes x_1$, the same three summands lie respectively in

$$V_{g\varepsilon^2} \otimes (X_1)_{gh}, \quad V_g \otimes (X_1)_{\varepsilon^2gh}, \quad V_{g\varepsilon} \otimes (X_1)_{\varepsilon gh}.$$

The identity summand is nonzero, so the image is nonzero and has degree εg^2h . Thus $X_2 \neq 0$. Since g^2h is central whereas εg^2h is not, these degrees belong to different conjugacy classes. If X_2 is simple, its degree- g^2h component must therefore vanish. Since $P_0 \neq 0$, (7.27) gives $\rho(g) = -1$. $\square$

Lemma 7.11. *Retain v, x_1 from Lemma 7.10. Assume $\Delta_3 = I$ and $\rho(g) = -1$, and let*

$$v_i = \varepsilon^i \triangleright v, \quad x_{1,i} = \varepsilon^i \triangleright x_1, \quad i \in \mathbb{Z}/3\mathbb{Z},$$

and

$$e_0 = v_2 \otimes x_{1,0}, \quad e_1 = v_0 \otimes x_{1,1}, \quad e_2 = v_1 \otimes x_{1,2}.$$

These vectors are linearly independent. There are nonzero scalars a, b, m_0, m_1, m_2 such that

$$x_2 := \varphi_2^\Phi(e_0) = e_0 + ae_1 + be_2 \quad (7.28)$$

and

$$\varepsilon \triangleright e_0 = m_0(ae_1), \quad \varepsilon \triangleright (ae_1) = m_1(be_2), \quad \varepsilon \triangleright (be_2) = m_2e_0. \quad (7.29)$$

They satisfy

$$m_2 = m_0\kappa_\Phi(\varepsilon)^{-1}. \quad (7.30)$$

Proof. The first tensor factors of e_0, e_1, e_2 have degrees $g\varepsilon^2, g, g\varepsilon$, respectively, which proves their linear independence. The vector and the cyclic action are computed in (E.8) and (E.9). The bar-complex comparison gives (E.18). $\square$

7.4.3 The (3, 1) case with $\dim \tau = 1$

Let $V = M(g, \rho)$ and $W = M(h, \tau)$, with $\dim \rho = \dim \tau = 1$. Equation (2.1), applied to (g, h) on W_h , gives $\rho(h)\tau(g)\tau(h) = \rho(h)\Phi_h(g, h)\tau(gh)$. Hence the last condition in (7.8) is not generally equivalent to $\rho(h)\tau(gh) = 1$. Choose nonzero vectors $v \in V_g$ and $w \in W_h$, and let $\tilde{x}_1 = v \otimes w$.

Lemma 7.12. *One has*

$$X_1 = 0 \iff \rho(h)\tau(g) = 1.$$

If $\rho(h)\tau(g) \neq 1$, then $X_1 \simeq M(gh, \rho_1)$ is simple, where

$$\rho_1(A) = \Phi^A(g, h)\rho(A)\tau(A) \quad (A \in C_G(gh)).$$

Under the same assumption, one has $X_2 \neq 0$. If X_2 is simple, then

$$(1 + \rho(g))(1 - \rho(g)\rho(h)\tau(g)) = 0. \quad (7.31)$$

Proof. The braiding formula gives $\varphi_1^\Phi(v \otimes w) = (1 - \rho(h)\tau(g))v \otimes w$, which proves the vanishing criterion. Assume $\rho(h)\tau(g) \neq 1$. The G -translates of $v \otimes w$ span $V \otimes W$, and φ_1^Φ is G -linear. Hence the translates of $\tilde{x}_1$ span X_1 , whose support is $(gh)^G$. Since h is central, Lemma 7.2 gives $C_G(gh) = C_G(g) = \langle g, h \rangle$, and (2.2) gives

$$A \triangleright \tilde{x}_1 = \Phi^A(g, h)\rho(A)\tau(A)\tilde{x}_1 \quad (A \in C_G(gh)).$$

Thus $(X_1)_{gh} = \mathbb{C}\tilde{x}_1$ affords the one-dimensional projective representation ρ_1 . Proposition 2.4 now gives $X_1 \simeq M(gh, \rho_1)$, and hence X_1 is simple.

For the second adjoint, the double braiding and the last recursive summand give, respectively,

$$\begin{aligned} c_{X_1, V} c_{V, X_1}(v \otimes \tilde{x}_1) &= \rho(g)^2 \rho(h)\tau(g) v \otimes \tilde{x}_1, \\ (\text{id}_V \otimes \varphi_1^\Phi) c_{1,2}^\Phi(v \otimes \tilde{x}_1) &= \rho(g)(1 - \rho(h)\tau(g))v \otimes \tilde{x}_1. \end{aligned}$$

Together with the identity summand, they give

$$\varphi_2^\Phi(v \otimes \tilde{x}_1) = (1 + \rho(g))(1 - \rho(g)\rho(h)\tau(g))v \otimes \tilde{x}_1. \quad (7.32)$$

The three recursive summands applied to $(\varepsilon^2 \triangleright v) \otimes \tilde{x}_1$ lie, respectively, in

$$V_{g\varepsilon^2} \otimes (X_1)_{gh}, \quad V_g \otimes (X_1)_{\varepsilon^2 gh}, \quad V_{g\varepsilon} \otimes (X_1)_{\varepsilon gh}.$$

These direct summands are distinct, and the identity summand is nonzero. Hence the image is nonzero and has degree $\varepsilon g^2 h$, so $X_2 \neq 0$. Since $g^2 h$ is central whereas $\varepsilon g^2 h$ is not, simplicity of X_2 forces the degree- $g^2 h$ vector in (7.32) to vanish. This gives (7.31). $\square$

Lemma 7.13. *Assume $\rho(h)\tau(g) \neq 1$, and let $x_{1,i} = \varepsilon^i \triangleright \tilde{x}_1$ and $v_i = \varepsilon^i \triangleright v$ for $i = 0, 1, 2$. Let*

$$e_0 = v_2 \otimes x_{1,0}, \quad e_1 = v_0 \otimes x_{1,1}, \quad e_2 = v_1 \otimes x_{1,2}.$$

These vectors lie in the component of degree $\varepsilon g^2 h$ and are linearly independent. There are $A, B \in \mathbb{C}^\times$ such that

$$x_2 := \varphi_2^\Phi(e_0) = e_0 + Ae_1 + Be_2. \quad (7.33)$$

Define $m_0, m_1, m_2 \in \mathbb{C}^\times$ by

$$\varepsilon \triangleright e_0 = m_0(Ae_1), \quad \varepsilon \triangleright (Ae_1) = m_1(Be_2), \quad \varepsilon \triangleright (Be_2) = m_2e_0. \quad (7.34)$$

These scalars satisfy

$$\begin{aligned}\frac{m_0}{m_1} &= \kappa_\Phi(\varepsilon) \frac{1 - \rho(h)\tau(g)}{\rho(g)^3 \rho(h)^2 \tau(g)^2}, \\ \frac{m_0}{m_2} &= -\frac{\kappa_\Phi(\varepsilon)}{\rho(g)^3 \rho(h)\tau(g)(1 - \rho(h)\tau(g))}, \\ \frac{m_1}{m_2} &= -\frac{\rho(h)\tau(g)}{(1 - \rho(h)\tau(g))^2}.\end{aligned}\tag{7.35}$$

Proof. Lemma E.5 shows that e_0, e_1, e_2 have degree $\varepsilon g^2 h$ and are linearly independent. It also gives nonzero scalars $\mathfrak{a}, \mathfrak{b}$ such that

$$\begin{aligned}A &= -\rho(g)^2 \rho(h)\tau(g)\mathfrak{a}, \\ B &= \rho(g)(1 - \rho(h)\tau(g))\mathfrak{b}.\end{aligned}$$

The same lemma computes the cyclic action of ε on the e_i . Since $\rho(h)\tau(g) \neq 1$, this proves (7.33) and shows that the scalars in (7.34) are well defined and nonzero.

Lemma E.6 gives nonzero scalars μ_0, μ_1, μ_2 satisfying (E.24). By Lemma E.4, the two chains in (E.16) are boundaries. Evaluating these boundaries by Φ gives the first two identities below. Multiplying the three identities in (E.24) gives $\mu_0\mu_1\mu_2 = m_0m_1m_2$. By (7.34), this is the scalar of three successive ε -actions on e_0 . Equation (2.1) on the degree- $\varepsilon g^2 h$ component gives the third identity:

$$\begin{aligned}\frac{\mu_0}{\mu_2} &= \kappa_\Phi(\varepsilon), \\ \mu_0^3 &= \Phi_{\varepsilon g^2 h}(\varepsilon, \varepsilon) \Phi_{\varepsilon g^2 h}(\varepsilon^2, \varepsilon) \kappa_\Phi(\varepsilon)^{-1}, \\ \mu_0\mu_1\mu_2 &= \Phi_{\varepsilon g^2 h}(\varepsilon, \varepsilon) \Phi_{\varepsilon g^2 h}(\varepsilon^2, \varepsilon).\end{aligned}$$

The first identity gives $\mu_2 = \mu_0 \kappa_\Phi(\varepsilon)^{-1}$. The other two and $\kappa_\Phi(\varepsilon)^3 = 1$ then give $\mu_1 = \mu_0 \kappa_\Phi(\varepsilon)^{-1}$. Substitution in (E.24) proves (7.35). The GAP program [GAP source](#) checks the two algebraic steps just used. Before $\kappa_\Phi(\varepsilon)^3 = 1$ is imposed, the quotient of the derived value of μ_1 by $\mu_0 \kappa_\Phi(\varepsilon)^{-1}$ is $\kappa_\Phi(\varepsilon)^3$. After the resulting values of μ_1 and μ_2 are substituted in (E.24), the program verifies the three identities in (7.35). $\square$

7.5 Vanishing of the cocycle obstruction

The preceding adjoint calculations give the following three results. Throughout this subsection, write $X_n = (\text{ad } V)^n(W)$.

Proposition 7.14. *Let $V = M(g, \rho)$ and $W = M(\varepsilon h, \sigma)$ have supports as in the (3, 2) case, with $\dim \rho = \dim \sigma = 1$. If X_1 and X_2 are nonzero and simple, then $\rho(g) = -1$ and $\kappa_\Phi(\varepsilon) = 1$.*

Proof. Simplicity of X_1 gives $\Delta_1 = 1$ by Lemma 7.7, so Lemma 7.8 applies and gives $(X_2)_{g^2 h} \neq 0$. Since $g^2 h$ is central and X_2 is simple, Proposition 2.4 gives $\text{supp } X_2 = \{g^2 h\}$. Thus $(X_2)_{\varepsilon g^2 h} = 0$, and Lemma 7.8 yields $\rho(g) = -1$.

Let $z_0 = \varphi_2^\Phi(P_0)$, and retain the scalars from Lemma 7.8. On the span of P_0, P_1, P_2 , define

$$T = \lambda^{-1}(\varepsilon \triangleright -), \quad \tilde{P}_0 = P_0, \quad \tilde{P}_1 = \lambda^{-1} t_0 P_1, \quad \tilde{P}_2 = \lambda^{-2} t_0 t_1 P_2.$$

Equations (7.20) and (7.22) give

$$T\tilde{P}_0 = \tilde{P}_1, \quad T\tilde{P}_1 = \tilde{P}_2, \quad T\tilde{P}_2 = \tilde{P}_0.$$

Furthermore,

$$z_0 = (1 + s)\tilde{P}_0 + \frac{c_1 \lambda}{t_0} \tilde{P}_1 + \frac{c_2 \lambda^2}{t_0 t_1} \tilde{P}_2 = (1 + s)\tilde{P}_0 + r\tilde{P}_1 + r\tilde{P}_2.$$

Equation (7.21) gives the last equality. The P_i lie in distinct direct summands, so the $\tilde{P}_i$ are linearly independent. Since $r^2 = s$, the coordinate matrix of z_0, Tz_0, T^2z_0 with respect to the ordered basis $\tilde{P}_0, \tilde{P}_1, \tilde{P}_2$ has determinant

$$\det \begin{pmatrix} 1+s & r & r \\ r & 1+s & r \\ r & r & 1+s \end{pmatrix} = ((1+s) + 2r)((1+s) - r)^2 = (1+r^3)^2.$$

Since g^2h is central and X_2 is G -stable, these three vectors belong to $(X_2)_{g^2h}$. If $\kappa_\Phi(\varepsilon) \neq 1$, the determinant is nonzero: if it vanished, then $r^3 = -1$, whereas $\kappa_\Phi(\varepsilon) = s^3 = r^6 = 1$. Hence $(X_2)_{g^2h}$ contains three linearly independent vectors.

Since g^2h is central and $G = \langle \varepsilon, g, g^2h \rangle$, the proof of Lemma 7.3, with h replaced by g^2h , shows that every irreducible Φ_{g^2h} -projective G -module has dimension at most two. By Proposition 2.4, this applies to the simple object X_2 with support $\{g^2h\}$, contradicting the preceding three-dimensional subspace. Hence $\kappa_\Phi(\varepsilon) = 1$. $\square$

Proposition 7.15. *Let $V = M(g, \rho)$ and $W = M(h, \tau)$ have supports as in the (3,1) case, with $\dim \rho = 1$ and $\dim \tau = 2$. If X_1 and X_2 are nonzero and simple, then $\Delta_3 = I$, $\rho(g) = -1$, and $\kappa_\Phi(\varepsilon) = 1$.*

Proof. Lemma 7.10 gives $\Delta_3 = I$ and $\rho(g) = -1$, so Lemma 7.11 applies. The definitions of e_0, e_1, e_2 , together with $g\varepsilon g^{-1} = \varepsilon^2$, show that all three have degree εg^2h . They are linearly independent, and (7.28) therefore gives $0 \neq x_2 \in (X_2)_{\varepsilon g^2h}$.

Since g^2 is central, Lemma 7.2 gives $C_G(\varepsilon g^2h) = C_G(\varepsilon h) = \langle \varepsilon, g^2, h \rangle$. This group is abelian, and Lemma 7.3 together with (7.13) shows that its $\Phi_{\varepsilon g^2h}$ -twisted group algebra is commutative. By Proposition 2.4, $(X_2)_{\varepsilon g^2h}$ is an irreducible module over this algebra and is therefore one-dimensional. Consequently, $(X_2)_{\varepsilon g^2h} = \mathbb{C}x_2$ and $\varepsilon \triangleright x_2 \in \mathbb{C}x_2$. Equations (7.28) and (7.29) give $\varepsilon \triangleright x_2 = m_2e_0 + m_0ae_1 + m_1be_2$. Since e_0, ae_1, be_2 are linearly independent, comparison of coefficients with x_2 gives $m_0 = m_1 = m_2$. As these scalars are nonzero, (7.30) yields $\kappa_\Phi(\varepsilon) = 1$. $\square$

Proposition 7.16. *Let $V = M(g, \rho)$ and $W = M(h, \tau)$ have supports as in the (3,1) case, with $\dim \rho = \dim \tau = 1$. If X_1 and X_2 are nonzero and simple, then $\kappa_\Phi(\varepsilon) = 1$.*

Proof. Since $X_1 \neq 0$, Lemma 7.12 gives $\rho(h)\tau(g) \neq 1$, so Lemma 7.13 applies. It shows that e_0, e_1, e_2 lie in degree εg^2h and are linearly independent. Hence (7.33) gives $0 \neq x_2 \in (X_2)_{\varepsilon g^2h}$.

Since g^2 is central, Lemma 7.2 gives $C_G(\varepsilon g^2h) = C_G(\varepsilon h) = \langle \varepsilon, g^2, h \rangle$. This group is abelian, and Lemma 7.3 and (7.13) show that its $\Phi_{\varepsilon g^2h}$ -twisted group algebra is commutative. By Proposition 2.4, $(X_2)_{\varepsilon g^2h}$ is therefore one-dimensional. Thus $(X_2)_{\varepsilon g^2h} = \mathbb{C}x_2$, and (7.33) and (7.34) give $\varepsilon \triangleright x_2 = m_2e_0 + m_0Ae_1 + m_1Be_2 \in \mathbb{C}x_2$. Since $A, B \neq 0$, comparison of coefficients gives $m_0 = m_1 = m_2$. The last equality in (7.35) now gives

$$1 - \rho(h)\tau(g) + \rho(h)^2\tau(g)^2 = 0.$$

Hence $\rho(h)\tau(g)(1 - \rho(h)\tau(g)) = 1$ and $\rho(h)^3\tau(g)^3 = -1$. The second equality in (7.35) gives $\kappa_\Phi(\varepsilon) = -\rho(g)^3$, while (7.31) gives either $\rho(g) = -1$ or $\rho(g)\rho(h)\tau(g) = 1$. In the latter case, $\rho(g) = (\rho(h)\tau(g))^{-1}$, so again $\rho(g)^3 = -1$. Therefore $\kappa_\Phi(\varepsilon) = 1$. $\square$

The preceding results also give the obstruction for a finite Cartan graph.

Proposition 7.17. *Let $P = (V, W)$ be in the (3,2) case with $\dim \rho = \dim \sigma = 1$, or in the (3,1) case with $\dim \rho = 1$ and $\dim \tau \in \{1, 2\}$. If P admits all reflections and its Cartan graph is finite, then $\kappa_\Phi(\varepsilon) = 1$.*

Proof. By (2.13) and braided indecomposability, $X_1 \neq 0$. Hence $1 \leq -a_{12}^{[P]}$, and Lemma 2.24 shows that X_1 is simple. In the (3,2) case, Lemma 7.7 gives $\Delta_1 = 1$, and Lemma 7.8(2) gives $X_2 \neq 0$. In the two (3,1) cases, the same conclusion follows from Lemmas 7.10 and 7.12, respectively. Thus $2 \leq -a_{12}^{[P]}$, and Lemma 2.24 also shows that X_2 is simple. Propositions 7.14, 7.15, and 7.16 now give $\kappa_\Phi(\varepsilon) = 1$ in the respective cases. $\square$

7.6 Cohomological trivialization and braided equivalence

For $x \in \Gamma_3$, write $\bar{x} = \pi(x)$, and let $\tilde{\Phi} = \pi^*\Phi$. The value of $\kappa_\Phi(\bar{\varepsilon})$ determines the cohomology class of this pullback.

We use the coboundary convention

$$(\delta J)(a, b, c) = \frac{J(b, c)J(a, bc)}{J(ab, c)J(a, b)}. \quad (7.36)$$

Proposition 7.18. *The class $[\tilde{\Phi}]$ is trivial in $H^3(\Gamma_3, \mathbb{C}^\times)$ if and only if $\kappa_\Phi(\bar{\varepsilon}) = 1$. In this case there is a normalized two-cochain $J : \Gamma_3^2 \rightarrow \mathbb{C}^\times$ satisfying $\tilde{\Phi} = \delta J$.*

Proof. Since $\mathbb{C}^\times$ is divisible, the Ext term in the universal coefficient theorem vanishes, and evaluation gives an isomorphism

$$H^3(\Gamma_3, \mathbb{C}^\times) \simeq \text{Hom}(H_3(\Gamma_3, \mathbb{Z}), \mathbb{C}^\times).$$

By Lemma 7.6, $H_3(\Gamma_3, \mathbb{Z}) \simeq \mathbb{Z}/3$ and $[K]$ is a generator. Moreover, (7.14) and (7.15) give $\langle \tilde{\Phi}, K \rangle = \kappa_\Phi(\bar{\varepsilon})$. Thus $[\tilde{\Phi}] = 0$ if and only if $\kappa_\Phi(\bar{\varepsilon}) = 1$.

If the class is trivial, choose a two-cochain J with $\tilde{\Phi} = \delta J$. Since $\tilde{\Phi}$ is normalized, the cases in which the first or second argument in (7.36) equals 1 give $J(1, a) = J(a, 1) = J(1, 1)$ for every $a \in \Gamma_3$. Dividing J by the constant $J(1, 1)$ leaves its coboundary unchanged and makes J normalized. $\square$

The cochain J trivializes $\pi^*\Phi$ on Γ_3 . The class of Φ on G may remain nontrivial. Inflation along π preserves the relevant Yetter–Drinfeld structures. We next compare objects over G with their inflations along π .

Proposition 7.19. *Let $x \in \Gamma_3$, and let ρ be an irreducible $\Phi_{\bar{x}}$ -projective representation of $C_G(\bar{x})$. Let*

$$\tilde{\rho} = \rho \circ \pi|_{C_{\Gamma_3}(x)}.$$

Then $\tilde{\rho}$ is an irreducible $\tilde{\Phi}_x$ -projective representation. Through the bijection $x^{\Gamma_3} \simeq \bar{x}^G$, the object $M_{\Gamma_3}(x, \tilde{\rho})$ is realized on the same underlying vector space as $M_G(\bar{x}, \rho)$, with grading and action given, for $a, y \in \Gamma_3$, by

$$(M_{\Gamma_3}(x, \tilde{\rho}))_y = \begin{cases} (M_G(\bar{x}, \rho))_{\pi(y)}, & y \in x^{\Gamma_3}, \\ 0, & y \notin x^{\Gamma_3}, \end{cases}$$

$$a \triangleright_{\Gamma_3} v = \pi(a) \triangleright_G v.$$

Under these identifications, the recursively defined adjoint maps for any two such objects coincide as linear maps. The Nichols algebras of their direct sum and of the resulting adjoint objects are therefore identified as multigraded vector spaces. In particular, finite-dimensionality and dimension are preserved.

Proof. Equation (7.11) and $\tilde{\Phi} = \pi^*\Phi$ give

$$\pi(C_{\Gamma_3}(x)) = C_G(\bar{x}), \quad \tilde{\Phi}_x(a, b) = \Phi_{\bar{x}}(\pi(a), \pi(b)) \quad (a, b \in C_{\Gamma_3}(x)).$$

Consequently, $\tilde{\rho}(a)\tilde{\rho}(b) = \rho(\pi(a))\rho(\pi(b)) = \Phi_{\bar{x}}(\pi(a), \pi(b))\rho(\pi(ab)) = \tilde{\Phi}_x(a, b)\tilde{\rho}(ab)$. Moreover, $\tilde{\rho}(C_{\Gamma_3}(x)) = \rho(C_G(\bar{x}))$. Thus ρ and $\tilde{\rho}$ have the same invariant subspaces, so $\tilde{\rho}$ is irreducible.

The bijection $x^{\Gamma_3} \simeq \bar{x}^G$ makes the displayed grading well defined. Since $\pi(aya^{-1}) = \pi(a)\pi(y)\pi(a)^{-1}$, the pulled-back action respects this grading, and the Yetter–Drinfeld identities follow from those over G . Its degree- x component affords $\tilde{\rho}$. Hence Proposition 2.4 identifies the resulting simple object with $M_{\Gamma_3}(x, \tilde{\rho})$.

Finally, $\tilde{\Phi}(a, b, c) = \Phi(\pi(a), \pi(b), \pi(c))$. Together with the displayed action, this shows that the corresponding associators, tensor actions, and braidings agree as linear maps. Therefore, under the above identifications,

$$\varphi_n^{\tilde{\Phi}} = \varphi_n^{\Phi} \quad (n \geq 1).$$

The categorical quantum symmetrizers on the corresponding tensor powers are consequently the same linear maps. Their kernels, and hence the Nichols ideals, agree. This proves the remaining assertions. $\square$

Assume now that $\kappa_{\Phi}(\bar{\epsilon}) = 1$, and fix a normalized J as in Proposition 7.18. For $x \in \Gamma_3$ and $a \in C_{\Gamma_3}(x)$, let $L_x(a) = J(a, x)/J(x, a)$.

For $a, b \in C_{\Gamma_3}(x)$, the definition of the local cocycle, $\tilde{\Phi} = \delta J$, and $ax = xa, bx = xb$ give

$$\begin{aligned} \tilde{\Phi}_x(a, b) &= \frac{(\delta J)(a, b, x)(\delta J)(x, a, b)}{(\delta J)(a, x, b)} \\ &= \frac{J(a, x)J(b, x)J(x, ab)}{J(x, a)J(x, b)J(ab, x)} = \frac{L_x(a)L_x(b)}{L_x(ab)}. \end{aligned} \quad (7.37)$$

For $\rho, \tilde{\rho}$ as in Proposition 7.19, put

$$\hat{\rho}_x(a) = L_x(a)^{-1}\tilde{\rho}(a) = \frac{J(x, a)}{J(a, x)}\rho(\pi(a)). \quad (7.38)$$

The cochain J gives the following braided monoidal equivalence.

Proposition 7.20. *There is a braided strong monoidal equivalence*

$$F_J : {}_{\Gamma_3}^{\Gamma_3}\mathcal{YD}^{\tilde{\Phi}} \longrightarrow {}_{\Gamma_3}^{\Gamma_3}\mathcal{YD}$$

which is the identity on underlying graded vector spaces and morphisms. Its tensorator is

$$(F_J)_2(u_a \otimes v_b) = J(a, b)^{-1}u_a \otimes v_b, \quad (7.39)$$

and its action on a homogeneous vector is

$$a \triangleright_J v_x = \frac{J(axa^{-1}, a)}{J(a, x)}(a \triangleright v_x). \quad (7.40)$$

It transports duals, recursively defined braided adjoint objects, all defined reflections, and Nichols algebras to their ordinary counterparts. Consequently, it preserves simplicity and the vanishing of adjoint objects, as well as the finite-dimensionality and dimension of Nichols algebras.

Proof. Since $\tilde{\Phi} = \delta J$, substitution into the defining action, tensor-product action, associator, and braiding formulas shows that (7.39) and (7.40) define a braided strong monoidal functor. The normalization of J gives the unit condition, and the reciprocal scalar factors define its inverse. If $a \in C_{\Gamma_3}(x)$, then the factor in (7.40) is $J(x, a)/J(a, x) = L_x(a)^{-1}$, so the action on the component of degree x is (7.38). The assertions about duals, adjoint objects, and reflections follow from [34, Proposition 6.3], and the assertion about Nichols algebras follows from (2.11). Since F_J is an equivalence and does not change the underlying graded vector spaces, it preserves simplicity, the vanishing of adjoint objects, and the finite-dimensionality and dimension of Nichols algebras. $\square$

For an inducing projective representation ξ , write $\hat{\xi}$ for the corresponding ordinary representation defined by (7.38). For $1 \leq i \leq 4$, let $\mathcal{P}_i$ denote, over $\mathbb{C}$, the ordinary class in row i of [26, Definition 7.1 and Table 2], with the central generator denoted there by z identified with h .

Proposition 7.21. *Assume $\kappa_\Phi(\bar{\varepsilon}) = 1$. For every J as in Proposition 7.18 and every $1 \leq i \leq 4$, a pair belongs to $\mathcal{P}_i^\Phi$ if and only if the ordinary pair obtained from it by inflation along π and application of F_J belongs to $\mathcal{P}_i$.*

Proof. Proposition 7.19 shows that inflation preserves simplicity, supports, and the dimensions of the inducing representations. Proposition 7.20 shows that the same is true after applying F_J . These assertions also hold in the reverse direction. It remains to compare the parameter conditions.

For commuting $x, y \in \Gamma_3$, the definition of L gives $L_x(y)L_y(x) = 1$, and $L_x(x) = 1$. Hence $\hat{\rho}(g) = \rho(g)$, $\hat{\sigma}(\varepsilon h) = \sigma(\varepsilon h)$, and $\hat{\tau}(h) = \tau(h)$.

In the (3, 2) case, inflation and F_J preserve simplicity of X_1 , which is equivalent to $\Delta_1 = 1$ by Lemma 7.7. The same first-adjoint calculation with $\Phi = 1$ applies to finite-dimensional objects over Γ_3 , whose centralizer $C_{\Gamma_3}(gh) = \langle g, h \rangle$ is abelian. It gives the ordinary simplicity criterion $\hat{\rho}(h)^2\hat{\sigma}(\varepsilon^2)^2\hat{\sigma}(g^2) = 1$. Since the hatted representations are characters and $\varepsilon^3 = 1$, this is $\hat{\rho}(h^2)\hat{\sigma}(\varepsilon g^2) = 1$.

In the two-dimensional (3, 1) case, (7.37) and $L_g(h)L_h(g) = 1$ give

$$\hat{\rho}(h)^2\hat{\tau}(g^2) = \frac{L_h(g)^2}{L_h(g^2)}\rho(h)^2\tau(g^2) = \tilde{\Phi}_h(g, g)\rho(h)^2\tau(g^2) = \Delta_3,$$

where the last equality uses $\tilde{\Phi}_h(g, g) = \Phi_{\bar{h}}(\bar{g}, \bar{g})$. In the one-dimensional (3, 1) case, the same relation gives $\hat{\rho}(h)\hat{\tau}(g) = \rho(h)\tau(g)$. Ordinary multiplicativity and $\hat{\tau}(h) = \tau(h)$ then give the last identity below. Thus

$$\Delta_1 = 1 \iff \hat{\rho}(h^2)\hat{\sigma}(\varepsilon g^2) = 1, \quad (7.41)$$

$$\Delta_3 = \hat{\rho}(h)^2\hat{\tau}(g^2), \quad (7.42)$$

$$\rho(h)\tau(g)\tau(h) = \hat{\rho}(h)\hat{\tau}(gh). \quad (7.43)$$

Suppose now that the pair is in the (3, 2) case, $\Delta_1 = 1$, and $\rho(g) = -1$. By Lemma 7.7, g acts by χ on $(X_1)_{gh}$. Its transported action coefficient is $L_{gh}(g)^{-1}\chi$ by (7.40). Moreover, $\hat{\rho}(gh) = L_g(gh)^{-1}\rho(gh)$. Since h is central, g commutes with gh . Thus $L_{gh}(g)L_g(gh) = 1$, and

$$s = -(L_{gh}(g)^{-1}\chi)\hat{\rho}(gh).$$

The untwisted definitions of a, b_0, χ give the right-hand side as $\hat{\rho}(g)^2\hat{\sigma}(\varepsilon)^{-2}$. Since $\hat{\rho}(g) = -1$ and $\hat{\sigma}(\varepsilon)^3 = 1$,

$$s = \hat{\sigma}(\varepsilon). \quad (7.44)$$

Inflation and F_J preserve supports and representation dimensions. Equations (7.41) and (7.44), together with the equalities in the first paragraph, translate the conditions defining $\mathcal{P}_1^\Phi$ and $\mathcal{P}_2^\Phi$. For $i = 2$, under the condition $\hat{\sigma}(\varepsilon) = 1$, the equality $\hat{\sigma}(\varepsilon h) = -1$ is equivalent to $\hat{\sigma}(h) = -1$. Equation (7.42) translates the conditions defining $\mathcal{P}_3^\Phi$. The equalities $\hat{\rho}(g) = \rho(g)$ and $\hat{\rho}(h)\hat{\tau}(g) = \rho(h)\tau(g)$, together with (7.43), translate the conditions defining $\mathcal{P}_4^\Phi$. For the first two families, $\hat{\rho}(g) = \rho(g)$ and (7.41) give the hypotheses of (7.44) in either direction. Hence all four comparisons are reversible. $\square$

7.7 Necessity and reflection closure

Proposition 7.22. *If $\dim \mathcal{B}(V \oplus W) < \infty$, then, after possibly interchanging V and W , generators $\varepsilon, g, h \in G$ inducing an epimorphism $\Gamma_3 \twoheadrightarrow G$ can be chosen so that the resulting ordered pair belongs to one of $\mathcal{P}_1^\Phi, \dots, \mathcal{P}_4^\Phi$.*

Proof. By Theorem 2.25, finite-dimensionality implies that the pair admits all reflections and that its Cartan graph is finite. Proposition 7.5 supplies the required interchange and generators. The support and dimension possibilities are

$$\begin{aligned} (3, 2) : \quad & \dim \rho = \dim \sigma = 1, \\ (3, 1) : \quad & \dim \rho = 1, \quad \dim \tau \in \{1, 2\}. \end{aligned}$$

Interchanging the entries preserves the preceding reflection properties. Proposition 7.17 therefore gives $\kappa_\Phi(\varepsilon) = 1$ for these generators.

Choose J as in Proposition 7.18, and let (V^J, W^J) be the ordinary pair over Γ_3 obtained by inflation along π and application of F_J . Propositions 7.19 and 7.20 preserve simplicity, Nichols algebras, and adjoint objects. Since the ground field is $\mathbb{C}$, the objects V^J, W^J are absolutely simple. By (2.13), the original first adjoint object is nonzero. Hence (V^J, W^J) is braided-indecomposable and

$$\dim \mathcal{B}(V^J \oplus W^J) = \dim \mathcal{B}(V \oplus W) < \infty$$

The support bijection in Proposition 7.19 and the fact that F_J does not change the grading give

$$\langle \text{supp}(V^J \oplus W^J) \rangle = \begin{cases} \langle g, g\varepsilon, \varepsilon h \rangle, & (3, 2), \\ \langle g, g\varepsilon, h \rangle, & (3, 1), \end{cases} = \Gamma_3.$$

Thus all the hypotheses of [26, Theorem 2.1] are satisfied.

For these two support types over $\mathbb{C}$, the alternatives in [26, Examples 1.9–1.11] are exactly the first four classes in [26, Definition 7.1 and Table 2]. The support sizes fix the ordering. Hence $(V^J, W^J) \in \mathcal{P}_i$ for some $1 \leq i \leq 4$. Proposition 7.21 now gives

$$(V^J, W^J) \in \mathcal{P}_i \iff (V, W) \in \mathcal{P}_i^\Phi,$$

which proves the assertion. $\square$

Proposition 7.23. *Let $P = (V, W)$ belong to one of the four families $\mathcal{P}_i^\Phi$. The reflections $R_1(P)$ and $R_2(P)$ are defined, and their support generators may be chosen as*

$$\begin{array}{c|ccc} & g' & \varepsilon' & h' \\ \hline R_1 & g'^{-1} & \varepsilon & g'^2 h \\ R_2 & gh & \varepsilon^{-1} & h^{-1}. \end{array} \tag{7.45}$$

Proof. By the observation following (7.8), every pair in the four families satisfies $\kappa_\Phi(\varepsilon) = 1$. Hence Proposition 7.21 places the inflated and untwisted pair in the corresponding ordinary family. Propositions 7.19 and 7.20 identify the reflected objects with their ordinary counterparts. For the four ordinary families, the reflected support representatives in [26, Corollaries 4.13–4.16, 5.10–5.11, and 6.7–6.8], with the central generator in that paper identified with h , give (7.45). In both rows, $(\varepsilon')^3 = 1$, $g'\varepsilon'(g')^{-1} = (\varepsilon')^{-1}$, and h' is central. They also generate Γ_3 : in the first row, $g = (g')^{-1}$, $\varepsilon = \varepsilon'$, and $h = (g')^2 h'$. In the second, $g = g'h'$, $\varepsilon = (\varepsilon')^{-1}$, and $h = (h')^{-1}$. Their images therefore give the required generator triples in G . $\square$

Proposition 7.24. *Every pair in one of $\mathcal{P}_1^\Phi, \dots, \mathcal{P}_4^\Phi$ admits all reflections. Every pair obtained by successive reflections again belongs to one of these four families, and every Cartan matrix in its Cartan graph is (7.9).*

Proof. By the observation following (7.8), one has $\kappa_\Phi(\varepsilon) = 1$ in every family. Hence Proposition 7.21 shows that inflation and F_J identify the pair with one in the corresponding ordinary family. Its reflection orbits and Cartan matrices are given by [26, Equations (7.1)–(7.2) and Lemmas 7.3–7.4]. Propositions 7.19 and 7.20 identify the reflected pairs and preserve the vanishing

of all iterated adjoint objects, hence all defined reflections and Cartan integers. Proposition 7.23 supplies a generator triple after either reflection. Since $\tilde{\Phi} = \delta J$ on all of Γ_3 , the same untwisting applies to each such triple and gives $\kappa_{\Phi}(\varepsilon') = 1$. Proposition 7.21 therefore gives

$$\mathcal{P}_1^{\Phi} \xleftrightarrow{R_1} \mathcal{P}_4^{\Phi}, \quad \mathcal{P}_2^{\Phi} \xleftrightarrow{R_1} \mathcal{P}_3^{\Phi}, \quad R_2(\mathcal{P}_j^{\Phi}) = \mathcal{P}_j^{\Phi} \quad (1 \leq j \leq 4).$$

Thus every successive reflection remains in the four families, and the preservation of the Cartan integers transfers the standard B_2 Cartan graph to the original pair. $\square$

7.8 Sufficiency and dimension formulas

Corollary 7.25. *Every pair in one of the four families has a finite-dimensional Nichols algebra, and its dimension is given by (7.10).*

Proof. Let $(V, W) \in \mathcal{P}_i^{\Phi}$, where $1 \leq i \leq 4$. By the observation following (7.8), one has $\kappa_{\Phi}(\varepsilon) = 1$. Choose J as in Proposition 7.18, and let (V^J, W^J) be the ordinary pair over Γ_3 obtained by inflation along π and application of F_J . Proposition 7.21 places this pair in $\mathcal{P}_i$.

Over $\mathbb{C}$, [26, Theorems 8.1–8.4] give finite-dimensional Nichols algebras of dimension 10368 for $i = 1, 4$ and 2304 for $i = 2, 3$. Propositions 7.19 and 7.20 preserve Nichols dimensions, so

$$\dim \mathcal{B}(V \oplus W) = \dim \mathcal{B}(V^J \oplus W^J).$$

This proves finite-dimensionality and (7.10). $\square$

Proof of Theorem 7.1. Proposition 7.22 proves necessity. Conversely, interchanging the two entries does not change their direct sum, so Corollary 7.25 proves sufficiency and (7.10). Proposition 7.24 gives the asserted standard Cartan graph of type B_2 . $\square$

8 The complete classification

The following theorem combines the four case classifications with the reflection criterion. Whenever the two entries are interchanged, all support representatives and parameters are recomputed for the resulting ordered tuple and the fixed normalized cocycle Φ . For $M = (M_1, M_2)$ and $\omega \in S_2$, write $M^{\omega} = (M_{\omega^{-1}(1)}, M_{\omega^{-1}(2)})$.

Theorem 8.1. *Let G be a finite non-abelian group, let $\Phi \in Z^3(G, \mathbb{C}^{\times})$ be normalized, and let $M = (V, W)$ be a tuple of finite-dimensional simple objects in ${}^G\mathcal{YD}^{\Phi}$. Assume that*

$$G = \langle \text{supp } V \cup \text{supp } W \rangle, \quad M \text{ is braided-indecomposable.} \quad (8.1)$$

Then the following statements are equivalent.

- (1) *The Nichols algebra $\mathcal{B}(V \oplus W)$ is finite-dimensional.*
- (2) *The pair M admits all reflections, its Cartan graph $\mathcal{G}(M)$ is finite, and*

$$\dim \mathcal{B}(Q_i) < \infty \quad \text{for every } Q = (Q_1, Q_2) \in \mathcal{F}_2(M) \text{ and } i \in \{1, 2\}.$$

- (3) *There exists $\omega \in S_2$ such that M^{ω} satisfies one of the following alternatives. All parameters are computed for this ordering, the indicated generators and support representatives, and the fixed cocycle Φ .*

- (i) *There are $g, h, \varepsilon \in G$ inducing an epimorphism $\Gamma_2 \twoheadrightarrow G$ such that*

$$M^{\omega} \simeq (M(g, \rho), M(h, \sigma)), \quad \text{supp } M(g, \rho) = \{g, \varepsilon g\}, \quad \text{supp } M(h, \sigma) = \{h, \varepsilon h\}.$$

Either (A₂) or (Γ₂-G₂) holds.

- (ii) There are $\varepsilon, g, h \in G$ inducing an epimorphism $\Gamma_3 \twoheadrightarrow G$ from (7.1) such that M^ω is isomorphic to an ordered pair satisfying one of (7.5)–(7.8).
- (iii) There are $g, h, \varepsilon \in G$ inducing an epimorphism $\Gamma_4 \twoheadrightarrow G$, such that

$$M^\omega \simeq (M(h, \rho), M(g, \sigma)),$$

$$\text{supp } M(h, \rho) = \{h, \varepsilon^{-1}h\}, \quad \text{supp } M(g, \sigma) = \{g, \varepsilon g, \varepsilon^2 g, \varepsilon^3 g\},$$

and (B₂) holds.

- (iv) There are $z, x_1, x_2 \in G$ inducing an epimorphism $T \twoheadrightarrow G$. With $x_3 = x_2 x_1 x_2^{-1}$ and $x_4 = x_1 x_2 x_1^{-1}$, the elements x_1, x_2, x_3, x_4 are distinct, $x_1^G = \{x_1, x_2, x_3, x_4\}$, and

$$M^\omega \simeq (M(z, \rho), M(x_1, \sigma)).$$

The conditions in (T-G₂) hold.

For an ordering M^ω satisfying (3), the Cartan type and the dimension are given in Table 1, with every symbol computed for that ordering.

In alternative (i), the one-dimensionality of ρ and σ is automatic by Lemma 4.5. In alternative (ii), Lemma 7.3 gives the automatic dimensions recorded before (7.5), $\dim \tau$ is the only non-automatic dimension condition and separates the two (3, 1) dimension branches. In alternative (iii), Lemma 5.3 makes $\dim \rho = 1$ automatic, whereas $\dim \sigma = 1$ remains part of (B₂). In alternative (iv), the finite-dimensional condition forces the one-dimensionality of ρ and σ .

Proof of Theorem 8.1. The equivalence of (1) and (2) is Theorem 2.25.

(1) $\Rightarrow$ (3). Under (8.1), statement (2) and Proposition 2.28 give a tuple

$$N = R_{i_r} \cdots R_{i_1}(M) \in \mathcal{F}_2(M)$$

with $1 \leq a_{12}^{[N]} a_{21}^{[N]} \leq 3$. Hence its rank-two Cartan matrix is indecomposable and of finite type. Choose $\omega_0 \in S_2$ so that

$$-a_{12}^{[N^{\omega_0}]} = 1, \quad 1 \leq -a_{21}^{[N^{\omega_0}]} \leq 3.$$

Thus

$$(\text{ad}(N^{\omega_0})_1)((N^{\omega_0})_2) \neq 0, \quad (\text{ad}(N^{\omega_0})_1)^2((N^{\omega_0})_2) = 0, \quad (\text{ad}(N^{\omega_0})_2)^4((N^{\omega_0})_1) = 0.$$

Equation (2.13) shows that N^{ω_0} is braided-indecomposable. Interchanging the entries does not change their support union, and Lemma 2.29 gives $G_{N^{\omega_0}} = G_N = G_M = G$. Hence Theorem 3.4 shows that the support quandle of N^{ω_0} is one of

$$Z_T^{4,1}, \quad Z_2^{2,2}, \quad Z_3^{3,1}, \quad Z_3^{3,2}, \quad Z_4^{4,2}.$$

The epimorphism supplied by that theorem provides the generators and support representatives used in the corresponding case section.

Choose $\nu \in S_2$ so that $P = N^\nu$ has the ordering used in that section. Repeated use of Lemma 2.26 shows that $\mathcal{B}(P_1 \oplus P_2)$ is finite-dimensional. Apply to P the appropriate one of Theorems 4.1, 7.1, 5.1, and 6.1. Absorb any permitted interchange into ν . Then P satisfies the corresponding alternative in (3).

The definitions give

$$(R_i Q)^\nu \simeq R_{\nu(i)}(Q^\nu) \quad (Q \in \mathcal{F}_2(M), i \in \{1, 2\}),$$

and hence

$$P \simeq R_{\nu(i_r)} \cdots R_{\nu(i_1)}(M^\nu), \quad M^\nu \simeq R_{\nu(i_1)} \cdots R_{\nu(i_r)}(P).$$

Here the first identity is used only when $R_i(Q)$ is defined. Statement (2) and successive applications of Lemma 2.16 show that every reflection in the inverse sequence is defined and give the second isomorphism. The assumptions in (8.1) are unchanged by permuting the entries. Applied successively, Lemma 4.13 in the Γ_2 case and Propositions 5.23, 6.24, and 7.24 in the other three cases, show that every tuple in the inverse sequence, and hence M^ν , satisfies the standing support assumptions of the same case section. Its Nichols algebra is finite-dimensional because permuting the two summands does not change their direct sum. Applying the corresponding case theorem to M^ν proves (3).

(3) $\Rightarrow$ (1). The corresponding case theorem gives $\dim \mathcal{B}((M^\omega)_1 \oplus (M^\omega)_2) < \infty$. Since permuting the two summands does not change their direct sum, (1) follows. Finally, the Cartan-type assertions follow from Propositions 4.15 and 4.24 and Theorems 7.1, 5.1, and 6.1. The dimension formulas follow from Corollaries 4.33, 4.30, and 5.32 and Theorems 7.1 and 6.1. $\square$

A Guide and common notation for the appendices

The appendices contain the longer calculations used in the proofs for the four cases. The table below shows where these calculations can be found. Links to the corresponding GAP programs are given next to the lemmas whose calculations they verify.

Case	Appendix	Calculations
Γ_2	Appendix B	first-adjoint coefficients, higher adjoints, reflection calculations, and diagonal braiding coefficients.
Γ_4	Appendix C	calculations of X_2 , Y_2 , and Y_3 , the simplicity criterion for Y_2 , the formulas used after reflections, and the local Γ_2 parameter calculations used in the sufficiency proof.
T	Appendix D	the braiding coefficients of W , the calculations of Y_2 , Y_3 , and Y_4 , the quaternion cocycle, and the four reflections used to exclude its nontrivial class.
Γ_3	Appendix E	bar-complex calculations proving $\kappa_\Phi(\varepsilon) = 1$ in the $(3, 2)$ case and in the two $(3, 1)$ cases.

Let G be a group and let Φ be a normalized 3-cocycle on G . For $a, b, c, d \in G$, let

$$\mathcal{C}_\Phi(a, b, c, d) := \frac{\Phi(b, c, d)\Phi(a, bc, d)\Phi(a, b, c)}{\Phi(ab, c, d)\Phi(a, b, cd)}. \quad (\text{A.1})$$

Thus the normalized 3-cocycle identity is $\mathcal{C}_\Phi(a, b, c, d) = 1$. We use this notation throughout the appendices.

B Calculations for the Γ_2 case

Retain the notation and assumptions of Section 4. In particular, $hg = \varepsilon gh$, $\varepsilon^2 = 1$, and $\varepsilon \in Z(G)$. Each cocycle quotient below is reduced to a product of powers $\mathcal{C}_\Phi(a_i, b_i, c_i, d_i)^{m_i}$, with $m_i \in \mathbb{Z}$. Long products are recorded in the linked GAP programs. In these reductions, group elements are written in the abstract normal form $\varepsilon^e g^a h^b$, where $e \in \mathbb{Z}/2\mathbb{Z}$ and $a, b \in \mathbb{Z}$.

B.1 Coefficient reduction for the simplicity of X_1

Retain $a = \zeta / \Phi_g(h, \varepsilon)$ and the scalars A_1, B_1 from (4.5) and (4.7).

Lemma B.1. *One has*

$$\frac{a^2 A_1 B_1}{\lambda' \Phi^h(g, h) \Phi^g(g, h)} = \Delta, \quad (\text{B.1})$$

where Δ is defined in (4.2).

Proof. After cancelling the scalars from the one-dimensional projective actions appearing on both sides of (B.1), the only remaining character scalar is ζ^2 . Replacing it by $\zeta^2 = \Phi_g(\varepsilon, \varepsilon)$, the quotient of the left-hand side of (B.1) by Δ becomes an expression involving only values of the cocycle.

The GAP program [GAP source](#) verifies the following factorization:

$$\frac{a^2 A_1 B_1}{\lambda' \Phi^h(g, h) \Phi^g(g, h) \Delta} = \prod_{i=1}^{15} \mathcal{C}_\Phi(a_i, b_i, c_i, d_i)^{m_i},$$

where the tuples (a_i, b_i, c_i, d_i) and the exponents m_i are given by

i	(a_i, b_i, c_i, d_i)	m_i	i	(a_i, b_i, c_i, d_i)	m_i
1	$(\varepsilon g, g, h, \varepsilon)$	1	9	$(g, \varepsilon g, h, \varepsilon)$	-1
2	(ε, g, g, h)	1	10	$(h, \varepsilon, g, \varepsilon h)$	1
3	$(\varepsilon g, \varepsilon, \varepsilon, g)$	1	11	$(g, h, \varepsilon, \varepsilon g)$	-1
4	(ε, h, g, g)	1	12	(h, ε, g, g)	-1
5	$(\varepsilon g, h, \varepsilon g, \varepsilon)$	-1	13	$(\varepsilon, \varepsilon g, \varepsilon, \varepsilon)$	-1
6	$(h, \varepsilon g, \varepsilon, \varepsilon g)$	1	14	(h, ε, h, g)	-1
7	$(g, \varepsilon h, \varepsilon g, \varepsilon)$	1	15	$(\varepsilon, g, \varepsilon h, g)$	-1
8	$(h, \varepsilon, \varepsilon g, \varepsilon)$	1			

Every factor on the right-hand side is equal to 1. Therefore the quotient is 1, proving (B.1). $\square$

B.2 Reduction of Ξ_V

Lemma B.2. For Ξ_V defined in (4.14), one has $\Xi_V = \Phi_g(h, g)^{-1}$.

Proof. Put

$$\Omega_V = \frac{\zeta^2 \Phi(g, \varepsilon g, \varepsilon h) \Phi_g(g, h) \Phi_g(\varepsilon, h)}{\Phi(\varepsilon g, g, \varepsilon h) \Phi^g(g, h) \Phi_g(\varepsilon, hg) \Phi_g(h, \varepsilon)^2}.$$

Apply Lemma 2.1 with $x = g, y = h, z = \varepsilon, w = g$, and then with $x = g, y = \varepsilon, z = h, w = g$. This gives

$$\begin{aligned} \Phi_g(\varepsilon, g) \Phi_g(h, \varepsilon g) &= \Phi_g(h, \varepsilon) \Phi_g(\varepsilon h, g), \\ \Phi_g(h, g) \Phi_g(\varepsilon, hg) &= \Phi_g(\varepsilon, h) \Phi_g(\varepsilon h, g). \end{aligned}$$

Eliminating $\Phi_g(\varepsilon h, g)$, we obtain

$$\Phi_g(\varepsilon, g) \Phi_g(h, \varepsilon g) = \frac{\Phi_g(h, \varepsilon) \Phi_g(h, g) \Phi_g(\varepsilon, hg)}{\Phi_g(\varepsilon, h)}.$$

Substitution in (4.14) gives $\Xi_V = \Omega_V / \Phi_g(h, g)$.

Using $\zeta^2 = \Phi_g(\varepsilon, \varepsilon)$, the GAP program [GAP source](#) expands all occurrences of Φ_x and Φ^x , records the exponent of each formal cocycle value $\Phi(a, b, c)$, and verifies the factorization

$$\Omega_V = \prod_{i=1}^{11} \mathcal{C}_\Phi(a_i, b_i, c_i, d_i)^{m_i},$$

where

i	(a_i, b_i, c_i, d_i)	m_i	i	(a_i, b_i, c_i, d_i)	m_i
1	$(\varepsilon g, \varepsilon, \varepsilon g, \varepsilon h)$	-1	7	$(\varepsilon, \varepsilon g, h, \varepsilon)$	-1
2	$(\varepsilon g, \varepsilon, \varepsilon g, h)$	1	8	$(\varepsilon, h, g, \varepsilon)$	1
3	$(\varepsilon g, \varepsilon, h, \varepsilon g)$	1	9	$(h, g, \varepsilon, \varepsilon)$	1
4	$(\varepsilon g, h, \varepsilon, \varepsilon g)$	-1	10	$(h, \varepsilon, g, \varepsilon)$	-1
5	(g, h, ε, g)	-1	11	$(\varepsilon g h, \varepsilon, \varepsilon, g)$	1
6	$(\varepsilon, \varepsilon g, h, g)$	-1			

Every factor on the right-hand side is 1 by (A.1). Hence $\Omega_V = 1$, and the asserted identity follows. $\square$

B.3 The X_2 -simplicity coefficient Θ_Φ

The coefficient comparison produces the quotient in the following lemma, where $p = \frac{1}{\Phi_g(h,g)}$ and $D_1 = \Phi^h(\varepsilon g, h)\Phi_g(h, h)\mu\lambda'$.

Lemma B.3. *Assume that $\Delta = 1$. Then*

$$\frac{\Phi^h(g, gh)A_1}{p^2\Phi^g(g, h)\Phi^h(\varepsilon g, \varepsilon gh)\Phi_g(h, h)\mu B_1 D_1} = \Theta_\Phi. \quad (\text{B.2})$$

Proof. The GAP program [GAP source](#) constructs the left-hand side of (B.2) and Θ_Φ separately and verifies

$$\frac{1}{\Theta_\Phi} \frac{\Phi^h(g, gh)A_1}{p^2\Phi^g(g, h)\Phi^h(\varepsilon g, \varepsilon gh)\Phi_g(h, h)\mu B_1 D_1} = \Delta^{-1} \frac{\zeta'^2}{\Phi_h(\varepsilon, \varepsilon)}.$$

Since $\varepsilon \triangleright w = \zeta'w$, relation (2.1) gives $\zeta'^2 = \Phi_h(\varepsilon, \varepsilon)$. Together with $\Delta = 1$, the verified identity proves (B.2). $\square$

B.4 Cocycle identities for X_3 and the reflected characters

The formulas for X_3 and for the one-dimensional projective representations of the reflected adjoint objects contain the following cocycle quotients, each of which equals one.

Lemma B.4. *Let d_3 and κ_3 be the scalars in (4.21) and (4.18). Then $d_3/(\lambda^2\kappa_3) = 1$.*

Proof. Substituting (4.21) and (4.18), and cancelling the common factors ζ , λ^2 , and $\Phi_g(\varepsilon, g)$, gives

$$\frac{d_3}{\lambda^2\kappa_3} = \frac{\Phi_g(h, g)\Phi(\varepsilon g, g, \varepsilon gh)\Phi_g(h, \varepsilon g)\Phi^g(g, gh)}{\Phi_g(\varepsilon g, g)\Phi_g(h, \varepsilon g^2)\Phi(g, \varepsilon g, \varepsilon gh)\Phi_g(g, h)}.$$

Apply Lemma 2.1 first with $(x, y, z, w) = (g, h, \varepsilon g, g)$. Since g commutes with εg and $h\varepsilon g = gh$, this gives

$$\Phi_g(\varepsilon g, g)\Phi_g(h, \varepsilon g^2) = \Phi_g(h, \varepsilon g)\Phi_g(gh, g).$$

Applying the same lemma with $(x, y, z, w) = (g, g, h, g)$ gives

$$\Phi_g(h, g)\Phi_g(g, hg) = \Phi_g(g, h)\Phi_g(gh, g).$$

Consequently,

$$\frac{\Phi_g(h, g)\Phi_g(h, \varepsilon g)}{\Phi_g(\varepsilon g, g)\Phi_g(h, \varepsilon g^2)\Phi_g(g, h)} = \frac{1}{\Phi_g(g, hg)}.$$

Since $g(gh)g^{-1} = \varepsilon gh = hg$, $(hg)g(hg)^{-1} = \varepsilon g$, the definitions of Φ^g and Φ_g give $\Phi^g(g, gh) = \Phi(g, hg, g)$ and $\Phi_g(g, hg) = \frac{\Phi(g, hg, g)\Phi(\varepsilon g, g, hg)}{\Phi(g, \varepsilon g, hg)}$. Using again $\varepsilon gh = hg$, we obtain

$$\frac{\Phi(\varepsilon g, g, \varepsilon gh)\Phi^g(g, gh)}{\Phi(g, \varepsilon g, \varepsilon gh)} = \Phi_g(g, hg).$$

Combining the preceding two reductions gives $\frac{d_3}{\lambda^2\kappa_3} = \frac{1}{\Phi_g(g, hg)}\Phi_g(g, hg) = 1$. $\square$

Lemma B.5. *The two cocycle quotients in (4.25) and (4.26) satisfy $K_1 = K_1^{\text{op}} = 1$.*

Proof. Apply Lemma 2.1 with $(x, y, z, w) = (g, h, \varepsilon, g)$. Since ε is central, this gives

$$\Phi_g(\varepsilon, g)\Phi_g(h, \varepsilon g) = \Phi_g(h, \varepsilon)\Phi_g(\varepsilon h, g).$$

Consequently,

$$K_1 = \frac{\Phi^{gh}(g, h)}{\Phi_h(g, h)\Phi_g(\varepsilon h, g)}.$$

The relations $hg = \varepsilon gh$ and $\varepsilon^2 = 1$ imply that conjugation by gh sends g to εg and h to εh . Thus the definitions of Φ^x and Φ_x give

$$\Phi^{gh}(g, h) = \frac{\Phi(gh, g, h)\Phi(\varepsilon g, \varepsilon h, gh)}{\Phi(\varepsilon g, gh, h)}, \quad \Phi_h(g, h) = \Phi(\varepsilon h, g, h), \quad \Phi_g(\varepsilon h, g) = \Phi(\varepsilon g, \varepsilon h, g).$$

Substitution shows that

$$K_1 = \frac{\Phi(gh, g, h)\Phi(\varepsilon g, \varepsilon h, gh)}{\Phi(\varepsilon g, gh, h)\Phi(\varepsilon h, g, h)\Phi(\varepsilon g, \varepsilon h, g)} = \mathcal{C}_\Phi(\varepsilon g, \varepsilon h, g, h)^{-1} = 1.$$

Finally, $gh = \varepsilon hg$. Interchanging g and h in the preceding calculation transforms (4.25) into (4.26). Hence $K_1^{\text{op}} = 1$. $\square$

B.5 Cocycle identities after interchanging V and W

Since $hg = \varepsilon gh$ and $\varepsilon^2 = 1$, one also has $gh = \varepsilon hg$. The preceding coefficient identities therefore apply to (W, V) after interchanging g, h and the corresponding representation parameters. Under this substitution, Δ and $\Theta_\Phi(g, h)$ become Δ' and $\Theta_\Phi(h, g)$, respectively. The GAP program [GAP source](#) also verifies the second-adjoint coefficient reduction for (W, V) by separately encoding the coefficient and $\Theta_\Phi(h, g)$.

B.6 Cocycle identities for the third and fourth adjoint objects

The following identities determine the value of the one-dimensional projective representation of X_3 and the coefficient of X_4 .

Lemma B.6. *The cocycle factor $\mathcal{K}_3$ in (4.37) is equal to one. Moreover, with d_4 and κ_4 defined by (4.35) and (4.33), respectively, one has $d_4 = \lambda^3 \kappa_4$.*

Proof. The GAP program [GAP source](#) reconstructs $\mathcal{K}_3$ from (4.37) and verifies that it is the product of the 13 cocycle relators in the following table.

(a, b, c, d)	m	(a, b, c, d)	m
(h, g, g, g)	-1	$(\varepsilon g, h, \varepsilon g, g)$	1
(h, g, ε, g)	-1	$(\varepsilon g, h, \varepsilon g^2, g)$	1
$(h, g, \varepsilon g, g)$	-1	$(\varepsilon g, g, h, g)$	-1
$(h, g, \varepsilon g^2, g)$	-1	$(\varepsilon g, \varepsilon h, g, h)$	-1
$(g, \varepsilon g, h, g)$	1	$(\varepsilon g, \varepsilon gh, g, gh)$	-1
$(\varepsilon g, h, g, g)$	1	$(\varepsilon g, \varepsilon g^2 h, g, g^2 h)$	-1
$(\varepsilon g, h, \varepsilon, g)$	1		

The second check in the same GAP program [GAP source](#) reconstructs $d_4/(\lambda^3 \kappa_4)$ from (4.35), (4.18), and (4.33). Using $\zeta^2 = \Phi_g(\varepsilon, \varepsilon)$, it verifies that this quotient is the product of the 18 cocycle relators in the following table.

(a, b, c, d)	m	(a, b, c, d)	m
(h, g, g^2, g)	-1	$(\varepsilon g, h, g, \varepsilon g)$	-2
$(g, \varepsilon, g, \varepsilon)$	1	$(\varepsilon g, h, \varepsilon g, g)$	2
$(g, \varepsilon, \varepsilon, g)$	-1	$(\varepsilon g, g, h, g)$	-2
$(g, \varepsilon g, h, g)$	2	$(\varepsilon g, g, \varepsilon, g)$	1
$(g, \varepsilon g, \varepsilon, g)$	-1	$(\varepsilon g, g, \varepsilon g h, \varepsilon g)$	-1
$(g, \varepsilon g, \varepsilon g h, \varepsilon g)$	1	$(\varepsilon g, \varepsilon g h, \varepsilon g, \varepsilon g)$	1
$(g, \varepsilon g h, g, \varepsilon g)$	-1	$(\varepsilon g h, g, \varepsilon g, \varepsilon g)$	-1
$(g, \varepsilon g h, \varepsilon g, g)$	1	$(\varepsilon g h, \varepsilon g, g, \varepsilon g)$	1
$(\varepsilon g, h, g, g^2)$	1	$(\varepsilon g h, \varepsilon g, \varepsilon g, g)$	-1

□

B.7 Determinant and second-reflection identities in type G_2

Let $x, y \in G$ satisfy $yx = \varepsilon xy$, and let $q = x^2 y$. Define the cocycle factor used in the determinant calculation by

$$\begin{aligned}
s_\Phi(x, y) := & \left(\frac{\Phi(x, \varepsilon x, \varepsilon xy) \Phi_x(x, y)}{\Phi_x(y, x) \Phi(\varepsilon x, x, \varepsilon xy) \Phi_x(\varepsilon, x) \Phi_x(y, \varepsilon x) \Phi^x(x, xy)} \right)^2 \\
& \times \frac{\Phi_x(\varepsilon, \varepsilon)^2 \Phi_x(y, \varepsilon) \Phi_y(\varepsilon, x^2)}{\Phi_y(\varepsilon x, x) \Phi_x(y, y)} \\
& \times \frac{\Phi^x(\varepsilon x, \varepsilon q) \Phi^y(\varepsilon x, \varepsilon q) \Phi_x(x, y) \Phi_q(x, x) \Phi_x(y, y) \Phi_q(y, x)}{\Phi^x(x, q) \Phi^y(x, q) \Phi_x(\varepsilon, x) \Phi_x(y, \varepsilon x) \Phi_q(\varepsilon x, y) \Phi_q(x, \varepsilon) \Phi_x(x, x)^2} \\
& \times \Phi^\varepsilon(x, xy) \Phi^\varepsilon(x, y) \Phi^{x^2}(x, xy) \Phi^{x^2}(x, y).
\end{aligned} \tag{B.3}$$

Lemma B.7. Let $x, y \in G$ satisfy $yx = \varepsilon xy$, and assume that $\varepsilon^2 = 1$ and $\varepsilon \in Z(G)$. Then the scalars defined in (B.3) and (4.15) satisfy

$$s_\Phi(x, y) = \Theta_\Phi(x, y)^2.$$

Proof. Substitute the definitions of Φ_z and Φ^z into $s_\Phi(x, y) / \Theta_\Phi(x, y)^2$, and reduce every group element to the form $\varepsilon^e x^a y^b$, using $yx = \varepsilon xy$, $\varepsilon^2 = 1$, and the centrality of ε . The GAP program [GAP source](#) computes the integer exponent vector of the resulting quotient. It also computes the exponent vectors of the 63 listed normalized 3-cocycle identities, with their specified integer powers, and checks coordinate by coordinate that the two vectors coincide. By (A.1), the quotient is 1. Hence $s_\Phi(x, y) = \Theta_\Phi(x, y)^2$. □

A further cocycle calculation gives the value of Θ_Φ after the second reflection.

Lemma B.8. Under the hypotheses of the preceding lemma,

$$\Theta_\Phi(yx, y^{-1}) = \Theta_\Phi(x, y). \tag{B.4}$$

Proof. Since $y^{-1}(yx) = x = \varepsilon(yx)y^{-1}$, the pair (yx, y^{-1}) satisfies the same relation as (x, y) . Substitute the definitions of Φ_z and Φ^z into $\Theta_\Phi(yx, y^{-1}) / \Theta_\Phi(x, y)$, and simplify the group elements using $yx = \varepsilon xy$, $\varepsilon^2 = 1$, and the centrality of ε .

The GAP program [GAP source](#) records the exponent of every value $\Phi(a, b, c)$ both in the remaining quotient and in a product of 94 powers $\mathcal{C}_\Phi(a_i, b_i, c_i, d_i)^{m_i}$, where $m_i \in \mathbb{Z}$, and compares these exponents. It verifies that the quotient is equal to this product. Since every factor $\mathcal{C}_\Phi(a_i, b_i, c_i, d_i)$ is equal to 1 by (A.1), the quotient is 1, proving (B.4). □

B.8 Cocycle identities for the G_2 positive-root factors

The following identities determine the values of the one-dimensional projective representations and the edge labels of the generalized Dynkin diagrams for the simple factors of type G_2 .

Lemma B.9. *The cocycle quotient K_2 in (4.48) is equal to one.*

Proof. Let K_1 be the quotient in (4.25). By Lemma B.5, $K_1 = 1$. Comparing the definitions of K_2 and K_1 , the common factors cancel. Moreover, $g^2h = hg^2$, the conjugation by hg and by g^2h sends g to εg , while conjugation by g^2h sends gh to εgh . Substituting the definitions of Φ_x and Φ^x therefore gives

$$\begin{aligned}\Phi_g(h, g) &= \Phi(\varepsilon g, h, g), & \Phi_g(g, g) &= \Phi(g, g, g), \\ \Phi_g(h, g^2) &= \frac{\Phi(h, g^2, g)\Phi(\varepsilon g, h, g^2)}{\Phi(h, g, g^2)}, & \Phi_{gh}(g, gh) &= \Phi(\varepsilon gh, g, gh), \\ \Phi^{g^2h}(g, gh) &= \frac{\Phi(g^2h, g, gh)\Phi(\varepsilon g, \varepsilon gh, g^2h)}{\Phi(\varepsilon g, g^2h, gh)}.\end{aligned}$$

Consequently,

$$\begin{aligned}K_2 &= K_1 \frac{\Phi_g(h, g)\Phi^{g^2h}(g, gh)}{\Phi_g(h, g^2)\Phi_g(g, g)\Phi_{gh}(g, gh)} \\ &= \frac{\Phi(\varepsilon g, h, g)\Phi(h, g, g^2)}{\Phi(h, g^2, g)\Phi(\varepsilon g, h, g^2)\Phi(g, g, g)} \frac{\Phi(g^2h, g, gh)\Phi(\varepsilon g, \varepsilon gh, g^2h)}{\Phi(\varepsilon g, g^2h, gh)\Phi(\varepsilon gh, g, gh)} \\ &= \mathcal{C}_\Phi(h, g, g, g)^{-1} \mathcal{C}_\Phi(\varepsilon g, h, g, g) \mathcal{C}_\Phi(\varepsilon g, \varepsilon gh, g, gh)^{-1}.\end{aligned}$$

Each factor $\mathcal{C}_\Phi(a, b, c, d)$ is equal to 1, and hence $K_2 = 1$. □

Lemma B.10. *For every pair $x, y \in G$ satisfying $yx = \varepsilon xy$, one has*

$$\Theta_\Phi(x, y) \frac{\Phi_x(\varepsilon, \varepsilon)\Phi_x(x, y)}{\Phi_x(\varepsilon, x)^2\Phi_x(y, \varepsilon x)} = 1. \quad (\text{B.5})$$

Proof. We prove the identity for $(x, y) = (g, h)$. Substitute the definition of Θ_Φ , expand every occurrence of Φ_x and Φ^x , and simplify the group elements using $hg = \varepsilon gh$, $\varepsilon^2 = 1$, and the centrality of ε . The GAP program [GAP source](#) records the exponent of every value $\Phi(a, b, c)$ both in the left-hand side of (B.5) and in the product of the twenty terms $\mathcal{C}_\Phi(a, b, c, d)^m$ listed below, and verifies that these exponents agree. Thus the left-hand side is exactly the following product.

(a, b, c, d)	m	(a, b, c, d)	m
$(\varepsilon, \varepsilon, \varepsilon, h)$	1	$(\varepsilon, \varepsilon, \varepsilon, \varepsilon g)$	-1
(ε, g, g, h)	-1	$(\varepsilon, g, h, \varepsilon)$	1
$(\varepsilon, g, \varepsilon h, g)$	1	$(\varepsilon, h, \varepsilon, \varepsilon g)$	-1
(ε, h, g, g)	-1	$(\varepsilon, \varepsilon h, g, \varepsilon)$	-1
$(g, \varepsilon, \varepsilon, \varepsilon g)$	1	(g, h, ε, gh)	-1
$(g, \varepsilon h, g, h)$	-1	$(h, \varepsilon, \varepsilon, \varepsilon h)$	1
$(h, \varepsilon, g, \varepsilon)$	-1	(h, ε, g, g)	1
(h, ε, g, h)	-1	$(h, \varepsilon g, \varepsilon, gh)$	1
$(\varepsilon g, \varepsilon, \varepsilon g, \varepsilon h)$	1	$(\varepsilon g, h, g, h)$	1
$(\varepsilon h, \varepsilon, \varepsilon g, \varepsilon h)$	-1	$(\varepsilon h, \varepsilon, \varepsilon h, g)$	1

Every term in the table is equal to 1 by the normalized 3-cocycle identity. Hence (B.5) holds for (g, h) . The calculation uses only $hg = \varepsilon gh$, $\varepsilon^2 = 1$, and the centrality of ε . Replacing (g, h) by (x, y) proves the identity whenever $yx = \varepsilon xy$. $\square$

Lemma B.11. *Let $a, b \in G$ satisfy $ba = \varepsilon ab$, where $\varepsilon^2 = 1$ and $\varepsilon \in Z(G)$. Let $\Theta_\Phi(a, b)$ be the scalar obtained from (4.15) after replacing (g, h) by (a, b) . Then*

$$\Theta_\Phi(a, b) = \Theta_\Phi(b, a). \quad (\text{B.6})$$

Proof. Since $ba = \varepsilon ab$ and $\varepsilon^2 = 1$, one also has $ab = \varepsilon ba$. Hence Lemma B.10 applies to both (a, b) and (b, a) , and gives

$$\Theta_\Phi(a, b) = \frac{\Phi_a(\varepsilon, a)^2 \Phi_a(b, \varepsilon a)}{\Phi_a(\varepsilon, \varepsilon) \Phi_a(a, b)}, \quad (\text{B.7})$$

$$\Theta_\Phi(b, a) = \frac{\Phi_b(\varepsilon, b)^2 \Phi_b(a, \varepsilon b)}{\Phi_b(\varepsilon, \varepsilon) \Phi_b(b, a)}. \quad (\text{B.8})$$

Expand the local cocycles in the quotient of (B.7) by (B.8), and simplify the group elements using $ba = \varepsilon ab$. The GAP program [GAP source](#) records the exponent of every value $\Phi(r, s, t)$ and verifies that the resulting quotient is a product of fourteen factors, each an instance of the normalized 3-cocycle identity (A.1) or its inverse. Every such factor is equal to 1. Hence the quotient is 1, proving (B.6). $\square$

C Calculations for the Γ_4 case

Retain the notation of Section 5. Unless a statement says otherwise, ρ and σ are one-dimensional throughout this section. Further assumptions such as $\Delta_4 = 1$, simplicity, or vanishing are stated in the individual lemmas.

Group elements are reduced to the abstract normal form $\varepsilon^i h^j g^k$, where $i \in \mathbb{Z}/4\mathbb{Z}$ and $j, k \in \mathbb{Z}$. These reductions depend only on the defining relations of Γ_4 and therefore hold in every finite quotient considered in Section 5.

C.1 Cocycle reduction for X_2

Lemma C.1. *Recall that $\alpha = \frac{\rho(\varepsilon^{-1})}{\Phi_h(g, \varepsilon^{-1})}$ and*

$$\kappa_X = \frac{\rho(\varepsilon^2)}{\Phi_h(\varepsilon^2, h) \Phi_h(g, \varepsilon^2 h)}.$$

For

$$\Xi_X = \frac{\alpha \Phi_h(h, g) \rho(\varepsilon^{-1} h)}{\rho(h) \Phi^h(h, g) \Phi_h(g, \varepsilon^{-1} h)} \frac{\Phi(h, \varepsilon^{-1} h, \varepsilon g)}{\Phi(\varepsilon^{-1} h, h, \varepsilon g)},$$

one has $\Xi_X = \kappa_X$.

Proof. The projective-action relation gives

$$\begin{aligned} \rho(\varepsilon^{-1})^2 &= \Phi_h(\varepsilon^{-1}, \varepsilon^{-1}) \rho(\varepsilon^2), \\ \rho(\varepsilon^{-1}) \rho(h) &= \Phi_h(\varepsilon^{-1}, h) \rho(\varepsilon^{-1} h). \end{aligned}$$

Substituting these identities into the definitions above gives

$$\frac{\Xi_X}{\kappa_X} = \frac{\Phi_h(\varepsilon^{-1}, \varepsilon^{-1}) \Phi_h(\varepsilon^2, h) \Phi_h(g, \varepsilon^2 h) \Phi(h, \varepsilon^{-1} h, \varepsilon g) \Phi_h(h, g)}{\Phi_h(g, \varepsilon^{-1}) \Phi_h(g, \varepsilon^{-1} h) \Phi_h(\varepsilon^{-1}, h) \Phi(\varepsilon^{-1} h, h, \varepsilon g) \Phi^h(h, g)}.$$

Apply Lemma 2.1 with $(x, y, z, w) = (h, g, \varepsilon^{-1}, h)$, (h, g, ε^2, h) , and $(h, g, \varepsilon^{-1}, \varepsilon^{-1})$. Since $g\varepsilon^{-1} = \varepsilon g$, this gives

$$\begin{aligned}\Phi_h(\varepsilon^{-1}, h)\Phi_h(g, \varepsilon^{-1}h) &= \Phi_h(g, \varepsilon^{-1})\Phi_h(\varepsilon g, h), \\ \Phi_h(\varepsilon^2, h)\Phi_h(g, \varepsilon^2h) &= \Phi_h(g, \varepsilon^2)\Phi_h(\varepsilon^2g, h), \\ \Phi_h(\varepsilon^{-1}, \varepsilon^{-1})\Phi_h(g, \varepsilon^2) &= \Phi_h(g, \varepsilon^{-1})\Phi_h(\varepsilon g, \varepsilon^{-1}).\end{aligned}$$

After these substitutions, the GAP program [GAP source](#) constructs Ξ_X/κ_X from the displayed definitions, applies the two projective-action identities and the three instances of Lemma 2.1 used above, and expands every remaining occurrence of Φ_h and Φ^h . Using only the defining relations of Γ_4 , it verifies that the resulting quotient is

$$\frac{\mathcal{C}_\Phi(h, g, h, \varepsilon^{-1})\mathcal{C}_\Phi(\varepsilon^{-1}h, \varepsilon g, \varepsilon^{-1}, h)\mathcal{C}_\Phi(\varepsilon^{-1}h, h, g, \varepsilon^{-1})}{\mathcal{C}_\Phi(h, g, \varepsilon^{-1}, h)\mathcal{C}_\Phi(\varepsilon^{-1}h, \varepsilon g, h, \varepsilon^{-1})\mathcal{C}_\Phi(h, \varepsilon^{-1}h, g, \varepsilon^{-1})}.$$

Every factor in this quotient is equal to 1 by (A.1). Hence $\Xi_X/\kappa_X = 1$, and therefore $\Xi_X = \kappa_X$. $\square$

C.2 The orbit representatives z_0 and z_1 for Y_2

Lemma C.2. *The coefficient of $w \otimes (w \otimes v_1)$ in z_0 is $-\beta(1 + \sigma(g))$. If $\Delta_4 = 1$ and $\sigma(g) = -1$, then $z_0 = 0$.*

Proof. Take $a = w$ in (5.7). By (5.6), the first term contributes $-\beta$ to the component $w \otimes (w \otimes v_1)$. The definition of φ_1^Φ and (2.1) give $\varphi_1^\Phi(w \otimes v_1) = w \otimes v_1 - \Phi_h(g, g)\rho(g^2)w_1 \otimes v$. Moreover, $g \triangleright w = \sigma(g)w$, and the cocycle quotient in the last term of (5.7) is one. Thus the last term contributes $-\beta\sigma(g)$. The first W -factor in each of the other two terms has degree $\varepsilon^{-1}g$, so neither contributes to this component. Hence its coefficient is $-\beta(1 + \sigma(g))$.

Now assume that $\Delta_4 = 1$ and $\sigma(g) = -1$. The GAP program [GAP source](#) starts from the induced actions on V and W , constructs y_1 using φ_1^Φ , and then constructs $z_0 = \varphi_2^\Phi(w \otimes y_1)$ from the recursive formula. It performs this calculation before imposing $\sigma(g) = -1$. It then collects the four distinct multi-homogeneous components, expands the definitions of Φ^g and Φ_g , and uses the defining relations of Γ_4 , the projective-action relation, and normalized 3-cocycle identities. After imposing $\sigma(g) = -1$, it verifies that two components vanish and that the remaining two have a factor $1 - \Delta_4$. Since $\Delta_4 = 1$, this gives $z_0 = 0$. $\square$

For the remaining orbit representative, the general formula (5.7) gives

$$\begin{aligned}z_1 &= w_2 \otimes y_1 - (gh \triangleright w_2) \otimes ((\varepsilon^2g) \triangleright y_1) \\ &\quad + A_1((\varepsilon^2g) \triangleright w_1) \otimes \varphi_1^\Phi(w_2 \otimes v) \\ &\quad - \beta B_1((\varepsilon^2g) \triangleright w) \otimes \varphi_1^\Phi(w_2 \otimes v_1),\end{aligned}\tag{C.1}$$

where

$$A_1 = \frac{\Phi(\varepsilon^2g, \varepsilon g, h)}{\Phi(\varepsilon^{-1}g, \varepsilon^2g, h)}, \quad B_1 = \frac{\Phi(\varepsilon^2g, g, \varepsilon^{-1}h)}{\Phi(g, \varepsilon^2g, \varepsilon^{-1}h)}.$$

C.3 Coordinates and line-stability parameters for Y_2

For the conjugacy classes of h and g , choose $p_h = p_g = 1$, $p_{\varepsilon^{-1}h} = g$, $p_{\varepsilon g} = h$, $p_{\varepsilon^2g} = \varepsilon$, and $p_{\varepsilon^{-1}g} = \varepsilon h$. For $M(x, \tau)$, set

$$d_x(a; y) = p_{aya^{-1}}^{-1}ap_y, \quad m_{x, \tau}(a; y) = \frac{\Phi_x(a, p_y)}{\Phi_x(p_{aya^{-1}}, d_x(a; y))}\tau(d_x(a; y)).$$

By (2.6), $m_{x,\tau}(a; y)$ is the action coefficient in the standard homogeneous basis. Write $R(a; y) = m_{h,\rho}(a; y)$ and $S(a; x) = m_{g,\sigma}(a; x)$.

Let $v_y \in V_y$ and $w_x \in W_x$ be the corresponding standard basis vectors, chosen so that $v_h = v$, $w_g = w$, $w_{\varepsilon g} = w_1$, and $w_{\varepsilon^2 g} = w_2$. For $0 \leq j \leq 7$, let E_j be the standard tensor with the multidegree assigned to $z_{1,j}$ in Lemma 5.14. Define

$$\begin{aligned} \mathcal{A}_0 &= -\frac{\Phi(\varepsilon^2 g, g, \varepsilon^{-1} h)}{\Phi(g, \varepsilon^2 g, \varepsilon^{-1} h)} R(\varepsilon g; h) S(\varepsilon^2 g; g) S(\varepsilon^{-1} h; \varepsilon g), \\ \mathcal{A}_1 &= -\Phi^{\varepsilon^2 g}(\varepsilon g, h) R(\varepsilon^2 g; h) S(\varepsilon^2 g; \varepsilon g) S(\varepsilon^{-1} h g; \varepsilon^2 g), \\ \mathcal{A}_2 &= -R(\varepsilon g; h) S(\varepsilon^{-1} h; \varepsilon g), \\ \mathcal{A}_3 &= -\frac{\Phi(\varepsilon^2 g, \varepsilon g, h)}{\Phi(\varepsilon^{-1} g, \varepsilon^2 g, h)} R(\varepsilon^2 g; h) S(\varepsilon^2 g; \varepsilon g) S(\varepsilon^{-1} h; \varepsilon^2 g), \\ \mathcal{A}_4 &= \frac{\Phi(\varepsilon^2 g, g, \varepsilon^{-1} h)}{\Phi(g, \varepsilon^2 g, \varepsilon^{-1} h)} R(\varepsilon g; h) R(\varepsilon^2 g; \varepsilon^{-1} h) S(h; \varepsilon^2 g) S(\varepsilon^2 g; g) S(\varepsilon^{-1} h; \varepsilon g), \\ \mathcal{A}_5 &= \Phi^{\varepsilon^2 g}(g, \varepsilon^{-1} h) R(\varepsilon g; h) R(\varepsilon^2 g; \varepsilon^{-1} h) S(\varepsilon^2 g; g) S(\varepsilon^{-1} h; \varepsilon g) S(\varepsilon^{-1} h g; \varepsilon^2 g), \\ \mathcal{A}_6 &= 1, \\ \mathcal{A}_7 &= \frac{\Phi(\varepsilon^2 g, \varepsilon g, h)}{\Phi(\varepsilon^{-1} g, \varepsilon^2 g, h)} S(\varepsilon^2 g; \varepsilon g). \end{aligned} \tag{C.2}$$

If $E_j = w_{x_j} \otimes (w_{y_j} \otimes v_{\ell_j})$, put

$$H_j = \Phi^h(x_j, y_j \ell_j) \Phi^h(y_j, \ell_j) S(h; x_j) S(h; y_j) R(h; \ell_j). \tag{C.3}$$

Lemma C.3. *One has $z_1 = \sum_{j=0}^7 \mathcal{A}_j E_j$, with all $\mathcal{A}_j \neq 0$, and $h \triangleright E_j = H_j E_{\pi(j)}$, where $\pi = (0 \ 1 \ 2 \ 3)(4 \ 5 \ 6 \ 7)$. With the notation of Lemma 5.14, one has*

$$z_{1,j} = \mathcal{A}_j E_j, \quad \lambda_j = \frac{H_j \mathcal{A}_j}{\mathcal{A}_{\pi(j)}}.$$

Consequently,

$$\Theta_{2,j} = \frac{H_j \mathcal{A}_j \mathcal{A}_1}{H_0 \mathcal{A}_0 \mathcal{A}_{\pi(j)}} \quad (1 \leq j \leq 7). \tag{C.4}$$

Proof. For homogeneous tensors, the associator convention gives

$$c_{1,2}^\Phi(w_x \otimes (w_y \otimes v_\ell)) = \frac{\Phi(x, y, \ell)}{\Phi(xy x^{-1}, x, \ell)} (x \triangleright w_y) \otimes (w_x \otimes v_\ell).$$

Substituting the chosen representatives and the coefficients R and S into (5.7), with first W -factor of degree $\varepsilon^2 g$, shows that the four terms in (C.1) contribute, in order, to the pairs (E_6, E_2) , (E_5, E_1) , (E_7, E_3) , and (E_0, E_4) . Expanding the corresponding action coefficients gives exactly (C.2). Every $\mathcal{A}_j$ is nonzero because it is a product or quotient of nonzero cocycle values and action coefficients.

The defining relations of Γ_4 show that conjugation by h permutes the eight displayed multidegrees according to $\pi = (0 \ 1 \ 2 \ 3)(4 \ 5 \ 6 \ 7)$. Applying (2.2) twice to each E_j gives $h \triangleright E_j = H_j E_{\pi(j)}$, with H_j as in (C.3). Since $z_{1,j} = \mathcal{A}_j E_j$, it follows that $\lambda_j = H_j \mathcal{A}_j / \mathcal{A}_{\pi(j)}$. Now $\pi(0) = 1$, so (5.9) gives (C.4).

The same GAP program [GAP source](#) independently reconstructs z_1 from the induced actions and the recursive maps. It confirms that z_1 has eight nonzero coordinates forming the two stated h -orbits, and computes the seven quotients in (C.4). It does not separately compare the eight displayed formulas for $\mathcal{A}_j$ and H_j . Those formulas follow from the substitutions above. $\square$

C.4 Vanishing of Y_3

Lemma C.4. *If $\sigma(g) = -1$, then*

$$-\frac{\sigma(\varepsilon^2 g)}{\sigma(\varepsilon^2 g^2)} \frac{\Phi(h, g, \varepsilon^2 g) \Phi(h, \varepsilon^2 g^2, g) \Phi(\varepsilon g, h, \varepsilon^2 g^2)}{\Phi(h, g, \varepsilon^2 g^2) \Phi(h, \varepsilon^2 g, g) \Phi(\varepsilon g, h, \varepsilon^2 g) \Phi(\varepsilon g, \varepsilon^2 h g, g)} = 1.$$

Proof. Since $\sigma(g) = -1$, projective multiplication gives $-\sigma(\varepsilon^2 g)/\sigma(\varepsilon^2 g^2) = \Phi_g(\varepsilon^2 g, g)$. Since $\varepsilon^2 g$ commutes with g , the definition of Φ_g gives $\Phi_g(\varepsilon^2 g, g) = \Phi(g, \varepsilon^2 g, g)$. Expanding the definitions and using $g\varepsilon^2 = \varepsilon^2 g$, $(\varepsilon g)h = hg$, and $h\varepsilon^2 = \varepsilon^2 h$, the expression in the statement is

$$\frac{\mathcal{C}_\Phi(h, g, \varepsilon^2 g, g)}{\mathcal{C}_\Phi(\varepsilon g, h, \varepsilon^2 g, g)}.$$

The GAP program [GAP source](#) applies the same projective multiplication relation and verifies this reduction exactly. Both factors are one by the normalized 3-cocycle identity. $\square$

Lemma C.5. *Assume that $\Delta_4 = 1$ and that Y_2 is simple. Then the vector $\varphi_3^\Phi(w \otimes z_1)$ is zero.*

Proof. Set $y_3 = \varphi_3^\Phi(w \otimes z_1)$, and let

$$\begin{aligned} p_0 &= \varphi_1^\Phi(w_2 \otimes v), & p_1 &= \varphi_1^\Phi(w_2 \otimes v_1), \\ q_1 &= (\varepsilon^2 g) \triangleright w_1, & r_1 &= (\varepsilon^2 g) \triangleright w. \end{aligned}$$

Thus (C.1) becomes

$$z_1 = w_2 \otimes y_1 - ((gh) \triangleright w_2) \otimes ((\varepsilon^2 g) \triangleright y_1) + A_1 q_1 \otimes p_0 - \beta B_1 r_1 \otimes p_1.$$

Lemma 5.15 gives $\sigma(g) = -1$, while Lemma C.2 gives $z_0 = 0$.

In the last summand of $w \otimes z_1$, the two W -factors interchanged by $c_{1,2}^\Phi$ both have degree g . The associator factor is therefore one, because $x = y = g$ in $\Phi(x, y, \ell)/\Phi(xy x^{-1}, x, \ell)$. The recursive formula for φ_3^Φ gives

$$\begin{aligned} y_3 &= w \otimes z_1 - (\varepsilon^2 h g^2 \triangleright w) \otimes (g \triangleright z_1) \\ &\quad + A_1 \frac{\Phi(g, \varepsilon^{-1} g, \varepsilon h g)}{\Phi(\varepsilon g, g, \varepsilon h g)} (g \triangleright q_1) \otimes \varphi_2^\Phi(w \otimes p_0) - \beta B_1 (g \triangleright r_1) \otimes \varphi_2^\Phi(w \otimes p_1). \end{aligned} \quad (\text{C.5})$$

Indeed, the term involving $\varphi_2^\Phi(w \otimes y_1)$ is z_0 . Moreover, $(\varepsilon^2 g) \triangleright w$ is a nonzero multiple of w . The G -linearity of φ_2^Φ therefore shows that $\varphi_2^\Phi(w \otimes ((\varepsilon^2 g) \triangleright y_1))$ is a multiple of $(\varepsilon^2 g) \triangleright z_0$, and is also zero.

The vectors p_0 and p_1 are homogeneous of degrees $\varepsilon h g$ and $\varepsilon^2 h g$, respectively. Both are nonzero, since the two terms defining each have distinct first W -degrees. The relations of Γ_4 give

$$\begin{aligned} \varepsilon g(\varepsilon^2 g) &= g, & \varepsilon g(hg) &= \varepsilon h g, \\ \varepsilon(\varepsilon^2 g) &= g, & \varepsilon(hg) &= \varepsilon^2 h g. \end{aligned}$$

By Lemma 5.9, the homogeneous components of Y_1 are one-dimensional. The preceding identities therefore give $(\varepsilon g) \triangleright y_1 \in \mathbb{C}^\times p_0$ and $\varepsilon \triangleright y_1 \in \mathbb{C}^\times p_1$, while both $(\varepsilon g) \triangleright w_2$ and $\varepsilon \triangleright w_2$ belong to $\mathbb{C}^\times w$. Applying the tensor-product action and the G -linearity of φ_2^Φ to $0 \neq z_1 = \varphi_2^\Phi(w_2 \otimes y_1)$ gives

$$\varphi_2^\Phi(w \otimes p_0) \in \mathbb{C}^\times (\varepsilon g \triangleright z_1) = \mathbb{C}^\times (g \triangleright z_1), \quad \varphi_2^\Phi(w \otimes p_1) \in \mathbb{C}^\times (\varepsilon \triangleright z_1) = \mathbb{C}^\times z_1.$$

The last equalities follow from the one-dimensionality of the homogeneous components of Y_2 , by Lemma 5.13. The action on the one-dimensional components of W is likewise invertible. Hence there are nonzero scalars μ_i, ν_i , where $i \in \{0, 1\}$, such that

$$\varphi_2^\Phi(w \otimes p_0) = \mu_0(g \triangleright z_1), \quad \varphi_2^\Phi(w \otimes p_1) = \mu_1 z_1,$$

and $g \triangleright q_1 = \nu_0(\varepsilon^2 h g^2 \triangleright w)$, $g \triangleright r_1 = \nu_1 w$. Substitution in (C.5) gives

$$\begin{aligned} y_3 &= (1 - \beta B_1 \nu_1 \mu_1) w \otimes z_1 + (-1 + \Xi_3)(\varepsilon^2 h g^2 \triangleright w) \otimes (g \triangleright z_1), \\ \Xi_3 &= A_1 \frac{\Phi(g, \varepsilon^{-1} g, \varepsilon h g)}{\Phi(\varepsilon g, g, \varepsilon h g)} \nu_0 \mu_0. \end{aligned} \quad (\text{C.6})$$

The GAP program [GAP source](#) reconstructs the induced actions and the recursive maps. With the one-dimensionality of the homogeneous components established above as a mathematical input, it verifies that $\beta B_1 \nu_1 \mu_1 = -\sigma(g)$ before imposing $\sigma(g) = -1$. For the second coefficient, after imposing $\sigma(g) = -1$ and applying the projective multiplication relation for σ , it identifies Ξ_3 with the left-hand side of Lemma C.4. The reduction of y_3 to the two coefficients in (C.6) is supplied by the preceding argument.

Since $\sigma(g) = -1$, $\beta B_1 \nu_1 \mu_1 = 1$, while Lemma C.4 gives $\Xi_3 = 1$. Thus both coefficients in (C.6) vanish, and hence $y_3 = 0$. $\square$

C.5 The action of $\varepsilon h g^2$ on Y_2

Lemma C.6. Assume that ρ and σ are one-dimensional, $\Delta_4 = 1$, $\rho(h) = \sigma(g) = -1$, and that $\mathbb{C}z_1$ is $C_G(\varepsilon h g^2)$ -stable. Then

$$(\varepsilon h g^2) \triangleright (z_{1,3} + z_{1,7}) = -(z_{1,2} + z_{1,6}).$$

Proof. For this calculation, write

$$\begin{aligned} p_0 &= \varphi_1^\Phi(w_2 \otimes v), & p_1 &= \varphi_1^\Phi(w_2 \otimes v_1), \\ q_1 &= (\varepsilon^2 g) \triangleright w_1, & r_0 &= (gh) \triangleright w_2, & r_1 &= (\varepsilon^2 g) \triangleright w, \end{aligned}$$

and

$$C_0 = -\beta B_1 r_1 \otimes p_1, \quad C_1 = -r_0 \otimes ((\varepsilon^2 g) \triangleright y_1), \quad C_2 = w_2 \otimes y_1, \quad C_3 = A_1 q_1 \otimes p_0.$$

Equation (C.1) and the ordered multidegrees in Lemma 5.14 give $z_1 = C_0 + C_1 + C_2 + C_3$ and $C_i = z_{1,i} + z_{1,i+4}$ for $0 \leq i < 4$. Define $K, \lambda_1, \lambda_2 \in \mathbb{C}^\times$ by

$$(\varepsilon h g^2) \triangleright C_3 = K C_2, \quad h \triangleright C_1 = \lambda_1 C_2, \quad h \triangleright C_2 = \lambda_2 C_3,$$

and set

$$\begin{aligned} L_a &= \frac{\Phi_{\varepsilon g}(\varepsilon h g^2, \varepsilon^2 g) \Phi_g((\varepsilon h g^2) \varepsilon^2 g, h) \sigma(\varepsilon h^2 g^3)}{\Phi_g(\varepsilon, \varepsilon h^2 g^3)}, & L_p &= \Phi^{\varepsilon h g^2}(\varepsilon^2 g, h) \frac{\Phi_g(\varepsilon h g^2, \varepsilon) \sigma(\varepsilon^2 g^2) \rho(\varepsilon h g^2)}{\Phi_g(h, \varepsilon^2 g^2)}, \\ L_{ha} &= \frac{\Phi_g(h, \varepsilon) \sigma(g^{-1})}{\Phi_g(\varepsilon^2 g h, g^{-1}) \Phi_g(\varepsilon^2 g, h)}, & L_{hy} &= \Phi^h(\varepsilon g, h) \frac{\Phi_g(h, h) \sigma(\varepsilon^{-1} h^2) \rho(h)}{\Phi_g(\varepsilon, \varepsilon^{-1} h^2)}, \\ L_{hr} &= \frac{\Phi_{\varepsilon^2 g}(h, gh) \Phi_g(hgh, \varepsilon) \sigma(\varepsilon h^2 g)}{\Phi_g(\varepsilon, \varepsilon h^2 g)}, \\ L_{hY} &= \beta \Phi^{\varepsilon^2 g}(g, \varepsilon^{-1} h) \Phi_h(\varepsilon^2 g, g) \rho(\varepsilon^2 g^2) \\ &\quad \cdot \Phi^h(g, h) \sigma(\varepsilon^2 g) \rho(h). \end{aligned}$$

We use $\varepsilon^{-1}(\varepsilon h g^2)(\varepsilon^2 g)h = \varepsilon h^2 g^3$. We also use $\varepsilon^{-1}h(g h)\varepsilon = \varepsilon h^2 g$. The projective-action relation and (2.2) then give

$$\begin{aligned} (\varepsilon h g^2) \triangleright q_1 &= L_a w_2, & (\varepsilon h g^2) \triangleright p_0 &= L_p y_1, \\ h \triangleright w_2 &= L_{ha} q_1, & h \triangleright y_1 &= L_{hy} p_0, \\ h \triangleright r_0 &= L_{hr} w_2, & h \triangleright ((\varepsilon^2 g) \triangleright y_1) &= -L_{hY} y_1. \end{aligned}$$

It follows that

$$\begin{aligned} K &= A_1 \Phi^{\varepsilon h g^2}(\varepsilon^{-1} g, \varepsilon h g) L_a L_p, \\ \lambda_1 &= \Phi^h(\varepsilon g, \varepsilon^{-1} h g) L_{hr} L_{hY}, \\ \lambda_2 &= A_1^{-1} \Phi^h(\varepsilon^2 g, h g) L_{ha} L_{hy}. \end{aligned}$$

Thus, after cancelling A_1 ,

$$\mathcal{K}_{Y_2} = \frac{\Phi^{\varepsilon h g^2}(\varepsilon^{-1} g, \varepsilon h g) \Phi^h(\varepsilon^2 g, h g) L_a L_p L_{ha} L_{hy}}{\rho(h) \sigma(g)^2 \Phi^h(\varepsilon g, \varepsilon^{-1} h g) L_{hr} L_{hY}} = \frac{K \lambda_2}{\rho(h) \sigma(g)^2 \lambda_1}. \quad (\text{C.7})$$

Starting from the displayed expressions for $L_a, L_p, L_{ha}, L_{hy}, L_{hr}, L_{hY}$, the GAP program [GAP source](#) forms the quotient on the right-hand side of (C.7). It expands the local cocycles and conjugation factors, applies the one-dimensional projective multiplication relations, and uses $\Delta_4 = 1$, $\rho(h) = \sigma(g) = -1$, and $\varepsilon^4 = 1$. The program verifies that the resulting expression is an integral product of normalized 3-cocycle identities. Hence $\mathcal{K}_{Y_2} = 1$. Since $\mathbb{C}z_1$ is $C_G(\varepsilon h g^2)$ -stable, comparison of the components in $h \triangleright z_1$ gives $\lambda_1 = \lambda_2$. Hence $K = \rho(h) \sigma(g)^2 = -1$. Since $C_3 = z_{1,3} + z_{1,7}$ and $C_2 = z_{1,2} + z_{1,6}$, the assertion follows. $\square$

C.6 The seven Y_2 -simplicity parameters after the first reflection

This subsection proves the seven parameter identities used in Proposition 5.23. The simplicity of the reflected second adjoint object then follows from Lemma 5.15.

Lemma C.7. *Assume (B₂). For $R_1(V, W) = (V^*, X_1)$, take $(\varepsilon_1, h_1, g_1) = (\varepsilon^{-1}, h^{-1}, h g)$. Write $V^* \simeq M(h_1, \rho_1)$ and $X_1 \simeq M(g_1, \sigma_1)$, and define $\Delta_4^{(1)}$ and $\Theta_{2,j}^{(1)}$ for this tuple by (5.3) and Lemma 5.14, respectively. Assume that ρ_1 and σ_1 are one-dimensional, $\Delta_4^{(1)} = 1$, and $\rho_1(h_1) = \sigma_1(g_1) = -1$. Then*

$$\Theta_{2,j}^{(1)} = 1 \quad (1 \leq j \leq 7).$$

Proof. The GAP program [GAP source](#) reconstructs the induced actions and the recursive maps for the initial and reflected tuples. It uses the stability and one-dimensionality of $\mathbb{C}x_1$, established in Lemma 5.7, when it reads the action on this line from one nonzero coordinate. It then reconstructs the eight nonzero coordinates and the seven quotients defining $\Theta_{2,j}^{(1)}$. After using $\rho(h) = \sigma(g) = -1$, $\Delta_4 = \Delta_4^{(1)} = 1$, $\rho_1(h_1) = \sigma_1(g_1) = -1$, and $\Theta_{2,j} = 1$ for $1 \leq j \leq 7$, the program expresses each resulting quotient as an integral product of normalized 3-cocycle identities. Hence $\Theta_{2,j}^{(1)} = 1$ for $1 \leq j \leq 7$. $\square$

C.7 The local Γ_2 parameters

With the notation of Lemma 5.28, the local parameter is

$$\Delta_{01}^R = \chi_R(\varepsilon^2) \psi_R(\varepsilon^2) \chi_R(x_R^2) \psi_R(x_R^2) \frac{\Phi_{b_R}(\varepsilon^2 x_R, x_R) \Phi_{x_R}(b_R, b_R)}{\Phi_{x_R}(b_R, \varepsilon^2) \Phi_{b_R}(\varepsilon^2, x_R^2)}. \quad (\text{C.8})$$

Lemma C.8. Assume that ρ and σ are one-dimensional, that $\Delta_4 = 1$, that Y_2 is simple, and that $Y_3 = 0$. Then

$$\Delta_{01}^W = 1.$$

Proof. Put $x_i = \varepsilon^i g$, with indices in $\mathbb{Z}/4\mathbb{Z}$. Retain $w_0 = w, w_1, w_2$ from above and choose $0 \neq w_3 \in W_{x_3}$. For $a \in G$, let $S(a; x_i)$ be the coefficient of $w_{ax_i a^{-1}}$ in $a \triangleright w_i$.

Lemma 5.11 gives $z_1 \neq 0$. Since Y_2 is simple, it follows that $Y_2 = \mathbb{C}G \triangleright z_1$. Lemma 5.15 gives $\sigma(g) = -1$ and $\dim Y_2 = 2$. The vectors z_1 and $g \triangleright z_1$ have the two distinct degrees in $\text{supp } Y_2$, and therefore form a homogeneous basis of Y_2 .

The parameter in (C.8) can be written as

$$\Delta_{01}^W = S(g; x_1)S(x_3; x_0)S(x_1; x_2)S(x_2; x_3). \quad (\text{C.9})$$

Indeed, the projective-action relation gives

$$\begin{aligned} S(x_3; x_0) &= \frac{\chi_W(\varepsilon^2)S(x_1; x_0)}{\Phi_g(x_1, \varepsilon^2)}, \\ S(x_1; x_2)S(x_1; x_0) &= \Phi_g(x_1, x_1)\chi_W(g^2), \\ S(x_2; x_3)S(g; x_1) &= \frac{\Phi_{x_1}(\varepsilon^2 g, g)\psi_W(\varepsilon^2)\psi_W(g^2)}{\Phi_{x_1}(\varepsilon^2, g^2)}. \end{aligned}$$

Multiplying these identities and cancelling $S(x_1; x_0)$ identifies the product in (C.9) with Δ_{01}^W in (C.8).

The GAP program [GAP source](#) reconstructs $y_1, z_1, g \triangleright z_1$ and the three recursive maps from the group law, the tensor actions, and the braidings, using the associator convention fixed in Lemma C.3. Before imposing $Y_3 = 0$, it obtains 192 nonzero coordinate expressions for $\varphi_3^\Phi(w_i \otimes z_1)$ and $\varphi_3^\Phi(w_i \otimes (g \triangleright z_1))$, and checks nineteen selected expressions for the two orders of the local pair. Since $Y_3 = 0$, the quotient of the two nonzero terms in each selected coordinate is -1 . The program expresses the inverse of the product in (C.9) as an integral product of these quotients and verifies that the total exponent is zero. Hence $\Delta_{01}^W = 1$. $\square$

D Calculations for the T case

Retain the notation and assumptions of Section 6. All tensor products are right-associated. For the coordinate calculations in this appendix only, whenever $\dim \rho = \dim \sigma = 1$, let $p = \rho(x_1)\sigma(z)$. The maps $\varphi_m^{Y, \Phi}$ are the original recursive adjoint maps fixed in Subsection 6.3, with V and W interchanged. By Lemma 2.8, we identify their restricted images with the objects Y_m . For the higher-adjoint calculations, we also assume the last equality in (T-G₂) and use the bases fixed before (6.15). Under this condition, $\sigma(x_1) = -1$ and $\varepsilon_\Phi(W) = 1$ give (6.13).

We first give the coefficient tables for the higher adjoint objects. The last two subsections specify the quaternion cocycle and the four-reflection calculation used to exclude its nontrivial class. The higher-adjoint tables use the cochain comparison of Lemma 6.3. The nontrivial-class calculation retains the cocycle in (6.17).

D.1 The coefficient tables for Y_2 and Y_3

Assume throughout this subsection that $\dim \rho = \dim \sigma = 1$, $\sigma(x_1) = -1$, $\varepsilon_\Phi(W) = 1$, and $p^2 - p + 1 = 0$, together with the last equality in (T-G₂). Write $w_i = e_i^W$ for the chosen nonzero vector in W_{x_i} , and use the vectors $e_{i_1 \dots i_m}$ and the recursion (6.15) fixed in Subsection 6.3.

Lemma D.1. *The object Y_2 has basis u_1, u_2, u_3, u_4 , where*

$$\begin{aligned} u_1 &= e_{23} - pe_{34} + (p-1)e_{42}, \\ u_2 &= e_{14} - pe_{43} + (p-1)e_{31}, \\ u_3 &= e_{12} - pe_{24} + (p-1)e_{41}, \\ u_4 &= e_{13} - pe_{32} + (p-1)e_{21}. \end{aligned} \tag{D.1}$$

Proof. Use the identification $Y_1 = W \otimes V$ from the proof of Lemma 6.11. The sixteen vectors e_{ij} then form a basis of $W \otimes Y_1$.

The derivation of (6.15) from the categorical maps was proved in Subsection 6.3. The GAP program [GAP source](#) applies this recursion to all sixteen vectors e_{ij} , using $p^2 = p - 1$. It verifies that the four diagonal vectors have zero image, that every other image is a scalar multiple of one of the vectors in (D.1), and that each u_i occurs as an image. Thus the image of $\phi_2^{Y,\Phi}$ is spanned by u_1, u_2, u_3, u_4 .

This image realizes Y_2 by Lemma 2.8. The four nonzero vectors u_1, u_2, u_3, u_4 involve disjoint sets of ordered basis tensors, so they are linearly independent. They therefore form a basis of Y_2 . $\square$

Lemma D.2. *For the vector y defined in (6.16), one has $Y_3 = \mathbb{k}y \neq 0$.*

Proof. By Lemma D.1, the sixteen vectors $w_i \otimes u_j$ form a basis of $W \otimes Y_2$. The GAP program [GAP source](#) substitutes (D.1) into (6.15) and evaluates $\phi_3^{Y,\Phi}(w_i \otimes u_j)$ for all i, j . It verifies that the image is zero when $i \neq j$, while the images of $w_i \otimes u_i$, for $i = 1, 2, 3, 4$, are respectively $py, py, -y, (1-p)y$.

The ordered tensors in (6.16) are distinct, and the coefficient of e_{123} is one, so $y \neq 0$. The preceding calculation shows that the image of $\phi_3^{Y,\Phi}$ is $\mathbb{k}y$. The assertion follows from Lemma 2.8. $\square$

D.2 The coefficient factorization for Y_4

Lemma D.3. *Under the hypotheses of the preceding subsection, $\phi_4^{Y,\Phi}(w_1 \otimes y) = 0$.*

Proof. The GAP program [GAP source](#) applies (6.15) to $w_1 \otimes y$ over $\mathbb{Z}[p]$, before using $p^2 - p + 1 = 0$, and verifies that the coefficient of every ordered tensor is a multiple of $p^2 - p + 1$. Hence $\phi_4^{Y,\Phi}(w_1 \otimes y) = 0$. $\square$

D.3 A cocycle on the quaternion group

Write an element of Q_8 as $(-1)^s i^a j^b$, with $s, a, b \in \mathbb{F}_2$. In these coordinates the multiplication and the automorphism cycling i, j, k are

$$(s, a, b)(t, c, d) = (s + t + ac + bd + bc, a + c, b + d), \quad A(s, a, b) = (s, b, a + b).$$

The following tables specify $F : Q_8^3 \rightarrow \mathbb{Z}/8$ for first argument $i, -1, -i$. The rows and columns give the second and third arguments, in the order $(1, i, j, k, -1, -i, -j, -k)$. Set $F(1, v, w) = 0$ and extend by $F(Au, Av, Aw) = F(u, v, w)$. The table with first argument -1 is invariant under simultaneous application of A to the row and column indices, so these rules define F on all of Q_8^3 .

$$\begin{aligned}
u = i : & \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 7 & 6 & 0 & 1 & 3 & 2 \\ 0 & 3 & 0 & 3 & 0 & 7 & 4 & 3 \\ 0 & 6 & 3 & 3 & 0 & 2 & 3 & 7 \\ 0 & 0 & 2 & 2 & 4 & 4 & 6 & 6 \\ 0 & 5 & 5 & 4 & 0 & 5 & 1 & 0 \\ 0 & 7 & 6 & 1 & 0 & 3 & 2 & 1 \\ 0 & 2 & 5 & 1 & 4 & 2 & 1 & 1 \end{pmatrix} & u = -1 : & \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 4 & 6 & 4 & 6 & 0 & 2 \\ 0 & 6 & 2 & 4 & 4 & 2 & 6 & 0 \\ 0 & 4 & 6 & 2 & 4 & 0 & 2 & 6 \\ 0 & 4 & 4 & 4 & 4 & 0 & 0 & 0 \\ 0 & 6 & 4 & 6 & 0 & 6 & 4 & 6 \\ 0 & 6 & 6 & 4 & 0 & 6 & 6 & 4 \\ 0 & 4 & 6 & 6 & 0 & 4 & 6 & 6 \end{pmatrix} \\
u = -i : & \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 7 & 4 & 0 & 3 & 3 & 0 \\ 0 & 1 & 6 & 7 & 0 & 5 & 2 & 7 \\ 0 & 6 & 1 & 1 & 4 & 6 & 5 & 1 \\ 0 & 4 & 2 & 2 & 0 & 4 & 2 & 2 \\ 0 & 3 & 1 & 6 & 4 & 7 & 1 & 6 \\ 0 & 5 & 4 & 1 & 4 & 5 & 4 & 5 \\ 0 & 2 & 7 & 7 & 4 & 2 & 3 & 7 \end{pmatrix}
\end{aligned}$$

Lemma D.4. *The function F is a normalized A -invariant additive three-cocycle, and*

$$F(i, j, k) = F(j, k, i) = F(k, i, j) = 3.$$

If $u = (-1)^s i^{a(u)} j^{b(u)}$, then

$$F(u, v, w) \equiv \sum_{r=0}^2 a(A^r u) a(A^r v) b(A^r w) \pmod{2}.$$

Proof. Normalization follows from the zero rows and columns, and A -invariance holds by construction. Substitution of the multiplication rule into the values determined by these tables gives

$$F(v, w, t) - F(uv, w, t) + F(u, vw, t) - F(u, v, wt) + F(u, v, w) = 0 \pmod{8}$$

for all $u, v, w, t \in Q_8$. The GAP program [GAP source](#) checks all 8^4 identities from the complete table in `certificates/T/q8_primitive8_cocycle_data.g`. It also checks normalization, A -invariance, the three stated values, and the parity identity for all 8^3 triples. Finally, the same program substitutes these entries into (2.6) over the normal form of T and verifies the three coefficients used in Lemma 6.4. $\square$

D.4 Four reflections for the nontrivial cocycle

Retain the group, cocycle, and pair in (6.17) and (6.18). The support elements are $x_1 = -t$, $x_2 = kt$, $x_3 = jt$, and $x_4 = it$, so $x_1 x_2 x_3 = 1$. Assume that the reflections R_2, R_1, R_2, R_1 are defined, and let (V_r, W_r) , $0 \leq r \leq 4$, be the resulting pairs.

Lemma D.5. *The dimensions, supports, and central actions of these pairs are given by the following table. The support columns denote sets. The ordering of their elements is immaterial.*

r	$\dim V_r$	$\text{supp } V_r$	$z _{V_r}$	$\text{supp } W_r$	$z _{W_r}$
0	1	$\{z\}$	p^{-1}	$\{x_i\}$	p
1	2	$\{z\}$	p^2	$\{x_i^{-1}\}$	p^{-1}
2	2	$\{z^{-1}\}$	p^{-2}	$\{z^2 x_i^{-1}\}$	-1
3	1	$\{z^{-1}\}$	p	$\{z^{-2} x_i\}$	-1
4	1	$\{z\}$	p^{-1}	$\{z^3 x_i\}$	$-p$

(D.2)

Each W_r has dimension four and diagonal self-braiding scalar -1 . For $0 \leq r \leq 3$, the successive adjoint dimensions, ending at the first zero object, are

r	$\dim(\mathrm{ad} V_r)^m(W_r), m = 1, 2, \dots$	$\dim(\mathrm{ad} W_r)^m(V_r), m = 1, 2, \dots$	
0	4, 0	4, 4, 2, 0	. (D.3)
1	4, 4, 0	4, 4, 1, 0	
2	4, 4, 0	4, 4, 1, 0	
3	4, 0	4, 4, 2, 0	

After rescaling the homogeneous basis vectors of W_4 , its H -action agrees with that of W_0 , and the H -action on V_4 is trivial.

Proof. In the induced basis corresponding to $1, i, k, j$, the coefficients of the braiding on W_0 are

$$(\beta_{ij}) = \begin{pmatrix} -1 & -1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix}.$$

The mixed braidings are $c_{V_0, W_0}(v \otimes w) = p w \otimes v$ and $c_{W_0, V_0}(w \otimes v) = v \otimes w$. We use the recursive images in Lemma 2.8. On a right-associated homogeneous word, crossing two adjacent factors of degrees a, b , followed by a tail of degree c , multiplies the corresponding two-factor braiding coefficient by

$$\frac{\Phi_*(a, b, c)}{\Phi_*(aba^{-1}, a, c)}.$$

The tensor actions are given by (2.2), and the dual actions are determined by the invariance of evaluation, with the same tensor-action factors. At each step the recursive map is restricted to the tensor product with its preceding image. The recursion gives (D.2) and (D.3).

The GAP program [GAP source](#) performs these calculations over $\mathbb{Q}[p]/(p^2 - p + 1)$. It constructs every restricted image and its induced action and computes the dimensions by exact elimination, including the stated zero images. It also checks the projective action identities, the central sixth powers, the mixed braid relations, and the invariant evaluation pairings. Thus the calculation retains all the Φ_* -factors in the preceding formulas and applies to both roots of $p^2 - p + 1$.

For the final pair, the same calculation identifies the H -action on W_4 with that on W_0 after a diagonal change of basis, for every element of H , and gives the trivial H -action on V_4 . $\square$

E Calculations for the Γ_3 case

Section 7 states the structural conclusions. This appendix gives the coefficient calculations, bar-chain reductions, and GAP verifications used in their proofs.

All bar-chain identities below are computed in the normalized integral bar complex of Γ_3 . Because they use only the normal form $\varepsilon^i g^m h^n$, where $i \in \mathbb{Z}/3\mathbb{Z}$ and $m, n \in \mathbb{Z}$, they descend to every finite non-abelian quotient of Γ_3 . Throughout this appendix, we continue to write Φ for its pullback $\pi^*\Phi$ to Γ_3 .

E.1 Normalized bar-complex setup

Define the normalized bar three-chains $L_x(a, b) = [a|b|x] + [{}^{ab}x|a|b] - [a|{}^b x|b]$, $U_a(x, y) = [a|x|y] + [{}^a x|{}^a y|a] - [{}^a x|a|y]$, and $\mathcal{R}(a; x, y) = [a|x|y] - [{}^a x|a|y]$. Their evaluations by a multiplicative cocycle are, respectively, $\Phi_x(a, b)$, $\Phi^a(x, y)$, and

$$R(a; x, y) = \frac{\Phi(a, x, y)}{\Phi({}^a x, a, y)}.$$

Suppose that a normalized three-chain C satisfies $\partial C = 0$ and $\omega_* C = \partial D$ for a normalized four-chain D over S_3 . The injectivity in Lemma 7.6 implies that C is a boundary over Γ_3 , so $\langle \Phi, C \rangle = 1$. This also applies when $\omega_* C = 0$, by taking $D = 0$.

E.2 The (3, 2) obstruction

Retain $V, W, v_0, w_0, v_i, w_1, E_0, E_1, u_0$ from Lemma 7.7. With indices in $\mathbb{Z}/3\mathbb{Z}$, set

$$p_i = g\varepsilon^i, \quad q_i = p_i h, \quad y_0 = \varepsilon h, \quad y_1 = \varepsilon^2 h.$$

Thus $v_i \in V_{p_i}, w_j \in W_{y_j}$. If ξ_x, η_y, ζ_z are homogeneous, the definition of $c_{1,2}^\Phi$ and the associativity constraint give

$$c_{1,2}^\Phi(\xi_x \otimes (\eta_y \otimes \zeta_z)) = R(x; y, z)(x \triangleright \eta_y) \otimes (\xi_x \otimes \zeta_z). \quad (\text{E.1})$$

Retain $a, b_0, \Delta_1, \chi, s$ from (7.16)–(7.19). The calculation in Lemma 7.7 gives $u_0 := \varphi_1^\Phi(E_0) = E_0 - aE_1$, together with $g \triangleright E_0 = b_0 E_1$. Assume henceforth that $\Delta_1 = 1$, and let $u_i = \varepsilon^i \triangleright u_0$. Then $(X_1)_{q_i} = \mathbb{C}u_i$. Set

$$\begin{aligned} \ell_0 &= \Phi^\varepsilon(p_2, y_0) \Phi_g(\varepsilon, \varepsilon^2) \sigma(\varepsilon), & k_2 &= \frac{\Phi_g(g, \varepsilon^2)}{\Phi_g(\varepsilon, g)} \rho(g), \\ \ell_1 &= \Phi^{\varepsilon^2}(p_1, y_1) \Phi_g(\varepsilon^2, \varepsilon) \frac{\Phi_{\varepsilon h}(\varepsilon^2, g)}{\Phi_{\varepsilon h}(g, \varepsilon)} \sigma(\varepsilon), & k_1 &= \frac{\Phi_g(g, \varepsilon)}{\Phi_g(\varepsilon^2, g)} \rho(g), \end{aligned}$$

and define

$$\begin{aligned} 1 - \chi \rho(gh) &= 1 + s, \\ c_1 &= R(g; p_2, y_0) \frac{k_2}{\ell_0}, \\ c_2 &= R(g; p_1, y_1) \frac{k_1}{\ell_1}. \end{aligned}$$

Lemma E.1. *Let $P_i = v_i \otimes u_i$. These tensors have total degree $g^2 h$, and*

$$\varphi_2^\Phi(P_0) = (1 + s)P_0 + c_1 P_1 + c_2 P_2, \quad (\text{E.2})$$

where $c_1 c_2 \neq 0$. Moreover, $v_2 \otimes u_0$ and $v_0 \otimes u_1$ have total degree $\varepsilon g^2 h$, and

$$\varphi_2^\Phi(v_2 \otimes u_0) - (1 + \rho(g))(v_2 \otimes u_0) \in \mathbb{C}(v_0 \otimes u_1). \quad (\text{E.3})$$

Proof. Since $p_i^2 = g^2$, we have $p_i q_i = g^2 h$ for every i . Moreover, $p_2 q_0 = p_0 q_1 = \varepsilon g^2 h$. This proves the assertions about the homogeneous degrees.

Under the assumption $\Delta_1 = 1$, Lemma 7.7 gives $g \triangleright u_0 = \chi u_0$. The G -linearity of φ_1^Φ , together with $\varepsilon \triangleright E_0 = \ell_0(v_0 \otimes w_0)$, $g \triangleright E_0 = b_0 E_1$, and $\varepsilon^2 \triangleright E_1 = \ell_1(v_0 \otimes w_1)$, gives $\varphi_1^\Phi(v_0 \otimes w_0) = \ell_0^{-1} u_1$, $\varphi_1^\Phi(E_1) = -a^{-1} u_0$, and $\varphi_1^\Phi(v_0 \otimes w_1) = -(a \ell_1)^{-1} u_2$.

Apply the defining three-term recursion for φ_2^Φ , using $u_0 = E_0 - aE_1$ and (E.1). The GAP program [GAP source](#) uses (2.1) and (2.2) to reconstruct the coefficients and checks that, on P_0 , the identity term, the double-braiding term, and the last summand are respectively P_0 , $-\chi \rho(gh)P_0$, and $c_1 P_1 + c_2 P_2$. For the three coefficient comparisons that do not cancel directly, it checks the corresponding explicit four-chain boundaries in the normalized bar complex. Since $s = -\chi \rho(gh)$, this proves (E.2). All factors in the definitions of c_1 and c_2 are nonzero, so $c_1 c_2 \neq 0$.

For $v_2 \otimes u_0$, the relations $p_2 q_0 = q_1$ and $q_1 p_2 = p_0$ show that the double-braiding term belongs to $\mathbb{C}(v_0 \otimes u_1)$. Likewise, $p_2 p_1 = p_0$ and $\varphi_1^\Phi(v_2 \otimes w_1) \in (X_1)_{q_1} = \mathbb{C}u_1$ show that the E_1 -term in the last summand belongs to the same line. The same GAP program [GAP source](#) checks these targets and verifies $R(p_2; p_2, y_0) = 1$ and $p_2 \triangleright v_2 = \rho(g)v_2$. Thus the identity term and the E_0 -term give the $v_2 \otimes u_0$ -coefficient $1 + \rho(g)$, proving (E.3). $\square$

Equation (2.2) gives $\varepsilon \triangleright P_0 = t_0 P_1$, $\varepsilon \triangleright P_1 = t_1 P_2$, and $\varepsilon \triangleright P_2 = t_2 P_0$, where

$$t_0 = \Phi^\varepsilon(p_0, q_0), \quad (\text{E.4})$$

$$t_1 = \Phi^\varepsilon(p_1, q_1) \Phi_g(\varepsilon, \varepsilon) \Phi_{gh}(\varepsilon, \varepsilon), \quad (\text{E.5})$$

$$t_2 = \Phi^\varepsilon(p_2, q_2) \Phi_g(\varepsilon, \varepsilon^2) \Phi_{gh}(\varepsilon, \varepsilon^2). \quad (\text{E.6})$$

Indeed, the formula for t_0 follows from $v_1 = \varepsilon \triangleright v_0$ and $u_1 = \varepsilon \triangleright u_0$. For the other two, (2.1) gives $\varepsilon \triangleright v_1 = \Phi_g(\varepsilon, \varepsilon) v_2$, $\varepsilon \triangleright u_1 = \Phi_{gh}(\varepsilon, \varepsilon) u_2$, $\varepsilon \triangleright v_2 = \Phi_g(\varepsilon, \varepsilon^2) v_0$, and $\varepsilon \triangleright u_2 = \Phi_{gh}(\varepsilon, \varepsilon^2) u_0$. Substitution in (2.2) gives t_1 and t_2 , all three scalars are nonzero.

Lemma E.2. Assume $\Delta_1 = 1$ and $\rho(g) = -1$, and retain s, λ, r from (7.19) and (7.21). Then the identities in (7.22) hold.

Proof. Define the following normalized bar three-chains:

$$\begin{aligned} A &= L_g(\varepsilon^2 h, \varepsilon^2) - L_g(\varepsilon, h) - L_{y_0}(g, \varepsilon^2), \\ B_0 &= U_g(p_2, y_0) + L_g(g, \varepsilon^2) - L_g(\varepsilon, g), \\ S &= B_0 - A - L_g(g, h) + L_{y_0}(\varepsilon, \varepsilon), \\ E &= L_{y_0}(\varepsilon, \varepsilon) + L_{y_0}(\varepsilon^2, \varepsilon), \\ C_1 &= \mathcal{R}(g; p_2, y_0) + L_g(g, \varepsilon^2) - L_g(\varepsilon, g) - U_\varepsilon(p_2, y_0) - L_g(\varepsilon, \varepsilon^2), \\ C_2 &= \mathcal{R}(g; p_1, y_1) + L_g(g, \varepsilon) - L_g(\varepsilon^2, g) - U_{\varepsilon^2}(p_1, y_1) - L_g(\varepsilon^2, \varepsilon) \\ &\quad - L_{y_0}(\varepsilon^2, g) + L_{y_0}(g, \varepsilon), \\ T_0 &= U_\varepsilon(p_0, q_0), \\ T_1 &= U_\varepsilon(p_1, q_1) + L_g(\varepsilon, \varepsilon) + L_{gh}(\varepsilon, \varepsilon), \\ T_2 &= U_\varepsilon(p_2, q_2) + L_g(\varepsilon, \varepsilon^2) + L_{gh}(\varepsilon, \varepsilon^2). \end{aligned}$$

Using the evaluations of $L, U, \mathcal{R}$ recorded above, (2.1), and $\rho(g) = -1$, the GAP program [GAP source](#) expands the formulas above for $a, b_0, \ell_i, k_i, c_i, t_i$ into values of Φ and verifies

$$\begin{aligned} s &= \sigma(\varepsilon)^{-2} \langle \Phi, S \rangle, & \sigma(\varepsilon)^3 &= \langle \Phi, E \rangle, \\ c_i &= -\sigma(\varepsilon)^{-1} \langle \Phi, C_i \rangle \quad (i = 1, 2), & t_i &= \langle \Phi, T_i \rangle \quad (i = 0, 1, 2). \end{aligned} \quad (\text{E.7})$$

Let K be the cycle in (7.15), and set

$$\begin{aligned} C_s &= 3S - 2E - K, \\ C_\lambda &= 3(C_1 + T_1 - C_2) - (T_0 + T_1 + T_2), \\ C_r &= 4C_1 + 2T_1 - 2C_2 - 2T_0 - S. \end{aligned}$$

The same GAP program [GAP source](#) checks coefficientwise that these three chains have zero normalized bar differential and that their images under $\omega : \Gamma_3 \rightarrow S_3$ are the boundaries of the explicit normalized four-chains recorded in the program.

By the argument in Subsection E.1, all three chains are boundaries over Γ_3 , and their evaluations by Φ are 1. By (E.7), (7.21), and $\langle \Phi, K \rangle = \kappa_\Phi(\varepsilon)$, these evaluations are respectively $s^3/\kappa_\Phi(\varepsilon)$, $\lambda^3/(t_0 t_1 t_2)$, and r^2/s . This proves (7.22). $\square$

E.3 The (3, 1) obstruction with $\dim \tau = 2$

Retain a_ε from (7.24), $b, \zeta_3, w, \bar{w}$ from Lemma 7.9, and v, x_1 from Lemma 7.10. Thus $x_1 = v \otimes (w - \rho(h)\bar{w})$. For $i \in \mathbb{Z}/3\mathbb{Z}$, put $v_i = \varepsilon^i \triangleright v$ and $x_{1,i} = \varepsilon^i \triangleright x_1$, and let $e_0 = v_2 \otimes x_{1,0}$, $e_1 = v_0 \otimes x_{1,1}$, and $e_2 = v_1 \otimes x_{1,2}$.

Lemma E.3. Assume that $\Delta_3 = I$ and $\rho(g) = -1$. The vectors e_0, e_1, e_2 lie in $(V \otimes X_1)_{\varepsilon g^2 h}$ and are linearly independent. There are nonzero scalars $\mathfrak{a}, \mathfrak{b}, m_0, m_1, m_2$ such that

$$x_2 := \varphi_2^\Phi(e_0) = e_0 + \mathfrak{a}e_1 + \mathfrak{b}e_2 \neq 0 \quad (\text{E.8})$$

and

$$\varepsilon \triangleright e_0 = m_0(\mathfrak{a}e_1), \quad \varepsilon \triangleright (\mathfrak{a}e_1) = m_1(\mathfrak{b}e_2), \quad \varepsilon \triangleright (\mathfrak{b}e_2) = m_2e_0. \quad (\text{E.9})$$

Proof. We first prove the cocycle identity

$$\frac{\Phi_g(h, \varepsilon^2)}{\Phi_g(\varepsilon^2, h)} = a_\varepsilon \Phi_h(\varepsilon, g). \quad (\text{E.10})$$

Let $R = L_g(h, \varepsilon^2) - L_g(\varepsilon^2, h) - 2L_h(\varepsilon, g) + L_h(g, \varepsilon^2) + L_h(\varepsilon, \varepsilon)$. The normalized bar differential of R and its image under the bar-chain map induced by ω are computed in [GAP source](#). The program verifies coefficientwise that R is a cycle and that its image is zero. It also expands the local cocycles from their definitions and verifies symbolically that

$$\langle \Phi, R \rangle = \frac{\Phi_g(h, \varepsilon^2)}{a_\varepsilon \Phi_g(\varepsilon^2, h) \Phi_h(\varepsilon, g)}.$$

The argument in Subsection E.1 gives $\langle \Phi, R \rangle = 1$, proving (E.10).

Set $C_0 = \Phi_h(\varepsilon, \varepsilon) a_\varepsilon^2 / \Phi^{\varepsilon^2}(g, h)$. We have $h \triangleright v_2 = \rho(h) \frac{\Phi_g(h, \varepsilon^2)}{\Phi_g(\varepsilon^2, h)} v_2$. Moreover, $x_{1,2} = \Phi^{\varepsilon^2}(g, h) v_2 \otimes (\varepsilon^2 \triangleright (w - \rho(h)\overline{w}))$, where

$$\varepsilon^2 \triangleright (w - \rho(h)\overline{w}) = \frac{a_\varepsilon^{-2}}{\Phi_h(\varepsilon, \varepsilon)} (\zeta_3^2 w - \rho(h) \zeta_3 \overline{w}).$$

Using the action table (7.25), (E.10), $\rho(h)^2 b = 1$, and $1 + \zeta_3 = -\zeta_3^2$ gives

$$\begin{aligned} \varphi_1^\Phi(v_2 \otimes w) &= \zeta_3 C_0 x_{1,2}, \\ \varphi_1^\Phi(v_2 \otimes \overline{w}) &= -\zeta_3^2 \rho(h)^{-1} C_0 x_{1,2}, \\ \varphi_1^\Phi(v_2 \otimes (w - \rho(h)\overline{w})) &= -C_0 x_{1,2}. \end{aligned} \quad (\text{E.11})$$

Using (E.10), $\rho(h)^2 b = 1$, and the action table, both sides are expanded in [GAP source](#), which verifies the three identities in (E.11) symbolically.

Define

$$\mathfrak{a} = \frac{\Phi_g(\varepsilon(g h), \varepsilon^2)}{\Phi_{gh}(\varepsilon, g)}, \quad (\text{E.12})$$

$$\mathfrak{b} = \frac{\Phi(\varepsilon g, g, h) \Phi_h(\varepsilon, \varepsilon) a_\varepsilon^2}{\Phi(\varepsilon^2 g, \varepsilon g, h) \Phi_g(\varepsilon, g) \Phi^{\varepsilon^2}(g, h)}. \quad (\text{E.13})$$

These scalars are nonzero. The first tensor factors of e_0, e_1, e_2 have degrees $g\varepsilon^2, g, g\varepsilon$, respectively, while their second tensor factors have degrees $gh, g\varepsilon h, g\varepsilon^2 h$, respectively. It follows that all three tensors have degree $\varepsilon g^2 h$. Their first tensor factors have distinct degrees, so they are linearly independent.

The identity summand in $\varphi_2^\Phi(e_0)$ is e_0 . Equation (2.1) gives

$$\begin{aligned} g \triangleright x_1 &= -\Phi^g(g, h) \rho(g) \rho(h)^{-1} x_1, \\ (\varepsilon g) \triangleright x_1 &= -\frac{\Phi^g(g, h) \rho(g)}{\rho(h) \Phi_{gh}(\varepsilon, g)} x_{1,1}, \\ \varepsilon(g h) \triangleright v_2 &= \Phi_g(\varepsilon(g h), \varepsilon^2) \rho(g h) v. \end{aligned}$$

Since g and h commute, $\Phi^g(g, h) = \Phi_g(g, h)$ and $\rho(gh) = \rho(g)\rho(h)/\Phi_g(g, h)$. Consequently, $-c_{X_1, V}c_{V, X_1}(e_0) = \mathfrak{a}e_1$.

For the last recursive summand, Equations (2.1), (E.1), and (E.11) give

$$\begin{aligned}(\varepsilon g) \triangleright v &= \rho(g)\Phi_g(\varepsilon, g)^{-1}v_1, \\ (\text{id}_V \otimes \varphi_1^\Phi)c_{1,2}^\Phi(e_0) &= \mathfrak{b}e_2.\end{aligned}$$

Starting from (2.1), (2.2), the braiding formula, (E.1), and (4.10), [GAP source](#) expands the coefficients of the three summands of $\varphi_2^\Phi(e_0)$ and verifies that they are 1, $\mathfrak{a}$, $\mathfrak{b}$. It also computes the degrees of e_0, e_1, e_2 and verifies that all three are $\varepsilon g^2 h$. This proves (E.8). Its right-hand side is nonzero by the linear independence already proved.

Equations (2.2) and (2.1) give

$$\begin{aligned}\varepsilon \triangleright e_0 &= \eta_0 e_1, & \eta_0 &= \Phi^\varepsilon(\varepsilon g, gh)\Phi_g(\varepsilon, \varepsilon^2), \\ \varepsilon \triangleright e_1 &= \eta_1 e_2, & \eta_1 &= \Phi^\varepsilon(g, {}^\varepsilon(gh))\Phi_{gh}(\varepsilon, \varepsilon), \\ \varepsilon \triangleright e_2 &= \eta_2 e_0, & \eta_2 &= \Phi^\varepsilon(\varepsilon^2 g, {}^{\varepsilon^2}(gh))\Phi_g(\varepsilon, \varepsilon)\Phi_{gh}(\varepsilon, \varepsilon^2).\end{aligned}\tag{E.14}$$

For example, $\varepsilon \triangleright v_2 = \Phi_g(\varepsilon, \varepsilon^2)v$ and $\varepsilon \triangleright x_{1,0} = x_{1,1}$, which give the first line. The other two follow in the same way from the definitions of v_i and $x_{1,i}$. Starting from these definitions, (2.1), and (2.2), [GAP source](#) computes the action of ε on e_0, e_1, e_2 and verifies both the three target vectors and the three coefficients in (E.14).

Finally, set $m_0 = \eta_0/\mathfrak{a}$, $m_1 = \mathfrak{a}\eta_1/\mathfrak{b}$, and $m_2 = \mathfrak{b}\eta_2$. These scalars are nonzero, and (E.9) follows directly from (E.14). $\square$

For use here and in the next subsection, define the following chains in the normalized bar complex. Let

$$E = L_h(\varepsilon, g) - L_h(g, \varepsilon^2) - L_h(\varepsilon, \varepsilon),$$

and set

$$\begin{aligned}A &= L_g({}^\varepsilon(gh), \varepsilon^2) - L_{gh}(\varepsilon, g), \\ B &= [\varepsilon g|g|h] - [\varepsilon^2 g|\varepsilon g|h] + 2E + L_h(\varepsilon, \varepsilon) - L_g(\varepsilon, g) - U_{\varepsilon^2}(g, h), \\ T_0 &= U_\varepsilon(\varepsilon g, gh) + L_g(\varepsilon, \varepsilon^2), \\ T_1 &= U_\varepsilon(g, {}^\varepsilon(gh)) + L_{gh}(\varepsilon, \varepsilon), \\ T_2 &= U_\varepsilon(\varepsilon^2 g, {}^{\varepsilon^2}(gh)) + L_g(\varepsilon, \varepsilon) + L_{gh}(\varepsilon, \varepsilon^2).\end{aligned}$$

Set

$$M_0 = T_0 - A, \quad M_1 = A + T_1 - B, \quad M_2 = B + T_2.\tag{E.15}$$

Retain K from (7.15), set $\Theta = L_{\varepsilon g^2 h}(\varepsilon, \varepsilon) + L_{\varepsilon g^2 h}(\varepsilon^2, \varepsilon)$, and define

$$C_{02} = M_0 - M_2 - K, \quad C_{\text{pow}} = 3M_0 - \Theta + K.\tag{E.16}$$

The definitions of L_x, U_x , together with (7.24), (E.12), (E.13), and (E.14), give

$$\langle \Phi, E \rangle = a_\varepsilon, \quad \langle \Phi, A \rangle = \mathfrak{a}, \quad \langle \Phi, B \rangle = \mathfrak{b}, \quad \langle \Phi, T_i \rangle = \eta_i \quad (0 \leq i \leq 2).\tag{E.17}$$

These identities depend only on Φ and the indicated group elements, and apply to both values of $\dim \tau$. Moreover, $\langle \Phi, \Theta \rangle = \Phi_{\varepsilon g^2 h}(\varepsilon, \varepsilon)\Phi_{\varepsilon g^2 h}(\varepsilon^2, \varepsilon)$ and $\langle \Phi, K \rangle = \kappa_\Phi(\varepsilon)$. The GAP program [GAP source](#) expands Φ_x and Φ^x from their definitions and verifies these evaluations symbolically.

Lemma E.4. *The chains C_{02} and C_{pow} in (E.16) are boundaries in the normalized bar complex of Γ_3 . If $\Delta_3 = I$, $\rho(g) = -1$, and m_0, m_2 are the scalars in Lemma E.3, then*

$$m_2 = m_0 \kappa_\Phi(\varepsilon)^{-1}.\tag{E.18}$$

Proof. Applying the normalized bar differential gives

$$\partial M_i = [\varepsilon | \varepsilon g^2 h] - [\varepsilon g^2 h | \varepsilon], \quad \partial \Theta = 3([\varepsilon | \varepsilon g^2 h] - [\varepsilon g^2 h | \varepsilon]).$$

Since K is a cycle, C_{02} and C_{pow} are cycles. These formulas are verified coefficientwise in [GAP source](#).

The same program contains two integral four-chains in the normalized bar complex of S_3 and verifies coefficientwise that their boundaries are, respectively, the images of C_{02} and C_{pow} under the bar-chain map induced by ω . The argument in Subsection E.1 therefore proves the first assertion.

Now assume that $\Delta_3 = I$ and $\rho(g) = -1$. Equations (E.17) and (E.15), together with the definitions of m_0, m_2 , give $\langle \Phi, M_0 \rangle = \eta_0 / \mathfrak{a} = m_0$ and $\langle \Phi, M_2 \rangle = \mathfrak{b} \eta_2 = m_2$. Evaluating the boundary C_{02} yields $m_0 / (m_2 \kappa_\Phi(\varepsilon)) = 1$, proving (E.18). $\square$

E.4 The $(3, 1)$ obstruction with $\dim \tau = 1$

For the remainder of this subsection, retain V, W, v, w and ρ_1 from Lemma 7.12, assume $\rho(h)\tau(g) \neq 1$, and retain a_ε from (7.24). Applying (2.1) to (ε, g) , $(\varepsilon, \varepsilon)$, and (g, ε^2) gives $(a_\varepsilon \tau(\varepsilon))\tau(g) = \tau(g)(a_\varepsilon \tau(\varepsilon))^2$. Since these scalars are nonzero, cancellation yields

$$a_\varepsilon \tau(\varepsilon) = 1. \quad (\text{E.19})$$

Equation (2.1) also gives

$$\tau(\varepsilon)^2 = \Phi_h(\varepsilon, \varepsilon)\tau(\varepsilon^2), \quad \frac{1}{\tau(\varepsilon^2)} = \Phi_h(\varepsilon, \varepsilon)a_\varepsilon^2. \quad (\text{E.20})$$

For $i \in \mathbb{Z}/3\mathbb{Z}$, put $v_i = \varepsilon^i \triangleright v$ and $x_{1,i} = \varepsilon^i \triangleright (v \otimes w)$, and let $e_0 = v_2 \otimes x_{1,0}$, $e_1 = v_0 \otimes x_{1,1}$, and $e_2 = v_1 \otimes x_{1,2}$.

Lemma E.5. *The vectors e_0, e_1, e_2 lie in the component of degree $\varepsilon g^2 h$ and are linearly independent. There are nonzero scalars $\mathfrak{a}, \mathfrak{b}, \eta_0, \eta_1, \eta_2$ such that*

$$x_2 := \varphi_2^\Phi(e_0) = e_0 - \rho(g)^2 \rho(h) \tau(g) \mathfrak{a} e_1 + \rho(g) (1 - \rho(h) \tau(g)) \mathfrak{b} e_2. \quad (\text{E.21})$$

Moreover, $\varepsilon \triangleright e_0 = \eta_0 e_1$, $\varepsilon \triangleright e_1 = \eta_1 e_2$, and $\varepsilon \triangleright e_2 = \eta_2 e_0$.

Proof. By (7.1) and the last relation in (2.1), the first tensor factors of e_0, e_1, e_2 have degrees $g\varepsilon^2, g, g\varepsilon$, respectively, whereas their second tensor factors have degrees $gh, \varepsilon^2 gh, \varepsilon gh$, respectively. Thus all three tensors have degree $\varepsilon g^2 h$. Equation (2.1) shows that their factors are nonzero. Their first tensor factors have distinct degrees, so the tensors are linearly independent.

Take $\mathfrak{a}, \mathfrak{b}$ from (E.12) and (E.13). These scalars are nonzero, and (E.20) identifies the factor $\Phi_h(\varepsilon, \varepsilon)a_\varepsilon^2$ in $\mathfrak{b}$ with $\tau(\varepsilon^2)^{-1}$. Starting from the definition of ρ_1 in Lemma 7.12, (2.1), (2.2), the braiding formula, (E.1), the G -linearity of φ_1^Φ , and $\varphi_1^\Phi(v \otimes w) = (1 - \rho(h)\tau(g))v \otimes w$, the GAP program [GAP source](#) expands every occurrence of Φ_x and Φ^x from their definitions and verifies, using (E.19) and (E.20) only as rewrite rules, that

$$c_{X_1, V^C V, X_1}(e_0) = \rho(g)^2 \rho(h) \tau(g) \mathfrak{a} e_1, \quad (\text{E.22})$$

$$(\text{id}_V \otimes \varphi_1^\Phi) c_{1,2}^\Phi(e_0) = \rho(g) (1 - \rho(h) \tau(g)) \mathfrak{b} e_2. \quad (\text{E.23})$$

Substitution of (E.22) and (E.23) into (4.10) proves (E.21). The assumption $\rho(h)\tau(g) \neq 1$ shows that both nonidentity coefficients are nonzero.

Finally, let η_0, η_1, η_2 be as in (E.14). Their calculation uses only the degrees and definitions of $v_i, x_{1,i}$, and e_i , together with (2.1) and (2.2). It therefore gives the asserted cyclic action in this case as well. The coefficients are nonzero. The GAP program [GAP source](#) also reconstructs the three target tensors and their coefficients from these definitions. $\square$

Lemma E.6. For the chains M_i in (E.15), let $\mu_i = \langle \Phi, M_i \rangle$. Then μ_0, μ_1, μ_2 are nonzero, and the scalars m_i in (7.34) satisfy

$$\begin{aligned} m_0 &= -\frac{\mu_0}{\rho(g)^2 \rho(h) \tau(g)}, \\ m_1 &= -\frac{\rho(g) \rho(h) \tau(g)}{1 - \rho(h) \tau(g)} \mu_1, \\ m_2 &= \rho(g) (1 - \rho(h) \tau(g)) \mu_2. \end{aligned} \tag{E.24}$$

Proof. Equations (E.17) and (E.15) give

$$\mu_0 = \frac{\eta_0}{a}, \quad \mu_1 = \frac{a\eta_1}{b}, \quad \mu_2 = b\eta_2.$$

All three scalars are nonzero. The cyclic action and (7.34) give $m_0 = \eta_0/A$, $m_1 = A\eta_1/B$, and $m_2 = B\eta_2$. Substituting $A = -\rho(g)^2 \rho(h) \tau(g) a$ and $B = \rho(g) (1 - \rho(h) \tau(g)) b$ gives (E.24). The GAP program [GAP source](#) expands the local cocycles from their definitions and verifies the chain evaluations and these three identities, using (E.19) and (E.20) only as rewrite rules. $\square$

Data and code availability

The GAP programs supporting the symbolic and finite-group calculations are linked at their first use. The complete supplementary repository is available at [commit 835a6ad3388e](#). Each certificate directory contains a README file, archived output, and a checksum manifest. These computations were performed with GAP 4.16.0.

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